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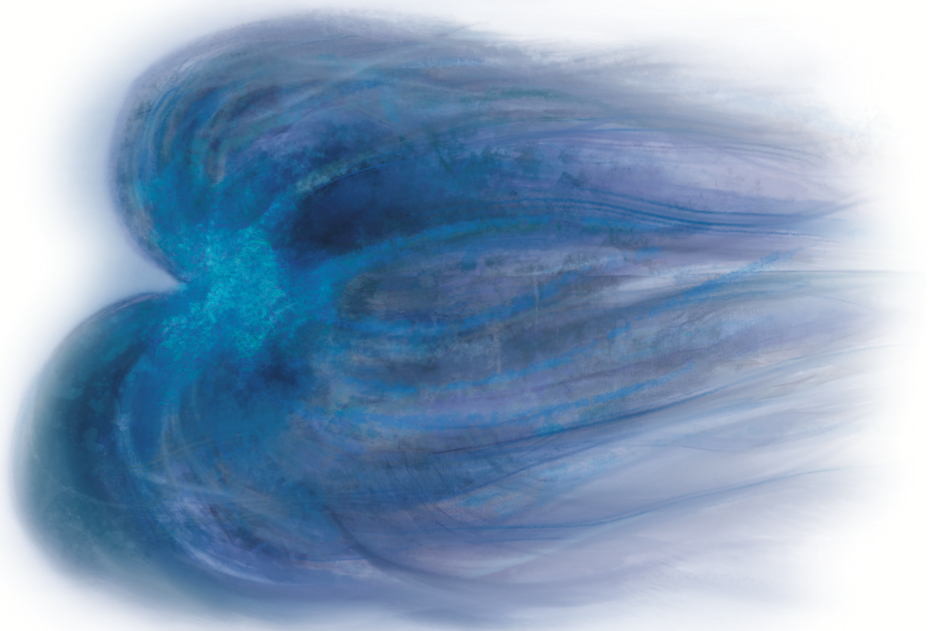
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LOCAL EQUILIBRIUM APPROACH TO QUANTUM TRANSPORT IN NORMAL FLUIDS

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*En mémoire d'Abuela,
Pour m'avoir aimé comme une mère le ferait,
Pour m'avoir montré la voie de l'amour,
De la sagesse et de la bonté,
Pour être partie bien trop tôt.*

PREFACE

Writing a master's thesis is an achievement but also a beginning. The achievement is that of five years of study. The beginning is that of a scientific life and career. Finalising a master's thesis is also an opportunity to take a step back from these past years and to appreciate the personal and intellectual accomplishments.

Studying physics is a beautiful and fascinating thing. It is about understanding the world around us, to try to fathom Nature at its most fundamental. It is a journey in all directions, a walk among the smallest and most invisible constituents of matter, a stroll among the largest and most unattainable objects of our universe.

Studying physics is also about dreaming, being immersed in space and time, contemplating the stars, colours, and particles in their never-ending, ceaseless and poetic ballet. Imagination becomes a precious tool helping us to tame abstraction. Did not the French philosopher Gaston Bachelard say that “l'imagination qui veut saisir tout le réel et qui trouve plus de réalité à ce qui se cache qu'à ce qui se montre”[†]? In the world of physics, the freedom to think, discuss or build is great. No one could be indifferent to it.

Finally, studying physics is also a lesson about life and humility. In a world and a society as dynamic as ours, in perpetual innovation and growth, it seems that it is becoming increasingly important to rethink the position and condition of human beings. They are only stardust and, as such, are subjected to the same physical laws as any other matter or living creatures inhabiting this universe. Physics thus reminds us that Nature exists without humans.

About five years ago, my primary motivation for studying physics was to understand the fundamental workings and mechanisms of Nature. I would not presume to say that this endeavour is completed. Nor, really, that it has begun. Learning and knowing also mean ignoring and knowing how to ignore. What could I say after these five years? That Nature is complex, mysterious, and enthralling. That the road to knowledge is long, strewn with pitfalls, but exciting. I must confess to a strange feeling of knowing less than I did five years ago. And yet, it can hardly be so. My way of reasoning, questioning, thinking, and approaching the world has profoundly changed. I cannot help but think back to those years, when I was knee-high to a grasshopper, dreaming of understanding Nature, the universe, and the hidden mysteries of the microscopic world. Today, I cannot help but think of those years ahead, which I hope will make some of those dreams come true.

I now invite the reader to discover my work, to be guided through the world of the smallest and imperceptible things, and to rise little by little to the surface of the visible world. I hope the journey will be worthwhile!

[†]Suggested translation: “the imagination that wants to grasp all reality and finds more reality in what is hidden than in what is shown.” From G. Bachelard. *La terre et les rêveries du repos*. Douzième réimpression. Paris: Librairie José Corti, 1948.

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My special thanks to Ludmila Karadjova for the cover illustration. I am very grateful that you have accepted my project to think about art in physics. I hope I have not been too insistent and demanding. I also thank Neige Lebel for the graphic editing work.

Finally, I thank Professors Glenn Barnich and Nathan Goldman for their reading of my master's thesis. I hope they enjoyed this work as well as found it inspiring.

ABSTRACT

Transport properties play a central role in irreversible processes dissipating energy and producing entropy at the macroscopic scale. Hydrodynamic equations describe the different phases of matter and are formulated in terms of transport properties through the so-called transport coefficients. For example, they include thermal conductivity, diffusion, viscosity, or thermodiffusion. A fundamental issue is to understand the origin of these transport properties in terms of the underlying quantum dynamics of particles moving in matter, as well as the microscopic basis of the hydrodynamic equations. To this end, the local equilibrium approach is formulated in the context of quantum mechanics. The entropy production and the dissipativeless and dissipative parts of the current densities are deduced, allowing to reconstruct the hydrodynamic equations as well as the quantum Green-Kubo formulas for the transport coefficients. The consequences of the spatial symmetries and the microreversibility of the system on the transport properties are investigated.

Keywords: *Transport properties, nonequilibrium statistical mechanics, hydrodynamic equations, quantum Green-Kubo formulas, local equilibrium statistical operator.*

RÉSUMÉ

Les propriétés de transport jouent un rôle central au sein des phénomènes irréversibles en dissipant de l'énergie et en produisant de l'entropie à l'échelle macroscopique. Les équations hydrodynamiques décrivent différentes phases de la matière et prennent en compte ces propriétés de transport au travers de coefficients dit de transport. Il s'agit par exemple de la conductivité thermique, diffusion, viscosité ou encore de la thermodiffusion. Une question fondamentale est de comprendre l'origine de ces propriétés de transport et celle des équations hydrodynamiques à partir de la dynamique quantique des particules qui composent le système. Pour ce faire, l'approche d'équilibre local est formulée dans le cadre de la mécanique quantique. La production d'entropie ainsi que les parties non-dissipatives et dissipatives des densités de courant sont déduites, permettant de reconstruire les équations hydrodynamiques ainsi que les formules quantiques de Green-Kubo pour les coefficients de transport. Les conséquences des symétries spatiales ainsi que de la microréversibilité du système sont étudiées pour les propriétés de transport.

Mots-clés : *Propriétés de transports, mécanique statistique de non-équilibre, équations hydrodynamiques, formules quantiques de Green-Kubo, opérateur statistique d'équilibre local.*

NOTATIONS & CONVENTIONS

Unless explicitly stated, the following notations and conventions will be used.

- Latin letters $a, b, c, d, \dots$ correspond to spatial coordinates x, y, z .
- Greek letters $\alpha, \beta, \gamma, \delta, \dots$ label the hydrodynamic variables.
- The indices $i, j, k, l, \dots = 1, 2, \dots, N$ label the particles, where N is the total number of particles. The letters $k, l, p = 1, 2, \dots, c$ will denote the particle species, where c is the total number of species.
- Vectors will be denoted by a bold italic letter $\mathbf{p}$ or in components p^a . Tensors will be denoted by a bold Latin letter $\mathbf{T}$ or in components $T_{a_1 a_2 \dots a_r}$, where r is the rank of the tensor. The author takes the liberty of switching from the vector to the component form without further notice. The upper or lower position of the spatial components does not matter since this work is fundamentally nonrelativistic.
- Operators/variables related to particles are denoted with the particle label as subscript. For instance, the a spatial component of the momentum operator of particle labelled i is denoted $\hat{p}_i^a$.
- Einstein's convention of summation over repeated spatial indices is adopted. Summations over particle species will always be made explicit.
- Partial derivatives with respect to time are denoted by $\partial_t \equiv \frac{\partial}{\partial t}$. Spatial derivatives with respect to the position r_i^a of the particle labelled i are written as $\partial_i^a \equiv \frac{\partial}{\partial r_i^a}$.
- Operators are written with a hat $\hat{A}$. The mean value of an operator is denoted by $\langle \hat{A} \rangle$.
- Dotless j , j or $\hat{j}$, will always refer to current (fluxes) densities.
- $\hat{\rho}$ always denotes a statistical operator.
- SI units are used.

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INTRODUCTION

Invitation. A multitude of transport phenomena are present all around us, on every spatial scale. There is no need to look very far to find some. For illustrative purposes, let us examine the first five minutes after someone wakes up in the morning. First, the light is switched on. An electric current flows through the wires and powers the incandescent light bulb. Almost instantly, the tungsten filament acts as a resistor, reducing the flow of current, and dissipating heat around it. Then, the kettle is placed on the gas cooker. Gradually, it starts to become uniformly hot, so that soon enough it can no longer be touched without exposure to burning. In a few minutes, the water begins to boil. A tea bag is poured into the kettle, which turns the water into a greenish colour. Within five minutes, a rich variety of transport phenomena were encountered: the transport of electric currents within electrical wires, the transport of heat from the tungsten filament and the stove respectively to the external environment and to the kettle, mass transport corresponding to tea particles diffusing in boiling water, or momentum transport associated with the transport of gas through pipes. It is amusing to notice the omnipresence of transport phenomena in the daily life. But of course, this example should not be taken too seriously as transport phenomena occur in many other areas and contexts: industrial area, nanoscale systems, aeronautics, astrophysics, chemistry, *etc.*

Some transport phenomena are at the center of economic consequences of physical origin. How can we envisage the transport of electric currents over long distances? Indeed, wires have a non-zero resistance and are therefore responsible of a loss of energy by Joule effect. How can high pressure gas be transported over long distances? Since the pipes have a fixed diameter, the velocity of the gas will only decrease near the solid walls, slowing down its overall flow.

A transport phenomenon is defined in terms of its causes and effects. Some effects have been illustrated in the small scenario above and should be considered as flows of a certain quantity. This could be a flow of matter, momentum, heat, or electric charges. What are the sources (*i.e.* the causes) of these flows? A particle flow is produced from a variation in particle density. Tea diffuses in boiling water since the density of the tea particles decreases away from the tea bag. A momentum flow is produced from a difference in velocity. A heat flow results from a spatial difference in temperature. A flow of electric charges is produced by variations of the electrical potential. In the following, we will say that thermodynamic forces or affinities produce these different flows. A central question is to understand to what extent the various causes influence the effects. In other words, what is the mathematical relation between thermodynamic forces and flows? It turns out that, to a first approximation, the relation is linear and defines a transport coefficient. Among these transport coefficients are the diffusion coefficient, the thermal conductivity, or the viscosity. Many of these coefficients appear in a multitude of physical equations: Navier-Stokes equations, heat equation, diffusion equation... At this stage, the various transport phenomena are considered from a macroscopic point of view.

What is nonequilibrium statistical mechanics? All around us, objects exchange heat, chemical reactions are constantly taking place, electric currents are flowing, Earth is irradiated by a large amount of energy from the Sun, and bombarded by a multitude of cosmic rays, oceans are crossed by powerful currents produced by the Earth's rotation, wind friction, and variation in salinity. Motion appears to be a law: a motionless Nature at rest seems almost inconceivable. All these systems cannot be in thermodynamic equilibrium. Heat exchanges are the result of a non-uniform temperature, chemical or physical reactions between elementary particles are consequences of an inhomogeneous particle density, local pressure variations around the globe implies the existence of atmospheric winds. The question that arises is how to describe these systems? The major problem is that thermodynamics as we know it describes reversible phenomena that are at all times at equilibrium. How can thermodynamics be applied to systems found in a nonequilibrium state? As motivated above, a system held out of equilibrium by external constraints reacts in producing flows. These flows are at the origin of irreversible and dissipative phenomena, related to a production of entropy. A nonequilibrium state is precisely a state in which a set of irreversible phenomena leads the system towards equilibrium or to a steady nonequilibrium state. How can these irreversible phenomena be understood on a microscopic scale? What are the microscopic mechanisms of relaxation towards equilibrium? It should be noted that outside equilibrium, only a limited number of general results are available.

Nonequilibrium statistical mechanics studies the processes of transport of particles, momentum, and energy in different systems, and is to be seen as a further development of equilibrium statistical mechanics. The overall purpose is to obtain general equations of nonequilibrium thermodynamics using the methods of statistical mechanics, and with the equations of quantum or classical mechanics as a start. A nonequilibrium statistical theory is essential to understand the world around us, but it is nevertheless difficult to establish. In 1981, the physicist R. Zwanzig, pioneering in the comprehension of the statistical mechanics of irreversible processes, has summarized the basic goals of nonequilibrium statistical mechanics as consisting of:

- Deriving transport equations and to grasp their structure.
- Understanding how the approach to equilibrium occurs in natural systems.
- Studying the properties of steady states.
- Calculating the instantaneous values and the temporal evolution of the physical quantities which specify the macroscopic state of the system.

Purposes of this work. Transport coefficients are at the heart of everyday life and are of major importance in many physical processes, in engineering sciences and in the industrial world. Their definition and meaning are well understood from a macroscopic point of view. In the

framework of nonequilibrium statistical mechanics, the question is to understand their origins from the microscopic nature and dynamics of particles. In other words, is it possible to give a deeper, more fundamental meaning to transport coefficients?

A fluid can be the subject of a variety of transport phenomena and as such must be considered a dynamic system. Locally, it can be described by physical fields, for example local mass, particle, momentum and energy densities. These fields are functions of time and of spatial coordinates. Their temporal evolution defines the hydrodynamic state of the fluid at any time. For instance, the continuity equation dictates the evolution of mass density, just as the Navier-Stokes equations dictate the evolution of the velocity field. Nonlinear effects often arise because of advection induced by the velocity field, as it is the case in fluid turbulence. Another central question is to understand whether these hydrodynamic equations can be reconstructed from the microscopic dynamics of the particles that constitute the system. Also, how can a fluid be described in a nonequilibrium state?

All these questions have been at the heart of nonequilibrium statistical mechanics since the second half of the 20th century. In 1957, the Green-Kubo formulas, discovered by M. S. Green and R. Kubo, that describe the transport coefficients as time integrals of correlation functions of microscopic currents, were born. In 1958, one year later, these formulas were generalised by H. Mori to quantum systems. By the early 1960s, it was understood that hydrodynamic equations and transport coefficients could be simultaneously deduced from the same formalism with the use of a local equilibrium probability distribution. More generally, it turns out that nonequilibrium statistical mechanics can be formulated from nonequilibrium ensembles, in the light of the traditional procedure of equilibrium statistical mechanics. The starting point is to note that the description of some nonequilibrium systems is based on the knowledge of a certain finite number of observables (or rather of their average value) called the relevant variables. This knowledge constitutes partial information about the system. The question that arises is how to construct a statistical operator presenting the best balance between the partial and the disregarded information? In other words, how to obtain the least biased judgement in terms of the given information, that is, the one which is maximally neutral in regard to the lost information? This problem was originally introduced by E. T. Jaynes in 1957. The corresponding statistical operator is constructed on the basis of entropy maximisation in the perspective that probability must be distributed as largely as possible without contradicting the available information.

The objective of this work is to make use of the nonequilibrium statistical ensemble formalism and to apply it to a normal quantum fluid. An adequate statistical operator will be constructed on the basis of the relevant variables of the problem and the entropy maximisation requirement. The relevant variables are chosen to be conserved observables as a result of the fundamental symmetries of the system. They thus satisfy local conservation equations from which the corresponding local current density operators can be identified. From this formalism,

the microscopic expressions of the dissipativeless part of the current densities of Eulerian type, local thermodynamics and the statistical-mechanical expression of the entropy production in terms of the dissipative part of the current densities will be obtained. The dissipative current densities can be obtained at leading order in the gradients of the macrofields, revealing the transport coefficients given by the Green–Kubo formulas. In this framework, the consequences of microreversibility and spatial symmetries will also be investigated. The originality of this work lies in the scope of the formalism as well as in the scope of the results. By extension, it provides a powerful method to obtain generalized hydrodynamic equations for many kinds of systems and the Green-Kubo formulas for the transport coefficients. The construction of a relevant statistical operator from the perspective of nonequilibrium quantum statistical mechanics is poorly discussed in the literature. The obtained results provide a first-principle understanding of the relevant statistical operator to hydrodynamics and nonequilibrium thermodynamics.

Structure. In [Chapter 1](#), an introduction to transport phenomena is proposed. The objective is to define and clearly explain some of the central notions appearing in this work. Why are some transport phenomena dissipative? Why are they associated with irreversible processes? [Chapter 2](#) introduces the essential framework and tools of quantum mechanics that will be used in the following chapters. A Hamiltonian operator is constructed. From it, various conserved observables will be identified and their corresponding local conservation equations obtained. [Chapter 3](#) goes into the heart of the matter. The relevant statistical operator will be constructed from the nonequilibrium statistical ensemble formalism, and various relations and properties will be demonstrated. In particular, we will show that the relevant statistical operator describes a local equilibrium configuration. An expression for entropy production, fundamental to this work, will be obtained in full generality. [Chapter 4](#) will study the dissipativeless part of the current densities and will provide a physical expression of the conjugate fields that appear in the expression of the relevant statistical operator. Finally, [Chapter 5](#) will focus on the dissipative part of the current densities from which the hydrodynamic equations and the quantum Green-Kubo formulas will be obtained. Here, particular attention will be given to the spatial and time-reversal symmetries of the system.

CHAPTER 1

AN INTRODUCTION TO TRANSPORT PHENOMENA

*“You should call it entropy, for two reasons.
[...] and more important, no one knows what
entropy really is, so in a debate you will
always have the advantage.”*

VON NEUMANN TO SHANNON IN A REPORTED
DISCUSSION.¹

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The purpose of this first chapter is twofold. It can be seen as an introduction but also as an invitation to this work. The chapter occupies a prime position, both in terms of the concepts discussed and the intuition it provides.

It is here question of presenting succinctly the current state of the art about transport phenomena. Particular attention will be given to providing answers to the following questions:

¹M. Tribus and E. C. McIrvine. “Energy and Information”. In: *Scientific American* 225.3 (Sept. 1971), 179–190. DOI: [10.1038/scientificamerican0971-179](https://doi.org/10.1038/scientificamerican0971-179).

- What are transport phenomena?
- How to define transport coefficients?
- What is the connection between transport phenomena and irreversibility?

In §1.1, an intuitive and pictorial approach to transport phenomena and transport coefficients is proposed. §1.2 is devoted to the study of conservation laws, in particular obtaining the mass, momentum and energy local conservation equations. Conservation laws will occupy a fundamental role throughout this work. In §1.3, the notion of entropy is reviewed as well as the local form of the second law of thermodynamics. §1.4 aims at giving a more general approach to the results obtained in the preceding sections. Finally, §1.5 provides a connection between transport phenomena and irreversible processes.

1.1. BASIC INTRODUCTION

The goal of this section is to develop intuition about transport phenomena and to introduce the basic definitions of transport coefficients.

As already motivated in the introduction, transport properties are common in everyday life: thermos flasks to stop heat conduction, mutual diffusion between water and grenadine syrup, lubricant grease reducing friction, electric current transport, *etc.* They appear in many physical equations: Navier-Stokes equations, heat equation, diffusion equation...

The major transport coefficients are defined by the Fourier law for heat conduction, the Fick law for diffusion, the Newton law for viscosity or the Ohm law. Let us look at what these laws have in common.

Fourier's law for heat conduction. The Fourier law for heat conduction simply states that in an isotropic medium, the flux of heat $\mathbf{q}$ is proportional to and goes in the inverse direction of the temperature gradient ∇T ,

$$\mathbf{q} = -\kappa \nabla T. \quad (1.1)$$

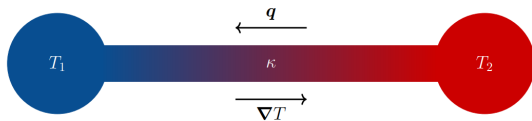


Figure 1.1: *Illustration of heat conductivity between two heat reservoirs. The temperature gradient produces a heat flux.*

The equation is only valid as long as the temperature gradient is not too large. The proportionality constant κ is called the thermal (or heat) conductivity. The heat flux has the unit $[\mathbf{q}] = \text{J.m}^{-2}.\text{s}^{-1}$ and the thermal conductivity $[\kappa] = \text{J.m}^{-1}.\text{s}^{-1}.\text{K}^{-1}$ that is an energy per unit length, time and temperature.

Consequently, κ is expected to be somehow related to the transport and dissipation of energy (in the form of heat). Our daily experience suggests that heat goes from hot to cold regions:

$\mathbf{q} \propto -\nabla T \Leftrightarrow \kappa > 0$. The Fourier law for heat conduction illustrates a transport property: heat is transported from one region to another as a response to the temperature gradient.

Usually, κ is taken as a constant as a first approximation. In practice, it is not true as it may depend on the local properties of the material (composition), temperature or pressure conditions (even if the medium is homogeneous). A system in thermal equilibrium with its environment is at uniform temperature, $\nabla T = 0$, and is thus not subjected to a heat flux. Consequently, only systems that are out of thermal equilibrium may be subjected to a heat flux. Because heat is a form of energy related to molecular motion, the source for the heat transport is to be considered at the molecular level.

Fick's law for diffusion. Let us here assume that a drop of dye is poured in a glass of water. Quickly, the dye is seen to diffuse within the water, its color (proportional to its concentration) fading over time. The process is stopped when the dye is uniformly mixed with the water: this again illustrates a transport property. Such binary mixture (solute and solvent) can be described by the Fick law for diffusion,

$$\mathbf{j} = -D \nabla n \quad (1.2)$$

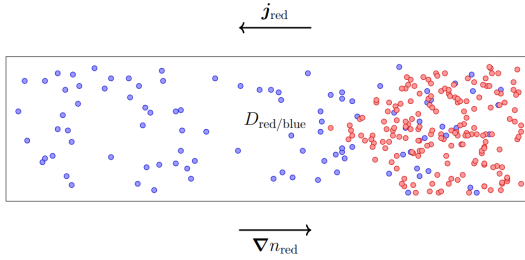


Figure 1.2: *Illustration of particle diffusion. Higher concentration of red particles on the right side of the box will produce diffusion towards the left side. The concentration gradient produces a particle flux.*

where $\mathbf{j}$ is the particle flux having the unit $[\mathbf{j}] = \# \cdot \text{m}^{-2} \cdot \text{s}^{-1}$ ($\#$ standing for the number of particles), n is the density of the substance (for instance the dye), $[n] = \# \cdot \text{m}^{-3}$, and D is the diffusion coefficient, $[D] = \text{m}^2 \cdot \text{s}^{-1}$. This relation remains valid as long as the density gradient is not too large. Again, the diffusion coefficient depends on the composition, temperature and pressure of the mixture. The unit of the diffusion coefficient suggests that it is somehow related to the transport

of particles (mass) over an area per unit time. A particle flux cannot exist in a uniformly mixed binary mixture. Only systems with an inhomogeneous particle density are subjected to a particle flux. The diffusion of particles can also be considered at the molecular level. The solute particles are much less abundant than the solvent particles and so solvent-solute collisions are much more frequent than solute-solute ones. A random walk model can be used to understand and explain the diffusion of the solute particles within the solvent (diffusive stochastic process).

Newton's law for viscosity. The Newton law for viscosity is in a sense less intuitive to catch than the two previous processes. As in figure 1.3, let us consider a simple shear flow $\mathbf{u} = u(y) \mathbf{e}_x$. The fluid located above some level $y = y^{\text{st}}$ exerts a force per unit area (*i.e.* a stress) on the fluid

immediately below and vice versa. The Newton law for viscosity (in its simplest form) gives an expression for the shear stress τ ,

$$\tau = \eta \frac{du}{dy} \quad (1.3)$$

and it is valid as long as the velocity gradient is small. η is the coefficient of viscosity with the unit $[\eta] = \text{kg.m}^{-1}.\text{s}^{-1}$, that is a momentum per unit area. Viscosity can be thought of as a sort of internal friction. Its units indicate that it is somehow related to the transport of momentum. This can be understood as follows. Assume that at $t = 0$, the bowl is rotated. At $t > 0$, we start to rotate the bowl. At first, it is noticed that the soup near the edge of the bowl rotates and then, progressively, the rotation is diffused towards the center of the bowl.

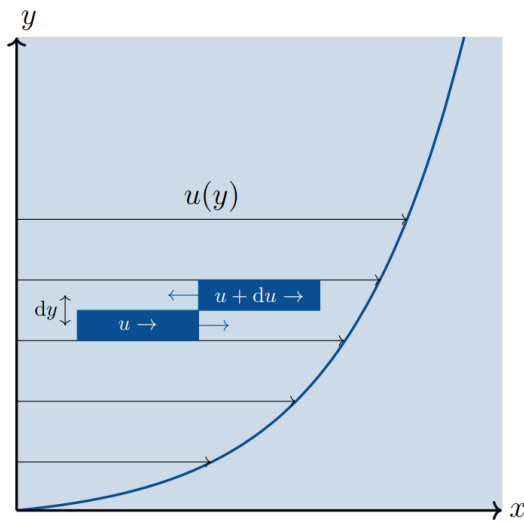


Figure 1.3: Illustration of the shear stress occurring between two neighbouring parcels of fluids. The shear stress is represented by a blue arrow.

If the same experiment is performed with water, we will notice that the rotation will be diffused much faster towards the center: soup is more viscous than water. The velocity gradient appearing in the Newton law is a driving force for momentum transport. In the absence of motion, no momentum can be transported nor diffused. As for the previous processes, the coefficient of viscosity depends on the composition, the temperature and the pressure. The viscosity can be considered at the molecular level. Take the level $y = c^{\text{st}}$. Just above it, the fluid elements move at a slightly faster velocity than the fluid elements below it. The fastest particles

transfer momentum to the slowest ones. It is reasonable to think that the amount of momentum transferred will be proportional to their relative motion (*i.e.* velocity gradient).

Ohm's law. If an electric field is applied to an electrically conducting material, electric charges that compose it move under the Lorentz force and produce a current. Denoting by $\mathbf{j}$ the current density (current intensity per unit area) and if the electric field is not too intense, the Ohm law states that

$$\mathbf{j} = \sigma \mathbf{E} = -\sigma \nabla \varphi \quad (1.4)$$

where φ is the electric potential. If no electric field is applied to the material, the current density vanishes. σ is the electrical conductivity and has the unit $[\sigma] = \text{C}^2.\text{s.m}^{-3}.\text{kg}^{-1}$, that is $(\text{charge})^2$ multiplied by a time per unit volume and mass. Consequently, σ is expected to be related to the transport of electric charges, independently of their sign (because the charge is squared). Again,

the correspondence between the micro and macroscopic levels of description can be established. The macroscopic version of the Ohm law is $I = \frac{1}{R} V$ as it is usually used in circuit analysis. Its microscopic version is as above, taking into account the motion of the electric charges.

Thermodiffusion & thermoelectricity. The structure of the four laws presented above suggests that the fluxes are created by the inhomogeneity of some quantity (temperature, particle density, *etc.*). These effects (fluxes) are said to be direct.

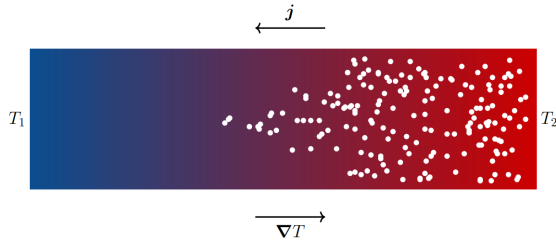


Figure 1.4: Illustration of the Soret effect. A gradient in temperature produces a particle flux towards the coolest region.

There also exist phenomena that produce fluxes as a response of the inhomogeneity of another, *a priori*, unrelated quantity. These effects are said to be crossed. For instance, the presence of an inhomogeneous temperature field $T(\mathbf{r}, t)$ can produce a particle flux (Soret effect or thermodiffusion). Conversely, the presence of an inhomogeneous particle density field $n(\mathbf{r}, T)$ can produce a heat flux

(Dufour effect) and is to be seen as the reciprocal of the Soret effect. Again, these phenomena can be described by linear laws between the fluxes and the gradients, the proportionality coefficient being the corresponding transport coefficient. In a similar way, in the context of thermoelectricity, an inhomogeneous temperature field can give rise to a flux of charged particles, *i.e.* an electric current (Seebeck effect). Conversely, an inhomogeneous scalar field $\varphi(\mathbf{r}, t)$ can give rise to a heat flux (Peltier effect).

Common characteristics. While presenting the several laws above, we have essentially repeated the same reasoning but in different physical contexts. Indeed, these laws feature common characteristics:

- Their structure is notable: the flux of the transported quantity is proportional to what caused it, *i.e.* the gradient of another quantity. The proportionality constant is precisely the corresponding transport coefficient.
- They are linear. Non-linearities are neglected because the validity of these laws is restricted to small gradients so that higher derivatives can be neglected.
- All these fluxes vanish when the system is at equilibrium.
- The transport coefficients can be given a microscopic explanation.

The structure of these laws can be generalised into *phenomenological equations* as it will be shown in §1.4. Before doing so, the mathematical description must be developed in more detail.

1.2. CONSERVATION LAWS

In this work, the notion of conserved quantity plays a fundamental role, if not crucial. The reason is that the conserved quantities are those which evolve the least rapidly under the time evolution of the system. It is therefore instructive and relevant to understand in detail what a balance equation is. Briefly, a balance equation relates the time evolution of a quantity to its fluxes and its sources.

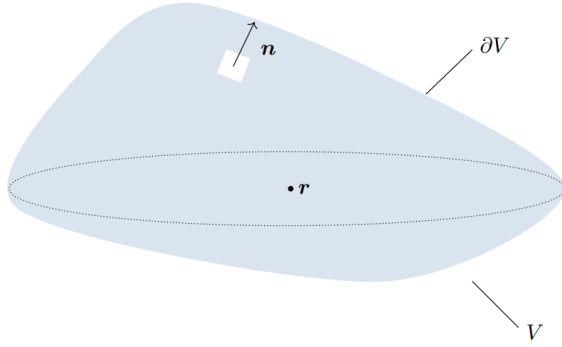


Figure 1.5: *An arbitrary fluid element.*

Consider an arbitrary fluid element centred around position $\mathbf{r}$, of volume V , and closed surface (boundary) ∂V as in figure 1.5. Let $X(t)$ be an extensive (see §1.3 for the definition of extensiveness) quantity whose time evolution is of interest. Assume that $X = X(t)$ can be locally described by a field $f = f(\mathbf{x}, t)$ such that $X(t) = \int_V dV f(\mathbf{x}, t)$. f is the density of X (per unit volume) and is an intensive quantity.

We ask the following question: what are the mechanisms affecting the time evolution of X ?

- If a given amount of X is contained within V , it can be transferred to another neighbouring fluid element. That is to say, X can flow outside or inside the volume V . Let us call $\mathbf{j}$ the flux density (or current density) of X through ∂V , describing the quantity of X flowing out or into V , and expressed per unit time and surface.
- Furthermore, X can be produced within the volume V . Let us call σ the production (or the source) of X within V , which is expressed per unit time and volume.

It is thus clear that a balance equation for X takes the form

$$\frac{dX}{dt} = - \int_{\partial V} d\mathbf{S} \cdot \mathbf{j} + \int_V dV \sigma \quad (1.5)$$

where $d\mathbf{S} = \mathbf{n} dS$ with $\mathbf{n}$ a unit vector normal to ∂V and pointing outside V . We note that if $\mathbf{j}$ is parallel to $d\mathbf{S}$, the flux density tends to decrease the amount of X inside V , so that there is a minus sign in front of its surface integral. With the use of the divergence theorem:

$$\frac{dX}{dt} = \int_V dV \partial_t f = \int_V dV (-\nabla \cdot \mathbf{j} + \sigma). \quad (1.6)$$

As the equality must hold for an arbitrary finite volume V , one obtains the local form of the balance equation for X ,

$$\partial_t f + \nabla \cdot \mathbf{j} = \sigma. \quad (1.7)$$

Two remarks:

- In the absence of a source term, (1.7) is referred to as a local conservation equation. In the absence of a flux term, the equation is usually referred to as a pure relaxation process. In its general form it is called a balance equation.
- A conservation equation is valid at any time and any point. In particular, when the system of interest is far from equilibrium [1].

Equilibrium thermodynamics relies on two fundamental laws: the first law expressing the conservation of energy and the second law related to the notion of entropy. These two laws also form the main pillars of nonequilibrium thermodynamics. The purpose for the remaining of this section and the following will be to obtain a local formulation of these laws.

Let us consider a normal fluid contained in an isolated system, without any interaction with the exterior environment. In the nonrelativistic limit, the total mass is conserved. In addition, in the absence of external forces, the total momentum is also conserved (by the Newton law of motion). However, the mass and momentum locally vary in space and time and we shall need the local formulation of their conservation.

A preliminary remark. In this work, and in particular for the equations appearing in this chapter, we will often switch between the Eulerian and the Lagrangian descriptions of a flow field. Let us briefly review the main differences and make explicit the connection between both.

In the Lagrangian description of a flow field, a fluid element is followed through space and time. In the Eulerian description, attention is given to a specific position in space by which the fluid flows over time. More explicitly, in the Lagrangian description, the flow field is given by a function $\mathbf{X} = \mathbf{X}(\mathbf{x}_0, t)$ labelling the position of the fluid element whose position at time $t = t_0$ was $\mathbf{x}_0$. In the Eulerian description, the flow field is a function of $\mathbf{x}$ and t , where $\mathbf{x}$ is a spatial coordinate in the frame of the observer.

Let f be a quantity of interest. In the Lagrangian description, $f = f(\mathbf{X}(\mathbf{x}_0, t), t)$. The connection between the two descriptions is established through the following relation,

$$\frac{df}{dt} = \partial_t f + \frac{d\mathbf{X}}{dt} \cdot \nabla f \quad (1.8)$$

being a simple consequence of the chain rule for derivatives. $\frac{d\mathbf{X}}{dt}$ is precisely understood as the velocity $\mathbf{u}$ of the flow field so that

$$\frac{df}{dt} = \partial_t f + \mathbf{u} \cdot \nabla f. \quad (1.9)$$

Mass conservation. In the light of equations (1.5)-(1.7), let us take $X(t) = M(t)$, the mass of a fluid element of arbitrary volume V and $f(\mathbf{x}, t) = \rho(\mathbf{x}, t)$ the mass density. Since mass conservation holds in nonrelativistic systems, we have that $\sigma = 0$. The flux of matter flowing

across ∂V is $\mathbf{j}_\rho = \rho \mathbf{u}$ where $\mathbf{u}$ is the velocity of the flow. Directly with equation (1.7), the local mass conservation equation is

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{u}) = 0 \quad (1.10)$$

or in the Lagrangian form

$$\frac{d\rho}{dt} + \rho \nabla \cdot \mathbf{u} = 0. \quad (1.11)$$

This equation is usually referred to as the continuity equation.

The case of multicomponent systems is to be discussed in further detail. Let a system be composed of $k = 1, 2, \dots, c$ different species. Let $N = \sum_{k=1}^c N_k$ be the total number of particles in the system. Its total mass is $M = \sum_{k=1}^c m_k N_k$ where m_k is the mass of a particle of species k . If no chemical reactions occur between the different species, equation (1.10) naturally leads to the form

$$\partial_t \rho_k + \nabla \cdot (\rho_k \mathbf{u}_k) = 0 \quad (1.12)$$

where $\rho = \sum_{k=1}^c \rho_k$ and $\mathbf{u} = \sum_{k=1}^c \mathbf{u}_k$ with $\mathbf{u}_k = \mathbf{u}_k(\mathbf{r}, t)$ the velocity field corresponding to each of the species, and defined such that

$$\mathbf{u} = \frac{1}{\rho} \sum_{k=1}^c \rho_k \mathbf{u}_k. \quad (1.13)$$

In this context, the velocity $\mathbf{u}$ is the barycentric (center of mass) velocity. Taking the sum over all the species in equation (1.12) gives back (1.10).

The diffusion flux of the species k is defined with respect to the barycentric motion,

$$\mathbf{j}_k \equiv \rho_k (\mathbf{u}_k - \mathbf{u}). \quad (1.14)$$

As a trivial consequence, $\sum_{k=1}^c \mathbf{j}_k = 0$. If the fluid is at rest, $\mathbf{u} = 0$ and $\mathbf{j}_k$ becomes the flux of the species k across ∂V . Equation (1.12) can be written in a more speaking form,

$$\partial_t \rho_k + \nabla \cdot (\rho_k \mathbf{u} + \mathbf{j}_k) = 0 \quad (1.15)$$

in terms of the flow field $\mathbf{u}$. The flux density is the sum of a convective part, $\rho_k \mathbf{u}$, and a conductive part, $\mathbf{j}_k$.

Alternatively, one could be interested in obtaining a local conservation equation for the particle density $n_k \equiv \frac{N_k}{V}$. The correspondence between the mass and particle densities is established as follows,

$$\rho_k = \frac{m_k N_k}{V} = \frac{M_{k,\text{mol}}}{\mathcal{N}_A} \frac{N_k}{V} = \frac{M_{k,\text{mol}}}{\mathcal{N}_A} n_k \quad (1.16)$$

with $M_{k,\text{mol}}$ the molar mass of the species k and $\mathcal{N}_A$ the Avogadro number, so that equation (1.15) now reads

$$\partial_t n_k + \nabla \cdot (n_k \mathbf{u} + \mathbf{j}_k) = 0 \quad (1.17)$$

but now with the diffusion flux $\mathbf{j}_k \equiv n_k (\mathbf{u}_k - \mathbf{u})$ satisfying $\sum_{k=1}^c m_k \mathbf{j}_k = 0$.

On the other hand, one could also be interested to study the mass fraction $c_k \equiv \frac{\rho_k}{\rho} = \frac{m_k N_k}{M}$, $\sum_{k=1}^c c_k = 1$. With the use of (1.15), the associated local conservation equation for c_k can be written in the form

$$\rho \frac{d}{dt} c_k + \nabla \cdot \mathbf{j}_k = 0. \quad (1.18)$$

Momentum conservation. The Newton law of motion can be written in a local form:

$$\int_V dV \rho \frac{d\mathbf{u}}{dt} = - \int_{\partial V} d\mathbf{S} \cdot \mathbf{P} = - \int_V dV \nabla \cdot \mathbf{P} \quad (1.19)$$

where external forces (forces per unit mass) were neglected and $\mathbf{P}$ is the pressure tensor (equal to minus the stress tensor). It is a tensor of rank two representing the forces acting per unit surface. The validity of this equation for a fluid element of arbitrary volume V allows to write

$$\rho \frac{d\mathbf{u}}{dt} + \nabla \cdot \mathbf{P} = 0. \quad (1.20)$$

In a viscous and isotropic fluid, the pressure tensor can be decomposed into one part due to hydrostatic pressure and another due to shear and bulk viscosities [2]:

$$\mathbf{P} = p \mathbf{1} + \boldsymbol{\Pi} \quad \text{with} \quad (\boldsymbol{\Pi})_{ij} = \Pi_{ij} = -\eta \left(\partial_i u_j + \partial_j u_i - \frac{2}{3} (\nabla \cdot \mathbf{u}) \delta_{ij} \right) - \xi (\nabla \cdot \mathbf{u}) \delta_{ij}. \quad (1.21)$$

As shown in appendix A, the conservation of the angular momentum density implies that the pressure tensor is symmetric. With the use of the continuity equation (1.11), the equation (1.20) becomes

$$\partial_t(\rho \mathbf{u}) + \nabla \cdot (\rho \mathbf{u} \mathbf{u} + \mathbf{P}) = 0 \quad (1.22)$$

where $\mathbf{u} \mathbf{u}$ is understood as a dyadic product $\mathbf{u} \otimes \mathbf{u}$.² The momentum flux density is $\rho \mathbf{u} \mathbf{u} + \mathbf{P}$ and is the sum of a convective part ($\rho \mathbf{u} \mathbf{u}$) and a conductive part $\mathbf{P}$.

Energy conservation. Energy is a term derived from the Greek $\epsilon\nu \epsilon\rho\gamma\omicron\zeta$, *en érgon*, translated as *in work*.³ Let U be the specific energy, *i.e.* energy per unit mass, of a small fluid element. U is defined as the sum of the specific kinetic energy $\frac{1}{2} \mathbf{u}^2$, the specific potential energy ψ from the external forces, and the specific internal energy e ,

$$U = \frac{1}{2} \mathbf{u}^2 + \psi + e. \quad (1.23)$$

²The divergence of a tensor $\mathbf{A} = A_{ij} \mathbf{e}_i \otimes \mathbf{e}_j$ of rank two is defined as

$$\nabla \cdot \mathbf{A} = \partial_i A_{ij} \mathbf{e}_j$$

where $\mathbf{e}_j$ is a unit vector in the j direction. In this case,

$$\begin{aligned} \nabla \cdot (\rho \mathbf{u} \mathbf{u}) &= \partial_i (\rho u_i u_j) = u_i \partial_i (\rho u_j) + \rho u_j \partial_i u_i \\ &= \mathbf{u} \cdot \nabla (\rho \mathbf{u}) + \rho \mathbf{u} \nabla \cdot \mathbf{u}. \end{aligned}$$

³H. G. Liddell and R. Scott. *A Greek-English Lexicon*. Ninth Edition. Oxford: Clarendon press, 1996.

From a macroscopic point of view, this can be considered as a definition of the specific internal energy: $e = U - \frac{1}{2}\mathbf{u}^2 - \psi$. From a microscopic point of view, e is the energy associated to thermal motion and to the short-range molecular interactions [3]. Because external forces are neglected, $\psi = 0$. The first law of thermodynamics can be expressed as [4]

$$\sigma_U = 0, \quad (1.24)$$

i.e. the source of energy vanishes and total energy is thus conserved. Alternatively, with (1.23),

$$\sigma_K + \sigma_e = 0 \quad (1.25)$$

where the K subscript refers to kinetic energy. To obtain the source of internal energy, we shall first obtain the source of kinetic energy. Taking the scalar product of equation (1.22) with $\mathbf{u}$, one obtains⁴

$$\partial_t(\rho K) + \nabla \cdot (\rho K \mathbf{u} + \mathbf{P} \cdot \mathbf{u}) = \mathbf{P} : \nabla \mathbf{u} \quad (1.26)$$

where $K \equiv \frac{1}{2}\mathbf{u}^2$. The source of kinetic energy is identified with $\sigma_K = \mathbf{P} : \nabla \mathbf{u} = -\sigma_e$. It has the units of a power per unit volume and is associated to the work performed by pressure per unit time and volume. The flux density of internal energy can be split into a convective part $\rho e \mathbf{u}$ and a conductive part $\mathbf{q}$ which is by definition the heat flux. The local balance equation for the internal energy density therefore reads

$$\partial_t(\rho e) + \nabla \cdot (\rho e \mathbf{u} + \mathbf{q}) = -\mathbf{P} : \nabla \mathbf{u} \quad (1.27)$$

or with use of the continuity equation (1.10)

$$\rho \frac{de}{dt} + \nabla \cdot \mathbf{q} + \mathbf{P} : \nabla \mathbf{u} = 0. \quad (1.28)$$

Equation (1.27) shows that the internal energy e is not a conserved quantity. Equations (1.26) and (1.27) can be combined to give the local conservation equation for the total energy density⁵ in the absence of external forces,

$$\partial_t(\rho U) + \nabla \cdot (\rho U \mathbf{u} + \mathbf{P} \cdot \mathbf{u} + \mathbf{q}) = 0 \quad (1.29)$$

which can be considered as a local formulation of the first law of thermodynamics.

1.3. NOTION OF ENTROPY

Second law of thermodynamics & entropy production. The entropy is by far one of the most central quantity of this work. Let us start by recalling the second law of thermodynamics.

⁴The double-dot product of two tensors $\mathbf{A}$ and $\mathbf{B}$ of rank two is defined as

$$\mathbf{A} : \mathbf{B} = A_{ij}B_{ij}.$$

This definition may contrast with the other definition $\mathbf{A} : \mathbf{B} = A_{ij}B_{ji}$, but this choice will be relevant for later purposes. In particular, the two definitions are equivalent as long as at least one of the two tensors is symmetric.

⁵ U is the specific energy but ρU is the energy density.

Precisely, it states [5] that there exists a function of state, called entropy (from the Greek $\epsilon\nu \tau\rho\omicron\pi\eta$, *en tropē*, translated as *in turning* or *in change*⁶) that possesses the following properties:

- The entropy of a system is an extensive property. If a system $\mathcal{C}$ can be decomposed into N subsystems $\mathcal{C}_i$ such that $\mathcal{C} = \cup_i^N \mathcal{C}_i$, then the total entropy S of the system is the sum of the entropies S_i of all the subsystems, $S = \sum_i^N S_i$.
- The variation of entropy can be split in two parts:

$$dS = d_e S + d_i S. \quad (1.30)$$

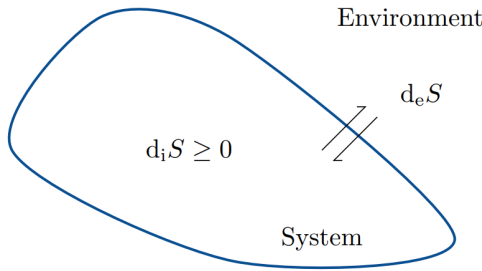


Figure 1.6: *Entropy variation for a system in contact with an external environment.*

The first term is associated to the flow of entropy occurring from the exchange with the external environment. The second term results from the internal production of entropy. The latter is always greater than or equal to zero,

$$d_i S \geq 0. \quad (1.31)$$

For a reversible process, the equality

holds. For an irreversible process, $d_i S > 0$ always.

The decomposition of the entropy variation as in (1.30) is a consequence of its extensiveness. If the system is closed, $d_e S = 0$. In the light of what was done in §1.2, the second law of thermodynamics can be written in the form

$$\partial_t(\rho s) + \nabla \cdot \mathbf{j}_s = \sigma_s \geq 0 \quad (1.32)$$

and the comparison with equation (1.30) suggests

$$\frac{d_i S}{dt} = \int_V dV \sigma_s \geq 0 \quad (1.33)$$

$$\frac{d_e S}{dt} = - \int_{\partial V} dS \cdot \mathbf{j}_s. \quad (1.34)$$

Here, s is the specific entropy. The approach can be further developed with the assumption that thermodynamic local equilibrium is achieved within small fluid elements. This assumption does not prevent the whole system to be out of equilibrium and provides a situation for which the macroscopic evaluation of entropy still remains possible. However, the deviation from equilibrium is assumed to be not too large.

Based on appendix B, the local equilibrium hypothesis for a multicomponent fluid takes the form:

$$\begin{cases} s = s(e, \rho, c_k) \\ ds = \frac{1}{T} de - \frac{p}{\rho^2 T} d\rho - \sum_{k=1}^c \frac{\mu_k}{T} dc_k. \end{cases} \quad (1.35)$$

⁶cf. footnote 3.

The local equilibrium hypothesis is valid for most familiar transport phenomena [3]. For a fluid element followed along its centre of gravity,

$$\frac{ds}{dt} = \frac{1}{T} \frac{de}{dt} - \frac{p}{\rho^2 T} \frac{d\rho}{dt} - \sum_{k=1}^c \frac{\mu_k}{T} \frac{dc_k}{dt} \quad (1.36)$$

which considering equations (1.11), (1.18) and (1.28) takes the form

$$\rho T \frac{ds}{dt} + \nabla \cdot \mathbf{q} + \Pi : \nabla \mathbf{u} - \sum_{k=1}^c \mu_k \nabla \cdot \mathbf{j}_k = 0, \quad (1.37)$$

where the subdivision (1.21) of the pressure tensor into an hydrostatic and dissipative part was used. Alternatively,

$$\partial_t(\rho s) + \nabla \cdot \left(\rho s \mathbf{u} + \frac{1}{T} \mathbf{q} - \sum_{k=1}^c \frac{\mu_k}{T} \mathbf{j}_k \right) = -\frac{1}{T^2} \mathbf{q} \cdot \nabla T - \frac{1}{T} \Pi : \nabla \mathbf{u} - \sum_{k=1}^c \mathbf{j}_k \cdot \nabla \frac{\mu_k}{T} \quad (1.38)$$

which allows to identify the entropy flux density as well as the entropy source,

$$\begin{cases} \mathbf{j}_s = \rho s \mathbf{u} + \frac{1}{T} \mathbf{q} - \sum_{k=1}^c \frac{\mu_k}{T} \mathbf{j}_k \\ \sigma_s = -\frac{1}{T^2} \mathbf{q} \cdot \nabla T - \frac{1}{T} \Pi : \nabla \mathbf{u} - \sum_{k=1}^c \mathbf{j}_k \cdot \nabla \frac{\mu_k}{T} \geq 0. \end{cases} \quad (1.39)$$

The flux density is again the sum of a convective and conductive part. The source of entropy is always positive as a result of the second law of thermodynamics: in an isolated system, entropy can only be produced and always increases. By what mechanisms is entropy constantly produced? Here, we will use the explicit form of Π (1.21), the Fourier law for heat conduction (1.1) and the Fick law for diffusion (1.2). First, one shows that

$$\begin{aligned} \Pi : \nabla \mathbf{u} &= \frac{1}{2} \Pi : (\nabla \mathbf{u} + (\nabla \mathbf{u})^T) \\ &= \frac{1}{2} \Pi_{ij} (\partial_i u_j + \partial_j u_i) \\ &= -\frac{\eta}{2} (\partial_i u_j + \partial_j u_i - \frac{2}{3} (\nabla \cdot \mathbf{u}) \delta_{ij})^2 - \xi (\nabla \cdot \mathbf{u})^2 \end{aligned} \quad (1.40)$$

where the tensor $\nabla \mathbf{u}$ was symmetrised to avoid any ambiguity of definition. Then, the Fick law must be rewritten in a different form. On isothermal conditions, one has

$$\nabla \rho_k = T \sum_{l=1}^c \left(\frac{\partial \rho_k}{\partial \mu_l} \right)_{T, \{\mu_j\}_{j \neq l}} \nabla \frac{\mu_l}{T}. \quad (1.41)$$

The Fick law (1.2) can thus be written in the more general form

$$\mathbf{j}_k = - \sum_{l=1}^c D_{kl} \nabla \rho_l \Leftrightarrow \mathbf{j}_k = -T \sum_{l,p=1}^c D_{kl} \left(\frac{\partial \rho_l}{\partial \mu_p} \right)_{T, \{\mu_j\}_{j \neq p}} \nabla \frac{\mu_p}{T} \quad (1.42)$$

where $\mathbf{j}_k$ is the particle flux of species k and D_{kl} is the diffusion coefficient of species k within species l . If $\mathcal{L}_{k,l} \equiv T \sum_{p=1}^c D_{kp} \left(\frac{\partial \rho_p}{\partial \mu_l} \right)_{T, \{\mu_j\}_{j \neq l}}$ is defined, (1.42) can be rewritten as

$$\mathbf{j}_k = - \sum_{l=1}^c \mathcal{L}_{k,l} \nabla \frac{\mu_l}{T}. \quad (1.43)$$

Since $\sum_{k=1}^c \mathbf{J}_k = 0$, we have that $\sum_{k=1}^c \mathcal{L}_{k,l} = 0, \forall l = 1, \dots, c$. Accordingly, all coefficients $\mathcal{L}_{k,l}$ are not independent. Furthermore, these coefficients satisfy the Onsager-Casimir reciprocal relation⁷ $\mathcal{L}_{k,l} = \mathcal{L}_{l,k}$. The number of these coefficients is therefore $c(c-1)/2$.

Consequently, the entropy source takes the form

$$\sigma_s = \frac{\kappa}{T^2} (\nabla T)^2 + \frac{\eta}{2T} \left(\partial_i u_j + \partial_j u_i - \frac{2}{3} (\nabla \cdot \mathbf{u}) \delta_{ij} \right)^2 + \frac{\xi}{T} (\nabla \cdot \mathbf{u})^2 + \sum_{k,l=1}^c \mathcal{L}_{k,l} \nabla \frac{\mu_l}{T} \nabla \frac{\mu_k}{T} \geq 0. \quad (1.44)$$

Accordingly, some important points must be outlined:

- The first term appearing in the expression of the entropy source is related to the transport of energy, the second and third to the transport of momentum, and the last to the transport of mass.
- The second law of thermodynamics implies that $\kappa, \eta, \xi, \mathcal{L}_{k,l} \geq 0$.
- Transport phenomena are responsible for the internal production of entropy (*cf.* remark below).
- If the system is at equilibrium, the entropy flux density and source vanish so that $s = c^{\text{st}}$ by equation (1.37).

It now becomes clearer that transport phenomena are often associated to dissipative processes.

An important remark. It is clear from the form of the entropy production (1.44) that heat and momentum transport are dissipative processes. However, there also exist non-dissipative transport phenomena as found in the quantum Hall effect [6], spin transport amongst semiconductors [7], chiral magnetic effect [8], as well as in some symmetry classes of crystals and liquid crystals [9] (anisotropic systems).

1.4. PHENOMENOLOGICAL LAWS

Some vocabulary. Let an isolated system consist of two subsystems. Let $\{A_i\}$ be a set of extensive and globally conserved quantities taking values A_i^1 and A_i^2 in both subsystems such that

$$A_i^{\text{tot}} = A_i^1 + A_i^2 = c^{\text{st}}. \quad (1.45)$$

From the perspective of the two subsystems, the A_i^1 and A_i^2 are seen as intensive quantities. The extensive property of entropy implies that

$$S_{\text{tot}} = S^1(\{A_i^1\}) + S^2(\{A_i^2\}). \quad (1.46)$$

⁷The Onsager-Casimir reciprocal relation will be discussed in the fifth chapter.

The local equilibrium value of A_i^1 and A_i^2 satisfies the condition

$$\left. \frac{\partial S_{\text{tot}}}{\partial A_i^1} \right|_{A_i^{\text{tot}}} = 0 \Leftrightarrow \frac{\partial S^1}{\partial A_i^1} - \frac{\partial S^2}{\partial A_i^2} = 0. \quad (1.47)$$

If one defines $\gamma_i^1 \equiv \frac{\partial S^1}{\partial A_i^1}$, which is an intensive quantity, it is clear that the difference $X_i \equiv \gamma_i^1 - \gamma_i^2$ controls how far from equilibrium the system is. In particular, if $X_i = 0$ the system is found at equilibrium. It is tempting to compare X_i as a *force*: it is usually called *thermodynamic force* or *affinity*. γ_i^1 is the *thermodynamic conjugate variable* to A_i^1 .

Such considerations can be extended to the case where the system is divided into a large number of subsystems, and by extension to the continuous case. In the latter, the distance from equilibrium will be controlled by the gradients of the conjugate variables, *i.e.* $\nabla \gamma_i(\mathbf{r}, t)$.

General form of entropy production. In equation (1.39), the general form of the entropy production was obtained with the use of the local equilibrium hypothesis. In (1.44), the entropy production was obtained with the use of the Fourier law for heat conduction, the explicit expression of the pressure tensor and the Fick law for diffusion. Such results are a direct consequence of the specific entropy differential (B.13). Let us generalise the form of this differential expression as follows,

$$ds = \gamma_i da_i \quad \text{or} \quad d(\rho s) = \gamma_i d(\rho a_i) \quad (1.48)$$

with $\gamma_i = \left(\frac{\partial S}{\partial a_i} \right)_{a_1, \dots, a_{i-1}, a_{i+1}, \dots}$ and where the generalised Euler equation $s = \gamma_i a_i$ was used to obtain the differential of the entropy density. By pure analogy, it is reasonable to expect the entropy flux density to be of the form [10]

$$J_s = \gamma_i J_i, \quad (1.49)$$

J_i being the flux density of the globally conserved quantities A_i . At the same time, the A_i quantities satisfy local conservation equations

$$\partial_t(\rho a_i) + \nabla \cdot J_i = 0. \quad (1.50)$$

The generalised entropy differential implies that

$$\partial_t(\rho s) = \gamma_i \partial_t(\rho a_i) \quad (1.51)$$

and in parallel,

$$\nabla \cdot J_s = \nabla \gamma_i \cdot J_i + \gamma_i \nabla \cdot J_i \quad (1.52)$$

so that the entropy production can be written as

$$\sigma_s = \nabla \gamma_i \cdot J_i. \quad (1.53)$$

This result is remarkable. Under the hypothesis of local equilibrium, the entropy production can always be written as the product of the affinities and the corresponding fluxes densities. The

affinities act as thermodynamic forces producing a flux of a_i . This equation is to be compared to what was obtained at (1.39) which can be written in a more condensed form

$$\sigma_s = \mathbf{q} \cdot \nabla \left(\frac{1}{T} \right) - \Pi : \frac{\nabla \mathbf{u}}{T} - \sum_{k=1}^c \mathbf{j}_k \cdot \nabla \frac{\mu_k}{T}. \quad (1.54)$$

The correspondence is clear: the affinity associated to the specific energy e is $\nabla \left(\frac{1}{T} \right)$ and is responsible for the production of an heat flux $\mathbf{q}$. The affinity associated to the momentum density is $\frac{\nabla \mathbf{u}}{T}$ and is responsible for the production of a momentum flux Π . The affinity associated to the mass fraction is $\nabla \frac{\mu_k}{T}$ and is responsible for the production of a particle flux. We notice that these affinities have different tensorial order. If one writes the affinity $X_i \equiv \nabla \gamma_i$, the entropy production takes the final form

$$\sigma_s = X_i j_i \quad (1.55)$$

with j_i and X_i being quantities of same tensorial order to produce a scalar under contraction.

Phenomenological laws. A general pattern seems to emerge: the affinities act as perturbations to the equilibrium configuration of a system. A system subjected to a non-vanishing affinity will immediately respond to it by the production of fluxes which tend to re-establish equilibrium. These fluxes vanish at the same time as the affinities. A central question is to understand how the fluxes depend on the affinities. The preliminary study in §1.1 suggested that the fluxes may depend on all the affinities, not only on the conjugate ones (cross effects). Thus, the flux j_i at point $\mathbf{r}$ and time t can be written as

$$j_i(\mathbf{r}, t) = j_i[X_1(\mathbf{r}, t), X_2(\mathbf{r}, t), \dots]. \quad (1.56)$$

Intuitively, we may expect these fluxes to be complex and highly non-linear functions of the X_i . Expanding them in Taylor series around the equilibrium yields ($\mathbf{r}$ and t dependence left)

$$j_i = j_i[X_{i,\text{eq}}] + \left(\frac{\partial j_i}{\partial X_j} \right)_{\text{eq}; X_1, \dots, X_{j-1}, X_{j+1}, \dots} (X_j - X_{j,\text{eq}}) + \dots \quad (1.57)$$

Because $X_{i,\text{eq}} = 0$ and $j_i[X_{i,\text{eq}}] = 0$, and if higher order terms are neglected,

$$j_i \simeq \mathcal{L}_{ij} X_j \quad (1.58)$$

with the $\mathcal{L}_{ij} \equiv \left(\frac{\partial j_i}{\partial X_j} \right)_{\text{eq}; X_1, \dots, X_{j-1}, X_{j+1}, \dots}$. The $\mathcal{L}_{ij}$ are referred to as the linear response coefficients and relations (1.58) as the phenomenological equations.

Some comments:

- What we learn here is that near equilibrium, the relations between the affinities and the fluxes are linear. This is justified as long as the mean free path of the particles in matter is shorter than the characteristic length associated with the gradients.

- The matrix $(\mathcal{L})_{ij} = \mathcal{L}_{ij}$ containing the linear response coefficients not necessarily is diagonal, illustrating the existence of cross phenomena for which a given flux may depend on several affinities of different thermodynamic nature.
- The laws presented in §1.1 can all be cast in the form (1.58), given that the transport coefficients are related to the linear response coefficients. For instance, the heat conductivity is given by $\kappa = \frac{\mathcal{L}_{ee}}{T^2}$.
- The entropy production (1.55) can thus be written as the quadratic expression

$$\sigma_s = \mathcal{L}_{ij} X_i X_j \geq 0. \quad (1.59)$$

As $X_i X_j$ is a symmetric tensor, only the symmetric part of the matrix $(\mathcal{L})_{ij} = \mathcal{L}_{ij}$ contributes. A sufficient condition for the quadratic form to be positive definite is that $\mathcal{L}_{ij} + \mathcal{L}_{ji} \geq 0$. Meaning that in the context of this work, $\kappa, \eta, \xi, D_{kl} \geq 0$ as previously obtained.

- The reader may ask why in §1.1 the transport coefficients are scalars while they appear as tensors in equation (1.58). This is related to the symmetries of the system. In an isotropic system, one shows that the linear response coefficients reduce to scalars. Their tensorial nature may subsist in anisotropic systems though. In addition, in a homogeneous system, they no longer are functions of $\mathbf{r}$ and t .

1.5. EXPLICIT IRREVERSIBILITY FOR A ONE-COMPONENT FLUID

In §1.2, local conservation equations for the mass (1.10), the momentum (1.22) and the internal energy (1.27) densities were obtained. This set of equations is not closed. Indeed, there is *a priori* no further knowledge on the form of $p(\mathbf{x}, T)$, $\Pi(\mathbf{x}, t)$, $\mathbf{u}(\mathbf{x}, t)$ and $\mathbf{q}(\mathbf{x}, t)$. This is a major difficulty that can be avoided by supplementing the set by the equations of state,

$$\begin{cases} p = p(\rho, T) \\ e = e(\rho, T) \end{cases} \quad (1.60)$$

where ρ and T are themselves functions of $\mathbf{x}$ and t . This closure operation can be seen as phenomenological and its validity relies on the basis of experimental evidences as well on the final equations obtained with such method [11].

Let us start with the local conservation equation for the momentum density (1.22). The second term on the left-hand side can be explicitly developed as follows:

$$\nabla \cdot (\rho \mathbf{u} \mathbf{u}) = (\nabla \rho \cdot \mathbf{u}) \mathbf{u} + \rho (\nabla \cdot \mathbf{u}) \mathbf{u} + \rho (\mathbf{u} \cdot \nabla) \mathbf{u} \quad (1.61)$$

$$\nabla p = \left(\frac{\partial p}{\partial T} \right)_\rho \nabla T + \left(\frac{\partial p}{\partial \rho} \right)_T \nabla \rho \quad (1.62)$$

$$\nabla \cdot \Pi = -\eta \nabla^2 \mathbf{u} - \left(\frac{\eta}{3} + \xi \right) \nabla (\nabla \cdot \mathbf{u}). \quad (1.63)$$

Combining all and assuming constant transport coefficients ($\nabla\kappa = \nabla\eta = \nabla\xi = 0$)⁸ with the use of (1.10), equation (1.22) thus becomes

$$\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \left(\frac{\partial p}{\partial T} \right)_\rho \nabla T - \frac{1}{\rho} \left(\frac{\partial p}{\partial \rho} \right)_T \nabla \rho + \frac{\eta}{\rho} \nabla^2 \mathbf{u} + \frac{\eta/3 + \xi}{\rho} \nabla(\nabla \cdot \mathbf{u}) \quad (1.64)$$

which are the familiar Navier-Stokes equations. Now, performing similar calculations with the local balance equation for the internal energy (1.27) yields,

$$\begin{aligned} \partial_t(\rho e) &= e \partial_t \rho + \rho \partial_t e \\ &= -e \nabla \cdot (\rho \mathbf{u}) - \rho \mathbf{u} \cdot \nabla e + \kappa \nabla^2 T - p \nabla \cdot \mathbf{u} - \Pi : \nabla \mathbf{u} \end{aligned} \quad (1.65)$$

where the Fourier law for heat conduction (1.1) was used. With a bit of algebra and remembering that $c_V \equiv \left(\frac{\partial e}{\partial T} \right)_\rho$ is the specific heat at constant volume,

$$\begin{aligned} \rho \partial_t e &= -\rho \mathbf{u} \cdot \nabla e + \kappa \nabla^2 T - p \nabla \cdot \mathbf{u} - \Pi : \nabla \mathbf{u} \\ c_V \partial_t T + \left(\frac{\partial e}{\partial \rho} \right)_T \partial_t \rho &= -\mathbf{u} \cdot \nabla e + \frac{\kappa}{\rho} \nabla^2 T - \frac{p}{\rho} \nabla \cdot \mathbf{u} - \frac{1}{\rho} \Pi : \nabla \mathbf{u} \\ c_V \partial_t T - \left(\frac{\partial e}{\partial \rho} \right)_T \rho \nabla \cdot \mathbf{u} &= -c_V (\mathbf{u} \cdot \nabla) T + \frac{\kappa}{\rho} \nabla^2 T - \frac{p}{\rho} \nabla \cdot \mathbf{u} - \frac{1}{\rho} \Pi : \nabla \mathbf{u} \\ \partial_t T &= \frac{\rho}{c_V} \left(\frac{\partial e}{\partial \rho} \right)_T \nabla \cdot \mathbf{u} - (\mathbf{u} \cdot \nabla) T + \frac{\kappa}{\rho c_V} \nabla^2 T - \frac{p}{\rho c_V} \nabla \cdot \mathbf{u} - \frac{1}{\rho c_V} \Pi : \nabla \mathbf{u}. \end{aligned} \quad (1.66)$$

With the use of (1.40),

$$\partial_t T + (\mathbf{u} \cdot \nabla) T = \frac{\rho}{c_V} \left(\frac{\partial e}{\partial \rho} \right)_T \nabla \cdot \mathbf{u} + \frac{\kappa}{\rho c_V} \nabla^2 T - \frac{p}{\rho c_V} \nabla \cdot \mathbf{u} + \frac{\eta}{2\rho c_V} (\partial_i u_j + \partial_j u_i - \frac{2}{3} (\nabla \cdot \mathbf{u}) \delta_{ij})^2 + \frac{\xi}{\rho c_V} (\nabla \cdot \mathbf{u})^2 \quad (1.67)$$

which is the familiar heat equation expressed in a moving medium. The set consisting of equations (1.10), (1.22), (1.27), (1.64) and (1.67) is now closed.

An interesting conclusion can be put forward by applying a time-inversion T such that

$$T : \begin{cases} t \rightarrow -t \\ \mathbf{u} \rightarrow -\mathbf{u} \end{cases} \quad (1.68)$$

to these equations. The reversibility is broken by the terms containing transport coefficients (dissipative terms). Hence, the hydrodynamic equations are not invariant under time inversion and therefore describe irreversible process. We note that the continuity equation possesses no dissipative part.

⁸In general, this is not true, at least for large scale systems. For instance, the shear viscosity is expected to depend on the pressure and the temperature, $\eta = \eta(p, T)$. Hence,

$$\nabla \eta = \left(\frac{\partial \eta}{\partial p} \right)_T \nabla p + \left(\frac{\partial \eta}{\partial T} \right)_p \nabla T.$$

Whenever the variation of the shear viscosity with the pressure and the temperature is small, it is reasonable to assume that $\nabla \eta \approx 0$. This would not be correct for large scale systems, for example the atmosphere, where the pressure variation can be very large. Furthermore, in heterogeneous systems, an explicit spatial dependence of η is expected, $\eta = \eta(p, T, \mathbf{x})$. Similar reasoning applies to the diffusion coefficient which is in general a function of the density and the temperature, $D = D(n, T)$. For dilute solutions with small temperature variation, $\nabla D \approx 0$.

CHAPTER 2

QUANTUM MECHANICS & MICROSCOPIC DYNAMICS

“Le savant n’étudie pas la nature parce que cela est utile ; il l’étudie parce qu’il y prend plaisir et il y prend plaisir parce qu’elle est belle. [...] je veux parler de cette beauté plus intime qui vient de l’ordre harmonieux des parties, et qu’une intelligence pure peut saisir.”

HENRI POINCARÉ¹

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This chapter aims at defining the framework and the concepts of this work. The objective is also to go through the basic definitions of the different concepts that will be used.

¹Suggested translation: “The scientist does not study nature because it is useful to do so; he studies it because he takes pleasure in it, and he takes pleasure in it because it is beautiful. [...] what I mean is that more intimate beauty which comes from the harmonious order of its parts, and which a pure intelligence can grasp.” From H. Poincaré. *Science et méthode*. Edition définitive. Paris: E. Flammarion, 1908.

The first chapter proposed to approach the notion of transport phenomena from a macroscopic perspective. The point of view of this chapter fundamentally changes and moves on to a microscopic description.

The following questions motivate the content of this chapter:

- What kinds of systems are we considering here?
- What are the tools needed for a microscopic description of the system? At what fundamental level is the system described?
- Is it possible to construct local conservation equations that take into account the microscopic dynamics?

In §2.1, the system that will be studied throughout this work will be properly defined. In §2.2, all the tools needed for a microscopic description are introduced as well as the fundamental equations. Here, the author wished to be concise because this part could prove to be elementary for some readers, although central for the rest of this work. §2.3 introduces the Hamiltonian operator describing the system of interest, allowing to identify the symmetries of the system and the related conserved quantities. In the first chapter, the notion of local conservation equation was introduced. Those conservation equations were understood from a macroscopic point of view. In §2.4, we will move on to a microscopic description and we will attempt to write conservation laws at this scale, for instance conservation equations for the mass, particle, momentum and energy densities. Finally, in §2.5 an even more fundamental approach on the basis of quantum field theory is proposed. In particular, the connection with the results obtained in the preceding section is highlighted.

2.1. DESCRIPTION OF THE SYSTEM

This section is devoted to the description of the system that will be studied in this work.

We consider a normal fluid composed of N particles with c different species and confined in a finite volume V . The mass of a particle of species k is m_k , $k = 1, \dots, c$. The number of particles of each species is N_k so that $N = \sum_{k=1}^c N_k$. The system is assumed to be isolated and thus does not exchange energy and matter with the external environment.

A normal fluid can be seen as a system composed of many atoms and molecules interacting together by forces of electrodynamic origin. The temperature is sufficiently high so that the system is not solid but liquid or gaseous and can thus undergo the typical flowing motion of a fluid. It is assumed isotropic, hypothesis that will be important in the context of this work. It possesses well-defined thermodynamic properties: located far enough from phase transitions, the thermodynamic variables are well defined and do not diverge. The N particles move near local thermodynamic equilibrium which is satisfied at least on small length scales. A normal fluid can also be defined by what it is not. At low temperature, phase transition may occur

from normal fluids to liquid crystals or crystalline solids in which some continuous symmetries may be broken. From this point of view, a normal fluid neither is a crystalline solid, nor a liquid crystal. Normal fluids are described as Newtonian fluids. Consequently, some mixtures of polymers or non-Newtonian fluids cannot be regarded as normal fluids.

Attention will here be given to nonrelativistic normal quantum fluids, which are fluids having the same properties as ordinary and classical normal fluids, but for which the transport coefficients should be determined by quantum mechanics. The nonrelativistic property is satisfied as long as the temperature T of the fluid is much smaller than the rest mass m_k of a particle of species k ,

$$k_B T \ll m_k c^2, \quad k = 1, 2, \dots, c. \quad (2.1)$$

where c in the left-hand side of the inequality denotes the speed of light.

The role of quantum mechanics can be assessed by considering the de Broglie wavelength $\lambda = h/p$ of the particles composing matter with h the Planck constant. At room temperature T , the linear momentum of a particle can be evaluated as $p \simeq \sqrt{mk_B T}$ in terms of the mass m of the particle and the Boltzmann constant k_B . The frontier between the classical and the quantum regimes can be assessed by comparing the de Broglie wavelength to the distance between particles. For a system of N particles and finite volume V , the ratio $(V/N)^{1/3} = n^{-1/3}$ is roughly the distance between particles. The classical regime applies as long as $\lambda \ll n^{-1/3}$. Quantum effects begin to appear once $\lambda \leq n^{-1/3}$ and dominate when $\lambda \geq n^{-1/3}$ as in the case of superfluid systems. Electrons being several thousand times lighter than the nuclei are found in deeply quantum regimes. Indeed, they form the atomic and molecular orbitals, which determine the interatomic forces between the nuclei. Instead, the nuclei have a de Broglie wavelength typically shorter than the mean distance between next neighbouring nuclei. In this regard, the motion of the nuclei could be ruled by classical mechanics. Nevertheless, the quantum effects can be important, for instance, for the contribution of molecular vibrations to the heat capacity. Therefore, the microscopic theory of transport properties should be based on quantum mechanics, even for matter at room temperature. The validity of this work ranges in between the classical to the slightly quantum regime.

The fundamental constituents of the system will be considered as point particles infinitely localised at a given spatial coordinate. The quantum machinery is introduced throughout the so-called first quantization, or canonical quantization, procedure. This approach is perhaps the most natural as the fundamental constituents of the normal fluid are particles. In §2.2, the axiomatic procedure of first quantization will be reviewed, though very briefly. An approach based on the method of nonrelativistic field theory (or second quantization) will be explored in §2.5 and shown to coincide with the various results obtained in the context of first quantization. We note that field theory is required for the study of superfluidity and superconductivity.

We insist on the point that the framework that will be presented is restricted to normal quantum fluids and does not extend to superfluids. For instance, a superfluid is characterised

by two velocities: one associated to the normal flow and the other to the superfluid flow. The superfluid flow has the property to flow without viscosity, in contrast to what happens in normal quantum fluids. Furthermore, a superfluid exhibits additional transport coefficients responsible for dissipation compared to the normal fluid case [2].

2.2. QUANTUM MECHANICS TOOLBOX

It is not a question here of reformulating quantum mechanics. The aim is to review the different steps of the first quantization procedure as well as to define the different objects that will be used throughout this work.

First quantization. In classical mechanics, the state of a system composed of a single particle (for simplicity) is described by the knowledge of its position $\mathbf{r}$ and momentum $\mathbf{p}$. The $\mathbf{r}$ and $\mathbf{p}$ variables are canonical variables as they satisfy the following Poisson bracket relations

$$\{r_i, r_j\} = 0, \quad \{p_i, p_j\} = 0 \quad \text{and} \quad \{r_i, p_j\} = \delta_{ij}. \quad (2.2)$$

The state space, *i.e.* the phase space, of the system consists of all the positions and momenta of that single particle. Classical dynamical quantities are represented by functions of the canonical variables (and eventually t).

In quantum mechanics, the formulation is significantly different. The state space is now a state space $\mathcal{H}$ defined as a complex and complete Euclidean space, which has infinite dimensionality. The state of a system is described by vectors of $\mathcal{H}$. Next, linear operators $\hat{A} : \mathcal{H} \rightarrow \mathcal{H}$ can be defined. The two most important linear operators are the position $\hat{\mathbf{r}}$ and momentum $\hat{\mathbf{p}}$ operators. They satisfy the following commutation relations

$$[\hat{r}_i, \hat{r}_j] = 0, \quad [\hat{p}_i, \hat{p}_j] = 0 \quad \text{and} \quad [\hat{r}_i, \hat{p}_j] = i\hbar \delta_{ij}, \quad (2.3)$$

the latter being the canonical commutation relation expressed in terms of the reduced Planck constant $\hbar$. Any physical observable is represented by a Hermitian operator. The position and momentum operators are Hermitian, *i.e.* $\hat{\mathbf{r}}^\dagger = \hat{\mathbf{r}}$ and $\hat{\mathbf{p}}^\dagger = \hat{\mathbf{p}}$ and are thus observables. If one denotes their eigenvectors as $|\mathbf{r}\rangle$ and $|\mathbf{p}\rangle$ such that

$$\hat{\mathbf{r}}|\mathbf{r}\rangle = \mathbf{r}|\mathbf{r}\rangle \quad (2.4)$$

$$\hat{\mathbf{p}}|\mathbf{p}\rangle = \mathbf{p}|\mathbf{p}\rangle, \quad (2.5)$$

then $\{|\mathbf{r}\rangle\}$ and $\{|\mathbf{p}\rangle\}$ define two possible basis of $\mathcal{H}$, respectively the position (giving rise to wave mechanics) and the momentum representation. The former representation will be used in this work.

In classical mechanics, the phase space of a multi-particle system is naturally the space of all the positions and momenta of all the particles.

In quantum mechanics, the approach is more subtle. Let $\mathcal{H}$ be a single particle Hilbert space. Then, the Hilbert space of a N -particle system is $\mathcal{H}_N = \otimes_{i=1}^N \mathcal{H}$. More generally, if the system consists of $k = 1, 2, \dots, c$ species, each consisting of N_k particles, then the Hilbert space of the total system is

$$\mathcal{H}_N = (\mathcal{H}^1)^{\otimes N_1} \otimes \dots \otimes (\mathcal{H}^c)^{\otimes N_c} \quad (2.6)$$

where $\mathcal{H}^k$ denotes the Hilbert space associated to the species k . $\otimes$ must be taken to be the symmetric tensor product for bosonic particles and the antisymmetric tensor product for fermionic particles. Such decomposition of the total Hilbert space allows to consider mixture of bosonic and fermionic particles as long as the tensor product is carefully chosen.

The first quantization procedure is summarised as follows:

- An adequate Hilbert space $\mathcal{H}$ is defined.
- To the classical position $\mathbf{r}$ and momentum $\mathbf{p}$ of the particle are associated the observables $\hat{\mathbf{r}}$ and $\hat{\mathbf{p}}$ satisfying the canonical commutation relations (2.3).
- Any observable describing a classical physical quantity is obtained, in the suitable Hermitian form, by replacing the canonical variables $\mathbf{r}$ and $\mathbf{p}$ by the observables $\hat{\mathbf{r}}$ and $\hat{\mathbf{p}}$ respectively.

Position representation. Let $|\psi\rangle \in \mathcal{H}$ be an arbitrary vector. In the position representation, the vector $|\psi\rangle$ is represented by its components in the $\{|\mathbf{r}\rangle\}$ basis, $\langle \mathbf{r} | \psi \rangle \equiv \psi(\mathbf{r})$ where $\psi(\mathbf{r})$ is the wave function of the system. The Hilbert space corresponding to the position representation is $\mathcal{L}^2(\mathbb{R}^3, d\mathbf{r})$, i.e. the space of the square-integrable functions in $\mathbb{R}^3$ defined as

$$\psi \in \mathcal{L}^2(\mathbb{R}^3, d\mathbf{r}) \Leftrightarrow \langle f | f \rangle \text{ exists and is finite} \quad (2.7)$$

with the inner product

$$\langle f | g \rangle = \int_{\mathbb{R}^3} d\mathbf{r} f^*(\mathbf{r}) g(\mathbf{r}). \quad (2.8)$$

In the position representation, the position operator becomes the physical position $\mathbf{r}$, while the momentum operator becomes $-i\hbar\nabla$, both acting over wave functions.

Let us now consider a N -particle system. The Hilbert space becomes $\mathcal{H}_N = \otimes_{i=1}^N \mathcal{L}^2(\mathbb{R}^3, d\mathbf{r}) = \mathcal{L}^2(\mathbb{R}^{3N}, d^N\mathbf{r})$ and a basis of $\mathcal{H}_N$ is $\{|\mathbf{r}_1\rangle \otimes |\mathbf{r}_2\rangle \otimes \dots \otimes |\mathbf{r}_N\rangle \equiv |\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N\rangle\}$ where each vector $|\mathbf{r}_i\rangle$ is an eigenvector of the position operator $\hat{\mathbf{r}}_i$ corresponding to the particle labelled i . An arbitrary vector $|\psi\rangle$ is now represented by its component $\psi(\mathbf{r}_1, \dots, \mathbf{r}_N) \equiv \psi(\mathbf{r}^N)$. The momentum operator $\hat{\mathbf{p}}_i$ becomes $-i\hbar\nabla_{\mathbf{r}_i}$. The general prescription to obtain the action of an operator $\hat{A}$ in the position representation is to calculate the matrix element $\langle \mathbf{r}_1, \dots, \mathbf{r}_N | \hat{A} | \psi \rangle$ for an arbitrary vector $|\psi\rangle$. If $\hat{A}_i$ is an operator acting only over the Hilbert space of the particle labelled i , it can always be extended to the full Hilbert space by setting $\hat{A} = \mathbf{1} \otimes \dots \otimes \hat{A}_i \otimes \dots \otimes \mathbf{1}$.

Heisenberg picture. A fundamental issue is to understand the time evolution of an arbitrary vector of $\mathcal{H}$. The time evolution of the vector $|\psi_t\rangle$ is given by the Schrödinger equation,

$$i\hbar \partial_t |\psi_t\rangle = \hat{H} |\psi_t\rangle. \quad (2.9)$$

in terms of the Hamiltonian operator $\hat{H}$. That corresponds to the Schrödinger picture of quantum mechanics: the evolution of the system is contained in the vector $|\psi_t\rangle$, while the observables, such as the position and the momentum observables, do not depend on time and remain unaffected by the time evolution. If the Hamiltonian operator is not an explicit function of time (*i.e.* no time-dependent external forces), the solution of the Schrödinger equation can be written as

$$|\psi_t\rangle = \hat{U}(t, t_0) |\psi_{t=0}\rangle \quad (2.10)$$

where $\hat{U}(t, t_0) \equiv e^{-\frac{i}{\hbar} \hat{H}(t-t_0)}$ is the evolution operator. Since the Hamiltonian operator is not an explicit function of time, it does not contribute to the time evolution of $|\psi_t\rangle$.²

Now, let us consider an observable $\hat{A}$ that is not an explicit function of the time t and let us construct the observable $\hat{A}_t$ such that

$$\hat{A}_t \equiv \hat{U}(t, t_0)^\dagger \hat{A} \hat{U}(t, t_0). \quad (2.11)$$

$\hat{A}_t$ now depends on time. Its time evolution can easily be found to be

$$i\hbar \partial_t \hat{A}_t = [\hat{A}_t, \hat{H}_t] \quad (2.12)$$

using that $\partial_t \hat{U}(t, t_0) = -\frac{i}{\hbar} \hat{H} \hat{U}(t, t_0)$. That corresponds to the Heisenberg picture: states are fixed at their initial condition and observables evolve in time. The correspondence between the two pictures can be made clearer by the following relation,

$$\underbrace{\langle \psi_t | \hat{A} | \psi_t \rangle}_{\text{S. p.}} = \underbrace{\langle \psi_{t=0} | \hat{A}_t | \psi_{t=0} \rangle}_{\text{H. p.}} \quad (2.13)$$

where S. p. stands for the Schrödinger picture and H. p. for the Heisenberg picture. As the Hamiltonian operator commutes with $\hat{U}(t, t_0)$, it takes the same form in the Schrödinger and Heisenberg pictures, $\hat{H}_t = \hat{H}$.

Quantum statistical mechanics. A quantum theory of statistical mechanics exists just as there is a classical theory of statistical mechanics. Quantum statistical mechanics occupies a prominent role in this work.

The very starting point of classical statistical mechanics is to assume that the state of a given system is not known for sure, but is instead distributed according some probability distribution. Consequently, statistics are expected to enter in quantum statistical mechanics in two ways:

²In the sense that it is $\hat{H}t$ that contributes to the time evolution.

first inherently to quantum mechanics and also because of the impossibility of knowing the state of the system with certainty.

Let a quantum system be described by a Hilbert space $\mathcal{H}$. Let us assume that the state of the system is $|\psi_1\rangle$ with probability p_1 , $|\psi_2\rangle$ with probability p_2 , *etc.* The probabilities naturally satisfy $\sum_i p_i = 1$. All states are normalised to unity, *i.e.* $\langle\psi_i|\psi_i\rangle = 1$. The statistical operator (or density operator) $\hat{\rho}$ is defined by

$$\hat{\rho} = \sum_i |\psi_i\rangle p_i \langle\psi_i| \quad (2.14)$$

and its eigenvectors are thus the states $|\psi_i\rangle$ with corresponding eigenvalues p_i . The average value of an observable $\hat{A}$ is given by

$$\langle\hat{A}\rangle = \text{Tr } \hat{\rho}\hat{A}. \quad (2.15)$$

It is indeed elementary to show that

$$\text{Tr } \hat{\rho}\hat{A} = \sum_i p_i \langle\psi_i|\hat{A}|\psi_i\rangle. \quad (2.16)$$

The statistical operator (2.14) satisfies the following properties, naturally arising from its definition

$$\begin{cases} \text{Tr } \hat{\rho} = 1 \\ \hat{\rho}^\dagger = \hat{\rho} \\ \hat{\rho} \text{ is positive semi-definite, i.e. eigenvalues } \geq 0. \end{cases} \quad (2.17)$$

The time evolution of the states $|\psi_i\rangle$ is given by the Schrödinger equation (2.9). Accordingly, the statistical operator (2.14) satisfies the equation

$$i\hbar \partial_t \hat{\rho}_t = [\hat{H}, \hat{\rho}_t] \quad (2.18)$$

called the von Neumann equation. It can also be written in the form

$$\partial_t \hat{\rho}_t = -\hat{\mathcal{L}} \hat{\rho}_t \quad (2.19)$$

in terms of the superoperator $\hat{\mathcal{L}} \cdot \equiv \frac{i}{\hbar} [\hat{H}, \cdot]$. The von Neumann equation is not the Heisenberg equation (2.12) for the statistical operator due to a crucial sign difference. In the Schrödinger picture, it is the states that evolve in time, and thus the statistical operator. In the Heisenberg picture, it is the observables that evolve in time, and thus the statistical operator is kept fixed. The correspondence between both pictures is established as follows. Let $\hat{A}$ be an observable. Then,

$$\begin{aligned} \langle\hat{A}\rangle_t &= \text{Tr } \hat{\rho}_t \hat{A} = \underbrace{\text{Tr } \hat{U}(t) \hat{\rho}_0 \hat{U}^\dagger(t) \hat{A}}_{\text{S. p.}} \\ &= \text{Tr } \hat{\rho}_0 \hat{U}(t)^\dagger \hat{A} \hat{U}(t) = \underbrace{\text{Tr } \hat{\rho}_0 \hat{A}_t}_{\text{H. p.}} \end{aligned} \quad (2.20)$$

where $\hat{U}(t) \equiv \hat{U}(t, t_0 = 0)$ to lighten notations and where the cyclic property of the trace was used. If the Hamiltonian does not explicitly depend on time, the von Neumann equation can be formally integrated

$$\hat{\rho}_t = e^{-\hat{\mathcal{L}}t} \hat{\rho}_0. \quad (2.21)$$

Let $\hat{B}$ be an operator. We have [12],

$$e^{[\hat{B}, \cdot]} = e^{\hat{B}} \cdot e^{-\hat{B}} \quad (2.22)$$

allowing to express (2.21) in the form

$$\hat{\rho}_t = e^{-\frac{i}{\hbar} \hat{H}t} \hat{\rho}_{t=0} e^{\frac{i}{\hbar} \hat{H}t}. \quad (2.23)$$

To better understand the role of the statistical operator, let us consider the grand-canonical ensemble. This ensemble is the statistical ensemble that is used to represent an open system of volume V at equilibrium that exchanges matter and energy with the external environment at fixed temperature T and fixed chemical potentials μ_k , $k = 1, 2, \dots, c$. A given state of the system is fully determined by its energy E and the number of particles N_k , $N = \sum_{k=1}^c N_k$, of each species it contains, the former depending on the latter. Let us assume $\hat{H}^{(N)}$ to be the Hamiltonian of the system and let $\{|n, j, N\rangle\}$ be a basis of the corresponding Hilbert space such that

$$\hat{H}^{(N)} |n, j, N\rangle = E_{n,j}^{(N)} |n, j, N\rangle \quad (2.24)$$

where n refers to the energy level and j the multiplicity of each level. The Hamiltonian is taken at fixed number of particles N (hence the superscript³). Indeed, for the Hamiltonian to have a discrete spectrum, the system has to be compactified, that is to say be momentarily closed. The probability for the system to be in the state $|n, j, N\rangle$ is

$$P_{nj}^{(N)} = \frac{1}{\Xi} e^{-\beta(E_{n,j}^{(N)} - \sum_{k=1}^c \mu_k N_k)} \quad (2.25)$$

where Ξ , the partition function of the grand canonical ensemble, and such that $\sum_{N=0}^{\infty} \sum_{n,j} P_{nj}^{(N)} = 1$. $\beta \equiv \frac{1}{k_B T}$ is the inverse temperature. Consequently, the statistical operator of the grand canonical ensemble is

$$\hat{\rho}_{\text{gc}} = \sum_{N=0}^{\infty} \sum_{n,j} |n, j, N\rangle P_{nj}^N \langle n, j, N| = e^{\beta(J - \hat{H} + \sum_{k=1}^c \mu_k \hat{N}_k)} \quad (2.26)$$

where $J = -k_B T \ln \Xi$ is the grand thermodynamic potential. In a local description, one would be interested in the particle $\hat{n}_k(\mathbf{r})$ and energy $\hat{\epsilon}(\mathbf{r})$ local densities defined such that

$$\hat{N}_k = \int d\mathbf{r} \hat{n}_k(\mathbf{r}) \quad (2.27)$$

$$\hat{H} = \int d\mathbf{r} \hat{\epsilon}(\mathbf{r}). \quad (2.28)$$

³In the following, the N superscript will be dropped, but recovered if needed.

The statistical operator of the grand canonical ensemble would then take the form

$$\hat{\rho}_{\text{gc}} = \exp \left\{ \beta \left(J - \int d\mathbf{r} \, \hat{\epsilon}(\mathbf{r}) + \int d\mathbf{r} \sum_{k=1}^c \mu_k \hat{n}_k(\mathbf{r}) \right) \right\}. \quad (2.29)$$

Such local expression of the statistical operator will play a fundamental role in the next chapter.

To any statistical operator $\hat{\rho}$ corresponds a statistical entropy S given by [13]

$$S = -k_B \text{Tr}(\hat{\rho} \ln \hat{\rho}), \quad (2.30)$$

and referred to as the von Neumann entropy. If the statistical operator is diagonal, the previous equation reduces to the Shannon entropy (though introduced after the von Neumann entropy was).

Time reversal in quantum mechanics. Time reversal is not as trivial in quantum mechanics as it is in the classical case. In the latter, one takes the transformation $t \rightarrow -t$ as in (1.68). In the former, it must be discussed in more detail.

Let $\hat{\Theta}$ denotes the time reversal operator. It is defined by its action over the position and momentum operators,

$$\hat{\Theta} \hat{r}_i \hat{\Theta}^{-1} = \hat{r}_i \quad (2.31)$$

$$\hat{\Theta} \hat{p}_i \hat{\Theta}^{-1} = -\hat{p}_i, \quad (2.32)$$

by pure analogy with the classical case. Hence, $\hat{\Theta}$ commutes with $\hat{r}_i$ and anticommutes with $\hat{p}_i$. Since the canonical commutation relation (2.3) must be preserved under the action of the time reversal operator, $\hat{\Theta}$ must be antiunitary, *i.e.* $\hat{\Theta}^\dagger = -\hat{\Theta}$. More generally, for any $|\psi_1\rangle, |\psi_2\rangle \in \mathcal{H}$ and $c_1, c_2 \in \mathbb{C}$,

$$\hat{\Theta} (c_1 |\psi_1\rangle + c_2 |\psi_2\rangle) = c_1^* \hat{\Theta} |\psi_1\rangle + c_2^* \hat{\Theta} |\psi_2\rangle. \quad (2.33)$$

Let us assume that the time-dependent family of kets $|\psi(t)\rangle$ with $t \in \mathbb{R}$ is a solution of the Schrödinger equation (2.9). Let $|\psi_r(t)\rangle \equiv \hat{\Theta} |\psi(-t)\rangle$ be the reversed state. Under the hypothesis that the time reversal operator commutes with the Hamiltonian operator, one has that

$$-i\hbar \partial_t \hat{\Theta} |\psi(t)\rangle = \hat{H} \hat{\Theta} |\psi(t)\rangle, \quad (2.34)$$

since $\hat{\Theta}$ commutes with ∂_t being just a derivative with respect to a time parameter that the state depends on. As $\partial_t |\psi(-t)\rangle = -\partial_t |\psi(-t)\rangle$, we see that the reversed state $|\psi_r(t)\rangle$ is a solution of the Schrödinger equation. Note that the two solutions $|\psi(t)\rangle$ and $|\psi_r(t)\rangle$ may or may not coincide, depending on whether their initial conditions coincide or not, up to a global phase. As a result of the antiunitary property of $\hat{\Theta}$, $\langle \mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N | \hat{\Theta} |\psi\rangle = \langle \mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N | \psi\rangle^*$, so that $\psi^*(\mathbf{r}, -t)$ is a solution of the Schrödinger equation in the representation basis. In that sense, the time reversal symmetry is not trivial in quantum mechanics.

2.3. HAMILTONIAN OPERATOR, SYMMETRIES & CONSERVED QUANTITIES

Hamiltonian operator. The very first step of any quantum description of a system starts by the introduction of an adequate Hamiltonian operator.

Let a system be composed of N nonrelativistic particles of masses $\{m_i\}_{i=1}^N$. The most general Hamiltonian operator for such a system is of the form,

$$\hat{H} = \sum_{i=1}^N \left(\frac{\hat{\mathbf{p}}_i^2}{2m_i} + V_{\text{ext}}(\hat{\mathbf{r}}_i, t) \right) + V(\hat{\mathbf{r}}_1, \dots, \hat{\mathbf{r}}_N, t). \quad (2.35)$$

The interaction potential is purely arbitrary and may contain binary, ternary or higher interactions. Spin-orbit terms are not considered. Every particle of the same species are identical and have the same mass. In this respect, their labels should be grouped together into sets $i \in \mathcal{S}_k$ such that $m_i = m_k$. The number of labels included in each set $\mathcal{S}_k$ is equal to the number N_k of particles of each species. The above Hamiltonian should thus be totally symmetric under the $N_k!$ permutations of the labels $i \in \mathcal{S}_k$. A summation over all the N particles can alternatively be written as $\sum_{k=1}^c \sum_{i \in \mathcal{S}_k}$.

Using several assumptions, the form of the Hamiltonian will be simplified:

Hypothesis 1. Because the system is isolated, it must somehow be contained in a vessel with solid walls, thus the presence of the external potential. However, if the dynamics is assumed to take place far from the walls, the external potential term can be dropped.

Hypothesis 2. Ternary or higher interaction terms are not considered. Only binary interactions are kept so that the potential term reduces to a two-body potential.

Hypothesis 3. The two-body potential is assumed to be independent of time and to be identical for every pair of particles of the same species, *i.e.* for $i \in \mathcal{S}_k$ and $j \in \mathcal{S}_l$, $V_{ij} \equiv V_{kl}$.

Hypothesis 4. The two-body potential is also assumed to depend on the distance between two particles, $r_{ij} \equiv \|\mathbf{r}_i - \mathbf{r}_j\|$.

With these hypothesis, it is clear that the Hamiltonian operator (2.35) reduces to

$$\hat{H} = \sum_{i=1}^N \frac{\hat{\mathbf{p}}_i^2}{2m_i} + \sum_{1 \leq i < j \leq N} V_{ij}(\|\mathbf{r}_i - \mathbf{r}_j\|) = \sum_{i=1}^N \frac{\hat{\mathbf{p}}_i^2}{2m_i} + \frac{1}{2} \sum_{i=1}^N \sum_{j(\neq i)}^N V_{ij}(r_{ij}). \quad (2.36)$$

Accordingly, with (2.31) and (2.32), $\hat{\Theta} \hat{H} \hat{\Theta}^{-1} = \hat{H}$ so that $[\hat{\Theta}, \hat{H}] = 0$.

Hamiltonian operator in the position representation. Previously, it was explained that $\{|\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N\rangle\}$ forms a basis of the N -particle Hilbert space that is here considered. The Hamiltonian operator (2.36) in the position representation can be obtained as follows. Let $|\psi\rangle$ be an arbitrary vector of $\mathcal{H}_N$. Then,

$$\langle \mathbf{r}_1, \dots, \mathbf{r}_N | \hat{H} | \psi \rangle = \sum_{i=1}^N \frac{1}{2m_i} \langle \mathbf{r}_1, \dots, \mathbf{r}_N | \hat{\mathbf{p}}_i^2 | \psi \rangle + \frac{1}{2} \sum_{i=1}^N \sum_{j(\neq i)}^N \langle \mathbf{r}_1, \dots, \mathbf{r}_N | V_{ij}(\hat{\mathbf{r}}_i, \hat{\mathbf{r}}_j) | \psi \rangle. \quad (2.37)$$

→ First term: we have

$$\begin{aligned}
 \hat{p}^2 &= \int d\mathbf{r} d\mathbf{p} |\mathbf{r}\rangle \langle \mathbf{r} | \hat{p}^2 | \mathbf{p}\rangle \langle \mathbf{p}| \\
 &= \frac{1}{(2\pi\hbar)^{3/2}} \int d\mathbf{r} d\mathbf{p} |\mathbf{r}\rangle \mathbf{p}^2 e^{\frac{i}{\hbar} \mathbf{p} \cdot \mathbf{r}} \langle \mathbf{p}| \\
 &= \frac{1}{(2\pi\hbar)^{3/2}} \int d\mathbf{r} d\mathbf{p} |\mathbf{r}\rangle (-\hbar^2 \nabla^2) e^{\frac{i}{\hbar} \mathbf{p} \cdot \mathbf{r}} \langle \mathbf{p}| \\
 &= \int d\mathbf{r} d\mathbf{p} |\mathbf{r}\rangle (-\hbar^2 \nabla^2) \langle \mathbf{r} | \mathbf{p}\rangle \langle \mathbf{p}| = \int d\mathbf{r} |\mathbf{r}\rangle (-\hbar^2 \nabla^2) \langle \mathbf{r}|, \tag{2.38}
 \end{aligned}$$

so that

$$\begin{aligned}
 \langle \mathbf{r} | \hat{p}^2 | \psi \rangle &= \int d\mathbf{r}' \langle \mathbf{r} | \mathbf{r}' \rangle (-\hbar^2 \nabla'^2) \langle \mathbf{r}' | \psi \rangle \\
 &= \int d\mathbf{r}' \delta(\mathbf{r} - \mathbf{r}') \{-\hbar^2 \nabla'^2 \psi(\mathbf{r}')\} = -\hbar^2 \nabla^2 \psi(\mathbf{r}). \tag{2.39}
 \end{aligned}$$

Then,

$$\begin{aligned}
 \langle \mathbf{r}_1, \dots, \mathbf{r}_i, \dots, \mathbf{r}_N | \hat{p}_i^2 | \psi \rangle &= \langle \mathbf{r}_1, \dots, \mathbf{r}_i, \dots, \mathbf{r}_N | \mathbf{1} \otimes \dots \otimes \hat{p}_i^2 \otimes \dots \otimes \mathbf{1} | \psi \rangle \\
 &= -\hbar^2 \nabla_{\mathbf{r}_i}^2 \langle \mathbf{r}_1, \dots, \mathbf{r}_i, \dots, \mathbf{r}_N | \psi \rangle \\
 &= -\hbar^2 \nabla_{\mathbf{r}_i}^2 \psi(\mathbf{r}^N). \tag{2.40}
 \end{aligned}$$

→ Second term: since

$$\hat{\mathbf{r}}_i | \mathbf{r}_i \rangle = \mathbf{r}_i | \mathbf{r}_i \rangle, \tag{2.41}$$

one has

$$\langle \mathbf{r}_1, \dots, \mathbf{r}_N | V_{ij}(\hat{\mathbf{r}}_i, \hat{\mathbf{r}}_j) | \psi \rangle = V_{ij}(\mathbf{r}_i, \mathbf{r}_j) \psi(\mathbf{r}^N). \tag{2.42}$$

Thus,

$$\langle \mathbf{r}_1, \dots, \mathbf{r}_N | \hat{H} | \psi \rangle = - \sum_{i=1}^N \frac{\hbar^2}{2m_i} \nabla_{\mathbf{r}_i}^2 \psi(\mathbf{r}^N) + \frac{1}{2} \sum_{i=1}^N \sum_{j(\neq i)}^N V_{ij}(\mathbf{r}_i, \mathbf{r}_j) \psi(\mathbf{r}^N). \tag{2.43}$$

The Hamiltonian operator in the position representation is given by

$$\hat{H} = - \sum_{i=1}^N \frac{\hbar^2}{2m_i} \nabla_{\mathbf{r}_i}^2 + \frac{1}{2} \sum_{i=1}^N \sum_{j(\neq i)}^N V_{ij}(\mathbf{r}_i, \mathbf{r}_j) = \sum_{i=1}^N \frac{\hat{\mathbf{p}}_i^2}{2m_i} + \frac{1}{2} \sum_{i=1}^N \sum_{j(\neq i)}^N V_{ij}(r_{ij}) \tag{2.44}$$

and acts over wave functions $\psi \in \mathcal{L}^2(\mathbb{R}^{3N}, d^N \mathbf{r})$. In order to make the following calculations clearer, the $-\hbar^2 \nabla_{\mathbf{r}_i}^2$ was replaced with $\hat{\mathbf{p}}_i^2$. In the last line, the hypothesis 4 on the form of the Hamiltonian was recovered. To lighten notations, $V_{ij}(r_{ij}) \equiv V_{ij}$. The advantage of working in position representation is to interpret $\mathbf{r}_i$ as the physical position of the particle labelled i .

Conserved quantities. The Hamiltonian (2.44) is independent of time, invariant under spatial translation and under rotations since the potential term depends only on r_{ij} . The total energy, momentum and orbital angular momentum of the system are thus expected to be conserved. Let us formalise this.

First, two consequences of the Heisenberg equation (2.12) are to be highlighted:

1. A time independent Hamiltonian $\hat{H}$ is always a conserved quantity. As stated previously, this property is interpreted by noting that the system is invariant under the continuous group of time translations. Indeed, two experiments performed at two different times must produce the same results if the Hamiltonian operator is independent of time. The total energy of the system is then conserved as a consequence of the invariance under time translations.
2. Any observable $\hat{A}$ that commutes with the Hamiltonian $[\hat{A}, \hat{H}] = 0$ is a conserved quantity.

We would like to show that the total momentum $\hat{\mathbf{P}} \equiv \sum_{i=1}^N \hat{\mathbf{p}}_i$ and the total orbital angular momentum $\hat{\mathbf{L}} \equiv \sum_{i=1}^N \mathbf{r}_i \times \hat{\mathbf{p}}_i$ are indeed conserved quantities.

Total momentum. To show that the total momentum is a conserved quantity, we must verify that $[\hat{\mathbf{P}}^a, \hat{H}] = 0$. We have

$$[\hat{\mathbf{P}}^a, \hat{H}] = \sum_{i=1}^N [\hat{\mathbf{p}}_i^a, \hat{H}] = \sum_{i,j=1}^N \frac{1}{2m_i} \underbrace{[\hat{\mathbf{p}}_i^a, \hat{\mathbf{p}}_j^2]}_{=0 \text{ (trivially)}} + \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \sum_{k(\neq j)}^N [\hat{\mathbf{p}}_i^a, \hat{V}_{jk}]. \quad (2.45)$$

In parallel, we have

$$[\hat{\mathbf{p}}_i^a, V_{ij}] = -i\hbar \partial_i^a V_{ij} = i\hbar F_{ij}^a \quad (2.46)$$

$$[\hat{\mathbf{p}}_k^a, V_{ij}] = -i\hbar \partial_i^a V_{ij} (\delta_{ik} - \delta_{jk}) = i\hbar F_{ij}^a (\delta_{ik} - \delta_{jk}) \quad (2.47)$$

where F_{ij} is the force exerted on the particle labelled i by the particle labelled j as a result of their binary interaction. Hence,

$$\begin{aligned} \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \sum_{k(\neq j)}^N [\hat{\mathbf{p}}_i^a, V_{jk}] &= \frac{i\hbar}{2} \sum_{i=1}^N \sum_{j=1}^N \sum_{k(\neq j)}^N F_{jk}^a \delta_{ji} - \frac{i\hbar}{2} \sum_{i=1}^N \sum_{j=1}^N \sum_{k(\neq j)}^N F_{jk}^a \delta_{ki} \\ &= \frac{i\hbar}{2} \sum_{i=1}^N \sum_{k(\neq i)}^N F_{ik}^a - \frac{i\hbar}{2} \sum_{j=1}^N \sum_{i(\neq j)}^N F_{ji}^a \\ &= \frac{i\hbar}{2} \sum_{i=1}^N \sum_{k(\neq i)}^N F_{ik}^a - \frac{i\hbar}{2} \sum_{i=1}^N \sum_{j(\neq i)}^N F_{ij}^a = 0 \end{aligned} \quad (2.48)$$

where in the third line the i and j labels were interchanged.

This property is interpreted by noting that the system is invariant under the continuous group of spatial translations. Indeed, an experiment carried out in a certain place will give identical results to another experiment carried out by translating all the experimental apparatus by a vector in $\mathbb{R}^3$. The total momentum of the system is thus conserved as a consequence of the invariance under spatial translations.

Total orbital angular momentum. Now, we must verify that $[\hat{\mathbf{L}}^a, \hat{H}] = 0$. The detailed calculations are carried out in appendix C.

Here, the property is interpreted by noting that the system is invariant under the continuous group of spatial rotations. An experiment carried out in a given place will give identical results to another experiment if this place is rotated by a certain angle around a certain axis. The total orbital angular momentum of the system is then conserved as a consequence of the invariance under rotations. If the particles carry a spin, it would also be true. Under such circumstance, the total spin be conserved.

We have therefore identified three conserved quantities: the total energy $\hat{H}$, the total momentum $\hat{\mathbf{P}}$ and the total orbital angular momentum $\hat{\mathbf{L}}$. Furthermore, in the nonrelativistic limit, the total mass is also conserved, and so the total number of particles. These conserved quantities play a central role because the temporal evolution preserves them whatever the level of description is. In the first chapter, conservation laws have been obtained for the mass, particle, momentum and energy densities. Two fundamental questions are now of interest:

- How are these quantities conserved at the microscopic scale?
- Can the macroscopic conservation laws be somehow recovered from the microscopic dynamics?

In the next section, conservation equations for the corresponding microscopic densities will be obtained.

2.4. MICROSCOPIC DENSITIES & CONSERVATION EQUATIONS

Microscopic densities. In the microscopic approach, the challenge is to obtain the macroscopic dynamics of the system from the intrinsic microscopic dynamics of the particles making it up. The very starting point is to define microscopic densities that depend on the position $\mathbf{r}_i$ and momentum $\mathbf{p}_i$ of each particles. In the quantum formulation, these microscopic densities become observable and should be described by Hermitian operators depending on the position $\mathbf{r} \in \mathbb{R}^3$ where they are measured.

The mass, particle, momentum and energy local (or microscopic) densities are defined by⁴,

$$\hat{\rho}(\mathbf{r}) \equiv \sum_{i=1}^N m_i \delta(\mathbf{r} - \mathbf{r}_i) \quad (2.49)$$

$$\hat{n}_k(\mathbf{r}) \equiv \sum_{i \in \mathcal{S}_k} \delta(\mathbf{r} - \mathbf{r}_i) \quad (2.50)$$

$$\hat{\mathbf{g}}(\mathbf{r}) \equiv \frac{1}{2} \sum_{i=1}^N \left\{ \hat{\mathbf{p}}_i \delta(\mathbf{r} - \mathbf{r}_i) + \delta(\mathbf{r} - \mathbf{r}_i) \hat{\mathbf{p}}_i \right\} \quad (2.51)$$

$$\hat{\varepsilon}(\mathbf{r}) \equiv \frac{1}{2} \sum_{i=1}^N \left\{ \hat{\varepsilon}_i \delta(\mathbf{r} - \mathbf{r}_i) + \delta(\mathbf{r} - \mathbf{r}_i) \hat{\varepsilon}_i \right\} \quad (2.52)$$

⁴To lighten notation, $\delta^3(\mathbf{r} - \mathbf{r}_i) \equiv \delta(\mathbf{r} - \mathbf{r}_i)$.

where $\hat{e}_i$ is the Hermitian operator giving the energy of the particle labelled i , $\hat{e}_i \equiv \frac{\hat{p}_i^2}{2m_i} + \frac{1}{2} \sum_{j(\neq i)}^N V_{ij}$. These operators may be considered in the Heisenberg picture. Note that the symmetrization ensures the microscopic densities to be Hermitian. With the use of the relation $\int d\mathbf{r} \delta(\mathbf{r} - \mathbf{r}_i) = 1$, we have

$$\int d\mathbf{r} \hat{\rho}(\mathbf{r}) = \sum_{i=1}^N m_i \int d\mathbf{r} \delta(\mathbf{r} - \mathbf{r}_i) = \sum_{k=1}^c m_k \hat{N}_k = M \quad (2.53)$$

$$\int d\mathbf{r} \hat{n}_k(\mathbf{r}) = \hat{N}_k \quad (2.54)$$

$$\int d\mathbf{r} \hat{\mathbf{g}}(\mathbf{r}) = \sum_{i=1}^N \hat{\mathbf{p}}_i = \hat{\mathbf{P}} \quad (2.55)$$

$$\int d\mathbf{r} \hat{e}(\mathbf{r}) = \sum_{i=1}^N \hat{e}_i = \hat{H} \quad (2.56)$$

giving back the conserved quantities.

Two remarks are to be made:

- The reader might wonder why the mass local density (and eventually the particle local density) is defined with a hat. The mass local density acts on wave functions in accordance to

$$\begin{aligned} \hat{\rho}(\mathbf{r})\psi(\mathbf{r}^N) &= \sum_{i=1}^N m_i \langle \mathbf{r}_1, \dots, \mathbf{r}_i, \dots, \mathbf{r}_N | \delta(\mathbf{r} - \hat{\mathbf{r}}_i) | \psi \rangle \\ &= \sum_{i=1}^N m_i \langle \mathbf{r}_1, \dots, \mathbf{r}_i, \dots, \mathbf{r}_N | \mathbf{1} \otimes \dots \otimes \delta(\mathbf{r} - \hat{\mathbf{r}}_i) \otimes \dots \otimes \mathbf{1} | \psi \rangle \\ &= \sum_{i=1}^N m_i \langle \mathbf{r}_1, \dots, \mathbf{r}_i, \dots, \mathbf{r}_N | \mathbf{1} \otimes \dots \otimes \delta(\mathbf{r} - \mathbf{r}_i) \otimes \dots \otimes \mathbf{1} | \psi \rangle = \rho(\mathbf{r})\psi(\mathbf{r}^N) \end{aligned} \quad (2.57)$$

and similarly for the particle local density. Moreover, the microscopic density $\hat{\rho}(\mathbf{r})$ should not be confused with the macroscopic density $\rho(\mathbf{r})$ introduced in the first chapter. The relation between the both will be developed in the fourth chapter.

- The reader might also wonder why the operators (2.49)-(2.52) are referred to as *densities*. In the classical case, it is clear that a density is obtained by dividing the corresponding macroscopic field by the volume of the considered system. Here, we must not forget that a Dirac delta distribution has inverse units of its argument. Thanks to the Dirac distribution, we can thus define microscopic densities in terms of the microscopic positions of the particles.

Local conservation equations. We would like to obtain local conservation equations in the form

$$\partial_t (\text{local density}) + \nabla \cdot (\text{local current density}) = 0.$$

For this purpose, the Heisenberg equation for the time evolution of the observables is used. The conservation equation for each local density is obtained by systematically calculating its commutator with the Hamiltonian. In order to avoid interrupting the reader, the calculations are carried out in appendix D. The final results are summarised below:

$$\partial_t \hat{\rho} + \partial^a \hat{j}_\rho^a = 0 \quad (2.58)$$

$$\partial_t \hat{n}_k + \partial^a \hat{j}_{n_k}^a = 0 \quad (2.59)$$

$$\partial_t \hat{g}^a + \partial^b \hat{j}_{g^a}^b = 0 \quad (2.60)$$

$$\partial_t \hat{e} + \partial^a \hat{j}_e^a = 0 \quad (2.61)$$

with the respective local current densities

$$\hat{j}_\rho^a = \hat{g}^a \quad (2.62)$$

$$\hat{j}_{n_k}^a = \frac{1}{2} \sum_{i \in \mathcal{S}_k} \left\{ \frac{\hat{p}_i^a}{m_i} \delta(\mathbf{r} - \mathbf{r}_i) + \delta(\mathbf{r} - \mathbf{r}_i) \frac{\hat{p}_i^a}{m_i} \right\} \quad (2.63)$$

$$\hat{j}_{g^b}^a = \sum_{i=1}^N \frac{1}{4m_i} \left\{ \hat{p}_i^b \hat{p}_i^a \delta(\hat{\mathbf{r}} - \hat{\mathbf{r}}_i) + \hat{p}_i^b \delta(\hat{\mathbf{r}} - \hat{\mathbf{r}}_i) \hat{p}_i^a + \text{h.c.} \right\} + \frac{1}{2} \sum_{i=1}^N \sum_{j(\neq i)}^N F_{ij}^a D^b(\mathbf{r}; \mathbf{r}_i, \mathbf{r}_j) \quad (2.64)$$

$$\begin{aligned} \hat{j}_e^a = & \sum_{i=1}^N \frac{1}{4m_i} \hat{e}_i \left\{ \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_i^a + \hat{p}_i^a \delta(\mathbf{r} - \mathbf{r}_i) \right\} \\ & + \frac{1}{16} \sum_{i=1}^N \sum_{j(\neq i)}^N \left\{ F_{ij}^b \left(\frac{\hat{p}_i^b}{m_i} + \frac{\hat{p}_j^b}{m_j} \right) + \left(\frac{\hat{p}_i^b}{m_i} + \frac{\hat{p}_j^b}{m_j} \right) F_{ij}^b \right\} D^a(\mathbf{r}; \mathbf{r}_i, \mathbf{r}_j) + \text{h.c.} \end{aligned} \quad (2.65)$$

The quantity $D^a(\mathbf{r}; \mathbf{r}_i, \mathbf{r}_j)$ is defined in (D.21). It may be supposed to be distributed along a straight line between $\mathbf{r}_i$ and $\mathbf{r}_j$. Under this condition, the momentum current density $\hat{j}_{g^b}^a$ can be shown to be symmetric (*cf.* appendix D). For the energy current density, the quantity D^a is not necessarily distributed along such a straight line.

The mass, particle and energy current densities are vectors while the momentum current density is a tensor of rank two. Equations (2.58)-(2.61) with their respective local current densities (2.62)-(2.65) occupy an important role in this work. To lighten notations, they will be written in the following form, with spatial and time dependence recovered,

$$\partial_t \hat{c}^\alpha(\mathbf{r}, t) + \partial^a \hat{j}_{c^\alpha}^a(\mathbf{r}, t) = 0 \quad (2.66)$$

where $\hat{c}^1 = \hat{\rho}$, $\hat{c}^2 = \hat{n}_k$, $\hat{c}^3 = \hat{g}^a$ and $\hat{c}^4 = \hat{e}$. A Greek letter will always refer to the various local densities.

These equations can also be considered in the classical case. In this context, the operator $\hat{p}_i$ reduces to the vector momentum of the particle labelled i and commutes with the position $\mathbf{r}_i$. It is easy to verify that equations (2.58)-(2.65) reduce to the expected classical ones [14].

In §2.3, it was shown that the total orbital angular momentum is a conserved quantity. The reader might ask why a conservation equation for its local density is not considered. In fact, such

equation would not have brought any additional knowledge since it constrains the momentum current density to be symmetric (as in the classical case, *cf.* first chapter).

2.5. APPROACH ON THE BASIS OF QUANTUM FIELD THEORY

Nonrelativistic quantum field theory offers a more fundamental approach and justification to the construction of the microscopic local densities, local current densities and their corresponding conservation equations. The formulation is no longer carried out in terms of particles but rather fields whose excitations give rise to particles. Nonrelativistic quantum field theory also has the benefit of not constraining the system to a fixed number of particles. In this section, the main ideas of this approach will be presented. The objects and the procedure of the second quantization will be taken as granted.

The local densities (2.49)-(2.52) are defined in the N -particle Hilbert space $\mathcal{H}_N$ defined in (2.6). Nonrelativistic quantum field theory is formulated in the Fock space $\mathcal{H}_F$ defined by

$$\mathcal{H}_F = \oplus_{N=0}^{\infty} \mathcal{H}_N \quad (2.67)$$

where $\mathcal{H}_0$ describes a system without any particle, so that this space contains a single state which is the vacuum state $|0\rangle$. The multi-particle states are constructed by applying creation operators to the vacuum state.

The annihilation and creation field operators, $\hat{\psi}(\mathbf{r}, \sigma)$ and $\hat{\psi}^\dagger(\mathbf{r}, \sigma)$, are introduced. They are defined at every space point and for every spin components $\sigma = -s, -s+1, \dots, s-1, s$, where s is the spin of the particle. They satisfy the following canonical commutation/anticommutation relations,

$$\hat{\psi}(\mathbf{r}, \sigma) \hat{\psi}^\dagger(\mathbf{r}', \sigma') - \theta \hat{\psi}^\dagger(\mathbf{r}', \sigma') \hat{\psi}(\mathbf{r}, \sigma) = \delta(\mathbf{r} - \mathbf{r}') \delta_{\sigma\sigma'}. \quad (2.68)$$

For bosons, $\theta = 1$ and $\theta = -1$ for fermions. We will assume that the system is a one-component fluid. For a multiple-component fluid, one should associate a quantum field to each species.

The total number of particles operator is given by

$$\begin{aligned} \hat{N} &= \sum_{\sigma=-s}^s \int d\mathbf{r} \hat{\psi}^\dagger(\mathbf{r}, \sigma) \hat{\psi}(\mathbf{r}, \sigma) \\ &\equiv \sum_{\sigma=-s}^s \int d\mathbf{r} \hat{n}(\mathbf{r}, \sigma) \end{aligned} \quad (2.69)$$

with $\hat{n}(\mathbf{r}, \sigma)$ the particle density. The Hamiltonian operator is given by

$$\begin{aligned} \hat{H} &= \sum_{\sigma=-s}^s \int d\mathbf{r} \hat{\psi}^\dagger(\mathbf{r}, \sigma) \hat{h}_0 \hat{\psi}(\mathbf{r}, \sigma) + \frac{1}{2} \sum_{\sigma, \sigma'=-s}^s \int d\mathbf{r} d\mathbf{r}' \hat{\psi}^\dagger(\mathbf{r}, \sigma) \hat{\psi}^\dagger(\mathbf{r}', \sigma') u(\mathbf{r}, \mathbf{r}', \sigma, \sigma') \hat{\psi}(\mathbf{r}', \sigma') \hat{\psi}(\mathbf{r}, \sigma) \\ &\equiv \hat{H}_0 + \hat{H}_{\text{int}} \end{aligned} \quad (2.70)$$

with $\hat{h}_0 \equiv -\frac{\hbar^2}{2m} \nabla^2 + u_{\text{ext}}(\mathbf{r}, \sigma)$ the Hamiltonian for a single particle of mass m in the external energy potential u_{ext} . $u(\mathbf{r}, \mathbf{r}', \sigma, \sigma')$ is a two-body interaction potential.

For the sake of simplicity, let us consider spinless bosons described by the fields $\hat{\psi}(\mathbf{r})$ obeying the commutation relation

$$[\hat{\psi}(\mathbf{r}), \hat{\psi}^\dagger(\mathbf{r}')] = \delta(\mathbf{r} - \mathbf{r}'). \quad (2.71)$$

We define, the mass, momentum and energy densities as

$$\hat{\rho}(\mathbf{r}) = m\hat{n}(\mathbf{r}) = m\hat{\psi}^\dagger(\mathbf{r})\hat{\psi}(\mathbf{r}) \quad (2.72)$$

$$\hat{\mathbf{g}}(\mathbf{r}) = \frac{1}{2} \left\{ \hat{\psi}^\dagger(\mathbf{r}) [\hat{\mathbf{p}}\hat{\psi}(\mathbf{r})] + [\hat{\mathbf{p}}\hat{\psi}(\mathbf{r})]^\dagger \hat{\psi}(\mathbf{r}) \right\} \quad (2.73)$$

$$\hat{e}(\mathbf{r}) = \frac{1}{2} \left\{ \hat{\psi}^\dagger(\mathbf{r}) [\hat{h}\hat{\psi}(\mathbf{r})] + [\hat{h}\hat{\psi}(\mathbf{r})]^\dagger \hat{\psi}(\mathbf{r}) \right\} \quad (2.74)$$

where $\hat{\mathbf{p}} = -i\hbar\nabla$ and $\hat{h} \equiv \hat{h}_0 + \frac{1}{2} \int d\mathbf{r}' \hat{\psi}^\dagger(\mathbf{r}') u(\mathbf{r}, \mathbf{r}') \hat{\psi}(\mathbf{r}')$ is the one-body Hamiltonian operator including the contribution of binary interactions.

The local densities (2.49), (2.51) and (2.52) can be deduced by considering the operators (2.72)-(2.74) inside the N -particle Hilbert space $\mathcal{H}_N$. A Fock state of N particles is constructed as

$$|\Psi\rangle = \frac{1}{\sqrt{N!}} \int d\mathbf{r}_1 \cdots d\mathbf{r}_N \Psi(\mathbf{r}^N) \hat{\psi}^\dagger(\mathbf{r}_N) \cdots \hat{\psi}^\dagger(\mathbf{r}_1) |0\rangle \quad (2.75)$$

where $\Psi(\mathbf{r}^N)$ is the N -body wavefunction as already introduced earlier. Since states of $\mathcal{H}_N$ are normalised to unity, *i.e.* $\langle\Psi|\Psi\rangle = 1$, we have that

$$\begin{aligned} \langle\Psi|\Psi\rangle &= \frac{1}{N!} \int d\mathbf{r}_1 d\mathbf{r}'_1 \cdots d\mathbf{r}_N d\mathbf{r}'_N \Psi^*(\mathbf{r}^N) \Psi(\mathbf{r}'^N) \langle 0 | \hat{\psi}(\mathbf{r}_1) \cdots \hat{\psi}(\mathbf{r}_N) \hat{\psi}^\dagger(\mathbf{r}'_N) \cdots \hat{\psi}^\dagger(\mathbf{r}'_1) | 0 \rangle \\ &= \frac{1}{N!} \int d\mathbf{r}_1 d\mathbf{r}'_1 \cdots d\mathbf{r}_N d\mathbf{r}'_N \Psi^*(\mathbf{r}^N) \Psi(\mathbf{r}'^N) \sum_{p \in S_N} \delta(\mathbf{r}_1 - \mathbf{r}'_{p(1)}) \cdots \delta(\mathbf{r}_N - \mathbf{r}'_{p(N)}) \\ &= \int d\mathbf{r}_1 \cdots d\mathbf{r}_N \Psi^*(\mathbf{r}^N) \Psi(\mathbf{r}^N) \\ &= \int d\mathbf{r}_1 \cdots d\mathbf{r}_N |\Psi(\mathbf{r}^N)|^2 \stackrel{!}{=} 1 \end{aligned} \quad (2.76)$$

where S_N is the discrete symmetric group with $N!$ elements. An element p of S_N is a permutation. In the third line, the symmetry of the N -body wavefunction under permutation of the positions was used. The normalisation of the states implies the N -body wavefunction to belong to $\mathcal{L}(\mathbb{R}^{3N}, d^N\mathbf{r})$. We have

$$\begin{aligned} &\langle\Psi|\hat{\rho}(\mathbf{r})|\Psi\rangle \\ &= \frac{m}{N!} \int d\mathbf{r}_1 d\mathbf{r}'_1 \cdots d\mathbf{r}_N d\mathbf{r}'_N \Psi^*(\mathbf{r}^N) \Psi(\mathbf{r}'^N) \langle 0 | \hat{\psi}(\mathbf{r}_1) \cdots \hat{\psi}(\mathbf{r}_N) \hat{\psi}^\dagger(\mathbf{r}) \hat{\psi}(\mathbf{r}) \hat{\psi}^\dagger(\mathbf{r}'_N) \cdots \hat{\psi}^\dagger(\mathbf{r}'_1) | 0 \rangle \\ &= \frac{m}{N!} \int d\mathbf{r}_1 d\mathbf{r}'_1 \cdots d\mathbf{r}_N d\mathbf{r}'_N \Psi^*(\mathbf{r}^N) \Psi(\mathbf{r}'^N) \sum_{i=1}^N \delta(\mathbf{r} - \mathbf{r}_i) \sum_{p \in S_N} \delta(\mathbf{r}_1 - \mathbf{r}'_{p(1)}) \cdots \delta(\mathbf{r}_N - \mathbf{r}'_{p(N)}) \\ &= \int d\mathbf{r}_1 \cdots d\mathbf{r}_N \left(m \sum_{i=1}^N \delta(\mathbf{r} - \mathbf{r}_i) \right) |\Psi(\mathbf{r}^N)|^2. \end{aligned} \quad (2.77)$$

In the expression above, the local mass density (2.49) defined in first quantization (and thus at fixed number of particles) is recognised. Similar relations hold for the densities (2.73) and

(2.74). In the passage from the second to the third line, we have used that

$$\begin{aligned}
 & \langle 0 | \hat{\psi}(\mathbf{r}_1) \cdots \hat{\psi}(\mathbf{r}_N) \hat{\psi}^\dagger(\mathbf{r}) \hat{\psi}(\mathbf{r}) \hat{\psi}^\dagger(\mathbf{r}'_N) \cdots \hat{\psi}^\dagger(\mathbf{r}'_1) | 0 \rangle \\
 &= \langle 0 | \hat{\psi}(\mathbf{r}_1) \cdots \hat{\psi}(\mathbf{r}_N) \hat{\psi}^\dagger(\mathbf{r}) \hat{\psi}^\dagger(\mathbf{r}'_{N-1}) \cdots \hat{\psi}^\dagger(\mathbf{r}'_1) | 0 \rangle \delta(\mathbf{r} - \mathbf{r}_N) \\
 &+ \langle 0 | \hat{\psi}(\mathbf{r}_1) \cdots \hat{\psi}(\mathbf{r}_N) \hat{\psi}^\dagger(\mathbf{r}) \hat{\psi}^\dagger(\mathbf{r}'_N) \hat{\psi}^\dagger(\mathbf{r}'_{N-2}) \cdots \hat{\psi}^\dagger(\mathbf{r}'_1) | 0 \rangle \delta(\mathbf{r} - \mathbf{r}_{N-1}) \\
 &+ \cdots \\
 &+ \langle 0 | \hat{\psi}(\mathbf{r}_1) \cdots \hat{\psi}(\mathbf{r}_N) \hat{\psi}^\dagger(\mathbf{r}) \hat{\psi}^\dagger(\mathbf{r}'_N) \cdots \hat{\psi}^\dagger(\mathbf{r}'_2) | 0 \rangle \delta(\mathbf{r} - \mathbf{r}_1) \\
 &= \sum_{i=1}^N \langle 0 | \hat{\psi}(\mathbf{r}_1) \cdots \hat{\psi}(\mathbf{r}_N) \hat{\psi}^\dagger(\mathbf{r}) \hat{\psi}^\dagger(\mathbf{r}'_N) \cdots \hat{\psi}^\dagger(\mathbf{r}'_{i+1}) \hat{\psi}^\dagger(\mathbf{r}'_{i-1}) \cdots \hat{\psi}^\dagger(\mathbf{r}'_1) | 0 \rangle \delta(\mathbf{r} - \mathbf{r}_i). \tag{2.78}
 \end{aligned}$$

Now, let us also show that the conservation equation for (2.72) is the same as the conservation equation (2.58) obtained in first quantization. The calculations are presented in appendix D. At the end, the equation

$$\partial_t \hat{\rho}(\mathbf{r}) + \nabla \cdot \hat{\mathbf{g}} = 0 \tag{2.79}$$

is recovered with $\hat{\mathbf{g}}(\mathbf{r}) = \frac{\hbar}{2i} (\hat{\psi}^\dagger(\mathbf{r}) \nabla \hat{\psi}(\mathbf{r}) - \nabla \hat{\psi}^\dagger(\mathbf{r}) \hat{\psi}(\mathbf{r}))$ which is similar to (2.73). We claim that the other conservation equations (2.60) and (2.61) can be obtained within this formalism. That was demonstrated by Zubarev [15] though without introducing the quantity (D.21). We note that the various densities (2.72)-(2.74) are quadratic so that the expression of the conservation equations are the same for bosonic and fermionic particles.

Hence, the procedure of first quantization carried in the previous sections finds its justification on a more fundamental level on the basis of nonrelativistic quantum field theory.

CHAPTER 3

NONEQUILIBRIUM STATISTICAL MECHANICS

*“Then all motion, of whatever nature,
creates?”*

EDGAR ALLAN POE¹

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This chapter brings us at the very heart of the subject. Having introduced transport phenomena in the first chapter, and discussed of the framework and the microscopic dynamics in the second, the journey towards the macroscopic hydrodynamic equations and the Green-Kubo formulas for the transport properties begins. To this purpose, it is first necessary to understand how can quantum and nonequilibrium statistical mechanics be combined together. The common thread of this chapter is to ask if anything on the microscopic dynamics of a system can be known from measurable macroscopic properties of the same system. Such consideration naturally leads to the concept of relevant and irrelevant variables, the relevant ones forming a set that is sufficient to describe the macroscopic state of a system. An effective macroscopic

¹From E. A. Poe. *The Power of Words*. 1845.

statistical operator, called relevant statistical operator, will be defined, and will remarkably describe a local equilibrium within small fluid elements.

In §3.1, the notion of relevant and irrelevant variables will be introduced and discussed in detail, the aim being to properly obtain the relevant statistical operator and to show that it describes a local equilibrium configuration. In §3.2, the time evolution of the relevant statistical operator will be discussed. Some useful results for the following sections and chapters will also be demonstrated. The §3.3 proposes an explicit expression for the deviation operator, which is defined as the difference between the exact and the relevant statistical operator. This operator occupies a central role in this work, as justified by the expression of the entropy production that will be obtained in §3.4. Finally, we will end by briefly discussing the correlation functions at local equilibrium in the §3.5.

3.1. RELEVANT STATISTICAL OPERATOR

Intuitive motivation. The total system that is here considered is composed of N particles, each carrying 3 distinct degrees of freedom, giving a total of $3N$ degrees of freedom. Yet, the microscopic dynamics is expected to be studied by means of four kinds of local densities, namely the mass, particle, momentum and energy local densities, hence reducing the description to much less variables. Clearly, there is something amiss here, raising the following questions:

- How can the microscopic dynamics be properly described by setting aside some degrees of freedom?
- Disregarding some degrees of freedom implies that the knowledge on the considered system can only be partial. How to make physical predictions with sufficient accuracy?
- Why are the mass, particle, momentum and energy local densities preferred over all the other possible local densities?

To this apparent paradox, we could object that the complexity of many-body systems exhibited at the microscopic level can sometimes be cast into rather simple equations of motion for the macroscopic variables, even if the microscopic system is composed of a great number of particles. For instance, the Navier-Stokes equations, forming a set of three coupled partial differential equations for the velocity field of a viscous fluid, describe very well a cubic meter of water that contains about 10^{28} water molecules. The same idea applies to the heat equation. Heat is a result of the kinetic energy of particles. Those particles collide each other and produce a chaotic and disordered motion. The heat equation describes by a single partial differential equation how heat evolves within a macroscopic piece of matter.

Though, it is important to mention that the previous comparisons established are performed between processes exhibiting very different time scales. The microscopic dynamics takes place over time scales of the order of the intercollision time (time between two successive collisions)

while the macroscopic dynamics over time scales of the order of the relaxation time, that is to say the time for the system to relax to equilibrium. To give an idea, the intercollision time of a dilute gas is roughly 10^{-10} s while its macroscopic relaxation time is roughly $10^0 - 10^3$ s. Going from the microscopic to the macroscopic point of view involves taking the average value of some dynamical variables.

Here, the initially stated problem was subtly circumvented. The local densities (2.49)-(2.52) are constructed from non-averaged dynamical variables and thus remain defined at the level of the microscopic dynamics. The crucial point is to note that the average values of these local densities are sufficient to describe the nonequilibrium macroscopic state of the system. As well, they can be experimentally measured, or at least numerically computed. Therefore, the question that is asked is not how going from the microscopic to the macroscopic dynamics but rather the reverse one: given the macroscopic state of a system, how to describe its microscopic state as precisely as possible? Let us be more explicit. Let $a = a(t)$ ² be a macroscopic variable describing the state of a nonequilibrium system. a results from the average of a dynamic variable (operator) $\hat{A}$ that takes into account the underlying microscopic dynamics, $a(t) = \text{Tr} \hat{\rho}_t \hat{A}$. The evolution equation for a takes the form

$$\partial_t a = -\frac{i}{\hbar} \text{Tr}([\hat{H}, \hat{\rho}_t] \hat{A}) = -\frac{i}{\hbar} \text{Tr}(\hat{\rho}_{t=0} [\hat{A}_t, \hat{H}]) \quad (3.1)$$

in the Schrödinger and Heisenberg pictures respectively. This equation can be solved only if the exact statistical operator $\hat{\rho}_t$, which is a solution of the von Neumann equation (2.18), is known (at least at $t = 0$). Here, the only knowledge we have is a which could, for example, come from the result of an experimental measurement. Under such circumstance, the exact statistical operator cannot be constructed: this would require the knowledge of a complete set of macroscopic variables [16]. Table 3.1 below summarizes the position of the problem.

Microscopic level	Macroscopic level
• Completely determined by the Hamiltonian operator. • With the Hamiltonian operator, the von Neumann equation can be solved to give the exact statistical operator $\hat{\rho}_t$. • The knowledge of $\hat{\rho}_t$ at any fixed time is enough to obtain all the properties of the system.	• Completely determined by a set of macroscopic variables. • The macroscopic variables derive from the average of dynamical operators. • The time evolution of those macroscopic variables yields the transport equations [17].

Table 3.1: Characteristics of each level of description in a nonequilibrium situation. The problem is made explicit: to obtain the time evolution of the macroscopic variables, the knowledge of the exact statistical operator is required.

²We assume that a does not explicitly depend on r : for the sake of simplicity, this dependence is left aside as the purpose is to give an overall idea of the problem.

The idea that will formally be discussed is to introduce an effective statistical operator which, taking into account the macroscopic variables, best describes the microscopic dynamics of the system that is studied.

Relevant variables. Let us assume that the finite set of observables $\{\hat{A}_1, \hat{A}_2, \dots, \hat{A}_i, \dots\}$, not needing to commute with each other, fully describes the macroscopic (or thermodynamic state) of a given system. It defines an *observation level* consisting, in general, of a small portion of all the possible observables of the system, since usually a few properties of the system are physically accessible and interesting. In the classical frame, operators would be replaced by classical variables. For instance, it is known that the set $\{T, V, N\}$ fully describes an ideal gas. The operators forming the set will be referred to as relevant variables.³ The choice of the set is fundamental as it determines the whole coherence of the formalism that will be exposed. It is usually dictated by experiments. Table 3.2 shows the most common relevant variables for different physical systems.

System	Nature of the relevant variable(s)	Relevant variable(s)
Classical kinetics: dilute gases	One-particle phase density	$n_1(\mathbf{r}, \mathbf{p}) \equiv \sum_{i=1}^N \delta(\mathbf{r} - \mathbf{r}_i) \delta(\mathbf{p} - \mathbf{p}_i)$
Quantum kinetics: weakly interacting gases of quasiparticles	Creation and annihilation operators	$\hat{a}$ and $\hat{a}^\dagger$
Hydrodynamics: one-component normal fluid	Slow-varying fields	$\hat{\rho}(\mathbf{r})$, $\hat{\mathbf{g}}(\mathbf{r})$ and $\hat{\varepsilon}(\mathbf{r})$
Broken symmetries: ordered systems	Order parameters	Depend on the symmetry that is broken
Chemical reactions: slow chemical reactions	Concentrations of the species	$[X_1], \dots, [X_n]$

Table 3.2: Some relevant variables for different physical systems.

Let us now consider the $a_i(\mathbf{r}, t) \equiv \langle \hat{A}_i(\mathbf{r}) \rangle_t = \text{Tr} \hat{\rho}_t \hat{A}_i(\mathbf{r}) = \text{Tr} \hat{\rho}_{t=0} \hat{A}_i(\mathbf{r}, t)$, the mean value of the relevant variables. As already explained, the knowledge of all the a_i is not sufficient to know the form of the exact statistical operator $\hat{\rho}_t$. It is here that the *relevant statistical operator* $\hat{\rho}_{\text{rel}}$ comes into play. First introduced by E. T. Jaynes in 1957 [18], it is defined by the following four axioms [16, 17]:

Axiom 1. $\hat{\rho}_{\text{rel}}$ is a statistical operator and thus satisfies (2.17).

Axiom 2. $\hat{\rho}_{\text{rel}}$ is completely specified by the a_i .

Axiom 3. The exact statistical operator $\hat{\rho}_t$ and the relevant one are macroequivalent, *i.e.*

$$a_i(\mathbf{r}, t) = \text{Tr} \hat{\rho}_t \hat{A}_i(\mathbf{r}) = \text{Tr} \hat{\rho}_{t=0} \hat{A}_i(\mathbf{r}, t)$$

³Relevant operators is never used in the literature.

$$= \text{Tr } \hat{\rho}_{\text{rel}} \hat{A}_i(\mathbf{r}, t). \quad (3.2)$$

Axiom 4. At $t = 0$, the exact and the relevant statistical operator coincide,

$$\hat{\rho}_{t=0} = \hat{\rho}_{\text{rel}}. \quad (3.3)$$

$\hat{\rho}_{\text{rel}}$ contains no further microscopic information than imposed by the axioms. The reason why $\hat{\rho}_t$ is referred to as the exact statistical operator is now clear: for any observable $\hat{B}$ that is not relevant, one has

$$\langle \hat{B} \rangle_t = \text{Tr } \hat{\rho}_t \hat{B} \quad (3.4)$$

while

$$\langle \hat{B} \rangle_t \neq \text{Tr } \hat{\rho}_{\text{rel}} \hat{B}_t \quad (3.5)$$

as the equality only stands for the relevant variables.

In general, there is not a single relevant statistical operator satisfying these four axioms, in particular (3.2). This is related to the fact that the knowledge of the relevant variables is not sufficient to describe the microscopic dynamics. Therefore, we claim that the only and more legitimate relevant statistical operator is the one maximising the uncertainty on the knowledge of the microscopic state of the system. This leads us to the principle of maximum entropy. To cite E. T. Jaynes, “there is nothing in the general laws of motion that can provide us with any additional information about the state of a system beyond what we have obtained from measurement”.

Principle of maximum entropy. Let X be a random variable and let $P(X = x_i) = p_i$, $i = 1, \dots, n$ be the probability for the possible events $\{x_i\}$ to occur. The probabilities satisfy $\sum_{i=1}^n p_i = 1$. Let us construct the quantity

$$S_S(\{p_i\}) = - \sum_{i=1}^n p_i \ln p_i \quad (3.6)$$

where the S subscript reminds us that S_S is the information entropy introduced by C. E. Shannon in 1948 [19] (up to an insignificant multiplicative constant). We insist that this is not the physical entropy and is not related to the thermodynamic properties such as the temperature. In fact, S_S was originally introduced in a whole different concept than statistical mechanics. It can be considered as a measure of the uncertainty in the information corresponding to the statistical distribution $\{p_i\}$. It takes its largest value when all the p_i are equals, and indeed, this limiting case corresponds to the greatest uncertainty, that is to say the smallest amount of information.

For later purposes, let us rather discuss of the principle of maximum entropy in the continuous case. Now, the possible events $\{x\}$ form a continuum and the probability they occur is $P(x < X < x + dx) = f(x)dx$, with $\int_{\mathbb{R}} dx f(x) = 1$. The Shannon entropy now takes the form

$$S_S[f] = - \int_{\mathbb{R}} dx f(x) \ln f(x) \quad (3.7)$$

and is seen as a functional of f . A central issue is to understand what would be the form required for the probability distribution f to maximise the Shannon entropy (3.7) under the condition that the mean value of some random variables be equal to known quantities. The resulting probability distribution is called the *extremal distribution* $f_c(x)$. The method used for constructing distributions of this type is interesting to discuss. We are searching for a probability distribution f that extremises (3.7) under the constraints

$$\begin{cases} \int_{\mathbb{R}} dx f(x) = 1 \\ a_m \equiv \langle A_m \rangle = \int_{\mathbb{R}} dx A_m(x) f(x) \end{cases} \quad (3.8)$$

for $m = 1, \dots, M$ given a_m quantities. As one might have guessed, the method of Lagrange multiplier will be used. To do so, the following Lagrange functional is constructed

$$\mathcal{L}[f; \lambda_0, \lambda_m] = - \int_{\mathbb{R}} dx f(x) \ln f(x) + \sum_{m=1}^M \lambda_m \left(a_m - \int_{\mathbb{R}} dx A_m(x) f(x) \right) + \lambda_0 \left(1 - \int_{\mathbb{R}} dx f(x) \right) \quad (3.9)$$

where the λ_0 and λ_m are Lagrange multipliers. The constrained extrema of f are the critical points of $\mathcal{L}$, i.e.

$$\left. \frac{\delta \mathcal{L}[f]}{\delta f(y)} \right|_{f_c} = -\ln f_c(y) - \sum_{m=1}^M \lambda_m A_m(y) - (1 + \lambda_0) \stackrel{!}{=} 0 \quad (3.10)$$

or alternatively

$$f_c(x) = \exp \left\{ -\Phi - \sum_{m=1}^M \lambda_m A_m(x) \right\} \quad (3.11)$$

where $\Phi \equiv 1 + \lambda_0$ with

$$\Phi = \ln \int_{\mathbb{R}} dx \exp \left\{ - \sum_{m=1}^M \lambda_m A_m(x) \right\}. \quad (3.12)$$

However, we still have to prove that $f_c(x)$ corresponds well to a maximum of the Shannon entropy (3.7). This will be done later. The partition function

$$Z = e^{\Phi} \quad (3.13)$$

arises from the normalisation condition $\int_{\mathbb{R}} dx f_c(x) = 1$. In particular, one sees that $f_c = f_c[A_1, \dots, A_M]$ and $\Phi = \Phi[A_1, \dots, A_M]$ are functionals of the relevant variables $A_m(x)$.

The above discussion remains rather suggestive and it is now time to include a more physical context.

Relevant statistical operator. Let us consider again a set of relevant variables $\{\hat{A}_1, \hat{A}_2, \dots, \hat{A}_i, \dots\}$. In the light of what was introduced before, we are now seeking an explicit form for the relevant statistical operator $\hat{\rho}_{\text{rel}}$. Let $\hat{\rho}$ be a trial relevant statistical operator. It has the property to maximise the von Neumann entropy (2.30)⁴ under the constraints

$$\begin{cases} \text{Tr } \hat{\rho}(t) = 1 \\ a_i(\mathbf{r}, t) = \text{Tr } \hat{\rho}(t) \hat{A}_i(\mathbf{r}). \end{cases} \quad (3.14)$$

⁴To lighten notations, the Boltzmann constant is dropped but will naturally appear back later.

To this purpose, the following Lagrange functional is constructed [20]

$$\mathcal{L}[\hat{\rho}; \lambda_0(t), \lambda_i(\mathbf{r}, t)] = -\text{Tr}(\hat{\rho} \ln \hat{\rho}) + \sum_i \int d\mathbf{r} \lambda_i(\mathbf{r}, t) (a_i(\mathbf{r}, t) - \text{Tr} \hat{\rho}(t) \hat{A}_i(\mathbf{r})) + \lambda_0(t) (1 - \text{Tr} \hat{\rho}) \quad (3.15)$$

and then extremized

$$\left. \frac{\delta \mathcal{L}[\hat{\rho}]}{\delta \hat{\rho}(t')} \right|_{\hat{\rho}_{\text{rel}}} = -\ln \hat{\rho}_{\text{rel}}(t') - \sum_i \int d\mathbf{r} \lambda_i(\mathbf{r}, t') \hat{A}_i(\mathbf{r}) - (\lambda_0(t') + 1) \stackrel{!}{=} 0 \quad (3.16)$$

to give

$$\hat{\rho}_{\text{rel}}(t) = \exp \left\{ -(\lambda_0(t) + 1) - \sum_i \int d\mathbf{r} \lambda_i(\mathbf{r}, t) \hat{A}_i(\mathbf{r}) \right\}. \quad (3.17)$$

Defining $\Omega \equiv 1 + \lambda_0(t)$ such that

$$\Omega[\lambda_i] = \ln \text{Tr} \exp \left\{ -\sum_i \int d\mathbf{r} \lambda_i(\mathbf{r}, t) \hat{A}_i(\mathbf{r}) \right\}, \quad (3.18)$$

the relevant statistical operator can be written in a neater form

$$\hat{\rho}_{\text{rel}}[\lambda_i] = \exp \left\{ -\Omega[\lambda_i] - \sum_i \int d\mathbf{r} \lambda_i(\mathbf{r}, t) \hat{A}_i(\mathbf{r}) \right\}. \quad (3.19)$$

The partition function of the theory is $Z[\lambda_i] = e^{\Omega[\lambda_i]}$.

Some comments:

- As the Lagrange multipliers depend on the position $\mathbf{r}$, an integral appears in the expression of $\hat{\rho}_{\text{rel}}$, in comparison to (3.11).
- The relevant statistical operator (3.19) is defined at time t only. Its time dependence comes from the Lagrange multipliers and its time evolution will be discussed later.
- The notation used for the relevant statistical operator will vary from one context to another. Sometimes, the functional dependence on the Lagrange multipliers will be made explicit while in other cases it might be forgot. Similarly for the time dependence.
- The Lagrange multipliers are obtained from the self-consistency conditions

$$a_i(\mathbf{r}, t) = \text{Tr} \hat{\rho}_t \hat{A}_i(\mathbf{r}) = \text{Tr} \hat{\rho}_{\text{rel}}(t) \hat{A}_i(\mathbf{r}), \quad (3.20)$$

and consequently, $\lambda_i = \lambda_i[a_i(\mathbf{r}, t)]$.

- An arbitrary statistical operator not meeting the criteria of maximum entropy would possess more information about the state of the system without further justification than is actually contained in the a_i .

Now, it must be demonstrated that (3.19) is indeed a maximum of the von Neumann entropy [21]. For this we need a preliminary result.

RESULT 3.1. Let $\hat{\rho} = e^{\hat{X}}$ with $\hat{X}$ a Hermitian operator. Then,

$$\text{Tr } \hat{\rho} \ln \hat{\rho} = \text{Tr } \hat{X} e^{\hat{X}}.$$

Proof: Let $\{|\psi_n\rangle\}$ be a basis diagonalising $\hat{\rho}$, $\hat{\rho}|\psi_n\rangle = p_n|\psi_n\rangle$. Hence, $\hat{\rho}$ can be written as $\hat{\rho} = \sum_n p_n |\psi_n\rangle \langle \psi_n|$. Let f be an arbitrary function that can, in a given domain, be written as a convergent series. Then,

$$\begin{aligned} \text{Tr } f(\hat{\rho}) &= \sum_n \langle \psi_n | f(\hat{\rho}) | \psi_n \rangle \\ &= \sum_n f(p_n). \end{aligned} \quad (3.21)$$

$\hat{X}$ has the same eigenstates than $\hat{\rho}$ but with eigenvalue $x_n = \ln p_n$. Thus,

$$\begin{aligned} \text{Tr } f(\hat{\rho}) &= \sum_n f(p_n) = \sum_n f(e^{x_n}) \\ &= \text{Tr } f(e^{\hat{X}}). \end{aligned} \quad (3.22)$$

Taking $f(x) = x \ln x$, we immediately obtain that

$$\begin{aligned} \text{Tr } \hat{\rho} \ln \hat{\rho} &= \sum_n e^{x_n} \ln e^{x_n} \\ &= \sum_n x_n e^{x_n} = \sum_n \langle \psi_n | \hat{X} e^{\hat{X}} | \psi_n \rangle = \text{Tr } \hat{X} e^{\hat{X}}. \end{aligned} \quad (3.23)$$

QUOD ERAT DEMONSTRANDUM

RESULT 3.2. Let $\hat{\rho}_{\text{rel}}$ be defined as in (3.19) and let $S[\hat{\rho}] = -\text{Tr } \hat{\rho} \ln \hat{\rho}$ be the entropy functional. If $\hat{\rho}$ denotes an arbitrary trial statistical operator satisfying the constraints (3.14), then

$$S[\hat{\rho}_{\text{rel}}] \geq S[\hat{\rho}].$$

Proof: With the result 3.1, we have that

$$\begin{aligned} \text{Tr } \hat{\rho}_{\text{rel}} \ln \hat{\rho}_{\text{rel}} &= -\Omega[\lambda_i] \text{Tr } \hat{\rho}_{\text{rel}} - \sum_i \int d\mathbf{r} \lambda_i(\mathbf{r}, t) a_i(\mathbf{r}, t) \\ &= -\Omega[\lambda_i] - \sum_i \int d\mathbf{r} \lambda_i(\mathbf{r}, t) a_i(\mathbf{r}, t), \end{aligned} \quad (3.24)$$

and

$$\text{Tr } \hat{\rho} \ln \hat{\rho}_{\text{rel}} = -\Omega[\lambda_i] - \sum_i \int d\mathbf{r} \lambda_i(\mathbf{r}, t) \text{Tr } \hat{\rho} \hat{A}_i(\mathbf{r}). \quad (3.25)$$

By hypothesis,

$$a_i(\mathbf{r}, t) = \text{Tr } \hat{\rho} \hat{A}_i(\mathbf{r}) = \text{Tr } \hat{\rho}_{\text{rel}} \hat{A}_i(\mathbf{r}) \quad (3.26)$$

so that we obtain the equality

$$\text{Tr } \hat{\rho}_{\text{rel}} \ln \hat{\rho}_{\text{rel}} = \text{Tr } \hat{\rho} \ln \hat{\rho}_{\text{rel}}. \quad (3.27)$$

Now, let $\{|\psi_n\rangle\}$ and $\{|\chi_n\rangle\}$ be two basis diagonalising $\hat{\rho}_{\text{rel}}$ and $\hat{\rho}$ respectively, i.e.

$$\hat{\rho}_{\text{rel}}|\psi_n\rangle = p_n|\psi_n\rangle, \quad 0 \leq p_n \leq 1, \quad \sum_n p_n = 1 \quad (3.28)$$

$$\hat{\rho}|\chi_n\rangle = q_n|\chi_n\rangle, \quad 0 \leq q_n \leq 1, \quad \sum_n q_n = 1. \quad (3.29)$$

In the basis $\{|\psi_n\rangle\}$, we have

$$\begin{aligned} \text{Tr } \hat{\rho} \ln \hat{\rho} - \text{Tr } \hat{\rho}_{\text{rel}} \ln \hat{\rho}_{\text{rel}} &= \text{Tr } \hat{\rho} \ln \hat{\rho} - \text{Tr } \hat{\rho} \ln \hat{\rho}_{\text{rel}} \\ &= \text{Tr } \hat{\rho} (\ln \hat{\rho} - \ln \hat{\rho}_{\text{rel}}) \\ &= \sum_{n,m} |\langle \psi_n | \chi_m \rangle|^2 q_m \ln \frac{q_m}{p_n} \\ &\geq \sum_{n,m} |\langle \psi_n | \chi_m \rangle|^2 (q_m - p_n) \\ &\geq \sum_m q_m - \sum_n p_n = 0 \end{aligned} \quad (3.30)$$

where the inequality $\ln(x) \geq 1 - 1/x, \forall x > 0$ and the completeness relations $\sum_n |\langle \psi_n | \chi_m \rangle|^2 = \sum_m |\langle \psi_n | \chi_m \rangle|^2 = 1$ were used. The last line yields the desired result. The equality holds if $p_n = q_n \forall n$ and $\langle \psi_n | \chi_m \rangle \neq 0$ implying that $\hat{\rho} = \hat{\rho}_{\text{rel}}$.

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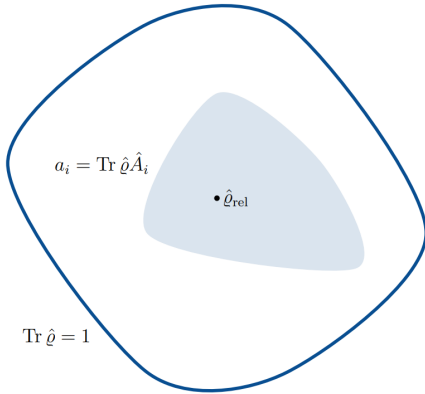


Figure 3.1: Abstract figure summarising the construction of the relevant statistical operator. Inspired from figure 5.1 of [21].

We note that if a new relevant variable is added to the existing set, say $\hat{A}_{i+1}(\mathbf{r})$, the expectation value $a_{i+1}(\mathbf{r}, t)$ can only bring new knowledge to the macroscopic state. Hence, the corresponding relevant statistical operator constructed on this extended set reduces entropy (or leave it unchanged) [21].

We conclude this discussion with figure 3.1 which offers an interesting intuitive representation of the construction of the relevant statistical operator. Amongst the space of all the statistical operators, we have considered

the operators satisfying the conditions (3.14). Amongst the space of the statistical operators satisfying (3.14), the relevant one is the only having the property to maximise entropy. In the next discussion, we will see how the relevant statistical operator describes a generalised local

equilibrium.

Equilibrium & local equilibrium. It is now time to understand the physics behind the relevant statistical operator. To do so, let us assume that the system described by this operator is at equilibrium. Consequently, the Lagrange multipliers become homogeneous in space and time and the relevant statistical operator reduces to

$$\hat{\rho}_{\text{rel}}[\lambda_i] = \exp \left\{ -\Omega[\lambda_i] - \sum_i \lambda_i \int d\mathbf{r} \hat{A}_i(\mathbf{r}) \right\}. \quad (3.31)$$

We claim here that the multicomponent normal fluid discussed in §2.1 can be described by the set $\{\hat{\rho}(\mathbf{r}), \hat{n}_k(\mathbf{r}), \hat{\mathbf{g}}(\mathbf{r}), \hat{e}(\mathbf{r})\}$ of relevant variables. However, the simultaneous choice of $\hat{\rho}(\mathbf{r})$ and $\hat{n}_k(\mathbf{r})$ may cause a conflict since they both describe the same transport phenomena but in different units, respectively mass per volume and particles per volume. In particular, from the various thermodynamic relations presented in appendix B, we learn that a choice in the unit of the chemical potentials must be made. Choosing unit of energy per particle for it, the set of relevant variables reduces to $\{\hat{n}_k(\mathbf{r}), \hat{\mathbf{g}}(\mathbf{r}), \hat{e}(\mathbf{r})\}$. At equilibrium

$$\hat{\rho}_{\text{rel}} = \exp \left\{ -\Omega - \sum_{k=1}^c \lambda_{n_k} \int d\mathbf{r} \hat{n}_k(\mathbf{r}) - \lambda_{\mathbf{g}} \cdot \int d\mathbf{r} \hat{\mathbf{g}}(\mathbf{r}) - \lambda_e \int d\mathbf{r} \hat{e}(\mathbf{r}) \right\} \quad (3.32)$$

which using (2.54)-(2.56) can be written as

$$\hat{\rho}_{\text{rel}} = \exp \left\{ -\Omega - \sum_{k=1}^c \lambda_{n_k} \hat{N}_k - \lambda_{\mathbf{g}} \cdot \hat{\mathbf{P}} - \lambda_e \hat{H} \right\}. \quad (3.33)$$

The temperature T , chemical potentials μ_k and velocity $\mathbf{u}$ are introduced as new Lagrange multipliers such that

$$\begin{cases} \lambda_{n_k} = -\frac{\mu_k}{k_B T} \\ \lambda_{\mathbf{g}} = -\frac{\mathbf{u}}{k_B T} \\ \lambda_e = \frac{1}{k_B T} \end{cases} \quad (3.34)$$

so that the relevant statistical operator takes the form

$$\hat{\rho}_{\text{rel}} = \frac{1}{Z} \exp \left\{ \beta \left(\sum_{k=1}^c \mu_k \hat{N}_k + \mathbf{u} \cdot \hat{\mathbf{P}} - \hat{H} \right) \right\}. \quad (3.35)$$

If the system is at rest in the laboratory frame, then $\mathbf{u} = 0$, and the grand-canonical statistical operator (2.26) is recovered. The total number of particle in species k , the total momentum and the Hamiltonian operator itself define conserved quantities as already discussed in the previous chapter. Hence, at equilibrium $[\hat{\rho}_{\text{rel}}, \hat{H}] = 0$, and $\hat{\rho}_{\text{rel}}$ is in this form a trivial solution of the von Neumann equation. More generally, a system in thermodynamic equilibrium is described by a statistical operator that only contains macroscopic constants of motion.

This result highlights several aspects:

- Given any set of relevant variables $\{\hat{A}_i(\mathbf{r})\}_{i=1,2,\dots}$ whose integration over space yields a conserved quantity, and if the Lagrange multipliers are homogenised over space and time, the relevant statistical operator reduces to an equilibrium statistical operator $\hat{\rho}_{\text{eq}}$.
- The equilibrium described by $\hat{\rho}_{\text{eq}}$ appears to be more general than what is described by the usual statistical ensembles. For this reason, $\hat{\rho}_{\text{eq}}$ is generally referred to as *generalised equilibrium statistical operator*.
- We are now interested in understanding the physical meaning of $\hat{\rho}_{\text{rel}}$ when the Lagrange multipliers remain inhomogeneous in space and time. In such a case, expressions (3.34) no longer holds (at least globally). One can assume the system to be divided into volume elements which are large enough for statistical correlations between them to be neglected, but are yet small enough for the variation of the Lagrange multipliers to be small. Whenever those Lagrange multipliers vary slowly in space and time, and were given a physical significance, $\hat{\rho}_{\text{rel}}$ describes a generalised local equilibrium [16, 17, 20, 22]. This is a fundamental assumption in this work. Consequently, $\hat{\rho}_{\text{rel}}$ will for the rest of this work be referred to as *local equilibrium statistical operator*, denoted by $\hat{\rho}_{\text{leq}}$ and abbreviated LESO.
- The system described by the LESO is locally at equilibrium but globally out of equilibrium as we will show later. It offers an interesting and natural extension of equilibrium statistical ensembles to nonequilibrium ones. In particular, we note that equilibrium statistical ensembles can be recovered using the principle of maximum entropy [17]. As a reminder, the traditional way of deriving the corresponding equilibrium ensembles is based on the postulate of microcanonical distribution, that is to say equal probabilities for all permissible dynamical states of an isolated system.

Relevant & irrelevant variables. What about the remaining irrelevant variables? Let $\mathcal{F}$ be the abstract space⁵ of all the operators acting over the corresponding Hilbert space $\mathcal{H}$. Let us assume that the relevant variables from the set $\{\hat{A}_1, \hat{A}_2, \dots, \hat{A}_i, \dots\}$ are linearly independent and span the subspace $\mathcal{F}_{\text{rel}} \subset \mathcal{F}$. It was already justified earlier that the relevant statistical operator is exact in $\mathcal{F}_{\text{rel}}$, cf. (3.2) What about the other subspace $\mathcal{F}_{\text{irr}} \subset \mathcal{F}$ defined such that $\mathcal{F}_{\text{rel}} \cup \mathcal{F}_{\text{irr}} = \mathcal{F}$?

Let $\hat{X}(\mathbf{r})$ be an observable of $\mathcal{F}$. The question we ask is how to divide $\hat{X}(\mathbf{r})$ into a relevant and an irrelevant part. To this purpose, one defines a projector $\hat{\mathbb{P}}$ ⁶ and its orthogonal counterpart $\hat{\mathbb{Q}} \equiv 1 - \hat{\mathbb{P}}$ whose roles are to project $\hat{X}(\mathbf{r})$ onto the respective subspaces $\mathcal{F}_{\text{rel}}$ and $\mathcal{F}_{\text{irr}}$. Many authors have proposed different explicit forms for the projector, amongst them R. Zwanzig, B. Robertson, D. Zubarev, H. Mori or H. Grabert. Here, the projection method of H. Grabert's method [23] is briefly reviewed.

⁵ $\mathcal{F}$ is usually referred to as the *Liouville space*.

⁶The double hat highlighting that it is a superoperator.

Given a macroscopic state described by a set of relevant variables, the mean value of $\hat{X}(\mathbf{r})$ with respect to the relevant statistical operator defined at time t is denoted by $\langle \overline{X(\mathbf{r}, t)} \rangle = \text{Tr } \hat{\rho}_{\text{leq}}(t) \hat{X}(\mathbf{r})$. At the same time, we introduce $\hat{\bar{X}}(\mathbf{r})$, the macroscopic part of the observable $\hat{X}(\mathbf{r})$ obtained according to $\langle \overline{X(\mathbf{r}, t)} \rangle = \text{Tr } \hat{\rho}_t \hat{\bar{X}}(\mathbf{r})$. Then, the projector $\hat{\bar{P}}$ is precisely defined, at time t , by

$$\hat{\bar{P}}_t \hat{X}(\mathbf{r}) = \hat{\bar{X}}(\mathbf{r}). \quad (3.36)$$

Hence,

$$\text{Tr } \hat{\rho}_{\text{leq}}(t) \hat{X}(\mathbf{r}) = \text{Tr } \hat{\rho}_t \hat{\bar{P}}_t \hat{X}(\mathbf{r}) \quad (3.37)$$

justifying that the reduced description of a system by a LESO corresponds to an exact description restricted to the macroscopic part of observables. The explicit expression of the projection superoperator introduced by H. Grabert is

$$\hat{\bar{P}}_t \hat{X}(\mathbf{r}) = \left(1 + \int d\mathbf{r} \delta \hat{A}_i(\mathbf{r}) \frac{\delta}{\delta a_i(\mathbf{r}, t)} \right) \text{Tr } \hat{\rho}_{\text{leq}}(t) \hat{X}(\mathbf{r}), \quad (3.38)$$

with $\delta \hat{A}_i(\mathbf{r}) \equiv \hat{A}_i(\mathbf{r}) - a_i(\mathbf{r}, t)$. Indeed, (3.37) is satisfied. This expression may be seen as a Taylor-series expansion with respect to powers of the fluctuations of the relevant variables $\delta \hat{A}_i(\mathbf{r})$ up to the linear order. One shows that $\hat{\bar{P}}_t \hat{\bar{P}}_t = \hat{\bar{P}}_t$ justifying that $\hat{\bar{P}}_t$ is indeed a projector. Since the right-hand side of (3.38) is a linear combination of the relevant variables, $\hat{\bar{P}}_t$ projects any observable into the subspace $\mathcal{F}_{\text{rel}}$. The orthogonal superoperator $\hat{\bar{Q}}_t$ defined earlier projects any observable onto the subspace of the remaining irrelevant variables $\mathcal{F}_{\text{irr}}$.

With the use of these two projectors, the exact statistical operator can be split into a relevant and irrelevant part as follows,

$$\begin{aligned} \hat{\rho}_t &= \hat{\rho}_t (\hat{\bar{P}}_t + \hat{\bar{Q}}_t) \\ &= \hat{\rho}_t \hat{\bar{P}}_t + \hat{\rho}_t \hat{\bar{Q}}_t \\ &\equiv \hat{\rho}_{\text{leq}}(t) + \hat{\Delta}_t. \end{aligned} \quad (3.39)$$

The $\hat{\Delta}_t$ operator, that we choose to call *deviation operator*, will be studied in §3.3. The irrelevant variables are somehow hidden in it. The mean value of any observable $\hat{X}(\mathbf{r})$ can be decomposed into its relevant (macroscopic) and irrelevant part according to

$$\begin{aligned} \langle \hat{X}(\mathbf{r}) \rangle_t &= \text{Tr } \hat{\rho}_t \hat{X}(\mathbf{r}) \\ &= \text{Tr } \hat{\rho}_{\text{leq}}(t) \hat{X}(\mathbf{r}) + \text{Tr } \hat{\Delta}_t \hat{X}(\mathbf{r}) \\ &= \langle \overline{X(\mathbf{r}, t)} \rangle + \mathcal{X}(\mathbf{r}, t). \end{aligned} \quad (3.40)$$

In the space of the relevant variables, the mean value of $\hat{X}(\mathbf{r})$ is obtained with the use of $\hat{\rho}_{\text{leq}}$ while it is obtained with $\hat{\Delta}$ in the space of the irrelevant variables.

Finally, one should keep in mind that the philosophy of the projection procedure is to be able eliminate the irrelevant variables that are not relevant and necessary for the description of the macroscopic state.

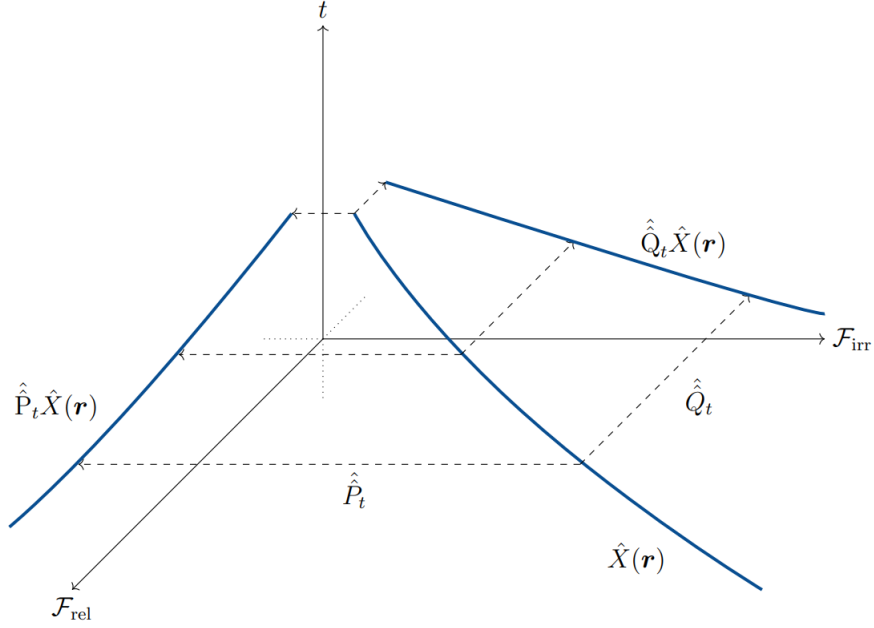


Figure 3.2: An illustrative representation of the projection procedure. Any observable can be projected onto the spaces of the relevant and irrelevant variables with the use of the $\hat{P}$ and $\hat{Q}$ projectors respectively.

3.2. NOTATIONS, TIME EVOLUTION OF THE LESO & SOME RESULTS

LESO in the context of this work. The system that is here considered is described by the set $\{\hat{n}_k(\mathbf{r}), \hat{g}(\mathbf{r}), \hat{e}(\mathbf{r})\}$ of relevant variables. From the previous section, the local equilibrium statistical operator thus takes the form

$$\hat{\rho}_{\text{leq}} = \exp \left\{ -\Omega - \sum_{k=1}^c \int d\mathbf{r} \lambda_{n_k}(\mathbf{r}, t) \hat{n}_k(\mathbf{r}) - \int d\mathbf{r} \lambda_g(\mathbf{r}, t) \hat{g}(\mathbf{r}) - \int d\mathbf{r} \lambda_e(\mathbf{r}, t) \hat{e}(\mathbf{r}) \right\} \quad (3.41)$$

or more compactly,

$$\hat{\rho}_{\text{leq}}[\lambda] = \exp \left\{ -\Omega[\lambda] - \int d\mathbf{r} \lambda^\alpha(\mathbf{r}) \hat{c}^\alpha(\mathbf{r}) \right\} \quad (3.42)$$

with

$$\Omega[\lambda] = \ln \text{Tr} \exp \left\{ - \int d\mathbf{r} \lambda^\alpha(\mathbf{r}) \hat{c}^\alpha(\mathbf{r}) \right\} \quad (3.43)$$

where $\hat{c}^1 = \hat{n}_k$, $\hat{c}^2 = \hat{g}^a$ and $\hat{c}^3 = \hat{e}$. The λ will from now on be referred to as *conjugate fields* (the reason for this will be clarified later). As a result of the conservation equations (2.59)-(2.61), $[\hat{\rho}_{\text{leq}}, \hat{H}] \neq 0$, suggesting that (3.42) is not a global equilibrium statistical operator. The $\Omega[\lambda]$ functional defines the partition function $Z[\lambda]$ of the system,

$$Z[\lambda] = e^{\Omega[\lambda]} \quad \text{or} \quad \Omega[\lambda] = \ln Z[\lambda]. \quad (3.44)$$

Hence, the functional $\Omega[\lambda]$ plays the role of a thermodynamic potential.

The LESO is a fundamental object, and a fundamental hypothesis, for this work. Before going further, we insist on the following points:

- $\hat{\rho}_{\text{leq}}$ is not a solution of the von Neumann equation (2.18).
- $\hat{\rho}_{\text{leq}}$ belongs to the subspace $\mathcal{F}_{\text{rel}} \subset \mathcal{F}$ and is therefore not exact for non relevant observables.
- In parallel, the exact statistical operator $\hat{\rho}_t$ is by definition a solution of the von Neumann equation and is exact for all observables.
- $\hat{\rho}_{\text{leq}}$ defines a generalised local equilibrium.
- $\hat{\rho}_{\text{leq}}$ is acting in the same Hilbert space as the local densities and the Hamiltonian do.

Notations. An arbitrary observable $\hat{A}$ can be considered in the Schrödinger picture $\hat{A} = \hat{A}(\mathbf{r})$ or in the Heisenberg picture $\hat{A} = \hat{A}(\mathbf{r}, t)$, both pictures being related by (2.11). In any case, the picture choice do not change anything in the evaluation of the mean values, cf. (2.13). Any mean value with respect to the LESO defined at time t will be written as

$$\langle \hat{A}(\mathbf{r}) \rangle_{\text{leq}, \lambda_t} = \text{Tr} \hat{\rho}_{\text{leq}}[\lambda_t] \hat{A}(\mathbf{r}). \quad (3.45)$$

Indeed, the time dependence of the LESO comes from the conjugate fields. The mean value with respect to the exact statistical operator at time t is written as

$$\langle \hat{A}(\mathbf{r}) \rangle_t = \text{Tr} \hat{\rho}_t \hat{A}(\mathbf{r}) \equiv A_t(\mathbf{r}). \quad (3.46)$$

For instance,

$$\mathbf{c}_t(\mathbf{r}) \equiv \langle \hat{\mathbf{c}}(\mathbf{r}) \rangle_t = \langle \hat{\mathbf{c}}(\mathbf{r}) \rangle_{\text{leq}, \lambda_t} \quad (3.47)$$

as a result of the third axiom defining the LESO. Tracing over the conservation equations (2.66), $\text{Tr} \hat{\rho}_t \cdot$, yields the conservation equations for the mean local densities,

$$\partial_t c_t^\alpha(\mathbf{r}) + \partial^a \langle j_{c^a}^\alpha(\mathbf{r}) \rangle_t = 0. \quad (3.48)$$

The quantity

$$\delta \hat{A} \equiv \hat{A} - \langle \hat{A} \rangle_{\text{leq}, \lambda} \quad (3.49)$$

expressing the deviation of the observable $\hat{A}$ from its local equilibrium mean value will appear many times. It will also prove to be useful to rewrite the LESO as

$$\hat{\rho}_{\text{leq}}[\lambda] = e^{-\hat{\sigma}[\lambda]} \quad (3.50)$$

with $\hat{\sigma}[\lambda] \equiv \Omega[\lambda] + \int d\mathbf{r} \lambda(\mathbf{r}) \cdot \hat{\mathbf{c}}(\mathbf{r})$. For the sake of brevity, $\hat{\sigma}[\lambda]$ will often be written $\hat{\sigma}$.

Time evolution of the LESO. The LESO (3.42) is seen as a functional of the conjugate fields. Its time evolution is dictated by the time evolution of the conjugate fields, whose time evolution is in turn dictated by the local densities according to (cf. result 3.8 below)

$$\lambda_t(\mathbf{r}) = \frac{1}{k_B} \frac{\delta S[c]}{\delta \mathbf{c}(\mathbf{r})} \Big|_{c_t(\mathbf{r})}, \quad (3.51)$$

where $S[c]$ is the entropy functional. Accordingly, the local densities evolve in time as a consequence of the conservation equation (3.48) for the mean densities. At each time t , the conjugate fields evolve such that (3.47) is satisfied.

The time evolution of the LESO cannot be unitary. As already justified earlier, the LESO reduces the description to a few relevant variables. Hence, the unitary time evolution for the total system becomes a non-unitary time evolution for the observables associated with the degrees of freedom that are not traced out, *i.e.* the relevant variables. In particular, it is clear that the equation for the time evolution of the LESO does not predict a unitary time evolution unless $\hat{Q}_t = 0$ [20, 24]. In such a case, the microscopic dynamics would be entirely described by the set of the relevant variables. In general, the exact statistical operator is expected to be related to the LESO by a non-unitary transformation

$$\hat{\rho}_t = \hat{T}_t \hat{\rho}_{\text{leq}}[\lambda_t], \quad (3.52)$$

which would imply that

$$A_t(\mathbf{r}) = \text{Tr } \hat{T}_t \hat{\rho}_{\text{leq}}[\lambda_t] \hat{A}(\mathbf{r}). \quad (3.53)$$

However, we cannot go further in this direction because the non-unitary transformation $\hat{T}_t$ cannot be expressed in terms of simple observables.

Some relations. Here, some useful relations will be proven.

RESULT 3.3.

$$c^\alpha(\mathbf{r}) = -\frac{\delta \Omega[\lambda]}{\delta \lambda^\alpha(\mathbf{r})}.$$

Proof: First, we have that

$$\begin{aligned} \frac{\delta}{\delta \lambda^\alpha(\mathbf{r})} \int d\mathbf{r}' \lambda(\mathbf{r}') \cdot \hat{\mathbf{c}}(\mathbf{r}') &= \int d\mathbf{r}' \frac{\delta}{\delta \lambda^\alpha(\mathbf{r})} \lambda^\beta(\mathbf{r}') \hat{c}^\beta(\mathbf{r}') \\ &= \int d\mathbf{r}' \delta^{\alpha\beta} \delta^{(3)}(\mathbf{r} - \mathbf{r}') \hat{c}^\beta(\mathbf{r}') \\ &= \hat{c}^\alpha(\mathbf{r}). \end{aligned} \quad (3.54)$$

Then,

$$\begin{aligned} -\frac{\delta \Omega[\lambda]}{\delta \lambda^\alpha(\mathbf{r})} &= -e^{-\Omega(\lambda)} \frac{\delta}{\delta \lambda^\alpha(\mathbf{r})} \text{Tr} \exp \left\{ - \int d\mathbf{r} \lambda(\mathbf{r}) \cdot \hat{\mathbf{c}}(\mathbf{r}) \right\} \\ &= e^{-\Omega(\lambda)} \text{Tr} \hat{c}^\alpha(\mathbf{r}) \exp \left\{ - \int d\mathbf{r} \lambda(\mathbf{r}) \cdot \hat{\mathbf{c}}(\mathbf{r}) \right\} \end{aligned}$$

$$= \langle \hat{c}^\alpha(\mathbf{r}) \rangle_{\text{leq}, \lambda} = c^\alpha(\mathbf{r}). \quad (3.55)$$

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RESULT 3.4.

$$\frac{\delta \hat{\rho}_{\text{leq}}}{\delta \lambda(\mathbf{r})} = - \int_0^1 dz e^{-z\hat{\sigma}} \delta \hat{c} e^{z\hat{\sigma}} \hat{\rho}_{\text{leq}}.$$

Proof: The proof closely follows [25]. By definition,

$$\delta e^{-\hat{\sigma}[\lambda]} = \lim_{\delta\lambda \rightarrow 0} (e^{-\hat{\sigma}[\lambda+\delta\lambda]} - e^{-\hat{\sigma}[\lambda]}). \quad (3.56)$$

Let $U(z)$ be defined such that $U(z) = e^{z\hat{\sigma}[\lambda]} e^{-\hat{\sigma}[\lambda+\delta\lambda]}$. We have

$$\partial_z U(z) = -e^{z\hat{\sigma}[\lambda]} (\hat{\sigma}[\lambda+\delta\lambda] - \hat{\sigma}[\lambda]) e^{-z\hat{\sigma}[\lambda]} U(z). \quad (3.57)$$

An integration from $z = 0$ to $z = 1$ yields,

$$e^{\hat{\sigma}[\lambda]} e^{-\hat{\sigma}[\lambda+\delta\lambda]} - 1 = - \int_0^1 dz e^{z\hat{\sigma}[\lambda]} (\hat{\sigma}[\lambda+\delta\lambda] - \hat{\sigma}[\lambda]) e^{-z\hat{\sigma}[\lambda]} U(z). \quad (3.58)$$

Alternatively,

$$\begin{aligned} e^{-\hat{\sigma}[\lambda+\delta\lambda]} - e^{-\hat{\sigma}[\lambda]} &= - \int_0^1 dz e^{(z-1)\hat{\sigma}[\lambda]} (\hat{\sigma}[\lambda+\delta\lambda] - \hat{\sigma}[\lambda]) e^{-z\hat{\sigma}[\lambda]} U(z) \\ &= - \int_0^1 dz' e^{-z'\hat{\sigma}[\lambda]} (\hat{\sigma}[\lambda+\delta\lambda] - \hat{\sigma}[\lambda]) e^{(z'-1)\hat{\sigma}[\lambda]} U(1-z') \end{aligned} \quad (3.59)$$

where $z' \equiv 1 - z$ was set. Hence,

$$\delta e^{-\hat{\sigma}[\lambda]} = - \int_0^1 dz e^{-z\hat{\sigma}[\lambda]} \delta \hat{\sigma}[\lambda] e^{z\hat{\sigma}[\lambda]} \hat{\rho}_{\text{leq}}[\lambda] \quad (3.60)$$

where the $U(z)$ operator reduces to the identity operator when taking the limit $\delta\lambda \rightarrow 0$. With the result 3.3,

$$\frac{\delta \hat{\sigma}[\lambda]}{\delta \lambda(\mathbf{r})} = -c(\mathbf{r}) + \hat{c}(\mathbf{r}) = \delta \hat{c}, \quad (3.61)$$

and we immediately obtain the desired result, *i.e.*

$$\frac{\delta \hat{\rho}_{\text{leq}}}{\delta \lambda(\mathbf{r})} = - \int_0^1 dz e^{-z\hat{\sigma}} \delta \hat{c} e^{z\hat{\sigma}} \hat{\rho}_{\text{leq}}[\lambda]. \quad (3.62)$$

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RESULT 3.5. Let $\hat{X}$ be any observable that does not depend on the conjugate fields λ . Then,

$$\frac{\delta \langle \hat{X} \rangle_{\text{leq}, \lambda}}{\delta \lambda^\alpha(\mathbf{r})} = - \int_0^1 dz \langle \hat{X} e^{-z\hat{\sigma}} \delta \hat{c}^\alpha e^{z\hat{\sigma}} \rangle_{\text{leq}, \lambda}.$$

Proof: With the result 3.4, we have

$$\begin{aligned}
 \frac{\delta \langle \hat{X} \rangle_{\text{leq}, \lambda}}{\delta \lambda^\alpha(\mathbf{r})} &= \frac{\delta}{\delta \lambda^\alpha(\mathbf{r})} \text{Tr} \hat{X} \hat{\rho}_{\text{leq}}[\lambda] \\
 &= \text{Tr} \hat{X} \frac{\delta \hat{\rho}_{\text{leq}}[\lambda]}{\delta \lambda^\alpha(\mathbf{r})} \\
 &= - \int_0^1 dz \text{Tr} \hat{X} e^{-z\hat{\sigma}} \delta \hat{c}^\alpha e^{z\hat{\sigma}} \hat{\rho}_{\text{leq}}[\lambda] \\
 &= - \int_0^1 dz \langle \hat{X} e^{-z\hat{\sigma}} \delta \hat{c}^\alpha e^{z\hat{\sigma}} \rangle_{\text{leq}, \lambda}.
 \end{aligned} \tag{3.63}$$

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RESULT 3.6. *The normalisation condition for the exact statistical operator $\hat{\rho}_t$ implies that*

$$\int d\mathbf{r} \partial^a \lambda^\alpha(\mathbf{r}) \langle \hat{f}_{c^\alpha}^a \rangle_{\text{leq}, \lambda} = 0.$$

Proof: First, with (2.21) and (2.23), and the fourth axiom of §3.1, we have that

$$\begin{aligned}
 \hat{\rho}_t &= e^{-\hat{\mathcal{L}}t} \hat{\rho}_0 = e^{-\hat{\mathcal{L}}t} \hat{\rho}_{\text{leq}}[\lambda_0] \\
 &= e^{-\frac{i}{\hbar} \hat{H}t} \hat{\rho}_{\text{leq}}[\lambda_0] e^{\frac{i}{\hbar} \hat{H}t} \\
 &= e^{-\Omega[\lambda_0] - \int d\mathbf{r} \lambda_0^\alpha(\mathbf{r}) e^{-\frac{i}{\hbar} \hat{H}t} \hat{c}^\alpha(\mathbf{r}) e^{\frac{i}{\hbar} \hat{H}t}} \\
 &= e^{-\Omega[\lambda_0] - \int d\mathbf{r} \lambda_0^\alpha(\mathbf{r}) \hat{c}^\alpha(\mathbf{r}, -t)}.
 \end{aligned} \tag{3.64}$$

Since the normalisation condition for $\hat{\rho}_t$ is $\text{Tr} \hat{\rho}_t = 1$, we have that $\frac{d}{dt} \text{Tr} \hat{\rho}_t = 0$. Hence,

$$\begin{aligned}
 \frac{d}{dt} \text{Tr} \hat{\rho}_t &= \frac{d}{dt} \text{Tr} e^{-\Omega[\lambda_0] - \int d\mathbf{r} \lambda_0^\alpha(\mathbf{r}) \hat{c}^\alpha(\mathbf{r}, -t)} \\
 &= -\text{Tr} \left\{ \left(\frac{d}{dt} \int d\mathbf{r} \lambda_0^\alpha(\mathbf{r}) \hat{c}^\alpha(\mathbf{r}, -t) \right) \hat{\rho}_t \right\} \\
 &= \text{Tr} \left\{ \int d\mathbf{r} \lambda_0^\alpha(\mathbf{r}) (\partial_t \hat{c}^\alpha)(\mathbf{r}, -t) \hat{\rho}_t \right\} \\
 &= -\text{Tr} \left\{ \int d\mathbf{r} \lambda_0^\alpha(\mathbf{r}) (\partial^a \hat{f}_{c^\alpha}^a)(\mathbf{r}, -t) \hat{\rho}_t \right\}
 \end{aligned} \tag{3.65}$$

where the conservation equations for the mean densities (2.66) were used. Evaluating at $t = 0$ yields

$$\begin{aligned}
 0 &= -\text{Tr} \left\{ \int d\mathbf{r} \lambda_0^\alpha(\mathbf{r}) (\partial^a \hat{f}_{c^\alpha}^a)(\mathbf{r}, 0) \hat{\rho}_0 \right\} \\
 &= -\text{Tr} \left\{ \int d\mathbf{r} \lambda_0^\alpha(\mathbf{r}) (\partial^a \hat{f}_{c^\alpha}^a)(\mathbf{r}, 0) \hat{\rho}_{\text{leq}}[\lambda_0] \right\} \\
 &= - \int d\mathbf{r} \lambda_0^\alpha(\mathbf{r}) \partial^a \langle \hat{f}_{c^\alpha}^a \rangle_{\text{leq}, \lambda_0} \\
 &= \int d\mathbf{r} \partial^a \lambda^\alpha(\mathbf{r}) \langle \hat{f}_{c^\alpha}^a \rangle_{\text{leq}, \lambda}
 \end{aligned} \tag{3.66}$$

where λ_0 was renamed λ and an integration by parts performed assuming that the conjugate fields vanish at $\mathbf{r} = \pm\infty$.

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3.3. DEVIATION OPERATOR

The fourth axiom defining the LESO says that it coincides, at time $t = 0$, with the exact statistical operator. However, there are no reasons that they coincide at times $t > 0$. To account for this feature, a deviation operator $\hat{\Delta}_t$, is introduced such that

$$\hat{\rho}_t = \hat{\rho}_{\text{leq}} + \hat{\Delta}_t \quad (3.67)$$

with $\hat{\Delta}_{t=0} = 0$. The deviation operator was previously encountered in equation (3.39). It cannot be treated as a statistical operator since

$$\begin{cases} \text{Tr } \hat{\Delta}_t = 0 \\ \hat{\Delta}_t^\dagger = \hat{\Delta}_t \end{cases} \quad (3.68)$$

and so does not satisfy the first property of (2.17).

Our purpose here is to obtain an explicit expression for $\hat{\Delta}_t$. The very starting point is the von Neumann equation for the exact statistical operator written in the form (2.19). Because the Hamiltonian function does not explicitly depend on time, a formal integration of this equation yields

$$\begin{aligned} \hat{\rho}_t &= e^{-\hat{\mathcal{L}}t} \hat{\rho}_{t=0} = e^{-\hat{\mathcal{L}}t} \hat{\rho}_{\text{leq}}[\lambda_0] \\ &= \hat{\rho}_{\text{leq}}[\lambda_t] - \int_0^t d\tau \partial_\tau \{e^{-\hat{\mathcal{L}}(t-\tau)} \hat{\rho}_{\text{leq}}[\lambda_\tau]\} \\ &= \hat{\rho}_{\text{leq}}[\lambda_t] - \int_0^t d\tau e^{-\hat{\mathcal{L}}(t-\tau)} \{\hat{\mathcal{L}} + \partial_\tau\} \hat{\rho}_{\text{leq}}[\lambda_\tau] \end{aligned} \quad (3.69)$$

so that

$$\hat{\Delta}_t = - \int_0^t d\tau e^{-\hat{\mathcal{L}}(t-\tau)} \{\hat{\mathcal{L}} + \partial_\tau\} \hat{\rho}_{\text{leq}}[\lambda_\tau]. \quad (3.70)$$

Two quantities have to be calculated: $\hat{\mathcal{L}} \hat{\rho}_{\text{leq}}[\lambda_\tau]$ and $\partial_\tau \hat{\rho}_{\text{leq}}[\lambda_\tau]$. Let us begin with the first one. Beforehand, the following result is needed.

RESULT 3.7. *Let $\hat{A}$ and $\hat{B}$ be two operators. Then,*

$$[\hat{A}, e^{-\alpha \hat{B}}] = e^{-\alpha \hat{B}} \int_0^\alpha dz e^{z \hat{B}} [\hat{B}, \hat{A}] e^{-z \hat{B}}$$

or alternatively

$$[\hat{A}, e^{-\alpha \hat{B}}] = \int_0^\alpha dz e^{-z \hat{B}} [\hat{B}, \hat{A}] e^{z \hat{B}} e^{-\alpha \hat{B}}.$$

Proof: Let us define the quantity $\hat{A}(z) \equiv e^{\alpha z \hat{B}} \hat{A} e^{-\alpha z \hat{B}}$ which obeys the equation

$$\partial_z \hat{A}(z) = \alpha [\hat{B}, \hat{A}(z)]. \quad (3.71)$$

Integrating this equation from $z = 0$ to $z = 1$ yields

$$e^{\alpha \hat{B}} \hat{A} e^{-\alpha \hat{B}} = \hat{A} + \alpha \int_0^1 dz [\hat{B}, \hat{A}(z)] \quad (3.72)$$

or,

$$\begin{aligned}\hat{A}e^{-\alpha\hat{B}} - e^{-\alpha\hat{B}}\hat{A} &= \alpha \int_0^1 dz e^{-\alpha\hat{B}}[\hat{B}, \hat{A}(z)] \\ &= \alpha e^{-\alpha\hat{B}} \int_0^1 dz e^{\alpha z\hat{B}}[\hat{B}, \hat{A}]e^{-\alpha z\hat{B}}.\end{aligned}\quad (3.73)$$

Setting $z' \equiv \alpha z$ yields the desired result,

$$[\hat{A}, e^{-\alpha\hat{B}}] = e^{-\alpha\hat{B}} \int_0^\alpha dz' e^{z'\hat{B}}[\hat{B}, \hat{A}]e^{-z'\hat{B}}, \quad (3.74)$$

and setting $u \equiv \alpha - z'$ yields the second desired result.

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Now, with the second identity of the result 3.7,

$$\begin{aligned}\hat{\mathcal{L}}\hat{\rho}_{\text{leq}}[\lambda_\tau] &= \hat{\mathcal{L}}e^{-\hat{\sigma}} \\ &= \frac{1}{\hbar}[\hat{H}, e^{-\hat{\sigma}}] \\ &= \frac{1}{\hbar} \int_0^1 dz e^{-z\hat{\sigma}}[\hat{\sigma}, \hat{H}]e^{z\hat{\sigma}}\hat{\rho}_{\text{leq}}[\lambda_\tau].\end{aligned}\quad (3.75)$$

In parallel, with the conservation equations (2.66), one has

$$[\hat{\sigma}, \hat{H}] = \int d\mathbf{r} \lambda(\mathbf{r}) \cdot [\hat{\mathbf{c}}(\mathbf{r}), \hat{H}] = i\hbar \int d\mathbf{r} \lambda(\mathbf{r}) \cdot \partial_\tau \hat{\mathbf{c}}(\mathbf{r}) = -i\hbar \int d\mathbf{r} \lambda^\alpha(\mathbf{r}) \partial^a f_{\hat{\mathbf{c}}^a}^a(\mathbf{r}) \quad (3.76)$$

so that an integration by parts yields,

$$\hat{\mathcal{L}}\hat{\rho}_{\text{leq}}[\lambda_\tau] = - \int_0^1 dz \int d\mathbf{r} e^{-z\hat{\sigma}} \left\{ \partial^a \lambda_\tau^\alpha(\mathbf{r}) f_{\hat{\mathbf{c}}^a}^a(\mathbf{r}) \right\} e^{z\hat{\sigma}} \hat{\rho}_{\text{leq}}[\lambda_\tau]. \quad (3.77)$$

Symbolically⁷, this expression can be written in vector form

$$\hat{\mathcal{L}}\hat{\rho}_{\text{leq}}[\lambda_\tau] = - \int_0^1 dz \int d\mathbf{r} e^{-z\hat{\sigma}} \left\{ \nabla \lambda(\mathbf{r}) : \hat{\mathbf{j}}(\mathbf{r}) \right\} e^{z\hat{\sigma}} \hat{\rho}_{\text{leq}}[\lambda_\tau]. \quad (3.78)$$

Finally, with the result 3.6, we have

$$\hat{\mathcal{L}}\hat{\rho}_{\text{leq}}[\lambda_\tau] = - \int_0^1 dz \int d\mathbf{r} e^{-z\hat{\sigma}} \left\{ \partial^a \lambda_\tau^\alpha(\mathbf{r}) \delta f_{\hat{\mathbf{c}}^a}^a(\mathbf{r}) \right\} e^{z\hat{\sigma}} \hat{\rho}_{\text{leq}}[\lambda_\tau] \quad (3.79)$$

with the notation (3.49).

Let us now calculate the second quantity. With the result 3.4,

$$\begin{aligned}\partial_\tau \hat{\rho}_{\text{leq}}[\lambda_\tau] &= \int d\mathbf{r} \frac{\delta \hat{\rho}_{\text{leq}}[\lambda_\tau]}{\delta \lambda_\tau(\mathbf{r})} \partial_\tau \lambda_\tau(\mathbf{r}) \\ &= - \int_0^1 dz \int d\mathbf{r} e^{-z\hat{\sigma}} \delta \hat{\mathbf{c}}_\tau(\mathbf{r}) \cdot \partial_\tau \lambda_\tau(\mathbf{r}) e^{z\hat{\sigma}} \hat{\rho}_{\text{leq}}[\lambda_\tau].\end{aligned}\quad (3.80)$$

⁷Indeed, the notation $\hat{\mathbf{j}}$ mixes indices of different nature: spatial indices with local densities indices.

Finally, using (2.22) for the $e^{-\hat{\mathcal{L}}(t-\tau)}$ term, we find that

$$\hat{\Delta}_t = \int_0^t d\tau \int_0^1 dz \int d\mathbf{r} e^{-\frac{i}{\hbar}\hat{H}(t-\tau)} e^{-z\hat{\sigma}} \left\{ \partial_\tau \lambda_\tau(\mathbf{r}) \cdot \delta \hat{\mathbf{c}}_\tau(\mathbf{r}) + \nabla \lambda_\tau(\mathbf{r}) : \delta \hat{\mathbf{j}}(\mathbf{r}, \tau) \right\} e^{z\hat{\sigma}} \hat{\rho}_{\text{leq}}[\lambda_\tau] e^{\frac{i}{\hbar}\hat{H}(t-\tau)}. \quad (3.81)$$

Accordingly, the mean value of any observable $\hat{A}(\mathbf{r})$ can be split up as

$$\begin{aligned} \langle \hat{A}(\mathbf{r}) \rangle_t &= \text{Tr} \hat{\rho}_t \hat{A}(\mathbf{r}) \\ &= \text{Tr} \hat{\rho}_{\text{leq}}[\lambda_t] \hat{A}(\mathbf{r}) + \text{Tr} \hat{\Delta}_t \hat{A}(\mathbf{r}) \\ &= \langle \hat{A}(\mathbf{r}) \rangle_{\text{leq}, \lambda_t} + \mathcal{A}(\mathbf{r}, t). \end{aligned} \quad (3.82)$$

Mean values with respect to the deviation operator will always be denoted by an upper case italic letter.

3.4. ENTROPY PRODUCTION

Expression of entropy. In the second chapter, it was explained that to any statistical operator is associated an entropy given by equation (2.30). Here, we intend to calculate the entropy associated to the LESO. As $\hat{\sigma} = -\Omega[\lambda] - \int d\mathbf{r} \lambda(\mathbf{r}) \cdot \hat{\mathbf{c}}(\mathbf{r})$ is Hermitian, result 3.1 applies and immediately yields

$$\begin{aligned} S &= -k_B \text{Tr}(\hat{\rho}_{\text{leq}} \ln \hat{\rho}_{\text{leq}}) \\ &= k_B \Omega[\lambda] + k_B \int d\mathbf{r} \lambda(\mathbf{r}) \cdot \mathbf{c}(\mathbf{r}). \end{aligned} \quad (3.83)$$

We have the following result.

RESULT 3.8. *Up to a multiplicative $-k_B$ constant, the entropy (3.83) is the Legendre transform of the Ω functional (3.43), hence justifying that the entropy is a functional of the mean local densities, $S = S[\mathbf{c}]$. In addition, the conjugate fields are given by the functional derivatives of the entropy functional,*

$$\lambda(\mathbf{r}) = \frac{1}{k_B} \frac{\delta S[\mathbf{c}]}{\delta \mathbf{c}(\mathbf{r})}.$$

Proof: A quick reminder on Legendre transforms is provided in appendix E. Let us calculate the Legendre transform of the $\Omega[\lambda]$ functional. With the result 3.3, $\frac{\delta \Omega[\lambda]}{\delta \lambda^\alpha(\mathbf{r})} = -c^\alpha(\mathbf{r})$. If $\frac{\delta^2 \Omega[\lambda]}{\delta \lambda^\alpha(\mathbf{r}) \delta \lambda^\beta(\mathbf{r}')} \neq 0$, one can express the mean local densities in terms of the conjugate fields, i.e., $\lambda = \lambda(\mathbf{c})$. Then, the Legendre transform of $\Omega[\lambda]$ can be calculated,

$$\begin{aligned} \Psi[\mathbf{c}] &= \left(- \int d\mathbf{r} \lambda(\mathbf{r}) \cdot \mathbf{c}(\mathbf{r}) - \Omega[\lambda] \right) \Big|_{\lambda=\lambda(\mathbf{c})} \\ &= - \int d\mathbf{r} \lambda(\mathbf{c}(\mathbf{r})) \cdot \mathbf{c}(\mathbf{r}) - \Omega[\lambda(\mathbf{c})]. \end{aligned} \quad (3.84)$$

So with (3.83), we see that $S[c] = -k_B \Psi[c]$ is the entropy associated to the LESO. It follows that

$$\lambda(\mathbf{r}) = \frac{1}{k_B} \frac{\delta S[c]}{\delta \mathbf{c}(\mathbf{r})}. \quad (3.85)$$

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Vice versa, the Legendre transform of the entropy gives back the Ω functional according to

$$\begin{aligned} \Omega[\lambda] &= - \int d\mathbf{r} \lambda(\mathbf{c}(\mathbf{r})) \cdot \mathbf{c}(\mathbf{r}) - \Psi[c] \\ &= - \int d\mathbf{r} \lambda(\mathbf{c}(\mathbf{r})) \cdot \mathbf{c}(\mathbf{r}) + \frac{1}{k_B} S[c]. \end{aligned} \quad (3.86)$$

We note that the inversions $\lambda = \lambda(\mathbf{c})$ or $\mathbf{c} = \mathbf{c}(\lambda)$ are possible only if, respectively, the second functional derivatives $\frac{\delta^2 \Omega[\lambda]}{\delta \lambda^\alpha(\mathbf{r}) \delta \lambda^\beta(\mathbf{r}')}$ and $\frac{\delta^2 S[c]}{\delta c^\alpha(\mathbf{r}) \delta c^\beta(\mathbf{r}')}$ are not singular, which is usually the case away from phase transitions (cf. §2.1).

We would now like to obtain an expression for the time evolution of the entropy.

Entropy production. The entropy can be evaluated at time t according to

$$S_t = k_B \Omega[\lambda_t] + k_B \int d\mathbf{r} \lambda_t(\mathbf{r}) \cdot \mathbf{c}_t(\mathbf{r}) \quad (3.87)$$

with λ_t understood as $\lambda(\mathbf{c}_t)$. As already explained in the first chapter, the entropy variation can be written as (1.30),

$$dS = d_i S + d_e S \quad (3.88)$$

d_i denoting an internal variation and the d_e variation occurring from the exchange with the external environment. As the system is assumed to be isolated, $d_e S = 0$. We have,

$$\frac{d_i S}{dt} = \frac{dS}{dt} = k_B \frac{d}{dt} \Omega[\lambda_t] + k_B \int d\mathbf{r} (\partial_t \lambda_t(\mathbf{r})) \cdot \mathbf{c}_t(\mathbf{r}) + k_B \int d\mathbf{r} \lambda_t(\mathbf{r}) \cdot \partial_t \mathbf{c}_t(\mathbf{r}). \quad (3.89)$$

With the result 3.3,

$$\begin{aligned} \frac{d}{dt} \Omega[\lambda_t] &= \int d\mathbf{r} \frac{\delta \Omega[\lambda_t]}{\delta \lambda_t} \cdot \partial_t \lambda_t \\ &= - \int d\mathbf{r} \mathbf{c}_t(\mathbf{r}) \cdot \partial_t \lambda_t \end{aligned} \quad (3.90)$$

so that

$$\frac{d_i S}{dt} = -k_B \int d\mathbf{r} \lambda^\alpha(\mathbf{r}) \partial^a \langle j_{c^\alpha}^a(\mathbf{r}) \rangle_t \quad (3.91)$$

with the conservation equations (3.48). Integrating by parts gives

$$\frac{d_i S}{dt} = k_B \int d\mathbf{r} \partial^a \lambda^\alpha(\mathbf{r}) \langle j_{c^\alpha}^a(\mathbf{r}) \rangle_t. \quad (3.92)$$

Let us now perform the split (3.82) of the local current densities,

$$\langle j_{c^\alpha}^a(\mathbf{r}) \rangle_t = \langle j_{c^\alpha}^a(\mathbf{r}) \rangle_{\text{leq}, \lambda_t} + \mathcal{J}_{c^\alpha}^a(\mathbf{r}, t). \quad (3.93)$$

Then, with the use of the result 3.6 , the entropy production takes its final form

$$\boxed{\frac{d_i S}{dt} = k_B \int d\mathbf{r} \partial^a \lambda^\alpha(\mathbf{r}) \mathcal{J}_{c^a}^a(\mathbf{r}, t).} \quad (3.94)$$

In the macroscopic limit, this quantity is expected to be non negative to satisfy the second law of thermodynamics. The local current densities $\langle \hat{\mathbf{j}}(\mathbf{r}) \rangle_{\text{leq}, \lambda_t}$ do not enter in the expression of the entropy production. They are associated to dissipativeless and reversible processes. Hence, the source of dissipation within the system is to be traced back to the $\mathcal{J}$ local current densities since they sole contribute to the entropy production. Were they vanish, the entropy would be conserved. It is very tempting to compare (3.94) with equation (1.55) giving the general expression of the entropy production in the classical frame. In the latter case, the entropy production is given by the product of the affinities (or thermodynamic forces) with the flux densities of globally conserved quantities. In this light, we expect the conjugate fields to be somehow related to the thermodynamic conjugate variables (*cf.* §1.4). This will be shown in the next chapter.

Equation (3.94) is fundamental because it demonstrates that even though the system is found at local equilibrium, it is in a global nonequilibrium state justifying why $\hat{\rho}_{\text{leq}}$ can be used to describe nonequilibrium states. The equation also suggests that we are on the right track to obtain the Green-Kubo formulas. Having identified the source of dissipation, and with the use of the conservation equations (3.48), the macroscopic hydrodynamic equations will be obtained, and the corresponding linear response coefficients identified.

3.5. CORRELATION FUNCTIONS AT LOCAL EQUILIBRIUM

Correlation functions. The second functional derivative of the functional $\Omega[\lambda]$ defines the matrix of correlation functions

$$\mathbf{C}(\mathbf{r}, \mathbf{r}') \equiv \frac{\delta^2 \Omega[\lambda]}{\delta \lambda^\alpha(\mathbf{r}) \delta \lambda^\beta(\mathbf{r}')}. \quad (3.95)$$

The justification can be developed as follows. We first begin by claiming the following relation,

$$\frac{\delta^2 \Omega[\lambda]}{\delta \lambda^\alpha(\mathbf{r}) \delta \lambda^\beta(\mathbf{r}')} = \int_0^1 dz \langle \hat{c}^\alpha(\mathbf{r}) e^{-z\hat{\sigma}} \delta \hat{c}^\beta(\mathbf{r}') e^{z\hat{\sigma}} \rangle_{\text{leq}, \lambda}. \quad (3.96)$$

If one defines

$$\delta \tilde{a}(\mathbf{r}) = \int_0^1 dz e^{-z\hat{\sigma}} \delta a(\mathbf{r}) e^{z\hat{\sigma}}, \quad (3.97)$$

then the right-hand side of (3.96) can be cast in a more speaking form which is directly related to the standard definition of correlation functions,

$$\frac{\delta^2 \Omega[\lambda]}{\delta \lambda^\alpha(\mathbf{r}) \delta \lambda^\beta(\mathbf{r}')} = \langle \delta \hat{c}^\alpha(\mathbf{r}) \delta \hat{c}^\beta(\mathbf{r}') \rangle_{\text{leq}, \lambda}. \quad (3.98)$$

This can be properly demonstrated with the use of results 3.3 and 3.5,

$$\frac{\delta^2 \Omega[\lambda]}{\delta \lambda^\alpha(\mathbf{r}) \delta \lambda^\beta(\mathbf{r}')} = -\frac{\delta c^\beta(\mathbf{r}')}{\delta \lambda^\alpha(\mathbf{r})} = -\frac{\delta c^\alpha(\mathbf{r})}{\delta \lambda^\beta(\mathbf{r}')}$$

$$= \int_0^1 dz \langle \hat{c}^\alpha(\mathbf{r}) e^{-z\hat{\sigma}} \delta \hat{c}^\beta(\mathbf{r}') e^{z\hat{\sigma}} \rangle_{\text{leq},\lambda} \quad (3.99)$$

$$= \langle \hat{c}^\alpha(\mathbf{r}) \delta \hat{c}^\beta(\mathbf{r}') \rangle_{\text{leq},\lambda} \\ = \langle \delta \hat{c}^\alpha(\mathbf{r}) \delta \hat{c}^\beta(\mathbf{r}') \rangle_{\text{leq},\lambda} + \langle \hat{c}^\alpha(\mathbf{r}) \rangle_{\text{leq},\lambda} \langle \delta \hat{c}^\beta(\mathbf{r}') \rangle_{\text{leq},\lambda} \quad (3.100)$$

where the last term vanishes because

$$\langle \delta \hat{c}^\beta(\mathbf{r}') \rangle_{\text{leq},\lambda} = \langle \hat{c}^\beta(\mathbf{r}') \rangle_{\text{leq},\lambda} - \langle \hat{c}^\beta(\mathbf{r}') \rangle_{\text{leq},\lambda} = 0. \quad (3.101)$$

This leads us to infer without any proof that the n -point correlation function is given by

$$\frac{\delta^n \Omega[\lambda]}{\delta \lambda^{\alpha_n}(\mathbf{r}_n) \cdots \delta \lambda^{\alpha_2}(\mathbf{r}_2) \delta \lambda^{\alpha_1}(\mathbf{r}_1)}. \quad (3.102)$$

It is tempting, though without any formal or rigorous justification, to establish the connection between with quantum field theory where such functional derivatives also produce the n -point correlation functions. In this light, the conjugate fields λ may be interpreted as source terms acting as thermodynamic forces. We note that the connection remains illustrative since the nature of the fields in both cases is radically different.

The corresponding inverse correlation matrix $\mathbf{C}^{-1}(\mathbf{r}, \mathbf{r}')$ is defined such that

$$\int d\mathbf{r}'' C_{\alpha\gamma}(\mathbf{r}, \mathbf{r}'') C_{\gamma\beta}^{-1}(\mathbf{r}'', \mathbf{r}') = \delta_{\alpha\beta} \delta(\mathbf{r} - \mathbf{r}'). \quad (3.103)$$

In fact, one has

$$C_{\gamma\beta}^{-1}(\mathbf{r}'', \mathbf{r}') = -\frac{1}{k_B} \frac{\delta^2 S[\mathbf{c}]}{\delta \mathbf{c}(\mathbf{r}) \delta \mathbf{c}(\mathbf{r}')} \quad (3.104)$$

as it may explicitly be verified,

$$\int d\mathbf{r}'' C_{\alpha\gamma}(\mathbf{r}, \mathbf{r}'') C_{\gamma\beta}^{-1}(\mathbf{r}'', \mathbf{r}') = -\frac{1}{k_B} \int d\mathbf{r}'' \frac{\delta^2 \Omega[\lambda]}{\delta \lambda^\alpha(\mathbf{r}) \delta \lambda^\gamma(\mathbf{r}'')} \frac{\delta^2 S[\mathbf{c}]}{\delta \mathbf{c}^\gamma(\mathbf{r}'') \delta \mathbf{c}^\beta(\mathbf{r}')} \quad (3.105)$$

$$= \int d\mathbf{r}'' \frac{\delta \mathbf{c}^\alpha(\mathbf{r})}{\delta \lambda^\gamma(\mathbf{r}'')} \frac{\delta \lambda^\gamma(\mathbf{r}'')}{\delta \mathbf{c}^\beta(\mathbf{r}')} \quad (3.106)$$

$$= \frac{\delta \mathbf{c}^\alpha(\mathbf{r})}{\delta \mathbf{c}^\beta(\mathbf{r}')} = \delta_{\alpha\beta} \delta(\mathbf{r} - \mathbf{r}'). \quad (3.107)$$

As a result of their definition, the correlation functions are connecting infinitesimal perturbations of the mean densities and conjugate fields according to

$$\delta \mathbf{c}(\mathbf{r}) = - \int d\mathbf{r}' \mathbf{C}(\mathbf{r}, \mathbf{r}') \cdot \delta \lambda(\mathbf{r}') \quad \text{and} \quad \delta \lambda(\mathbf{r}) = - \int d\mathbf{r}' \mathbf{C}^{-1}(\mathbf{r}, \mathbf{r}') \cdot \delta \mathbf{c}(\mathbf{r}'). \quad (3.108)$$

In normal fluids, such correlation functions are typically decaying as $C(r) \simeq r^{-1} e^{-r/l_c}$ with $r = \|\mathbf{r} - \mathbf{r}'\|$ and l_c the correlation length.

CHAPTER 4

DISSIPATIVELESS EQUATIONS

“Il faut faire de l’ordre avec le désordre.”¹

MARC CAUSSIDIÈRE AS PREFECT OF POLICE
DURING THE FEBRUARY 1848 FRENCH
REVOLUTION TAKING PLACE IN PARIS.

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The previous chapter laid the foundations of this work: introduction of the LESO, definition of the deviation operator, but also the expression of the entropy production. The latter showed that the entropy is conserved if the dissipative parts of the current densities are vanishing, meaning that the mean current densities with respect to the LESO are not involved in the entropy production. They are thus related to dissipativeless processes. This chapter is devoted to the study of these dissipativeless current densities, the objective being to obtain the dissipativeless hydrodynamic equations as well as the equations for the time evolution of the conjugate fields.

The conjugate fields introduced in the previous chapter were so far depicted as abstract fields (Lagrange multipliers) entering into the expression of the LESO. They require to be given

¹Suggested translation: “You have to make order out of disorder.”

a physical significance and this will be done in §4.1. The normal fluid system that is here considered is not static but described by a velocity field $\mathbf{u} = \mathbf{u}(\mathbf{r}, t)$ that is expected to appear, for instance, in the local conservation equations. In §4.2, a frame transformation will be performed to extract the dependence of the local densities and the dissipativeless current densities on the velocity field. Thereafter, in §4.3, the dissipativeless hydrodynamic equations will naturally emerge from the formalism using the microscopic conservation equations discussed in the previous chapter. Finally, in §4.4, the equations for the time evolution of the conjugate fields will be obtained, two of them recognised as the Navier-Stokes and the heat equations for dissipativeless fluids.

4.1. LOCAL THERMODYNAMICS & CONJUGATE FIELDS

In the §3.1 and §3.2, the LESO was introduced and discussed in detail. It was presented as a functional of some conjugate fields. These conjugate fields play an important role in many expressions obtained in the previous chapter, for instance in the deviation operator (3.81) or the entropy production (3.94). So far, the definition of these fields remains abstract and it is now time to give them a physical significance. To do that, we will use the result 3.8 obtained earlier, *i.e.*

$$\lambda(\mathbf{r}) = \frac{1}{k_B} \frac{\delta S[\mathbf{c}]}{\delta \mathbf{c}(\mathbf{r})}. \quad (4.1)$$

In general, the entropy functional $S[\mathbf{c}]$ may be expressed as the space integration of some function of the mean densities and their spatial derivatives [14], that is to say

$$S[\mathbf{c}] = \int d\mathbf{r} s(c^\alpha, \partial^a c^\alpha, (\partial^a c^\alpha)^2, \dots). \quad (4.2)$$

The function s can itself be expanded as

$$s(c^\alpha, \partial^a c^\alpha, (\partial^a c^\alpha)^2, \dots) = s(c^\alpha) + \frac{\partial s}{\partial c^\alpha} \partial^a c^\alpha + \frac{1}{2} \frac{\partial^2 s}{\partial c^\alpha \partial c^\beta} \partial^a c^\alpha \partial^b c^\beta + \dots \quad (4.3)$$

so that

$$S[\mathbf{c}] = \int d\mathbf{r} \left(s(c^\alpha) + \frac{\partial s}{\partial c^\alpha} \partial^a c^\alpha + \frac{1}{2} \frac{\partial^2 s}{\partial c^\alpha \partial c^\beta} \partial^a c^\alpha \partial^b c^\beta + \dots \right). \quad (4.4)$$

As explained in the §2.3, the system that is here considered is fundamentally invariant under the continuous group of spatial rotations. A special case of spatial rotation is the reflection symmetry $\mathbf{r} \rightarrow -\mathbf{r}$ leaving the measure $d\mathbf{r}$ invariant. Hence, the second term of (4.4) vanishes, allowing to write the entropy functional at leading order in the form

$$S[\mathbf{c}] = \int d\mathbf{r} s(\mathbf{c}) + \mathcal{O}((\nabla \mathbf{c})^2) \quad (4.5)$$

where $(\nabla \mathbf{c})^2$ is the vector notation for $\partial^a c^\alpha \partial^b c^\beta$. The inclusion of higher order terms may be important to describe inhomogeneous fluids or phase transitions. The same expansion applies to the normalisation functional,

$$\Omega[\lambda] = \int d\mathbf{r} \omega(\lambda) + \mathcal{O}((\nabla \lambda)^2). \quad (4.6)$$

In the following, the notation $\mathcal{O}(\nabla^2)$ stands for $\mathcal{O}((\nabla c)^2)$ or $\mathcal{O}((\nabla \lambda)^2)$ depending on the context. We further note that the densities $s(c)$ and $\omega(\lambda)$ are respectively functions of the mean densities and the conjugate fields. Since we have the relation $c^\alpha(\mathbf{r}) = -\frac{\delta \Omega[\lambda]}{\delta \lambda^\alpha(\mathbf{r})}$, we find that $c^\alpha = -\frac{\partial \omega}{\partial \lambda^\alpha}$. Similarly, since we have the relation $\lambda^\alpha(\mathbf{r}) = \frac{1}{k_B} \frac{\delta S[c]}{\delta c^\alpha(\mathbf{r})}$, we find that $\lambda^\alpha = \frac{1}{k_B} \frac{\partial s}{\partial c^\alpha}$. These relations can be rewritten as

$$d\omega = -c^\alpha d\lambda^\alpha \quad (4.7)$$

$$ds = k_B \lambda^\alpha dc^\alpha \quad (4.8)$$

$$s = k_B (\lambda^\alpha c^\alpha + \omega), \quad (4.9)$$

the last equation holding since the entropy and normalisation functionals are related by a Legendre transform. To each of the relations (4.7)-(4.9) corresponds a thermodynamic relation of the appendix B. Equation (4.7) is to be related to the Gibbs-Duhem relation for the Massieu density (B.21), equation (4.8) to the fundamental thermodynamic relation (B.18) for the entropy density, and (4.9) to the Euler relation (B.17). The thermodynamic relations of appendix B hold in every element of matter as the processes involved by the dissipativeless current densities are considered as reversible. They no longer hold if the dissipative part of the current densities is considered at next order of the expansion in the gradients ∇c or $\nabla \lambda$.

This being said, the conjugate fields can easily be obtained at leading order in the gradients,

$$\lambda_{n_k}(\mathbf{r}, t) = \frac{1}{k_B} \frac{\delta S[c]}{\delta n_k(\mathbf{r}, t)} = \frac{1}{k_B} \left(\frac{\partial s}{\partial n_k} \right)_{e, g^a} \quad (4.10)$$

$$\lambda_{g^a}(\mathbf{r}, t) = \frac{1}{k_B} \frac{\delta S[c]}{\delta g^a(\mathbf{r}, t)} = \frac{1}{k_B} \left(\frac{\partial s}{\partial g^a} \right)_{n_k, e} \quad (4.11)$$

$$\lambda_e(\mathbf{r}, t) = \frac{1}{k_B} \frac{\delta S[c]}{\delta e(\mathbf{r}, t)} = \frac{1}{k_B} \left(\frac{\partial s}{\partial e} \right)_{n_k, g^a}. \quad (4.12)$$

Using the fundamental thermodynamic relation² expressed as (B.18) in the frame moving with the fluid element, we obtain

$$\lambda_{n_k}(\mathbf{r}, t) = -\frac{\mu_k(\mathbf{r}, t)}{k_B T(\mathbf{r}, t)} + \mathcal{O}(\nabla^2) \equiv -\beta(\mathbf{r}, t) \mu_k(\mathbf{r}, t) + \mathcal{O}(\nabla^2) \quad (4.13)$$

$$\lambda_{g^a}(\mathbf{r}, t) = -\beta(\mathbf{r}, t) u^a(\mathbf{r}, t) + \mathcal{O}(\nabla^2) \quad (4.14)$$

$$\lambda_e(\mathbf{r}, t) = \beta(\mathbf{r}, t) + \mathcal{O}(\nabla^2) \quad (4.15)$$

as in the classical framework [9]. This result highlights several aspects:

- The name *conjugate field* finds its justification from the expression (4.1). In the §1.4, any quantity γ defined as the derivative of the entropy with respect to another quantity A was said to be conjugate to A . For instance, λ_{n_k} is conjugate to the number density, λ_{g^a} conjugate to the momentum density, and λ_e to the energy density, hence the name.

²In the appendix B, the densities (B.14) were denoted by a fraktur letter. To be consistent with the notations introduced in the previous chapter, densities will now be written by normal lower case letters.

- With the expressions (4.13)-(4.15) of the conjugate fields, the LESO reduces to the grand-canonical statistical operator in the frame moving at the local velocity $\mathbf{u}(\mathbf{r}, t)$, as already forecast in the §3.1.
- The LESO was obtained through the extremization of the Lagrange functional (3.15). In particular, the Boltzmann constant k_B was dropped but is here naturally recovered by equation (4.1).

4.2. TRANSFORMATION TO THE MOVING FRAME

A normal fluid is described by a velocity field $\mathbf{u} = \mathbf{u}(\mathbf{r}, t)$ defined as the barycentric velocity (1.13). Each fluid particle moves at a local velocity $\mathbf{u}_i \equiv \mathbf{u}(\mathbf{r}_i, t)$ with respect to the laboratory frame. The transformation from the laboratory to the moving frame (see figure 4.1) is carried out by the change

$$\hat{\mathbf{p}}_i \rightarrow \hat{\mathbf{p}}_{i0} + m_i \mathbf{u}_i, \quad (4.16)$$

where $\hat{\mathbf{p}}_i$ is the momentum operator associated to the particle labelled i seen in the laboratory frame and $\hat{\mathbf{p}}_{i0}$ is the momentum operator associated to the particle labelled i seen in the frame moving at the local velocity $\mathbf{u}(\mathbf{r}, t)$, in accordance with the principle of Galilean relativity.

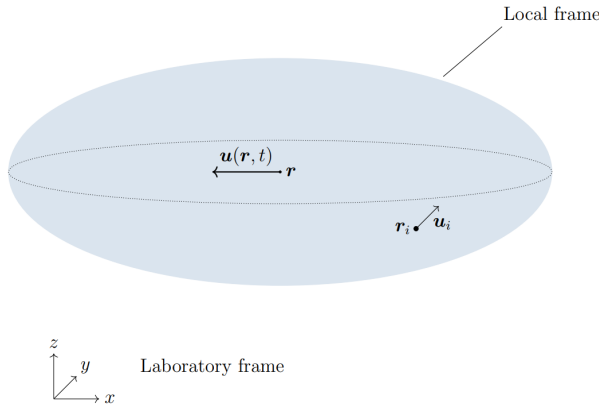


Figure 4.1: *Illustration of the laboratory and the moving frame. The moving frame designates a fluid element moving at the local velocity $\mathbf{u}(\mathbf{r}, t)$.*

This substitution procedure has the benefit to highlight the dependence on the velocity field of various quantities, in particular the local densities and the dissipativeless part of the current densities. The idea behind finds its origin in that the transport coefficients are defined at local equilibrium in a frame that accompanies the fluid elements. A further motivation relies on the observation that the microscopic conservation equations (2.66) do not depend on the velocity field while the macroscopic conservation equations presented in the first chapter do.

How is the validity of this substitution justified? In a nonrelativistic normal fluid, the value of the transport coefficients do not depend on the reference frame in which they are measured. This results from the fact that the hydrodynamic equations are invariant under Galilean transformations. Indeed, a normal fluid (with a single component for simplicity) is precisely described by three local fields ((2.49), (2.51) and (2.52)), each satisfying a local conservation equation as the result of the invariance of the system with respect to the Galilean

group [20]. Consequently, one expects, for instance, the Navier-Stokes or the heat equations to be Galilean invariant.

The purpose of this section is to obtain the dependence of the mean values of the local densities and the dissipativeless current densities on the velocity field. The calculations are no more difficult than performing a simple substitution, though paying attention that the momentum operator does not commute with $\mathbf{u}_i$ since it is a function of the position $\mathbf{r}_i$. Their commutator is given by

$$[\hat{p}_i^a, u_j^b] = -i\hbar \partial_i^a u_j^b \delta_{ij}. \quad (4.17)$$

Transformation of the local densities. The mass and number densities $\hat{\rho}$ and $\hat{n}_k$ are not modified as their definition does not contain any momentum operator.

Dropping the spatial dependence, the momentum local density transforms as

$$\begin{aligned} \hat{g}^a &\rightarrow \frac{1}{2} \sum_{i=1}^N \left\{ (\hat{p}_{i0}^a + m_i u_i^a) \delta(\mathbf{r} - \mathbf{r}_i) + \delta(\mathbf{r} - \mathbf{r}_i) (\hat{p}_{i0}^a + m_i u_i^a) \right\} \\ &= \hat{g}_0^a + \sum_{i=1}^N m_i u_i^a \delta(\mathbf{r} - \mathbf{r}_i) \\ &= \hat{g}_0^a + u^a \sum_{i=1}^N m_i \delta(\mathbf{r} - \mathbf{r}_i) \\ &= \hat{g}_0^a + \hat{\rho} u^a. \end{aligned} \quad (4.18)$$

The calculation for the transformation of the energy local density is very similar and is done in appendix F. The result is

$$\hat{e} \rightarrow \hat{e}_0 + \frac{1}{2} \hat{\rho} u^2 + u^a \hat{g}_0^a. \quad (4.19)$$

Transformation of the local current densities. The mass current density transforms as the momentum density, that is to say,

$$\hat{j}_\rho^a \rightarrow \hat{j}_{\rho_0}^a + \hat{\rho} u^a \quad (4.20)$$

with $\hat{j}_{\rho_0}^a = \hat{g}_0^a$. Similarly, the number current density transforms as

$$\hat{j}_{n_k}^a \rightarrow \hat{j}_{n_{k0}}^a + \hat{n}_k u^a. \quad (4.21)$$

Before going further, let us notice that:

→ The quantity $D(\mathbf{r}; \mathbf{r}_i, \mathbf{r}_j)$ defined in (D.21) does not transform as it does not contain any momentum operator.

→ Regarding the force F_{ij} between particles labelled i and j , we have that

$$\begin{aligned} [\hat{p}_k^a, V_{ij}] &= i\hbar F_{ij}^a(\delta_{ik} - \delta_{jk}) \rightarrow [\hat{p}_{k0}^a + m_k u^a, V_{ij}] = [\hat{p}_{k0}^a, V_{ij}] + m_k [u^a, V_{ij}] \\ &= i\hbar F_{ij0}^a(\delta_{ik} - \delta_{jk}). \end{aligned} \quad (4.22)$$

To calculate the transformation of the momentum current density, it is first decomposed as the sum of a kinetic (D.26) and potential part (D.27). We first focus on the transformation of the kinetic part. The explicit calculation is done as follows. A simple substitution (dropping the dependence of $\mathbf{u}$ in $\mathbf{r}$) yields

$$\begin{aligned} j_{g^b}^a \text{ kin.} &\rightarrow \sum_{i=1}^N \frac{1}{4m_i} \{ (\hat{p}_{i0}^a + m_i u_i^a)(\hat{p}_{i0}^b + m_i u_i^b) \delta(\hat{\mathbf{r}} - \hat{\mathbf{r}}_i) + (\hat{p}_{i0}^a + m_i u_i^a) \delta(\hat{\mathbf{r}} - \hat{\mathbf{r}}_i) (\hat{p}_{i0}^b + m_i u_i^b) + \text{h.c.} \} \\ &= j_{g_0^b}^a \text{ kin.} + \hat{\rho} u^a u^b + \frac{1}{4} \sum_{i=1}^N \{ 2\hat{p}_{i0}^a \delta(\mathbf{r} - \mathbf{r}_i) u_i^b + u_i^a \hat{p}_{i0}^b \delta(\mathbf{r} - \mathbf{r}_i) + u_i^a \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_{i0}^b + \text{h.c.} \}. \end{aligned} \quad (4.23)$$

Now, we note that

$$f(\mathbf{r}_i) \delta(\mathbf{r} - \mathbf{r}_i) = f(\mathbf{r}) \delta(\mathbf{r} - \mathbf{r}_i) \quad (4.24)$$

every times there is some function $f(\mathbf{r}_i)$ next to a Dirac distribution $\delta(\mathbf{r} - \mathbf{r}_i)$. Once we have some function $f(\mathbf{r})$ of the position $\mathbf{r}$, it no longer depends on the particle positions $\mathbf{r}_i$, so that it commutes with all the momentum operators $\hat{p}_{i0}$ and it can be factorized out of the sum $\sum_{i=1}^N$.

If the hermitian conjugate part of the terms in curly brackets is included and (4.17) used, one has

$$\begin{aligned} &2\hat{p}_{i0}^a \delta(\mathbf{r} - \mathbf{r}_i) u_i^b + 2u_i^b \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_{i0}^a + u_i^a \hat{p}_{i0}^b \delta(\mathbf{r} - \mathbf{r}_i) + \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_{i0}^b u_i^a + u_i^a \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_{i0}^b + \hat{p}_{i0}^b \delta(\mathbf{r} - \mathbf{r}_i) u_i^a \\ &= 2\hat{p}_{i0}^a \delta(\mathbf{r} - \mathbf{r}_i) u_i^b + 2u_i^b \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_{i0}^a + \hat{p}_{i0}^b u_i^a \delta(\mathbf{r} - \mathbf{r}_i) + i\hbar \partial_i^b u_i^a \delta(\mathbf{r} - \mathbf{r}_i) + \delta(\mathbf{r} - \mathbf{r}_i) u_i^a \hat{p}_{i0}^b \\ &\quad - i\hbar \delta(\mathbf{r} - \mathbf{r}_i) \partial_i^b u_i^a + u_i^a \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_{i0}^b + \hat{p}_{i0}^b \delta(\mathbf{r} - \mathbf{r}_i) u_i^a. \end{aligned}$$

If the sum is recovered as in line (4.23),

$$\begin{aligned} &\frac{1}{4} \sum_{i=1}^N \{ 2\hat{p}_{i0}^a \delta(\mathbf{r} - \mathbf{r}_i) u_i^b + u_i^a \hat{p}_{i0}^b \delta(\mathbf{r} - \mathbf{r}_i) + u_i^a \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_{i0}^b + \text{h.c.} \} \\ &= \hat{g}_0^a u^b + u^a \hat{g}_0^b. \end{aligned} \quad (4.25)$$

Hence, the momentum current density transforms as

$$j_{g^b}^a \rightarrow j_{g_0^b}^a + \hat{\rho} u^a u^b + u^a \hat{g}_0^b + \hat{g}_0^a u^b. \quad (4.26)$$

The calculation of the transformation of the energy current density is done in appendix F. The result is

$$j_e^a \rightarrow j_{e_0}^a + \hat{e}_0 u^a + j_{g_0^b}^a u^b + \hat{g}_0^b u^b u^a + \frac{1}{2} (\hat{g}_0^a + \hat{\rho} u^a) u^2 - (\hat{\Omega}^a + \hat{\Omega}'^a) \quad (4.27)$$

with

$$\Omega^a \equiv \frac{1}{2} \sum_{i=1}^N \sum_{j(\neq i)}^N F_{ij0}^b \left(u^b - \frac{1}{2} (u_i^b + u_j^b) \right) D^a(\mathbf{r}; \mathbf{r}_i, \mathbf{r}_j) \quad (4.28)$$

$$\hat{\Omega}'^a \equiv \frac{\hbar^2}{4} \partial^b \left\{ (\partial^a u^b) \sum_{i=1}^N \frac{1}{m_i} \delta(\mathbf{r} - \mathbf{r}_i) \right\}. \quad (4.29)$$

The latter quantity does not appear in the classical framework. Ω^a vanishes if the velocity field $\mathbf{u}$ is uniform. We have the following result:

RESULT 4.1. *The quantities Ω^a and $\hat{\Omega}'^a$ behave as $\mathcal{O}(\nabla^2)$, so that they are negligible in the expansion in powers of the gradients.*

Proof: First, it is clear that $\hat{\Omega}'^a = \mathcal{O}(\nabla^2)$ because it is defined with two gradients.

For Ω^a , let us rewrite the quantity $D^a(\mathbf{r}; \mathbf{r}_i, \mathbf{r}_j)$ as follows. Considering $\mathbf{R}_{ij} = \lambda \mathbf{r}_i + (1 - \lambda) \mathbf{r}_j$ which is a straight line between $\mathbf{r}_i$ and $\mathbf{r}_j$, and defining $r_{ij}^a \equiv r_i^a - r_j^a$ and $\mathbf{r}_{1/2} = \frac{1}{2}(\mathbf{r}_i + \mathbf{r}_j)$, one has

$$\begin{aligned} D^a(\mathbf{r}; \mathbf{r}_i, \mathbf{r}_j) &= r_{ij}^a \int_0^1 d\lambda \delta(\mathbf{r} - \mathbf{r}_j - \lambda \mathbf{r}_{ij}) \\ &= r_{ij}^a \int_0^1 d\lambda \delta(\mathbf{r} - \mathbf{r}_{1/2} - (\lambda - 1/2) \mathbf{r}_{ij}) \\ &= r_{ij}^a \int_{-1/2}^{1/2} d\tau \delta(\mathbf{r} - \mathbf{r}_{1/2} - \tau \mathbf{r}_{ij}) \end{aligned} \quad (4.30)$$

where $\tau \equiv \lambda - 1/2$. The Dirac delta distribution can be expanded in powers of τ according to

$$\begin{aligned} \delta(\mathbf{r} - \mathbf{r}_{1/2} - \tau \mathbf{r}_{ij}) &= \sum_{n=0}^{\infty} \frac{1}{n!} (-\tau \mathbf{r}_{ij} \cdot \nabla)^n \delta(\mathbf{r} - \mathbf{r}_{1/2}) \\ &= \delta(\mathbf{r} - \mathbf{r}_{1/2}) - \tau r_{ij}^a \partial^a \delta(\mathbf{r} - \mathbf{r}_{1/2}) + \frac{\tau^2}{2} r_{ij}^a r_{ij}^b \partial^a \partial^b \delta(\mathbf{r} - \mathbf{r}_{1/2}) + \dots \end{aligned} \quad (4.31)$$

Hence,

$$D^a(\mathbf{r}; \mathbf{r}_i, \mathbf{r}_j) = r_{ij}^a \left\{ \delta(\mathbf{r} - \mathbf{r}_{1/2}) + \frac{1}{24} r_{ij}^b r_{ij}^c \partial^b \partial^c \delta(\mathbf{r} - \mathbf{r}_{1/2}) + \dots \right\}. \quad (4.32)$$

Substituting into Ω^a , we obtain

$$\begin{aligned} \Omega^a &= \frac{1}{2} \sum_{i=1}^N \sum_{j(\neq i)}^N F_{ij0}^b \left(u^b - \frac{1}{2} (u_i^b + u_j^b) \right) r_{ij}^a \left(\delta(\mathbf{r} - \mathbf{r}_{1/2}) + \mathcal{O}(\nabla^2) \right) \\ &= \frac{1}{2} \sum_{i=1}^N \sum_{j(\neq i)}^N F_{ij0}^b \left(u^b(\mathbf{r}) - \frac{1}{2} u^b(\mathbf{r}_i) - \frac{1}{2} u^b(\mathbf{r}_j) \right) r_{ij}^a \delta(\mathbf{r} - \mathbf{r}_{1/2}) + \mathcal{O}(\nabla^2) \\ &= \frac{1}{2} \sum_{i=1}^N \sum_{j(\neq i)}^N F_{ij0}^b \left(u^b(\mathbf{r}_{1/2}) - \frac{1}{2} u^b(\mathbf{r}_i) - \frac{1}{2} u^b(\mathbf{r}_j) \right) r_{ij}^a \delta(\mathbf{r} - \mathbf{r}_{1/2}) + \mathcal{O}(\nabla^2). \end{aligned} \quad (4.33)$$

Now, using that

$$\mathbf{r}_i = \mathbf{r}_{1/2} + \frac{1}{2} \mathbf{r}_{ij} \quad \text{and} \quad \mathbf{r}_j = \mathbf{r}_{1/2} - \frac{1}{2} \mathbf{r}_{ij}, \quad (4.34)$$

we have that

$$\begin{aligned} &u^b(\mathbf{r}_{1/2}) - \frac{1}{2} u^b(\mathbf{r}_i) - \frac{1}{2} u^b(\mathbf{r}_j) \\ &= u^b(\mathbf{r}_{1/2}) - \frac{1}{2} u^b\left(\mathbf{r}_{1/2} + \frac{1}{2} \mathbf{r}_{ij}\right) - \frac{1}{2} u^b\left(\mathbf{r}_{1/2} - \frac{1}{2} \mathbf{r}_{ij}\right) \end{aligned}$$

$$\begin{aligned}
 &= u^b(\mathbf{r}_{1/2}) - \frac{1}{2} \left\{ u^b(\mathbf{r}_{1/2}) + \frac{1}{2} r_{ij}^c \partial^c u^b(\mathbf{r}_{1/2}) + \frac{1}{8} r_{ij}^c r_{ij}^d \partial^c \partial^d u^b(\mathbf{r}_{1/2}) + \dots \right\} \\
 &\quad - \frac{1}{2} \left\{ u^b(\mathbf{r}_{1/2}) - \frac{1}{2} r_{ij}^c \partial^c u^b(\mathbf{r}_{1/2}) + \frac{1}{8} r_{ij}^c r_{ij}^d \partial^c \partial^d u^b(\mathbf{r}_{1/2}) + \dots \right\} \quad (4.35)
 \end{aligned}$$

$$\begin{aligned}
 &= -\frac{1}{8} r_{ij}^c r_{ij}^d \partial^c \partial^d u^b(\mathbf{r}_{1/2}) + \dots \\
 &= \mathcal{O}(\nabla^2). \quad (4.36)
 \end{aligned}$$

Hence, we also find that

$$\Omega^a = \mathcal{O}(\nabla^2) + \mathcal{O}(\nabla^2) = \mathcal{O}(\nabla^2). \quad (4.37)$$

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Consequently, $\langle \Omega^a + \hat{\Omega}'^a \rangle_{\text{leq}, \lambda}$ can be neglected because it contributes to the square of the gradients.

Dependence on the velocity field of the mean values. We have the following result:

RESULT 4.2. *The mean value of odd powers of the momentum operator $\hat{\mathbf{p}}_{i0}$ vanish after averaging with respect to the LESO, i.e.*

$$\langle (\hat{\mathbf{p}}_{i0})^n \rangle_{\text{leq}, \lambda} = 0, \quad n \text{ odd}.$$

Proof: With the use of the expression of the conjugate fields (4.13)-(4.15), let us take back the expression of the LESO,

$$\hat{\rho}_{\text{leq}} = \exp \left\{ -\Omega + \sum_{k=1}^c \int d\mathbf{r} \beta(\mathbf{r}, t) \mu_k(\mathbf{r}, t) \hat{n}_k(\mathbf{r}) + \int d\mathbf{r} \beta(\mathbf{r}, t) u^a(\mathbf{r}, t) \hat{g}^a(\mathbf{r}) - \int d\mathbf{r} \beta(\mathbf{r}, t) \hat{e}(\mathbf{r}) \right\}. \quad (4.38)$$

In the moving frame, we have

$$\begin{aligned}
 &\sum_{k=1}^c \int d\mathbf{r} \beta(\mathbf{r}, t) \mu_k(\mathbf{r}, t) \hat{n}_k(\mathbf{r}) + \int d\mathbf{r} \beta(\mathbf{r}, t) u^a(\mathbf{r}, t) \hat{g}^a(\mathbf{r}) - \int d\mathbf{r} \beta(\mathbf{r}, t) \hat{e}(\mathbf{r}) \\
 &= \sum_{k=1}^c \int d\mathbf{r} \beta(\mathbf{r}, t) \left(\mu_{k0}(\mathbf{r}, t) - \frac{1}{2} m_k \mathbf{u}^2(\mathbf{r}, t) \right) \hat{n}_k(\mathbf{r}) + \int d\mathbf{r} \beta(\mathbf{r}, t) u^a(\mathbf{r}, t) \left(\hat{g}_0^a(\mathbf{r}) + \hat{\rho}(\mathbf{r}) u^a(\mathbf{r}, t) \right) \\
 &\quad - \int d\mathbf{r} \beta(\mathbf{r}, t) \left(\hat{e}_0(\mathbf{r}) + \frac{1}{2} \hat{\rho}(\mathbf{r}) \mathbf{u}^2(\mathbf{r}, t) + u^a(\mathbf{r}, t) \hat{g}_0^a(\mathbf{r}) \right) \\
 &= \int d\mathbf{r} \beta(\mathbf{r}, t) \left(\sum_{k=1}^c \mu_{k0}(\mathbf{r}, t) \hat{n}_k(\mathbf{r}) - \hat{e}_0(\mathbf{r}) \right) \quad (4.39)
 \end{aligned}$$

since $\sum_{k=1}^c m_k \hat{n}_k(\mathbf{r}) = \hat{\rho}(\mathbf{r})$ and where we have used that $\mu_k(\mathbf{r}, t) = \mu_{k0}(\mathbf{r}, t) - \frac{1}{2} m_k \mathbf{u}^2(\mathbf{r}, t)$ (cf. appendix B). The LESO thus takes the form

$$\hat{\rho}_{\text{leq}} = \exp \left\{ -\Omega + \int d\mathbf{r} \beta(\mathbf{r}, t) \left(\sum_{k=1}^c \mu_{k0}(\mathbf{r}, t) \hat{n}_k(\mathbf{r}) - \hat{e}_0(\mathbf{r}) \right) \right\} \quad (4.40)$$

which is symmetric under $\hat{\mathbf{p}}_{i0} \rightarrow -\hat{\mathbf{p}}_{i0}$. Consequently,

$$\langle (\hat{\mathbf{p}}_{i0})^n \rangle_{\text{leq}} = \text{Tr} \hat{\rho}_{\text{leq}} (\hat{\mathbf{p}}_{i0})^n$$

$$= (-1)^n \text{Tr } \hat{\rho}_{\text{leq}} (\hat{\mathbf{p}}_{i0})^n = (-1)^n \langle (\hat{\mathbf{p}}_{i0})^n \rangle_{\text{leq}}, \quad (4.41)$$

hence the result.

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With the result 4.2, $\langle \hat{g}_0 \rangle_{\text{leq}, \lambda} = \langle \hat{j}_{\rho_0}^a \rangle_{\text{leq}, \lambda} = \langle \hat{j}_{n_{k0}}^a \rangle_{\text{leq}, \lambda} = \langle \hat{j}_{e_0}^a \rangle_{\text{leq}, \lambda} = 0$. Also,

$$\langle \hat{j}_{\hat{g}_0^b}^a \rangle_{\text{leq}, \lambda} = p \delta^{ab} \quad (4.42)$$

since hydrostatic pressure is an equilibrium property that exists in the absence of macroscopic velocity. With the notations

$$\langle \hat{\rho} \rangle_{\text{leq}, \lambda} \equiv \rho \quad (4.43)$$

$$\langle \hat{n}_k \rangle_{\text{leq}, \lambda} \equiv n_k \quad (4.44)$$

$$\langle \hat{e}_0 \rangle_{\text{leq}, \lambda} \equiv e_0, \quad (4.45)$$

we therefore have

$$\langle \hat{g}^a \rangle_{\text{leq}, \lambda} = \rho u^a \quad (4.46)$$

$$e \equiv \langle \hat{e} \rangle_{\text{leq}, \lambda} = e_0 + \frac{1}{2} \rho \mathbf{u}^2 \quad (4.47)$$

$$\langle \hat{j}_{\rho}^a \rangle_{\text{leq}, \lambda} = \rho u^a \quad (4.48)$$

$$\langle \hat{j}_{n_k}^a \rangle_{\text{leq}, \lambda} = n_k u^a \quad (4.49)$$

$$\langle \hat{j}_{\hat{g}^b}^a \rangle_{\text{leq}, \lambda} = \rho u^a u^b + p \delta^{ab} \quad (4.50)$$

$$\langle \hat{j}_e^a \rangle_{\text{leq}, \lambda} = \left(e_0 + p + \frac{1}{2} \rho \mathbf{u}^2 \right) u^a + \mathcal{O}(\mathbf{V}^2) \quad (4.51)$$

valid within the fluid element located around position $\mathbf{r}$. With (4.46), $u^a = \langle \hat{g}^a \rangle_{\text{leq}, \lambda} / \rho$ so that the velocity field is the ratio between the mean momentum and mean mass densities. In equations (4.46)-(4.51), some terms vanish with the velocity. These terms are due to advection of the corresponding quantity by the motion of the fluid element at the velocity $\mathbf{u}$.

4.3. DISSIPATIVELESS HYDRODYNAMIC EQUATIONS

To obtain the dissipativeless hydrodynamic equations, let us take back the microscopic conservation equations (2.66) averaged with respect to the LESO,

$$\partial_t c_t^\alpha + \partial^a \langle j_{c^\alpha}^a \rangle_{\text{leq}, \lambda_t} = 0. \quad (4.52)$$

For the mass density ($\alpha = 1$), (4.52) becomes

$$\partial_t \langle \hat{\rho} \rangle_{\text{leq}, \lambda} + \partial^a \langle \hat{j}_{\rho}^a \rangle_{\text{leq}, \lambda_t} = 0 \quad (4.53)$$

or with the notation (4.43) and the equation (4.48),

$$\partial_t \rho + \partial^a (\rho u^a) = 0. \quad (4.54)$$

In vector form, the previous equation reads

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{u}) = 0, \quad (4.55)$$

or in the Lagrangian form,

$$\frac{d\rho}{dt} + \rho \nabla \cdot \mathbf{u} = 0. \quad (4.56)$$

The same procedure can be performed, without any difficulty, to the other densities. The final conservation equations in the Eulerian and Lagrangian forms respectively read

$\partial_t \rho + \nabla \cdot (\rho \mathbf{u}) = 0$	$\frac{d\rho}{dt} + \rho \nabla \cdot \mathbf{u} = 0 \quad (4.57)$
$\partial_t n_k + \nabla \cdot (n_k \mathbf{u}) = 0$	$\frac{dn_k}{dt} + n_k \nabla \cdot \mathbf{u} = 0 \quad (4.58)$
$\partial_t (\rho \mathbf{u}) + \nabla \cdot (\rho \mathbf{u} \mathbf{u} + p \mathbf{1}) = 0$	$\rho \frac{d\mathbf{u}}{dt} + \nabla p = 0 \quad (4.59)$
$\partial_t (e_0 + \frac{1}{2} \rho \mathbf{u}^2) + \nabla \cdot \left[\left(e_0 + \frac{1}{2} \rho \mathbf{u}^2 + p \right) \mathbf{u} \right] = 0$	$\frac{de_0}{dt} + (e_0 + p) \nabla \cdot \mathbf{u} = 0. \quad (4.60)$

Some remarks can be made:

- These equations describe adiabatic processes (*i.e.* isentropic processes) and thus leave the entropy constant. Indeed, the $\langle j_{c^a}^a \rangle_{\text{leq}, \lambda}$ do not enter in the expression of the entropy production (3.94). We note that discontinuities due to shock waves may already produce entropy within the framework of dissipativeless fluid for which viscosity and heat conductivity are assumed to be vanishing [2]. Such discontinuities are not described if the conjugate fields are smooth functions.
- In principle, equations (4.57)-(4.59) are exact for dissipativeless (reversible) processes as no approximations have been done. However, equation (4.60) is an approximation in the sense that the terms in $\mathcal{O}(\nabla^2)$ have been neglected.
- The equation (4.57) is the continuity equation (1.10) in its final form, and so does not contain any dissipative term.

Equation (4.58) is the same as (1.17) but without the dissipative part (diffusion flux).

Equation (4.59) is the Euler equation for a viscousless fluid (and in the absence of external forces such as gravity). It is the same as equation (1.22) without the dissipative part of the pressure tensor.

Finally, equation (4.60) is known as the energy flux equation for a dissipativeless fluid (again in the absence of external forces) where e_0 is here the internal energy. It is the same as equation (1.29) but without the energy flux (heat) which is associated to dissipative processes.

We highlight here that the macroscopic equations (4.57)-(4.60) were obtained from the intrinsic microscopic dynamics of the particles making up the normal fluid, which have to be compared to the equations (1.10), (1.17), (1.22) and (1.29) which were obtained from macroscopic considerations. This is a good progress yet. In the next chapter, the dissipative contributions will be added within these equations.

4.4. EQUATIONS FOR THE TIME EVOLUTION OF THE CONJUGATE FIELDS

Derivation of the equations. The time evolution of the LESO is dictated by the time evolution of the conjugate fields (*cf.* §3.2). Also, the expression (3.81) for the deviation operator contains time derivatives of the conjugate fields, whence the necessity to obtain the time evolution of these fields. For this purpose, some thermodynamics is first needed.

Let us work in the frame moving with the fluid element ($\mathbf{u} = 0$). The fundamental thermodynamic relation (B.16) can be rewritten as

$$ds = \frac{1}{T} de_0 - \sum_{k=1}^c \frac{\mu_{k0}}{T} dn_k \quad (4.61)$$

giving

$$\beta = \frac{1}{k_B} \left(\frac{\partial s}{\partial e_0} \right)_{n_1, \dots, n_c} \quad (4.62)$$

$$\beta \mu_{k0} = -\frac{1}{k_B} \left(\frac{\partial s}{\partial n_k} \right)_{e_0, \{n_j\}_{j \neq k}} \quad (4.63)$$

from which we deduce the Maxwell relations

$$\left(\frac{\partial \beta}{\partial n_k} \right)_{e_0, \{n_j\}_{j \neq k}} = \frac{1}{k_B} \left(\frac{\partial^2 s}{\partial n_k \partial e_0} \right) = - \left(\frac{\partial (\beta \mu_{k0})}{\partial e_0} \right)_{n_1, \dots, n_c} \quad (4.64)$$

$$\left(\frac{\partial (\beta \mu_{k0})}{\partial n_l} \right)_{e_0, \{n_j\}_{j \neq k, l}} = -\frac{1}{k_B} \left(\frac{\partial^2 s}{\partial n_l \partial n_k} \right) = \left(\frac{\partial (\beta \mu_{l0})}{\partial n_k} \right)_{e_0, \{n_j\}_{j \neq k, l}}. \quad (4.65)$$

In addition, the thermodynamic relation (B.22) in the frame moving with the fluid element reads

$$dp = -\frac{(e_0 + p)}{\beta} d\beta + \sum_{k=1}^c \frac{n_k}{\beta} d(\beta \mu_{k0}), \quad (4.66)$$

so that

$$\left(\frac{\partial p}{\partial \beta} \right)_{\beta \mu_{10}, \dots, \beta \mu_{c0}} = -\frac{(e_0 + p)}{\beta} \quad (4.67)$$

$$\left(\frac{\partial p}{\partial (\beta \mu_{k0})} \right)_{\beta, \{\beta \mu_{j0}\}_{j \neq k}} = \frac{n_k}{\beta}. \quad (4.68)$$

From the relation (4.61), $s = s(e_0, n_1, \dots, n_c)$ so that with (4.62) and (4.63), $\beta = \beta(e_0, n_1, \dots, n_c)$ and $\beta \mu_{k0} = \beta \mu_{k0}(e_0, n_1, \dots, n_{k-1}, n_{k+1}, \dots, n_c)$. Now, using equations (4.58) and (4.60) expressed in their Lagrangian form, as well as equations (4.64), (4.67) and (4.68), one has

$$\frac{d\beta}{dt} = \left(\frac{\partial \beta}{\partial e_0} \right)_{n_1, \dots, n_c} \frac{de_0}{dt} + \sum_{k=1}^c \left(\frac{\partial \beta}{\partial n_k} \right)_{e_0, \{n_j\}_{j \neq k}} \frac{dn_k}{dt}$$

$$\begin{aligned}
 &= -\left\{\left(\frac{\partial\beta}{\partial e_0}\right)_{n_1,\dots,n_c}(e_0+p) + \sum_{k=1}^c \left(\frac{\partial\beta}{\partial n_k}\right)_{e_0,\{n_j\}_{j\neq k}} n_k\right\} \nabla \cdot \mathbf{u} \\
 &= \beta \left\{\left(\frac{\partial\beta}{\partial e_0}\right)_{n_1,\dots,n_c} \left(\frac{\partial p}{\partial \beta}\right)_{\beta\mu_{10},\dots,\beta\mu_{c0}} - \sum_{k=1}^c \left(\frac{\partial\beta}{\partial n_k}\right)_{e_0,\{n_j\}_{j\neq k}} \left(\frac{\partial p}{\partial(\beta\mu_{k0})}\right)_{\beta,\{\beta\mu_{j0}\}_{j\neq k}}\right\} \nabla \cdot \mathbf{u} \\
 &= \beta \left\{\left(\frac{\partial\beta}{\partial e_0}\right)_{n_1,\dots,n_c} \left(\frac{\partial p}{\partial \beta}\right)_{\beta\mu_{10},\dots,\beta\mu_{c0}} + \sum_{k=1}^c \left(\frac{\partial(\beta\mu_{k0})}{\partial e_0}\right)_{n_1,\dots,n_c} \left(\frac{\partial p}{\partial(\beta\mu_{k0})}\right)_{\beta,\{\beta\mu_{j0}\}_{j\neq k}}\right\} \nabla \cdot \mathbf{u}. \quad (4.69)
 \end{aligned}$$

Consequently,

$$\boxed{\frac{d\beta}{dt} = \beta \left(\frac{\partial p}{\partial e_0}\right)_{n_1,\dots,n_c} \nabla \cdot \mathbf{u}.} \quad (4.70)$$

Following the same reasoning, and using the same equations plus the equation (4.65), we obtain that

$$\begin{aligned}
 \frac{d(\beta\mu_{k0})}{dt} &= \left(\frac{\partial(\beta\mu_{k0})}{\partial e_0}\right)_{n_1,\dots,n_c} \frac{de_0}{dt} + \sum_{l=1}^c \left(\frac{\partial(\beta\mu_{k0})}{\partial n_l}\right)_{e_0,\{n_j\}_{j\neq k,l}} \frac{dn_l}{dt} \\
 &= -\left\{\left(\frac{\partial(\beta\mu_{k0})}{\partial e_0}\right)_{n_1,\dots,n_c}(e_0+p) + \sum_{l=1}^c \left(\frac{\partial(\beta\mu_{k0})}{\partial n_l}\right)_{e_0,\{n_j\}_{j\neq k,l}} n_l\right\} \nabla \cdot \mathbf{u} \\
 &= \beta \left\{\left(\frac{\partial(\beta\mu_{k0})}{\partial e_0}\right)_{n_1,\dots,n_c} \left(\frac{\partial p}{\partial \beta}\right)_{\beta\mu_{10},\dots,\beta\mu_{c0}} - \sum_{l=1}^c \left(\frac{\partial(\beta\mu_{k0})}{\partial n_l}\right)_{e_0,\{n_j\}_{j\neq l}} \left(\frac{\partial p}{\partial(\beta\mu_{l0})}\right)_{\beta,\{\beta\mu_{j0}\}_{j\neq l}}\right\} \nabla \cdot \mathbf{u} \\
 &= -\beta \left\{\left(\frac{\partial\beta}{\partial n_k}\right)_{e_0,\{n_j\}_{j\neq k}} \left(\frac{\partial p}{\partial \beta}\right)_{\beta\mu_{10},\dots,\beta\mu_{c0}} + \sum_{l=1}^c \left(\frac{\partial(\beta\mu_{l0})}{\partial n_k}\right)_{e_0,\{n_j\}_{j\neq k}} \left(\frac{\partial p}{\partial(\beta\mu_{l0})}\right)_{\beta,\{\beta\mu_{j0}\}_{j\neq l}}\right\} \nabla \cdot \mathbf{u}. \quad (4.71)
 \end{aligned}$$

Consequently,

$$\boxed{\frac{d(\beta\mu_{k0})}{dt} = -\beta \left(\frac{\partial p}{\partial n_k}\right)_{e_0,\{n_j\}_{j\neq k}} \nabla \cdot \mathbf{u}.} \quad (4.72)$$

With (4.59), (4.67) and (4.68), we also have

$$\begin{aligned}
 \rho \frac{d\mathbf{u}}{dt} &= -\nabla p \\
 &= -\left(\frac{\partial p}{\partial \beta}\right)_{\beta\mu_{10},\dots,\beta\mu_{c0}} \nabla \beta - \sum_{k=1}^c \left(\frac{\partial p}{\partial(\beta\mu_{k0})}\right)_{\beta,\{\beta\mu_{j0}\}_{j\neq k}} \nabla(\beta\mu_{k0}) \\
 &= \frac{(e_0+p)}{\beta} \nabla \beta - \sum_{k=1}^c \frac{n_k}{\beta} \nabla(\beta\mu_{k0}), \quad (4.73)
 \end{aligned}$$

so that combining (4.70) and (4.73) yields

$$\boxed{\frac{d(\beta\mathbf{u})}{dt} = \beta \left(\frac{\partial p}{\partial e_0}\right)_{n_1,\dots,n_c} \mathbf{u}(\nabla \cdot \mathbf{u}) + \frac{(e_0+p)}{\rho} \nabla \beta - \sum_{k=1}^c \frac{n_k}{\rho} \nabla(\beta\mu_{k0}).} \quad (4.74)$$

Discussion. First, the set of equations (4.70), (4.72) and (4.74) is valid up to corrections of order $\mathcal{O}(\nabla^2)$ since the equation (4.60) was systematically used. We notice that the divergence of $\mathbf{u}$ appears since the fluid is compressible in general.

An equation for the time evolution of the chemical potentials can be obtained by combining (4.70) and (4.72),

$$\frac{d\mu_{k0}}{dt} = - \left[\left(\frac{\partial p}{\partial n_k} \right)_{e_0, \{n_j\}_{j \neq k}} + \mu_{k0} \left(\frac{\partial p}{\partial e_0} \right)_{n_1, \dots, n_c} \right] \nabla \cdot \mathbf{u}. \quad (4.75)$$

Equation (4.73) is the same as equation (1.64) for a dissipativeless fluid ($\kappa = \eta = \xi = 0$) except that here the pressure is seen as a functional of $(\beta, \beta\mu_{10}, \dots, \beta\mu_{c0})$ and not of (ρ, T) as it was in the first chapter. The internal energy of a mixture of c different monoatomic ideal gases is given by $e_0 = \frac{3}{2} \sum_{k=1}^c n_k k_B T$ and its pressure by $p = \sum_{k=1}^c n_k k_B T$, whence the well-known formula $p = \frac{2}{3} e_0$. In this case, (4.73) takes the form

$$\rho \frac{d\mathbf{u}}{dt} = \sum_{k=1}^c n_k k_B T \left(\frac{5}{2} \nabla \beta - \nabla(\beta\mu_{k0}) \right). \quad (4.76)$$

Equation (4.70) is the heat equation for a dissipativeless fluid. It can be written as

$$\frac{dT}{dt} = -T \left(\frac{\partial p}{\partial e_0} \right)_{n_1, \dots, n_c} \nabla \cdot \mathbf{u}. \quad (4.77)$$

For a mixture of c different monoatomic ideal gases it takes the form

$$\frac{dT}{dt} = -\frac{2T}{3} \nabla \cdot \mathbf{u}. \quad (4.78)$$

For instance, if $\nabla \cdot \mathbf{u} < 0$, the volume of the fluid element is compressed and its temperature increases, as expected.

CHAPTER 5

DISSIPATIVE EQUATIONS

“Notre durée est irréversible. Nous ne saurions en revivre une parcelle, car il faudrait commencer par effacer le souvenir de tout ce qui a suivi.”

HENRI BERGSON¹

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The previous chapter was devoted to the study of the dissipativeless current densities. Naturally, this fifth and final chapter studies the dissipative current densities, the ones entering in the expression of the entropy production. The aim of this chapter, and the culminating point of this work, is to obtain the Green-Kubo formulas for the linear response coefficients, in particular for the heat conductivity, diffusion, thermodiffusion, shear and bulk viscosities. Also, the dissipative hydrodynamic equations will be obtained. We stress that all these results are

¹Suggested translation: “Our duration is irreversible. We cannot relive any part of it, because we would have to start by erasing the memory of everything that followed.” From H. Bergson. *L’Evolution créatrice*. Paris: F. Alcan, 1907.

obtained from the intrinsic microscopic dynamics of the particles making up the system, which is non-trivial and remarkable. In the §5.1, the entropy production is rewritten with the help of the expression of the conjugate fields. The expression obtained is the same as in the framework of classical nonequilibrium thermodynamics. The integrand of the deviation operator will be rewritten in the §5.2, again with the help of the expression of the conjugate fields. Next, the dissipative current densities are obtained in the §5.3. A crucial result related to the symmetry of the system is presented in the §5.4, and is required for the final expression of the linear response coefficients. Finally, the Green-Kubo formulas and the dissipative hydrodynamic equations are obtained in the last section §5.5 of this work.

5.1. ENTROPY PRODUCTION: EXPRESSION WITH THE CONJUGATE FIELDS

With the help of expressions (4.13)-(4.15) for the conjugate fields, the entropy production (3.94) can be written more explicitly. We have

$$\begin{aligned} \frac{d_i S}{dt} &= k_B \int d\mathbf{r} \partial^a \lambda^a \mathcal{J}_{c^a}^a \\ &= k_B \int d\mathbf{r} \left(\sum_{k=1}^c \partial^a \lambda_{n_k} \mathcal{J}_{n_k}^a + \partial^a \lambda_{g^b} \mathcal{J}_{g^b}^a + \partial^a \lambda_e \mathcal{J}_e^a \right) \\ &= k_B \int d\mathbf{r} \left(- \sum_{k=1}^c \partial^a (\beta \mu_k) \mathcal{J}_{n_k}^a - \partial^a (\beta u^b) \mathcal{J}_{g^b}^a + \partial^a \beta \mathcal{J}_e^a \right) \end{aligned} \quad (5.1)$$

up to higher order terms in the gradients expansion. Expanding the gradient $\partial^a (\beta u^b)$ and expressing the chemical potentials in the frame moving with the fluid element, we obtain

$$\frac{d_i S}{dt} = k_B \int d\mathbf{r} \left(- \sum_{k=1}^c \partial^a (\beta \mu_{k0}) \mathcal{J}_{n_k}^a - \beta \partial^a u^b \mathcal{J}_{g^b}^a + \partial^a \beta (\mathcal{J}_e^a - \mathcal{J}_{g^b}^a u^b) \right) \quad (5.2)$$

since $\sum_{k=1}^c m_k \mathcal{J}_{n_k}^a = 0$ because of mass conservation. Accordingly, we can identify the diffusive current density of the species k as $\mathcal{J}_{n_k}^a$, the dissipative part of the pressure tensor as

$$\Pi_{ab} \equiv \mathcal{J}_{g^b}^a, \quad (5.3)$$

and the heat current density as

$$\mathcal{J}_q^a \equiv \mathcal{J}_e^a - \mathcal{J}_{g^b}^a u^b. \quad (5.4)$$

In order to put the entropy production into a more familiar form, the tensors $\partial^a u^b$ and Π_{ab} must be decomposed as the sum of a traceless and trace part,

$$\Pi = \overset{\circ}{\Pi} + \Pi \mathbf{1} \quad (5.5)$$

where $\text{Tr} \overset{\circ}{\Pi} = 0$ so that $\Pi = \frac{1}{3} \text{Tr} \Pi \mathbf{1}$. Similarly,

$$\nabla \mathbf{u} = \overset{\circ}{\nabla} \mathbf{u} + \frac{1}{3} (\nabla \cdot \mathbf{u}) \mathbf{1}, \quad (5.6)$$

where $\text{Tr } \nabla \mathbf{u} = \nabla \cdot \mathbf{u} = \text{Tr } \overset{\circ}{\nabla} \mathbf{u} + \nabla \cdot \mathbf{u} \Rightarrow \text{Tr } \overset{\circ}{\nabla} \mathbf{u} = 0$ as required. Furthermore, the traceless part of the $\nabla \mathbf{u}$ tensor can be decomposed as the sum of a symmetric and an antisymmetric part,

$$\overset{\circ}{\nabla} \mathbf{u} = (\overset{\circ}{\nabla} \mathbf{u})^S + (\overset{\circ}{\nabla} \mathbf{u})^A \quad (5.7)$$

with

$$(\overset{\circ}{\nabla} \mathbf{u})^S = \frac{1}{2}(\nabla \mathbf{u} + (\nabla \mathbf{u})^\top) - \frac{1}{3}(\nabla \cdot \mathbf{u})\mathbf{1} \quad (5.8)$$

$$(\overset{\circ}{\nabla} \mathbf{u})^A = \frac{1}{2}(\nabla \mathbf{u} - (\nabla \mathbf{u})^\top). \quad (5.9)$$

Consequently,

$$\begin{aligned} \nabla \mathbf{u} : \Pi &= (\overset{\circ}{\nabla} \mathbf{u} + \frac{1}{3}(\nabla \cdot \mathbf{u})\mathbf{1}) : (\overset{\circ}{\Pi} + \Pi\mathbf{1}) \\ &= \overset{\circ}{\nabla} \mathbf{u} : \overset{\circ}{\Pi} + (\nabla \cdot \mathbf{u})\Pi \\ &= (\overset{\circ}{\nabla} \mathbf{u})^S : \overset{\circ}{\Pi} + (\overset{\circ}{\nabla} \mathbf{u})^A : \overset{\circ}{\Pi} + (\nabla \cdot \mathbf{u})\Pi \\ &= \frac{1}{2}(\nabla \mathbf{u} + (\nabla \mathbf{u})^\top - \frac{2}{3}(\nabla \cdot \mathbf{u})\mathbf{1}) : \overset{\circ}{\Pi} + (\nabla \cdot \mathbf{u})\Pi. \end{aligned} \quad (5.10)$$

With these notations, the entropy production finally becomes

$$\frac{d_i S}{dt} \simeq \int d\mathbf{r} \left(- \sum_{k=1}^c \partial^a \left(\frac{\mu_{k0}}{T} \right) \mathcal{J}_{n_k}^a - \frac{1}{2T} (\partial^a u^b + \partial^b u^a - \frac{2}{3} \partial^a u^a) \overset{\circ}{\Pi}^{ab} - \frac{1}{T} \partial^a u^a \Pi + \partial^a \left(\frac{1}{T} \right) \mathcal{J}_q^a \right) \quad (5.11)$$

which coincides with the expression found in the framework of nonequilibrium thermodynamics for normal fluids without chemical reactions and external forces. In particular, it corresponds exactly to the expression (1.39) of the entropy production (entropy source). Indeed, the first term $\partial_t(\rho s)$ on the left-hand side of (1.38) is the partial time derivative of the entropy density. An integration over space thus gives $\frac{dS}{dt}$. Since the system is isolated, the entropy flux density, which is the second term on the left-hand side of (1.38), vanishes. We are thus left with

$$\frac{d_i S}{dt} = \int d\mathbf{r} \partial_t(\rho s) = \int d\mathbf{r} \left(- \sum_{k=1}^c \partial^a \left(\frac{\mu_k}{T} \right) j_k^a - \frac{1}{T} \partial^a u^b : \Pi^{ab} + \partial^a \left(\frac{1}{T} \right) q^a \right) \quad (5.12)$$

which corresponds exactly to (5.11) provided that:

- $\mathcal{J}_q \leftrightarrow q$.
- The Π and $\nabla \mathbf{u}$ tensors be symmetrised as before.
- $j_k \leftrightarrow \mathcal{J}_{n_k}$.

We also note that in the framework of (macroscopic) nonequilibrium thermodynamics exposed in the first chapter, the equality (5.12) is strict while it is an approximation in (5.11), which is to trace back to the gradients expansion of the entropy and normalisation functionals. This difference does not find its source in the quantum approach since such approximation already

appears in the classical framework [9]. Nonetheless, the correspondence between the classical and quantum approaches is to be highlighted, both yielding the same result for the entropy production. Hence, we claim that the latter expression is not modified in account of the quantum dynamics of the particles.

5.2. INTEGRAND OF THE DEVIATION OPERATOR

The expression of the deviation operator $\hat{\Delta}_t$ is given by (3.81). In particular, the integrand is expressed in terms of the time and spatial derivatives of the conjugate fields. Since these fields are known at leading order in the gradients expansion, an explicit expression of the integrand can be obtained. The integrand is

$$\partial_\tau \lambda_\tau(\mathbf{r}) \cdot \delta \hat{\mathbf{c}}_\tau(\mathbf{r}) + \nabla \lambda_\tau(\mathbf{r}) : \delta \hat{\mathbf{j}}(\mathbf{r}, \tau), \quad (5.13)$$

or in components,

$$\partial_\tau \lambda_\tau^\alpha(\mathbf{r}) \delta \hat{c}_\tau^\alpha(\mathbf{r}) + \partial^a \lambda_\tau^\alpha(\mathbf{r}) \delta j_{c^a}^\alpha(\mathbf{r}, \tau) \quad (5.14)$$

up to higher order terms in the expansion in powers of the gradients. With the expressions (4.13)-(4.15) of the conjugate fields, the time derivative term can be written as

$$\begin{aligned} \partial_\tau \lambda_\tau^\alpha \delta \hat{c}^\alpha &= - \sum_{k=1}^c \partial_\tau (\beta \mu_k) \delta \hat{n}_k - \partial_\tau (\beta u^a) \delta \hat{g}^a + \partial_\tau \beta \delta \hat{e} \\ &= - \sum_{k=1}^c \partial_\tau (\beta \mu_k) \delta \hat{n}_k + \partial_\tau \beta \{ \delta \hat{e} - u^a \delta \hat{g}^a \} - \beta \partial_\tau u^a \delta \hat{g}^a \end{aligned} \quad (5.15)$$

in the laboratory frame. If the chemical potentials are expressed in the moving frame (as usual in hydrodynamics, cf. equations (4.72) or (4.74)),

$$\begin{aligned} \partial_\tau \lambda_\tau^\alpha \delta \hat{c}^\alpha &= - \sum_{k=1}^c \partial_\tau \left\{ \beta \left(\mu_{k0} - \frac{1}{2} m_k \mathbf{u}^2 \right) \right\} \delta \hat{n}_k + \partial_\tau \beta \{ \delta \hat{e} - u^a \delta \hat{g}^a \} - \beta \partial_\tau u^a \delta \hat{g}^a \\ &= - \sum_{k=1}^c \partial_\tau (\beta \mu_{k0}) \delta \hat{n}_k + \partial_\tau \beta \left\{ \delta \hat{e} - u^a \delta \hat{g}^a + \frac{1}{2} \mathbf{u}^2 \delta \hat{\rho} \right\} - \beta \partial_\tau u^a (\delta \hat{g}^a - u^a \delta \hat{\rho}) \end{aligned} \quad (5.16)$$

with $\sum_{k=1}^c m_k \hat{n}_k = \hat{\rho}$. Similarly, the spatial derivative term can be written as

$$\begin{aligned} \partial^a \lambda_\tau^\alpha \delta j_{c^a}^\alpha &= - \sum_{k=1}^c \partial^a (\beta \mu_k) \delta j_{n_k}^\alpha + \partial^a \beta \{ \delta j_{e^a}^\alpha - \delta j_{g^b}^\alpha u^b \} - \beta \partial^a u^b \delta j_{g^b}^\alpha \\ &= - \sum_{k=1}^c \partial^a (\beta \mu_{k0}) \delta j_{n_k}^\alpha + \partial^a \beta \left\{ \delta j_{e^a}^\alpha - \delta j_{g^b}^\alpha u^b + \frac{1}{2} \mathbf{u}^2 \delta \hat{g}^a \right\} - \beta \partial^a u^b (\delta j_{g^b}^\alpha - \hat{g}^a u^b) \end{aligned} \quad (5.17)$$

with $\hat{g}^a = j_\rho^a = \sum_{k=1}^c m_k j_{n_k}^a$. Next, the equations (4.70), (4.72) and (4.73) for the time evolution of the conjugate fields are used to replace in equation (5.16) the partial derivatives with respect to time by gradients,

$$\partial_\tau \lambda_\tau^\alpha \delta \hat{c}^\alpha$$

$$\begin{aligned}
 &= - \sum_{k=1}^c \partial^a (\beta \mu_{k0}) \left\{ -u^a \delta \hat{n}_k - \frac{n_k}{\rho} (\delta \hat{g}^a - u^a \delta \hat{\rho}) \right\} \\
 &\quad - \beta \partial^a u^b \left\{ - \sum_{k=1}^c \left(\frac{\partial p}{\partial n_k} \right)_{e_0, \{n_j\}_{j \neq k}} \delta \hat{n}_k \delta^{ab} - \left(\frac{\partial p}{\partial e_0} \right)_{n_1, \dots, n_c} \left(\delta \hat{e} - u^c \delta \hat{g}^c + \frac{1}{2} \mathbf{u}^2 \delta \hat{\rho} \right) \delta^{ab} - u^a (\delta \hat{g}^b - u^b \delta \hat{\rho}) \right\} \\
 &\quad + \partial^a \beta \left\{ -u^a \left(\delta \hat{e} - u^c \delta \hat{g}^c + \frac{1}{2} \mathbf{u}^2 \delta \hat{\rho} \right) - \frac{e_0 + p}{\rho} (\delta \hat{g}^a - u^a \delta \hat{\rho}) \right\}, \tag{5.18}
 \end{aligned}$$

so that

$$\begin{aligned}
 &\partial_\tau \lambda_\tau^\alpha \delta \hat{c}^\alpha + \partial^a \lambda_\tau^\alpha \delta \hat{j}_{c^a}^a \\
 &= - \sum_{k=1}^c \partial^a (\beta \mu_{k0}) \left\{ \delta \hat{j}_{n_k}^a - u^a \delta \hat{n}_k - \frac{n_k}{\rho} (\delta \hat{g}^a - u^a \delta \hat{\rho}) \right\} \\
 &\quad - \beta \partial^a u^b \left\{ \delta \hat{j}_{g^b}^a - \hat{g}^a u^b - \sum_{k=1}^c \left(\frac{\partial p}{\partial n_k} \right)_{e_0, \{n_j\}_{j \neq k}} \delta \hat{n}_k \delta^{ab} - \left(\frac{\partial p}{\partial e_0} \right)_{n_1, \dots, n_c} \left(\delta \hat{e} - u^c \delta \hat{g}^c + \frac{1}{2} \mathbf{u}^2 \delta \hat{\rho} \right) \delta^{ab} \right. \\
 &\quad \left. - u^a (\delta \hat{g}^b - u^b \delta \hat{\rho}) \right\} + \partial^a \beta \left\{ \delta \hat{j}_e^a - \delta \hat{j}_{g^b}^a u^b + \frac{1}{2} \mathbf{u}^2 \delta \hat{g}^a - u^a \left(\delta \hat{e} - u^c \delta \hat{g}^c + \frac{1}{2} \mathbf{u}^2 \delta \hat{\rho} \right) - \frac{e_0 + p}{\rho} (\delta \hat{g}^a - u^a \delta \hat{\rho}) \right\}. \tag{5.19}
 \end{aligned}$$

In the frame moving with the fluid element, $u^a = 0$, we find that²

$$\partial_\tau \lambda_\tau^\alpha \delta \hat{c}^\alpha + \partial^a \lambda_\tau^\alpha \delta \hat{j}_{c^a}^a = - \sum_{k=1}^c \partial^a (\beta \mu_{k0}) \delta \hat{j}_{n_k}^a - \beta \partial^a u^b \delta \hat{j}_{g^b}^a + \partial^a \beta \delta \hat{j}_e^a \tag{5.20}$$

with the following fluctuations of the current densities

$$\delta \hat{j}_{n_k}^a = \delta \hat{j}_{n_k}^a - \frac{n_k}{\rho} \delta \hat{g}^a \tag{5.21}$$

$$\delta \hat{j}_{g^b}^a = \delta \hat{j}_{g^b}^a - \left\{ \sum_{k=1}^c \left(\frac{\partial p}{\partial n_k} \right)_{e_0, \{n_j\}_{j \neq k}} \delta \hat{n}_k + \left(\frac{\partial p}{\partial e_0} \right)_{n_1, \dots, n_c} \delta \hat{e} \right\} \delta^{ab} \tag{5.22}$$

$$\delta \hat{j}_e^a = \delta \hat{j}_e^a - \frac{e_0 + p}{\rho} \delta \hat{g}^a. \tag{5.23}$$

This result can be expressed as

$$\begin{aligned}
 \partial_\tau \lambda_\tau^\alpha \delta \hat{c}^\alpha + \partial^a \lambda_\tau^\alpha \delta \hat{j}_{c^a}^a &= \mathcal{A}_{n_k}^a \delta \hat{j}_{n_k}^a + \mathcal{A}_{g^b}^a \delta \hat{j}_{g^b}^a + \mathcal{A}_e^a \delta \hat{j}_e^a \\
 &\equiv \mathcal{A} : \delta \mathbf{j}' \tag{5.24}
 \end{aligned}$$

in terms of the affinities defined by

$$\mathcal{A}_{n_k}^a \equiv -\partial^a (\beta \mu_{k0}) \tag{5.25}$$

$$\mathcal{A}_{g^b}^a \equiv -\beta \partial^a u^b \tag{5.26}$$

$$\mathcal{A}_e^a \equiv \partial^a \beta. \tag{5.27}$$

The last line of (5.24) is to be understood symbolically with $\mathcal{A} = (\mathcal{A}_{n_k}, \mathcal{A}_g, \mathcal{A}_e)$. As explained in the §1.4, the affinities, defined as the gradients of the conjugate variables, control how far from equilibrium the system is.

²In the §4.2, the quantities expressed in the moving frame were denoted by a 0 subscript. When evaluating $u^a = 0$, these quantities are equal to the corresponding ones without a 0 subscript.

5.3. DISSIPATIVE CURRENT DENSITIES

The dissipativeless current densities were studied in the previous chapter. Now, the dissipative current densities, appearing in the expression of the entropy production (3.94), are studied. The Green-Kubo formulas for the linear response coefficients are ready to be obtained since they act as proportionality constants between the dissipative current densities and the affinities (*cf.* (1.58)).

Expression of the dissipative current densities. The dissipative current densities are defined by

$$\begin{aligned}\mathcal{J}_{c^a}^a(\mathbf{r}, t) &= \text{Tr } \hat{\Delta}_t \hat{j}_{c^a}^a(\mathbf{r}) \\ &= \text{Tr } \hat{\Delta}_t \delta \hat{j}_{c^a}^a(\mathbf{r})\end{aligned}\quad (5.28)$$

where $\text{Tr } \hat{\Delta}_t = 0$ was used. The vector notation $\mathcal{J}(\mathbf{r}, t) = \text{Tr } \hat{\Delta}_t \delta \hat{\mathbf{j}}(\mathbf{r})$ will be preferred in the following. Using the expression of the deviation operator (3.81) and the relation (5.24), the dissipative current densities can be written as

$$\mathcal{J}(\mathbf{r}, t) = \int_0^t d\tau \int_0^1 dz \int d\mathbf{r}' \text{Tr} e^{-\frac{i}{\hbar} \hat{H}(t-\tau)} e^{-z\hat{\sigma}} \mathcal{A}(\mathbf{r}', \tau) : \delta \hat{\mathbf{j}}'(\mathbf{r}') e^{z\hat{\sigma}} \hat{\rho}_{\text{leq}}[\boldsymbol{\lambda}_\tau] e^{\frac{i}{\hbar} \hat{H}(t-\tau)} \delta \hat{\mathbf{j}}(\mathbf{r}). \quad (5.29)$$

The affinities can come out of the trace. Using the cyclic property of the trace,

$$\begin{aligned}\text{Tr} e^{-\frac{i}{\hbar} \hat{H}(t-\tau)} e^{-z\hat{\sigma}} \delta \hat{\mathbf{j}}'(\mathbf{r}') e^{z\hat{\sigma}} \hat{\rho}_{\text{leq}}[\boldsymbol{\lambda}_\tau] e^{\frac{i}{\hbar} \hat{H}(t-\tau)} \delta \hat{\mathbf{j}}(\mathbf{r}) \\ = \text{Tr} e^{-z\hat{\sigma}} \delta \hat{\mathbf{j}}'(\mathbf{r}') e^{z\hat{\sigma}} \hat{\rho}_{\text{leq}}[\boldsymbol{\lambda}_\tau] e^{\frac{i}{\hbar} \hat{H}(t-\tau)} \delta \hat{\mathbf{j}}(\mathbf{r}) e^{-\frac{i}{\hbar} \hat{H}(t-\tau)} \\ = \text{Tr} \hat{\rho}_{\text{leq}}[\boldsymbol{\lambda}_\tau] \delta \hat{\mathbf{j}}(\mathbf{r}, t-\tau) e^{-z\hat{\sigma}} \delta \hat{\mathbf{j}}'(\mathbf{r}', 0) e^{z\hat{\sigma}} \\ = \langle \delta \hat{\mathbf{j}}(\mathbf{r}, t-\tau) e^{-z\hat{\sigma}} \delta \hat{\mathbf{j}}'(\mathbf{r}', 0) e^{z\hat{\sigma}} \rangle_{\text{leq}, \boldsymbol{\lambda}_\tau}\end{aligned}\quad (5.30)$$

so that

$$\mathcal{J}(\mathbf{r}, t) = \int_0^t d\tau \int_0^1 dz \int d\mathbf{r}' \langle \delta \hat{\mathbf{j}}(\mathbf{r}, t-\tau) e^{-z\hat{\sigma}} \delta \hat{\mathbf{j}}'(\mathbf{r}', 0) e^{z\hat{\sigma}} \rangle_{\text{leq}, \boldsymbol{\lambda}_\tau} : \mathcal{A}(\mathbf{r}', \tau). \quad (5.31)$$

This preliminary result is interesting as it already suggests that the dissipative current densities can be written as integrals of time and spatial correlation functions. The correlation functions³ have the form $C(\mathbf{r}, \mathbf{r}', t, \tau) = \langle \hat{X}(\mathbf{r}, t-\tau) \hat{X}(\mathbf{r}', 0) \rangle$ where $\hat{X}$ is an abstract operator defined for convenience. Only the combination $t-\tau$ appears and since the system is isotropic, only the distance between two separate points matters, *i.e.* their relative position and not their absolute position. Hence, the form of the correlation functions can be rewritten as $C(\|\mathbf{r}-\mathbf{r}'\|, t')$ with $t' \equiv t-\tau$. The smallest time t_c for which $C(\|\mathbf{r}-\mathbf{r}'\|, t') \simeq 0$ is defined as the correlation time. Similarly, the smallest distance l_c for which the correlation functions decay to zero is the correlation length. In a process of relaxation towards equilibrium, it is expected that the correlation functions decay exponentially, typically as $C(\|\mathbf{r}-\mathbf{r}'\|, t') \sim e^{-t'/t_c} e^{-\|\mathbf{r}-\mathbf{r}'\|/l_c}$.

³Or more precisely autocorrelation functions in this case.

These correlation functions vary very quickly in space and time compared to typical macroscales. They are typically zero as soon as $\|\mathbf{r} - \mathbf{r}'\| > l_c$ and $t' > t_c$. To give an idea, l_c is of the order of the interatomic distance. Thus for liquids, $l_c \sim 0.1$ nm and about $1 - 10$ nm for dilute gases. t_c is of the order of the intercollision time, $t_c \sim l_c/v_{th}$ where $v_{th} = \sqrt{\frac{2k_B T}{m}}$ is the thermal velocity. For a liquid, $t_c \sim 10^{-12}$ s and $t_c \sim 10^{-10}$ s for diluted gases at atmospheric pressure. These numbers have to be understood far from phase transitions where the correlation length and time are known to diverge. They have to be compared with typical macroscales, $l \sim 1$ m and $t \sim 1$ s, defined in function of the system that is considered or the experiment performed.

The affinities and the conjugate fields are macroscopic fields that vary extremely slowly at the length scale of l_c and time scale of t_c . Consequently, in the frame moving with the fluid element, the conjugate fields can be taken to be quasi uniform

$$\lambda_{n_k}(\mathbf{r}, t) = -\beta \mu_{k0} \quad (5.32)$$

$$\lambda_{g^a}(\mathbf{r}, t) = 0 \quad (5.33)$$

$$\lambda_e(\mathbf{r}, t) = \beta, \quad (5.34)$$

up to higher order terms in the gradients expansion, so that the LESO reduces to the equilibrium grand canonical statistical operator as explained in the §3.1,

$$\hat{\rho}_{leq} \rightarrow \hat{\rho}_{eq} = e^{-\hat{\sigma}_{eq}} = \frac{1}{\Xi} \exp \left\{ \beta \left(\sum_{k=1}^c \mu_{k0} \hat{N}_k - \hat{H} \right) \right\} \quad (5.35)$$

with

$$\Xi = e^{\Omega}, \quad \Omega = \ln \text{Tr} \exp \left\{ \beta \left(\sum_{k=1}^c \mu_{k0} \hat{N}_k - \hat{H} \right) \right\}. \quad (5.36)$$

The particle number in species k and the Hamiltonian operator itself define conserved quantities. Hence, at equilibrium $[\hat{\rho}_{eq}, \hat{H}] = 0$, in particular, $[\hat{N}_k, \hat{H}] = 0$. Expression (5.31) becomes

$$\mathcal{J}(\mathbf{r}, t) \simeq \int_0^t d\tau \int_0^1 dz \int d\mathbf{r}' \langle \delta \hat{\mathbf{j}}(\mathbf{r}, t - \tau) e^{-z\hat{\sigma}_{eq}} \delta \hat{\mathbf{j}}'(\mathbf{r}', 0) e^{z\hat{\sigma}_{eq}} \rangle_{eq} : \mathcal{A}(\mathbf{r}', \tau). \quad (5.37)$$

The affinities are macroscopic fields and thus vary extremely slowly at the microscopic scale. Since $\mathbf{r} - \mathbf{r}'$ and $t - \tau$ are associated to microscopic spatial distances and times, $\mathcal{A}(\mathbf{r}', \tau)$ can be replaced by $\mathcal{A}(\mathbf{r}, t)$ in (5.37) giving

$$\mathcal{J}(\mathbf{r}, t) \simeq \int_0^t d\tau \int_0^1 dz \int d\mathbf{r}' \langle \delta \hat{\mathbf{j}}(\mathbf{r}, \tau) e^{-z\hat{\sigma}_{eq}} \delta \hat{\mathbf{j}}'(\mathbf{r}', 0) e^{z\hat{\sigma}_{eq}} \rangle_{eq} : \mathcal{A}(\mathbf{r}, t) \quad (5.38)$$

where the change $t - \tau \rightarrow \tau$ was carried out in the integration over time. In addition, the correlation functions decay to zero quickly for times larger than the correlation time: the $\int_0^t d\tau$ integral can be replaced by a $\int_0^\infty d\tau$ integral,

$$\mathcal{J}(\mathbf{r}, t) \simeq \int_0^\infty d\tau \int_0^1 dz \int d\mathbf{r}' \langle \delta \hat{\mathbf{j}}(\mathbf{r}, \tau) e^{-z\hat{\sigma}_{eq}} \delta \hat{\mathbf{j}}'(\mathbf{r}', 0) e^{z\hat{\sigma}_{eq}} \rangle_{eq} : \mathcal{A}(\mathbf{r}, t). \quad (5.39)$$

We see that the time dependence of $\mathcal{J}$ only comes from the affinities. Over spatial scales much larger than the distance between next-neighbouring atoms, the material properties can be defined by averaging over space. In this regard, the microscopic total currents are introduced as the integral over the system of the fluctuations of the current densities,

$$\delta\hat{\mathbf{J}}(t) \equiv \int_V d\mathbf{r} \delta\mathbf{j}'(\mathbf{r}, t). \quad (5.40)$$

With the use of equations (5.21)-(5.23) and (2.54)-(2.56), the total microscopic currents are given by

$$\delta\hat{j}_{n_k}^a = \int_V d\mathbf{r} \delta j_{n_k}^a - \frac{n_k}{\rho} \delta\hat{P}^a \quad (5.41)$$

$$\delta\hat{j}_{g^b}^a = \int_V d\mathbf{r} \delta j_{g^b}^a - \left\{ \sum_{k=1}^c \left(\frac{\partial p}{\partial n_k} \right)_{e_0, \{n_j\}_{j \neq k}} \delta\hat{N}_k + \left(\frac{\partial p}{\partial e_0} \right)_{n_1, \dots, n_c} \delta\hat{H} \right\} \delta^{ab} \quad (5.42)$$

$$\delta\hat{j}_e^a = \int_V d\mathbf{r} \delta j_e^a - \frac{e_0 + p}{\rho} \delta\hat{P}^a. \quad (5.43)$$

At this stage, two points must be clarified:

- The LESO was reduced to an equilibrium operator. But is the nonequilibrium state of the system lost? In the expression (5.31) of the dissipative current densities, the average is taken over the LESO, in accordance with the nonequilibrium state of the system. In the expression (5.39), the average is taken over an equilibrium ensemble. However, it is the fluctuations of the current densities around the equilibrium value that are considered, so that the nonequilibrium state of the system remains well accounted. The transformation from $\hat{\rho}_{\text{leq}}$ to $\hat{\rho}_{\text{eq}}$ is truly performed since the conjugate fields are quasi uniform at the spatial and time microscopic scale. It is at these scales that the transformation must be understood, not at the macroscopic one. Nonetheless, the link between equilibrium and nonequilibrium statistical mechanics is exhibited by the *fluctuation-dissipation theorem* that gives a general relation between the linear response of a system and its fluctuations around equilibrium, see for example [10, 11, 26, 27].
- Why is the system described by an equilibrium grand canonical ensemble if it is by definition closed? First, the equilibrium ensemble in (5.39) arises from the local equilibrium ensemble allowing for fluctuations in the particle numbers. The total particle number in each species, the total momentum and total energy are all constants of motion and so do not fluctuate in time. At equilibrium, their fluctuation vanishes. However, they are still random variables because the initial conditions in phase space may give different values to these quantities. Indeed, the initial conditions depend on the number of particles in each species and on the temperature of the system. Hence, the average appearing in the expression of the dissipative current densities is to be understood as an average in the grand canonical equilibrium ensemble. The closed system is thus described by a sequence

of Hamiltonians corresponding to the filling of the system by an increasing number of particles and fixed temperature as initial conditions.

As $\langle \delta J_{c^a}^a \rangle_{\text{eq}} = 0$, they may be added into the equilibrium correlation functions. Consequently, the unprime quantities can be replaced by the prime ones in the correlation functions. Integrating both sides of (5.39) over $\mathbf{r}$, we have

$$\int d\mathbf{r} \mathcal{J}(\mathbf{r}, t) \simeq \int_0^\infty d\tau \int_0^1 dz \int d\mathbf{r} \langle \delta \hat{\mathbf{j}}(\mathbf{r}, \tau) e^{-z\hat{\sigma}_{\text{eq}}} \delta \hat{\mathbf{j}}(0) e^{z\hat{\sigma}_{\text{eq}}} \rangle_{\text{eq}} : \mathcal{A}(\mathbf{r}, t). \quad (5.44)$$

If the volume V of the system is assumed to be much larger than the characteristic microscopic scales but smaller than the typical distances over which the macroscopic fields vary, then it is reasonable to assume that $\mathcal{J}(\mathbf{r}, t)$ is constant in V . This step is only formal since at the end, the linear response coefficients are defined at $V \rightarrow \infty$. In such a case, the macroscopic fields will no longer vary and become strictly constants. We thus find,

$$\mathcal{J}(\mathbf{r}, t) \simeq \frac{1}{V} \int_0^\infty d\tau \int_0^1 dz \langle \delta \hat{\mathbf{j}}(\tau) e^{-z\hat{\sigma}_{\text{eq}}} \delta \hat{\mathbf{j}}(0) e^{z\hat{\sigma}_{\text{eq}}} \rangle_{\text{eq}} : \mathcal{A}(\mathbf{r}, t). \quad (5.45)$$

From the grand canonical to the canonical equilibrium ensemble. The Green-Kubo formulas are expressed in the canonical equilibrium ensemble with a fixed number of particles N . In the limit where $N \gg 1$, expression (5.45) obtained in the grand canonical equilibrium ensemble can be reduced to the canonical equilibrium ensemble. For simplicity, we will assume the system to be a one-component fluid, the idea of the calculation being the same for a multicomponent fluid. We will use that $\{|n, j, N\rangle\}$ forms a basis of the corresponding Hilbert space and that $\langle n, j, N | \delta \hat{\mathbf{j}}(\tau) | n_1, j_1, N_1 \rangle = \langle n, j, N | \delta \hat{\mathbf{j}}(\tau) | n_1, j_1, N \rangle \delta_{N N_1}$ since $\delta \hat{\mathbf{j}}$ is an operator defined at fixed particle number. We have,

$$\begin{aligned} & \langle \delta \hat{\mathbf{j}}(\tau) e^{-z\hat{\sigma}_{\text{eq}}} \delta \hat{\mathbf{j}}(0) e^{z\hat{\sigma}_{\text{eq}}} \rangle_{\text{eq}} \\ &= \frac{1}{\Xi} \text{Tr} e^{\beta(\mu_0 \hat{N} - \hat{H}^{(N)})} \delta \hat{\mathbf{j}}(\tau) e^{-z\beta(\hat{H}^{(N)} - \mu_0 \hat{N})} \delta \hat{\mathbf{j}}(0) e^{z\beta(\hat{H}^{(N)} - \mu_0 \hat{N})} \\ &= \frac{1}{\Xi} \sum_{\substack{N=0 \\ N_1=0}}^{\infty} \sum_{\substack{n, j \\ n_1, j_1}} \langle n, j, N | e^{\beta(\mu_0 \hat{N} - \hat{H}^{(N)})} \delta \hat{\mathbf{j}}(\tau) | n_1, j_1, N_1 \rangle \langle n_1, j_1, N_1 | e^{-z\beta(\hat{H}^{(N)} - \mu_0 \hat{N})} \delta \hat{\mathbf{j}}(0) e^{z\beta(\hat{H}^{(N)} - \mu_0 \hat{N})} | n, j, N \rangle \\ &= \frac{1}{\Xi} \sum_{\substack{N=0 \\ N_1=0}}^{\infty} \sum_{\substack{n, j \\ n_1, j_1}} e^{\beta(\mu_0 N - E_{n,j}^{(N)})} \langle n, j, N | \delta \hat{\mathbf{j}}(\tau) | n_1, j_1, N_1 \rangle e^{-z\beta(E_{n_1, j_1}^{(N_1)} - \mu_0 N_1)} \langle n_1, j_1, N_1 | \delta \hat{\mathbf{j}}(0) | n, j, N \rangle e^{z\beta(E_{n,j}^{(N)} - \mu_0 N)} \\ &= \frac{1}{\Xi} \sum_{N=0}^{\infty} \sum_{\substack{n, j \\ n_1, j_1}} e^{\beta(\mu_0 N - E_{n,j}^{(N)})} \langle n, j, N | \delta \hat{\mathbf{j}}(\tau) | n_1, j_1, N \rangle e^{-z\beta E_{n_1, j_1}^{(N)}} \langle n_1, j_1, N | \delta \hat{\mathbf{j}}(0) | n, j, N \rangle e^{z\beta E_{n,j}^{(N)}} \\ &= \frac{1}{\Xi} \sum_{N=0}^{\infty} \sum_{n, j} e^{\beta\mu_0 N} \langle n, j, N | e^{-\beta \hat{H}^{(N)}} \delta \hat{\mathbf{j}}(\tau) \left(\sum_{n_1, j_1} | n_1, j_1, N \rangle e^{-z\beta E_{n_1, j_1}^{(N)}} \langle n_1, j_1, N | \right) \delta \hat{\mathbf{j}}(0) e^{z\beta \hat{H}^{(N)}} | n, j, N \rangle \\ &= \frac{1}{\Xi} \sum_{N=0}^{\infty} \sum_{n, j} e^{\beta\mu_0 N} \langle n, j, N | e^{-\beta \hat{H}^{(N)}} \delta \hat{\mathbf{j}}(\tau) e^{-z\beta \hat{H}^{(N)}} \delta \hat{\mathbf{j}}(0) e^{z\beta \hat{H}^{(N)}} | n, j, N \rangle. \end{aligned} \quad (5.46)$$

Let Z_N be the partition function of the canonical equilibrium ensemble with N particles. We have that

$$\Xi = \sum_{N=0}^{\infty} e^{\beta\mu_0 N} Z_N = \sum_{N=0}^{\infty} e^{\beta(\mu_0 N - F_N)} \quad (5.47)$$

where $F_N = F(T, V, N)$ is the Helmholtz free energy. In the limit $N \gg 1$, the sum can be approximated by an integral,

$$\Xi \simeq \int_0^{\infty} dN e^{\beta(\mu_0 N - F_N)} \quad (5.48)$$

which can in turn be evaluated by the steepest descent method. The maximum value of the integrand satisfies

$$\frac{\partial}{\partial N} e^{\beta(\mu_0 N - F_N)} = 0 \quad \Leftrightarrow \quad \mu_0 = \left(\frac{\partial F_N}{\partial N} \right)_{T,V}, \quad (5.49)$$

the latter equality fixing the mean number of particles $\bar{N}$. For convenience, let us write $\psi(N) \equiv \beta(\mu_0 N - F_N)$ the argument of the exponential in (5.48). It can be expanded around $\bar{N}$ according to

$$\psi(N) = \psi(\bar{N}) + \left. \frac{\partial \psi}{\partial N} \right|_{N=\bar{N}} (N - \bar{N}) + \frac{1}{2} \left. \frac{\partial^2 \psi}{\partial N^2} \right|_{N=\bar{N}} (N - \bar{N})^2 + \dots \quad (5.50)$$

where the second term in the right-hand side vanishes as a result of the equality (5.49). Hence,

$$\begin{aligned} \Xi &\simeq \int_0^{\infty} dN \exp \left\{ \psi(\bar{N}) + \frac{1}{2} \left. \frac{\partial^2 \psi}{\partial N^2} \right|_{N=\bar{N}} (N - \bar{N})^2 + \dots \right\} \\ &= \exp \left\{ \beta(\mu_0 \bar{N} - F_{\bar{N}}) \right\} \int_0^{\infty} dN \exp \left\{ -\beta \frac{1}{2} \left. \frac{\partial^2 F_N}{\partial N^2} \right|_{N=\bar{N}} (N - \bar{N})^2 \right\} \\ &= \exp \left\{ \beta(\mu_0 \bar{N} - F_{\bar{N}}) \right\} \sqrt{\frac{2\pi}{\beta \left. \frac{\partial^2 F_N}{\partial N^2} \right|_{N=\bar{N}}}} \end{aligned} \quad (5.51)$$

where $\int_{-\bar{N}}^{\infty} dN$ was approximated by $\int_{-\infty}^{\infty} dN$ since $\bar{N} \gg 1$.

Now, the numerator of (5.46) has to be discussed. Let us denote by $\hat{X}$ the operator $\delta \hat{\mathbf{J}}(\tau) e^{-z\beta \hat{H}^{(N)}} \delta \hat{\mathbf{J}}(0) e^{z\beta \hat{H}^{(N)}}$. The numerator of (5.46) can be rewritten as

$$\begin{aligned} &\sum_{N=0}^{\infty} \sum_{n,j} e^{\beta\mu_0 N} \langle n, j, N | e^{-\beta \hat{H}^{(N)}} \hat{X} | n, j, N \rangle \\ &= \sum_{N=0}^{\infty} e^{\beta\mu_0 N} Z_N \frac{1}{Z_N} \text{Tr} e^{-\beta \hat{H}^{(N)}} \hat{X} \\ &= \sum_{N=0}^{\infty} e^{\beta\mu_0 N} Z_N \langle \hat{X} \rangle_N \\ &= \sum_{N=0}^{\infty} e^{\beta(\mu_0 N - F_N)} \langle \hat{X} \rangle_N \\ &\simeq \int_0^{\infty} dN e^{\beta(\mu_0 N - F_N)} \langle \hat{X} \rangle_N \\ &= \exp \left\{ \beta(\mu_0 \bar{N} - F_{\bar{N}}) \right\} \sqrt{\frac{2\pi}{\beta \left. \frac{\partial^2 F_N}{\partial N^2} \right|_{N=\bar{N}}}} \langle \hat{X} \rangle_{\bar{N}}. \end{aligned} \quad (5.52)$$

We note that the mean value of $\hat{X}$ was evaluated at $N = \bar{N}$ since $\langle \hat{X} \rangle_N$ is not expected to behave as an exponential of N but rather as some power of N . For multicomponent fluids, the calculations are very similar, the difference being that $\{|n, j, N\rangle\}$ with $N = (N_1, \dots, N_k)$ now forms a basis of the corresponding Hilbert space. The integrals are then replaced by $\int_0^\infty dN_1 \dots dN_k$ yielding multi-dimensional Gaussian integrals.

The purpose of the calculations (5.51) and (5.52) is that (5.46) can be evaluated to be

$$\frac{1}{Z_{\bar{N}}} \text{Tr} e^{-\beta \hat{H}(\bar{N})} \delta \hat{\mathbf{J}}(\tau) e^{-z \beta \hat{H}(\bar{N})} \delta \hat{\mathbf{J}}(0) e^{z \beta \hat{H}(\bar{N})} \quad (5.53)$$

allowing to rewrite (5.45) as

$$\mathcal{J}(\mathbf{r}, t) \simeq \frac{1}{V} \int_0^\infty d\tau \int_0^1 dz \langle \delta \hat{\mathbf{J}}(\tau) e^{-z \beta \hat{H}} \delta \hat{\mathbf{J}}(0) e^{z \beta \hat{H}} \rangle_{\text{eq}} : \mathcal{A}(\mathbf{r}, t) \quad (5.54)$$

where the mean value is understood over the canonical equilibrium ensemble at fixed number of particles $\bar{N}$. Performing a variable change $\beta z \rightarrow z$ and remembering that

$$\delta \hat{\mathbf{J}}(t) = e^{\frac{i}{\hbar} \hat{H} t} \delta \hat{\mathbf{J}}(0) e^{-\frac{i}{\hbar} \hat{H} t}, \quad (5.55)$$

we obtain a final expression for the dissipative current densities

$$\boxed{\mathcal{J}(\mathbf{r}, t) \simeq \frac{k_B T}{V} \int_0^\infty d\tau \int_0^\beta dz \langle \delta \hat{\mathbf{J}}(\tau) \delta \hat{\mathbf{J}}(i\hbar z) \rangle_{\text{eq}} : \mathcal{A}(\mathbf{r}, t)} \quad (5.56)$$

up to higher order terms in the gradients expansion. If the indices related to the various local densities are recovered, the dissipative current densities can be expressed as

$$\mathcal{J}_\alpha(\mathbf{r}, t) \simeq \mathcal{L}_{\alpha\gamma} : \mathcal{A}_\gamma(\mathbf{r}, t) \quad (5.57)$$

in terms of the linear response coefficients

$$\boxed{\mathcal{L}_{\alpha\gamma} \equiv \frac{k_B T}{V} \int_0^\infty d\tau \int_0^\beta dz \langle \delta \hat{\mathbf{J}}_\alpha(\tau) \delta \hat{\mathbf{J}}_\gamma(i\hbar z) \rangle_{\text{eq}}}, \quad (5.58)$$

in the frame where $\mathbf{u} = 0$ and $\mathcal{J}_q = \mathcal{J}_e$.

5.4. CURIE'S PRINCIPLE

The phenomenological laws relate the tensorial fluxes to the tensorial thermodynamic forces. In principle, the fluxes can be linear functions of all the thermodynamic forces since there is *a priori* no expected form for the matrix $(\mathcal{L})_{ij} = \mathcal{L}_{ij}$ that contains the linear response coefficients. As we shall see, the presence of symmetries may have the effect to restrict the structure of the $\mathcal{L}$ matrix. In particular, we will show that in an isotropic system, the fluxes and thermodynamic forces of different tensorial degree never couple, and furthermore, that the linear response coefficients all reduce to scalars.

Historical approach. After having discovered piezoelectricity and studied magnetism, Pierre Curie was the first to stress the importance of the symmetry principle. As he wrote in [28], “Lorsque certaines causes produisent certains effets, les éléments de symétrie des causes doivent se retrouver dans les effets produits. Lorsque certains effets révèlent une certaine dissymétrie, cette dissymétrie doit se retrouver dans les causes qui leur ont donné naissance.”⁴ Here, the causes are understood as external or constrain fields for instance. Curie’s symmetry principle can be decomposed in two parts

1. If certain causes yield the known effects, the symmetry elements of the causes should be contained in the generated effects.
2. If the known effects manifest certain asymmetry, this latter should be contained in the causes which have generated those effects.

However, the Curie principle in this simple form only applies if the considered effect is unique. If this is not so, given causes may have fewer or equal symmetry than the effects they produce, as in the case of broken symmetries. In the linear domain, the simple formulation of Curie’s principle is sufficient however. When leaving the linear domain, additional complications such as bifurcations may appear. In such a case, two solutions can emerge and converge towards an attractor for example.

Prigogine [29] introduced the symmetry principle into nonequilibrium thermodynamics, and pointed that in general, irreversible processes of different tensorial nature do not couple [30]. In an isotropic system, the statement is strict: irreversible processes of different tensorial nature never couple as demonstrated by [3]. This can be understood intuitively by noting that tensors of different rank transform differently under rotations.

Result. We would now like to show that in an isotropic medium, the fluxes and thermodynamic forces of different tensorial degree do not couple. Notice that the entropy production (1.39) already couples quantities having the same tensorial order.

As a prerequisite, the definition of a polar and axial tensor needs to be recalled. Let $\mathbf{A} \in O(3)$ be an orthogonal matrix, $\det \mathbf{A} = \pm 1$. Under an orthogonal transformation, a tensor $\mathbf{T}$ of rank r transforms as

$$T'_{a_1 a_2 \dots a_r} = (\det \mathbf{A})^\epsilon \sum_{b_1, b_2, \dots, b_r=1}^d A_{a_1 b_1} A_{a_2 b_2} \dots A_{a_r b_r} T_{b_1 b_2 \dots b_r}, \quad (5.59)$$

where d is the dimension of the considered space ($d = 3$ in this work). For a polar tensor, $\epsilon = 0$ and $\epsilon = 1$ for an axial tensor. This formula can be symbolically written as

$$\mathbf{T}' = (\det \mathbf{A})^\epsilon \mathbf{A}^r(\cdot) \mathbf{T}. \quad (5.60)$$

⁴Suggested translation: “When certain causes produce certain effects, the elements of symmetry of the causes must be found in the effects produced. When certain effects reveal a certain asymmetry, this asymmetry must be found in the causes that gave rise to them.”

For instance, under an inversion $\mathbf{I} = \text{diag}(-1, -1, -1)$ with $\det \mathbf{I} = -1$, which is a particular case of an orthogonal transformation, an axial vector is left invariant.

Before proving the main result of this section, a preliminary result is needed.

RESULT 5.1. *In $\mathbb{R}^3$, the components of an antisymmetric tensor of rank two transform like the components of an axial vector.*

Consequently, in $\mathbb{R}^3$, any antisymmetric tensor of rank two can be represented by an axial vector. This result will not be proven, see [31] for a proof. We now present and demonstrate the following result [3]:

RESULT 5.2. *In an isotropic system, the fluxes and thermodynamic forces of different tensorial degree do not couple.*

Proof: In the expression (1.39) derived in the first chapter, no fluxes and thermodynamic forces occurred with a tensorial rank higher than two. We shall therefore restrict the following proof to an entropy production having the form

$$\sigma = j_i X_i + \mathbf{j}_i \cdot \mathbf{X}_i + \mathbf{J}_i : \mathbf{X}_i \quad (5.61)$$

where j_i and X_i are scalars, $\mathbf{j}_i$ and $\mathbf{X}_i$ vectors, $\mathbf{J}_i$ and $\mathbf{X}_i$ tensors of rank two.

Let $\mathbf{T}$ and $\mathbf{V}$ be two tensors of rank two. A tensor of rank two can always be decomposed in the form

$$\mathbf{T} = \frac{1}{3}(\text{Tr } \mathbf{T})\mathbf{1} + \mathbf{T}^A + \overset{\circ}{\mathbf{T}}^S \quad (5.62)$$

where $\mathbf{T}^A$ is its antisymmetric part and $\overset{\circ}{\mathbf{T}}^S$ the symmetric part of $\mathbf{T} - \frac{1}{3}(\text{Tr } \mathbf{T})\mathbf{1}$ (and so traceless). The double-dot product of $\mathbf{T}$ and $\mathbf{V}$ is therefore

$$\begin{aligned} \mathbf{T} : \mathbf{V} &= \left(\frac{1}{3}(\text{Tr } \mathbf{T})\mathbf{1} + \mathbf{T}^A + \overset{\circ}{\mathbf{T}}^S \right) : \left(\frac{1}{3}(\text{Tr } \mathbf{V})\mathbf{1} + \mathbf{V}^A + \overset{\circ}{\mathbf{V}}^S \right) \\ &= \frac{1}{3}(\text{Tr } \mathbf{T})(\text{Tr } \mathbf{V}) + \mathbf{T}^A : \mathbf{V}^A + \overset{\circ}{\mathbf{T}}^S : \overset{\circ}{\mathbf{V}}^S \\ &= \frac{1}{3}(\text{Tr } \mathbf{T})(\text{Tr } \mathbf{V}) + \mathbf{T}^a \cdot \mathbf{V}^a + \overset{\circ}{\mathbf{T}}^S : \overset{\circ}{\mathbf{V}}^S \end{aligned} \quad (5.63)$$

where the last line is a consequence of the result 5.1 with $\mathbf{T}^a$ and $\mathbf{V}^a$ the corresponding axial vectors. From (5.63), it is clear that only four types of terms can occur in the entropy production: product of scalars $j^s X^s$, dot products of polar vectors $\mathbf{j}^p \cdot \mathbf{X}^p$, dot products of axial vectors $\mathbf{j}^a \cdot \mathbf{X}^a$ and double dot products of traceless symmetric rank two tensors $\mathbf{j}^t : \mathbf{X}^t$. Taking a single term of each category, the entropy production is written as

$$\sigma = j^s X^s + \mathbf{j}^p \cdot \mathbf{X}^p + \mathbf{j}^a \cdot \mathbf{X}^a + \mathbf{j}^t : \mathbf{X}^t. \quad (5.64)$$

The phenomenological equations (1.58) will have the general form

$$\mathbf{j}^s = \mathcal{L}^{ss} X^s + \mathcal{L}^{sp} \cdot \mathbf{X}^p + \mathcal{L}^{sa} \cdot \mathbf{X}^a + \mathcal{L}^{st} : \mathbf{X}^t \quad (5.65)$$

$$\mathbf{j}^p = \mathcal{L}^{ps} X^s + \mathbf{L}^{pp} \cdot \mathbf{X}^p + \mathbf{L}^{pa} \cdot \mathbf{X}^a + \mathbf{L}^{pt} : \mathbf{X}^t \quad (5.66)$$

$$\mathbf{j}^a = \mathcal{L}^{as} X^s + \mathbf{L}^{ap} \cdot \mathbf{X}^p + \mathbf{L}^{aa} \cdot \mathbf{X}^a + \mathbf{L}^{at} : \mathbf{X}^t \quad (5.67)$$

$$\mathbf{j}^t = \mathbf{L}^{ts} X^s + \mathbf{L}^{tp} \cdot \mathbf{X}^p + \mathbf{L}^{ta} \cdot \mathbf{X}^a + \mathbf{L}^{tt} : \mathbf{X}^t. \quad (5.68)$$

The tensorial nature of each linear response coefficient is summarised in the table below.

Tensorial order	0	1	2	3	4
Polar tensor	$\mathcal{L}^{ss}$	$\mathcal{L}^{sp}, \mathcal{L}^{ps}$	$\mathbf{L}^{st}, \mathbf{L}^{ts}, \mathbf{L}^{pp}, \mathbf{L}^{aa}$	$\mathbf{L}^{pt}, \mathbf{L}^{tp}$	$\mathbf{L}^{tt}$
Axial tensor	$\times$	$\mathcal{L}^{sa}, \mathcal{L}^{as}$	$\mathbf{L}^{pa}, \mathbf{L}^{ap}$	$\mathbf{L}^{at}, \mathbf{L}^{ta}$	$\times$

Table 5.1: Tensorial nature of the linear response tensors coupling the fluxes to the thermodynamic forces.

The key points are the followings:

- If the system exhibits no symmetry, the linear response tensors of order 1 and higher (scalars do not transform) will in general be different under an orthogonal transformation.
- On the other hand, if the system has a symmetry property, then the linear response tensors of order 1 and higher are expected to remain identical under an orthogonal transformation associated to the symmetry.

Let us look at what happens in the case of an isotropic system. The linear response tensors are expected to remain invariant under a spatial inversion I . In this case, symbolic equation (5.60) gives

$$\mathbf{L} = (\det I)^\epsilon I^r(\cdot) \mathbf{L} \quad \text{or alternatively} \quad \mathbf{L} = (-1)^{\epsilon+r} \mathbf{L} \quad (5.69)$$

where $\mathbf{L}$ denotes the linear response tensors of order 1 and higher. From this equation, we learn that the linear response tensors having odd $\epsilon + r$ must vanish. This eliminates $\mathcal{L}^{sp}, \mathcal{L}^{ps}, \mathbf{L}^{pt}, \mathbf{L}^{tp}, \mathbf{L}^{pa}$ and $\mathbf{L}^{ap}$.

Under an arbitrary rotation R , $\det R = 1$, the properties of an isotropic system remain invariant. Equation (5.60) takes the form

$$\mathbf{L} = R^r(\cdot) \mathbf{L}. \quad (5.70)$$

For the scalar linear response coefficients, this equation is trivially satisfied. For the vectors $\mathcal{L}^{sa}$ and $\mathcal{L}^{as}$,

$$\mathcal{L}^{sa} = R \mathcal{L}^{sa}, \quad (5.71)$$

and similarly for $\mathcal{L}^{as}$, which is satisfied only if $\mathcal{L}^{sa} = \mathcal{L}^{as} = 0$. At this stage, the table 5.1 reduces to the table 5.2.

We now show that $\mathbf{L}^{st}, \mathbf{L}^{ts}, \mathbf{L}^{pp}$ and $\mathbf{L}^{aa}$ are scalars. A scalar is invariant under an arbitrary rotation. In particular, $\mathbf{L}^{pp} : \mathbf{ab}$ with $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$ is a scalar and thus satisfies $\mathbf{L}^{pp} : \mathbf{ab} = \mathbf{L}^{pp} : \mathbf{a'b'}$ under an arbitrary rotation. Since $\mathbf{L}^{pp}$ must remain invariant,

$$\mathbf{L}^{pp} : \mathbf{ab} = \mathbf{L}^{pp} : \mathbf{a'b'}. \quad (5.72)$$

Tensorial order	0	1	2	3	4
Polar tensor	$\mathcal{L}^{ss}$	$\times$	$\mathbf{L}^{st}, \mathbf{L}^{ts}, \mathbf{L}^{pp}, \mathbf{L}^{aa}$	$\times$	$\mathbf{L}^{tt}$
Axial tensor	$\times$	$\times$	$\times$	$\mathbf{L}^{at}, \mathbf{L}^{ta}$	$\times$

Table 5.2: The remaining possible linear response tensors.

An orthogonal transformation preserves the inner product. Hence, the only possible form for $\mathbf{L}^{pp}$ is

$$\mathbf{L}^{pp} = \mathcal{L}^{pp} \mathbf{1} \quad (5.73)$$

because $\mathbf{L}^{pp} : \mathbf{a}\mathbf{b} = \mathcal{L}^{pp}(\mathbf{a} \cdot \mathbf{b})$ and where $\mathcal{L}^{pp}$ is a scalar quantity. Similarly,

$$\mathbf{L}^{aa} = \mathcal{L}^{aa} \mathbf{1} \quad (5.74)$$

$$\mathbf{L}^{st} = \mathcal{L}^{st} \mathbf{1} \quad (5.75)$$

$$\mathbf{L}^{ts} = \mathcal{L}^{ts} \mathbf{1}. \quad (5.76)$$

We note that

$$\text{Tr} \mathbf{L}^{pp} = 3\mathcal{L}^{pp} \quad (5.77)$$

$$\text{Tr} \mathbf{L}^{aa} = 3\mathcal{L}^{aa} \quad (5.78)$$

$$\text{Tr} \mathbf{L}^{st} = 3\mathcal{L}^{st} = 0 \quad (5.79)$$

$$\text{Tr} \mathbf{L}^{ts} = 3\mathcal{L}^{ts} = 0, \quad (5.80)$$

the last two equations are satisfied only if $\mathbf{L}^{st} = \mathbf{L}^{ts} = 0$.

A similar reasoning can be applied to the tensors $\mathbf{L}^{at}$ and $\mathbf{L}^{ta}$. The following scalar can be constructed,

$$\mathbf{L}_{abc}^{at} a_a \mathbf{T}_{bc} = \mathbf{L}_{abc}^{at} a'_a \mathbf{T}'_{bc} \quad (5.81)$$

with $\mathbf{a}$ and $\mathbf{T}$ as above. The only way to satisfy this equation is to construct a rank three invariant tensor under an arbitrary rotation. The only invariant has the form $\varepsilon_{abc} a_a \mathbf{T}_{bc}$, but of course, this form would vanish if the antisymmetric part of $\mathbf{T}$ is not taken, $\varepsilon_{abc} a_a \mathbf{T}_{bc}^A$. Consequently, $\mathbf{L}^{at}$ is expected to have the form

$$\mathbf{L}_{abc}^{at} = \mathcal{L}^{at} \varepsilon_{abc}. \quad (5.82)$$

This is a contradiction since the two last spatial components of $\mathbf{L}^{at}$ are symmetric because of (5.67). Hence, $\mathbf{L}^{at} = 0$, and similarly $\mathbf{L}^{ta} = 0$.

Finally, (5.65)-(5.68) become

$$\mathbf{J}^s = \mathcal{L}^{ss} \mathbf{X}^s \quad (5.83)$$

$$\mathbf{J}^p = \mathcal{L}^{pp} \mathbf{X}^p \quad (5.84)$$

$$\mathbf{J}^a = \mathcal{L}^{aa} \mathbf{X}^a \quad (5.85)$$

$$\mathbf{J}^t = \mathbf{L}^{tt} : \mathbf{X}^t. \quad (5.86)$$

It was well shown that in an isotropic system, the fluxes and thermodynamic forces of different tensorial degree do not couple.

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In the proof above, the form of the tensors $\mathbf{L}^{aa}$ and $\mathbf{L}^{pp}$ was discussed, and showed to be $\mathbf{L}^{aa} = \mathcal{L}^{aa} \mathbf{1}$ for both of them. To complete the proof, let us discuss of the form of the rank four tensor $\mathbf{L}^{tt}$. To do so, the following scalar is constructed,

$$L_{abcd}^{tt} T_{ab} V_{cd} = L_{abcd}^{tt} T'_{ab} V'_{cd}. \quad (5.87)$$

The tensor $T_{ab} V_{cd}$ is expected to be isotropic so to be invariant under an arbitrary rotation. If each tensor is decomposed as in (5.62), the contraction of $\mathbf{T}$ and $\mathbf{V}$ implies that

$$L_{abcd}^{tt} = \frac{c_1}{3} \delta_{ab} \delta_{cd} + \frac{c_2}{2} (\delta_{ac} \delta_{bd} - \delta_{ad} \delta_{bc}) + \frac{c_3}{2} \left(\delta_{ac} \delta_{bd} + \delta_{ad} \delta_{bc} - \frac{2}{3} \delta_{ab} \delta_{cd} \right) \quad (5.88)$$

with $c_1, c_2, c_3 \in \mathbb{R}^3$ and in accordance with (5.63). The above expression is essentially justified because the product $\mathbf{T}^A : \mathbf{V}^A$ can be written in components as

$$\begin{aligned} \frac{1}{4} (T_{ab} - T_{ba}) (V_{ab} - V_{ba}) &= \frac{1}{2} (T_{ab} V_{ab} - T_{ab} V_{ba}) \\ &= \frac{1}{2} (\delta_{ac} \delta_{bd} - \delta_{ad} \delta_{bc}) T_{ab} V_{cd} \end{aligned} \quad (5.89)$$

and because the product $\mathbf{T}^{\circ} : \mathbf{V}^{\circ}$ can be written in components as

$$\begin{aligned} \left(\frac{1}{2} (T_{ab} + T_{ba}) - \frac{1}{3} \text{Tr} \mathbf{T} \delta_{ab} \right) \left(\frac{1}{2} (V_{ab} + V_{ba}) - \frac{1}{3} \text{Tr} \mathbf{V} \delta_{ab} \right) &= \frac{1}{2} (T_{ab} V_{ab} + T_{ab} V_{ba}) - \frac{1}{3} \text{Tr} \mathbf{T} \text{Tr} \mathbf{V} \\ &= \frac{1}{2} \left(\delta_{ac} \delta_{bd} + \delta_{ad} \delta_{bc} - \frac{2}{3} \delta_{ab} \delta_{cd} \right) T_{ab} V_{cd}. \end{aligned} \quad (5.90)$$

On the other hand, the tensor $\mathbf{L}^{tt}$ is symmetric and traceless in its two first and two last components, *i.e.*

$$L_{abcd}^{tt} = L_{bacd}^{tt} = L_{abdc}^{tt} \quad (5.91)$$

$$L_{aacd}^{tt} = L_{abcc}^{tt} = 0. \quad (5.92)$$

One easily shows that the condition (5.91) implies that $c_2 = 0$ and (5.92) that $c_1 = 0$. Hence,

$$L_{abcd}^{tt} = \mathcal{L}^{tt} \left(\delta_{ac} \delta_{bd} + \delta_{ad} \delta_{bc} - \frac{2}{3} \delta_{ab} \delta_{cd} \right) \quad (5.93)$$

where $\frac{c_3}{2}$ was renamed $\mathcal{L}^{tt}$. With this form, the phenomenological equation (5.86) becomes

$$\mathbf{J}_{ab}^t = \mathcal{L}^{tt} \mathbf{X}_{ab}^t. \quad (5.94)$$

Finally:

- In an isotropic system, the fluxes and thermodynamic forces of different tensorial degree do not couple. Consequently, the transport of particles and heat can be coupled (both having vectorial fluxes) while the momentum transport (tensorial flux of rank two) cannot couple to any other transported quantity.
- In general, the transport coefficients are described by tensors of arbitrary rank. In an isotropic system, the transport coefficients always are scalar quantities.
- If more than one vectorial transport phenomena exist, the several vectorial processes will be coupled through the cross linear response coefficients. These coefficients will also be scalar quantities as long as the system is isotropic.
- Since the pressure tensor (1.21) and also the momentum current density (2.64) are symmetric tensors, the j^a vectorial flux appearing in (5.85) vanishes.

5.5. FINAL RESULTS: GREEN-KUBO FORMULAS & DISSIPATIVE HYDRODYNAMIC EQUATIONS

This last section is the culmination of this work. First, the quantum Green-Kubo formulas for the linear response coefficients will be derived. Then, the dissipative hydrodynamic equations will be obtained.

Green-Kubo formulas. With what was discussed in the previous section, equation (5.57) reads

$$\mathcal{J}_{n_k}^a = \sum_{l=1}^c \mathcal{L}_{k,l} \mathcal{A}_{n_l}^a + \mathcal{L}_{k,e} \mathcal{A}_e^a \quad (5.95)$$

$$\mathcal{J}_{g^b}^a = \mathcal{L}_{g^b g^d}^{ac} \mathcal{A}_{g^d}^c \quad (5.96)$$

$$\mathcal{J}_e^a = \mathcal{L}_{e,e} \mathcal{A}_e^a + \sum_{l=1}^c \mathcal{L}_{e,k} \mathcal{A}_{n_k}^a \quad (5.97)$$

or with the expression of the affinities

$$\mathcal{J}_{n_k}^a = - \sum_{l=1}^c \frac{\mathcal{L}_{k,l}}{k_B} \partial^a \frac{\mu_{l0}}{T} + \frac{\mathcal{L}_{k,e}}{k_B} \partial^a \frac{1}{T} \quad (5.98)$$

$$\mathcal{J}_{g^b}^a = -\beta \mathcal{L}_{g^b g^d}^{ac} \partial^c u^d \quad (5.99)$$

$$\mathcal{J}_e^a = \frac{\mathcal{L}_{e,e}}{k_B} \partial^a \frac{1}{T} - \sum_{k=1}^c \frac{\mathcal{L}_{e,k}}{k_B} \partial^a \frac{\mu_{k0}}{T}, \quad (5.100)$$

up to higher order terms in the gradients expansion. Particle and energy transport are vectorial processes. From the result 5.1, we know that the corresponding linear response tensor has the form of a scalar times the unit matrix. For instance, the linear response tensor associated to energy transport is $\mathcal{L}_{e,e} = \mathcal{L}_{e,e} \mathbf{1}$. Taking the trace on both sides, we get $\mathcal{L}_{e,e} = \frac{1}{3} \text{Tr} \mathcal{L}_{e,e} = \frac{1}{3} (\mathcal{L}_{e,e})^{aa}$

or alternatively, $\mathcal{L}_{e,e} = (\mathcal{L}_{e,e})^{xx} = (\mathcal{L}_{e,e})^{yy} = (\mathcal{L}_{e,e})^{zz}$. Now, using (5.58), the linear response coefficients are given by the following Green-Kubo formulas

$$\mathcal{L}_{k,l} = \frac{k_B}{3} \frac{T}{V} \int_0^\infty d\tau \int_0^\beta dz \langle \delta \hat{j}_{n_k}^a(\tau) \delta \hat{j}_{n_l}^a(i\hbar z) \rangle_{\text{eq}} \quad (5.101)$$

$$\mathcal{L}_{k,e} = \frac{k_B}{3} \frac{T}{V} \int_0^\infty d\tau \int_0^\beta dz \langle \delta \hat{j}_{n_k}^a(\tau) \delta \hat{j}_e^a(i\hbar z) \rangle_{\text{eq}} \quad (5.102)$$

$$\mathcal{L}_{e,k} = \frac{k_B}{3} \frac{T}{V} \int_0^\infty d\tau \int_0^\beta dz \langle \delta \hat{j}_e^a(\tau) \delta \hat{j}_{n_k}^a(i\hbar z) \rangle_{\text{eq}} \quad (5.103)$$

$$\mathcal{L}_{e,e} = \frac{k_B}{3} \frac{T}{V} \int_0^\infty d\tau \int_0^\beta dz \langle \delta \hat{j}_e^a(\tau) \delta \hat{j}_e^a(i\hbar z) \rangle_{\text{eq}}. \quad (5.104)$$

As already noted in the first chapter, the heat conductivity κ is such that $\kappa = \frac{1}{T^2} \frac{\mathcal{L}_{e,e}}{k_B}$, while the diffusion coefficient is related to the corresponding linear response coefficient by $\frac{\mathcal{L}_{k,p}}{k_B T} = \sum_{p=1}^c D_{kp} \left(\frac{\partial n_p}{\partial \mu_{i0}} \right)_{T, \{\mu_i\}_{i \neq l}}$.

The mass conservation implies that $\sum_{k=1}^c m_k \mathcal{J}_{n_k}^a = 0$ which in turn implies that $\sum_{k=1}^c m_k \mathcal{L}_{k,l} = 0$ and $\sum_{k=1}^c m_k \mathcal{L}_{k,e} = 0$ separately. Hence, these coefficients are not all independent. A diffusion process can thus be reduced to the coefficients describing mutual diffusion of solute within solvent species. For a one-component fluid, $c = 1$, $m_1 \mathcal{J}_{n_1}^a = 0$ which expresses that diffusion never occurs in a single component fluid.

Regarding the transport of momentum, with (5.93), the viscosity tensor takes the form

$$\beta \mathcal{L}_{g^b g^d}^{ac} \equiv \eta_{abcd} = \eta \left(\delta_{ac} \delta_{bd} + \delta_{ad} \delta_{bc} - \frac{2}{3} \delta_{ab} \delta_{cd} \right) + \xi \delta_{ab} \delta_{cd} \quad (5.105)$$

so that

$$\mathcal{J}_{g^b}^a = -\eta \left(\partial^a u^b + \partial^b u^a - \frac{2}{3} (\nabla \cdot \mathbf{u}) \delta^{ab} \right) - \xi (\nabla \cdot \mathbf{u}) \delta^{ab}. \quad (5.106)$$

Also,

$$\eta_{abab} = 10\eta + 3\xi \quad (5.107)$$

$$\eta_{aabb} = 9\xi \quad (5.108)$$

so that $\eta = \frac{1}{10} \left(\eta_{abab} - \frac{1}{3} \eta_{aabb} \right)$. Alternatively,

$$\eta = \eta_{xyxy} \quad (5.109)$$

$$\xi - \frac{2}{3} \eta = \eta_{xxyy}. \quad (5.110)$$

The corresponding Green-Kubo formulas for the shear and bulk viscosities, thus read

$$\eta = \frac{1}{V} \int_0^\infty d\tau \int_0^\beta dz \langle \delta \hat{j}_{g^y}^x(\tau) \delta \hat{j}_{g^y}^x(i\hbar z) \rangle_{\text{eq}} \quad (5.111)$$

$$\xi = \frac{1}{9} \frac{1}{V} \int_0^\infty d\tau \int_0^\beta dz \langle \delta \hat{j}_{g^a}^a(\tau) \delta \hat{j}_{g^b}^b(i\hbar z) \rangle_{\text{eq}}. \quad (5.112)$$

In thermal systems and at large distance, the current correlation functions $\langle \delta \hat{j}(\mathbf{r}, \tau) e^{-Z \hat{\sigma}_{\text{eq}}} \delta \hat{j}'(\mathbf{r}', 0) e^{Z \hat{\sigma}_{\text{eq}}} \rangle_{\text{eq}}$ are expected to vanish due to the absence of correlations between

far enough spatial regions. Accordingly, the time correlation functions $\langle \delta \hat{\mathbf{J}}(\tau) \delta \hat{\mathbf{J}}(i\hbar z) \rangle_{\text{eq}}$ of the microscopic total currents are extensive. Consequently, the linear response coefficients are intensive quantities.

Onsager-Casimir reciprocal relations. The linear response coefficients are given by (5.58). Apart from their spatial indices, they also possess indices α and γ related to the local densities. What is the symmetry exhibited by the linear response coefficients under the change $\alpha \leftrightarrow \gamma$? To answer this, the consequences of time reversal symmetry will be explored. The time reversal operator was defined in the context of quantum mechanics in §2.2. We begin with the following result:

RESULT 5.3. Let $\hat{\Theta} \hat{A}(\tau) \hat{\Theta}^{-1} = \epsilon_A \hat{A}(-\tau)$ and $\hat{\Theta} \hat{B} \hat{\Theta}^{-1} = \epsilon_B \hat{B}$, with $\epsilon_{A/B} = \pm 1$. Let us define

$$\varphi_{BA}(\tau) \equiv \int_0^\beta dz \langle \hat{A}(\tau) \hat{B}(i\hbar z) \rangle_{\text{eq}}$$

the relaxation function. Then,

1. $\varphi_{BA}(\tau) = \varphi_{AB}(-\tau)$.
2. Reciprocity law: $\varphi_{BA}(\tau) = \epsilon_A \epsilon_B \varphi_{BA}(-\tau) = \epsilon_A \epsilon_B \varphi_{AB}(\tau)$.
3. $\varphi_{BA}(\tau)$ is real.

Proof: We have,

$$\begin{aligned} \varphi_{BA}(\tau) &= \int_0^\beta dz \langle \hat{A}(\tau) \hat{B}(i\hbar z) \rangle_{\text{eq}} \\ &= \frac{1}{Z} \int_0^\beta dz \text{Tr} e^{-\beta \hat{H}} \hat{A}(\tau) \hat{B}(i\hbar z) \\ &= \frac{1}{Z} \int_0^\beta dz \text{Tr} e^{-\beta \hat{H}} \hat{A}(\tau - i\hbar z) \hat{B}(0) \\ &= \frac{1}{Z} \int_0^\beta dz \text{Tr} e^{-\beta \hat{H}} \hat{A}(\tau - i\hbar(\beta - z)) \hat{B}(0) \\ &= \frac{1}{Z} \int_0^\beta dz \text{Tr} e^{-\beta \hat{H}} e^{\frac{i}{\hbar} \hat{H} \tau} e^{(\beta - z) \hat{H}} \hat{A}(0) e^{-\frac{i}{\hbar} \hat{H} \tau} e^{-(\beta - z) \hat{H}} \hat{B}(0) \\ &= \frac{1}{Z} \int_0^\beta dz \text{Tr} e^{\frac{i}{\hbar} \hat{H} \tau} e^{-z \hat{H}} \hat{A}(0) e^{z \hat{H}} e^{-\beta \hat{H}} e^{-\frac{i}{\hbar} \hat{H} \tau} \hat{B}(0) \\ &= \frac{1}{Z} \int_0^\beta dz \text{Tr} e^{-z \hat{H}} \hat{A}(0) e^{z \hat{H}} e^{-\beta \hat{H}} \hat{B}(-\tau) \\ &= \frac{1}{Z} \int_0^\beta dz \text{Tr} e^{-\beta \hat{H}} \hat{B}(-\tau) e^{-z \hat{H}} \hat{A}(0) e^{z \hat{H}} \\ &= \int_0^\beta dz \langle \hat{B}(-\tau) \hat{A}(i\hbar z) \rangle_{\text{eq}} \\ &= \varphi_{AB}(-\tau) \end{aligned} \tag{5.113}$$

which proves the first assertion. In the third line, we have used the invariance of the equilibrium correlation functions under time translation. In the fourth line, the change $z \rightarrow \beta - z$ was performed. Now,

$$\begin{aligned}
 \varphi_{BA}(\tau) &= \frac{1}{Z} \int_0^\beta dz \operatorname{Tr} e^{-\beta \hat{H}} \hat{A}(\tau) \hat{B}(i\hbar z) \\
 &= \frac{1}{Z} \int_0^\beta dz \operatorname{Tr} e^{-\beta \hat{H}} e^{\frac{i}{\hbar} \hat{H} \tau} \hat{A}(0) e^{-\frac{i}{\hbar} \hat{H} \tau} e^{-z \hat{H}} \hat{B}(0) e^{z \hat{H}} \\
 &= \frac{1}{Z} \int_0^\beta dz \operatorname{Tr} e^{-\beta \hat{H}} e^{\frac{i}{\hbar} \hat{H} \tau} \hat{A}(0) e^{-\frac{i}{\hbar} \hat{H} \tau} \hat{\Theta}^{-1} \hat{\Theta} e^{-z \hat{H}} \hat{B}(0) e^{z \hat{H}} \\
 &= \frac{1}{Z} \int_0^\beta dz \operatorname{Tr} \hat{\Theta} e^{-z \hat{H}} \hat{B}(0) e^{z \hat{H}} e^{-\beta \hat{H}} \hat{\Theta}^{-1} \hat{\Theta} e^{\frac{i}{\hbar} \hat{H} \tau} \hat{A}(0) e^{-\frac{i}{\hbar} \hat{H} \tau} \hat{\Theta}^{-1} \\
 &= \epsilon_A \epsilon_B \frac{1}{Z} \int_0^\beta dz \operatorname{Tr} e^{-z \hat{H}} \hat{B}(0) e^{z \hat{H}} e^{-\beta \hat{H}} e^{-\frac{i}{\hbar} \hat{H} \tau} \hat{A}(0) e^{\frac{i}{\hbar} \hat{H} \tau} \\
 &= \epsilon_A \epsilon_B \frac{1}{Z} \int_0^\beta dz \operatorname{Tr} e^{-\beta \hat{H}} e^{-\frac{i}{\hbar} \hat{H} \tau} \hat{A}(0) e^{\frac{i}{\hbar} \hat{H} \tau} e^{-z \hat{H}} \hat{B}(0) e^{z \hat{H}} \\
 &= \epsilon_A \epsilon_B \frac{1}{Z} \int_0^\beta dz \operatorname{Tr} e^{-\beta \hat{H}} \hat{A}(-\tau) \hat{B}(i\hbar z) \\
 &= \epsilon_A \epsilon_B \varphi_{BA}(-\tau) \\
 &= \epsilon_A \epsilon_B \varphi_{AB}(\tau)
 \end{aligned} \tag{5.114}$$

where in the fifth line we have used that $[\hat{H}, \hat{\Theta}] = 0 \Leftrightarrow \hat{\Theta} \hat{H} \hat{\Theta}^{-1} = \hat{H}$, since the Hamiltonian operator ruling the microscopic dynamics has the time-reversal symmetry. In the last line, the result (5.113) was used.

Finally,

$$\begin{aligned}
 \varphi_{BA}^*(\tau) &= \frac{1}{Z} \int_0^\beta dz \operatorname{Tr} \left(e^{-\beta \hat{H}} \hat{A}(\tau) \hat{B}(i\hbar z) \right)^* \\
 &= \frac{1}{Z} \int_0^\beta dz \operatorname{Tr} \left(e^{-\beta \hat{H}} \hat{A}(\tau) \hat{B}(i\hbar z) \right)^\dagger \\
 &= \frac{1}{Z} \int_0^\beta dz \operatorname{Tr} \hat{B}(-i\hbar z) \hat{A}(\tau) e^{-\beta \hat{H}} \\
 &= \frac{1}{Z} \int_0^\beta dz \operatorname{Tr} \hat{B}(0) \hat{A}(\tau + i\hbar(\beta - z)) e^{-\beta \hat{H}} \\
 &= \frac{1}{Z} \int_0^\beta dz \operatorname{Tr} \hat{A}(-i\hbar z) \hat{B}(-\tau) e^{-\beta \hat{H}} \\
 &= \frac{1}{Z} \int_0^\beta dz \operatorname{Tr} \hat{A}(\tau) \hat{B}(i\hbar z) e^{-\beta \hat{H}} \\
 &= \varphi_{BA}(\tau).
 \end{aligned} \tag{5.115}$$

QUOD ERAT DEMONSTRANDUM

The third point of the result 5.3 implies that the transport coefficient given by the Green-Kubo formulas are real. As an immediate consequence of this same result, the Onsager-Casimir

reciprocal relation follows

$$\mathcal{L}_{\alpha\gamma} = \epsilon_\alpha \epsilon_\gamma \mathcal{L}_{\gamma\alpha} \quad (5.116)$$

where $\epsilon_\alpha = \pm 1$ if $\delta \mathbf{j}_\alpha$ is even or odd under time reversal. In the context of this work,

$$\hat{\Theta} \hat{\rho} \hat{\Theta}^{-1} = \hat{\rho} \quad \hat{\Theta} \hat{j}_\rho^a \hat{\Theta}^{-1} = -\hat{j}_\rho^a \quad (5.117)$$

$$\hat{\Theta} \hat{n}_k \hat{\Theta}^{-1} = \hat{n}_k \quad \hat{\Theta} \hat{j}_{n_k}^a \hat{\Theta}^{-1} = -\hat{j}_{n_k}^a \quad (5.118)$$

$$\hat{\Theta} \hat{g}^a \hat{\Theta}^{-1} = -\hat{g}^a \quad \hat{\Theta} \hat{j}_{g^b}^a \hat{\Theta}^{-1} = \hat{j}_{g^b}^a \quad (5.119)$$

$$\hat{\Theta} \hat{e} \hat{\Theta}^{-1} = \hat{e} \quad \hat{\Theta} \hat{j}_e^a \hat{\Theta}^{-1} = -\hat{j}_e^a, \quad (5.120)$$

justifying that

$$\hat{\Theta} \delta \hat{j}_{n_k}^{a'} \hat{\Theta}^{-1} = -\delta \hat{j}_{n_k}^{a'} \quad (5.121)$$

$$\hat{\Theta} \delta \hat{j}_{g^b}^{a'} \hat{\Theta}^{-1} = \delta \hat{j}_{g^b}^{a'} \quad (5.122)$$

$$\hat{\Theta} \delta \hat{j}_e^{a'} \hat{\Theta}^{-1} = -\delta \hat{j}_e^{a'}. \quad (5.123)$$

Consequently, the Onsager-Casimir reciprocal relation imply that the linear response coefficients (5.101)-(5.103) satisfy

$$\mathcal{L}_{k,l} = \mathcal{L}_{l,k} \quad (5.124)$$

$$\mathcal{L}_{k,e} = \mathcal{L}_{e,k}. \quad (5.125)$$

We also note that if the momentum and energy transport were to couple, the corresponding linear response coefficient would satisfy,

$$\mathcal{L}_{e,g^a} = -\mathcal{L}_{g^a,e}. \quad (5.126)$$

Dissipative hydrodynamic equations. Taking back the conservation equation (3.48) for the mean local densities, performing the split (3.93) of the mean current densities, using the expressions (4.48)-(4.51) of the dissipativeless mean current densities and the expressions (5.98)-(5.100) of their dissipative part, the dissipative hydrodynamic equations are naturally obtained,

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{u}) = 0 \quad (5.127)$$

$$\partial_t n_k + \nabla \cdot (n_k \mathbf{u} + \mathcal{J}_{n_k}) = 0 \quad (5.128)$$

$$\partial_t (\rho \mathbf{u}) + \nabla \cdot (\rho \mathbf{u} \mathbf{u} + p \mathbf{1} + \mathbf{\Pi}) = 0 \quad (5.129)$$

$$\partial_t (e_0 + \frac{1}{2} \rho \mathbf{u}^2) + \nabla \cdot \left\{ (e_0 + \frac{1}{2} \rho \mathbf{u}^2 + p) \mathbf{u} + \mathbf{\Pi} \cdot \mathbf{u} + \mathcal{J}_q \right\} = 0. \quad (5.130)$$

For the last equation, (5.4) was used. In the absence of an external force, the last equation can be rewritten as $\partial_t U + \nabla \cdot \left\{ (U + p) \mathbf{u} + \mathbf{\Pi} \cdot \mathbf{u} + \mathcal{J}_q \right\} = 0$ with U the energy density. As expected, the continuity equation (5.127) contains no dissipative part. These equations coincide with

the equations (1.10), (1.17), (1.22) and (1.29) obtained in the first chapter from a macroscopic perspective on the basis of classical nonequilibrium thermodynamics. Therefore, the microscopic approach based on the LESO formalism and the expansion in powers of the gradients is in accordance with the classical approach of nonequilibrium thermodynamics for the linear transport properties. Alternatively,

$$\frac{d\rho}{dt} + \rho \nabla \cdot \mathbf{u} = 0 \quad (5.131)$$

$$\frac{dn_k}{dt} + n_k \nabla \cdot \mathbf{u} - \sum_{l=1}^c \frac{\mathcal{L}_{k,l}}{k_B} \nabla^2 \left(\frac{\mu_{l0}}{T} \right) + \frac{\mathcal{L}_{k,e}}{k_B} \nabla^2 \left(\frac{1}{T} \right) = 0 \quad (5.132)$$

$$\rho \frac{d\mathbf{u}}{dt} + \nabla p - \eta \nabla^2 \mathbf{u} - \left(\frac{\eta}{3} + \xi \right) \nabla (\nabla \cdot \mathbf{u}) = 0 \quad (5.133)$$

$$\begin{aligned} \frac{de_0}{dt} + (e_0 + p) \nabla \cdot \mathbf{u} - \frac{\eta}{2} \left(\nabla \mathbf{u} + (\nabla \mathbf{u})^T - \frac{2}{3} (\nabla \cdot \mathbf{u}) \mathbf{1} \right)^2 - \xi (\nabla \cdot \mathbf{u})^2 \\ - \sum_{k=1}^c \frac{\mathcal{L}_{e,k}}{k_B} \nabla^2 \left(\frac{\mu_{k0}}{T} \right) - \kappa \nabla^2 T = 0 \end{aligned} \quad (5.134)$$

if the transport coefficients are assumed to be independent of the spatial coordinates. The corresponding Green-Kubo formulas for the transport coefficients read,

$$\mathcal{L}_{k,l} = \mathcal{L}_{l,k} = \lim_{V \rightarrow \infty} \frac{k_B T}{V} \int_0^\infty d\tau \int_0^\beta dz \langle \delta \hat{j}_{n_k}^x(\tau) \delta \hat{j}_{n_l}^x(i\hbar z) \rangle_{\text{eq}} \quad (5.135)$$

$$\mathcal{L}_{k,e} = \mathcal{L}_{e,k} = \lim_{V \rightarrow \infty} \frac{k_B T}{V} \int_0^\infty d\tau \int_0^\beta dz \langle \delta \hat{j}_{n_k}^x(\tau) \delta \hat{j}_e^x(i\hbar z) \rangle_{\text{eq}} \quad (5.136)$$

$$\kappa = \lim_{V \rightarrow \infty} \frac{1}{TV} \int_0^\infty d\tau \int_0^\beta dz \langle \delta \hat{j}_e^x(\tau) \delta \hat{j}_e^x(i\hbar z) \rangle_{\text{eq}} \quad (5.137)$$

$$\eta = \lim_{V \rightarrow \infty} \frac{1}{V} \int_0^\infty d\tau \int_0^\beta dz \langle \delta \hat{j}_{g^y}^x(\tau) \delta \hat{j}_{g^y}^x(i\hbar z) \rangle_{\text{eq}} \quad (5.138)$$

$$\xi - \frac{2}{3} \eta = \lim_{V \rightarrow \infty} \frac{1}{V} \int_0^\infty d\tau \int_0^\beta dz \langle \delta \hat{j}_{g^x}^x(\tau) \delta \hat{j}_{g^y}^y(i\hbar z) \rangle_{\text{eq}}. \quad (5.139)$$

The system that was studied has a finite volume V . The Green-Kubo formulas may be subjected to finite-size effects encountered in molecular dynamics simulations. For this reason, the limit $V \rightarrow \infty$ is taken at constant values of the chemical potentials in accordance with the hypothesis 1 in §2.3.

The dissipative hydrodynamic equations (5.131)-(5.134), together with (5.135)-(5.139) constitute the final achievement of this work.

Discussion. Taking $\mathcal{L}_{k,l} = \mathcal{L}_{k,e} = \kappa = \eta = \xi = 0$ in the hydrodynamic equations (5.131)-(5.134) gives exactly (as expected) the dissipativeless hydrodynamic equations (4.57)-(4.60).

Since the macroscopic equations (5.131)-(5.134) were derived from the microscopic dynamics, the transport coefficients now have a microscopic foundation given by the Green-Kubo formulas (5.135)-(5.139). These formulas express the transport coefficients as integrals of time

correlation functions, with respect to the canonical equilibrium ensemble, of the fluctuations of microscopic total currents around their local equilibrium value.

The expressions of the transport coefficients given by the Green-Kubo formulas are exact in the sense that the corrections scaling as $\mathcal{O}(\nabla^2)$ do not affect these formulas. Indeed, with the use of equation (1.58), the linear response coefficients are precisely defined as the numerical coefficient relating the dissipative current densities to the affinities. The idea is the same as saying that in the Taylor expansion of a function f around $x = 0$, $f(x) = f(0) + f'(0)x + \mathcal{O}(x^2)$, the higher corrections do not modify the coefficient $f'(0)$ of the linear term.

On the other hand, it is true to say that the $\mathcal{O}(\nabla^2)$ corrections may affect the expressions of the dissipative fluxes (5.98)-(5.100) beyond the linear regime of hydrodynamics. The origin of these corrections can be traced back to the two main approximations that were previously carried out. First, the expansion of the entropy functional (4.5) at leading order propagates a correction scaling as $\mathcal{O}(\nabla c)^2$. Then, neglecting the two terms of the result 4.1 implies the equation (4.60) to be approximate which in turn implies the equations (4.70) and (4.72) for the time evolution of the conjugate fields to be approximate. These corrections thus propagate in the calculation of the integrand (5.24), thus in the dissipative current densities (5.98)-(5.100) and so in the final dissipative hydrodynamic equations. We note that these two approximations are hard to trace back from the final results (5.131)-(5.134) since they mix and interact in a complex way throughout this work.

Furthermore, at large scales compared to the mean free path of particles, the $\mathcal{O}(\nabla^2)$ corrective terms become negligible. These corrective terms come into play if the spatial or temporal scales are of the same size as the characteristic scales of microscopic dynamics such as the mean free path. At larger and larger scales, the hydrodynamic equations (5.131)-(5.134) therefore also become exact in three-dimensional or higher dimensional spaces. Indeed, in $d \geq 3$, d denoting the dimension of the space, the time integrals of the Green-Kubo formulas are known to converge [32]. However, it was shown [32] that for $d \leq 2$, the time correlation functions in normal fluids diverge. Therefore, normal transport properties and thus hydrodynamic equations can no longer be formulated in $d = 1$ and $d = 2$ spaces.

CONCLUSION & OUTLOOK

In this work, we have shown that the macroscopic equations describing the behaviour of normal quantum fluids can be derived in a unified microscopic approach based on the local equilibrium statistical operator.

First, the notion of transport phenomena in the classical framework was introduced and explored, as well as the notion of transport coefficients that follows from it. Three key elements emerged from this preliminary study and served as basis tools for classical nonequilibrium statistical mechanics: conservation equations, entropy production, and phenomenological equations defining the linear response coefficients. In particular, we have shown that for normal fluid systems, the reversibility of the hydrodynamic equations is broken by terms containing transport coefficients. Then, the framework of this work as well as the main tools of quantum mechanics were introduced. The observables of mass, particle, momentum, and energy local densities were defined. These alone form a set of relevant variables in the description of the macroscopic state of normal quantum fluids, hence their use. For each of these densities, a local conservation equation was constructed and the expression of the respective local current density operator could be obtained. An important part was devoted to the construction of the local equilibrium statistical operator. It is constructed from two ingredients: a set of relevant variables and the requirement to maximise entropy. This operator is to be considered as a functional of certain conjugate fields which initially act as Lagrange multipliers. Later on, the local equilibrium statistical operator could be understood as the projection of the exact statistical operator on the abstract space of the relevant observables. The projection over the remaining irrelevant variables defines the deviation operator.

Next, the expression of the entropy production in terms of the conjugate fields was obtained and constitutes a first notable result of this work. It is expressed from the mean value of the current density operators taken with respect to the deviation operator, and takes the same form as in the classical framework. The currents appearing in the expression of entropy production were associated with dissipative currents. An exact expression of the deviation operator for quantum systems was also obtained. It is noted that, at this stage, the scope of the results is broad and is not restricted to the choice of microscopic local densities.

The last two chapters were devoted to the dissipativeless and dissipative equations. The starting point was to give a physical meaning to the conjugate fields. To do this, the entropy functional was developed in powers of the gradients of the mean densities. The conjugate fields appeared to be the conjugate thermodynamic variables to their respective densities. The time evolution of the conjugate fields could be obtained from the dissipativeless conservation equations, the latter obtained by including only the dissipativeless part of the current densities in the conservation equations. All these equations explicitly show the dependence of the mean values on the local velocity of the fluid, since a transformation from the laboratory to the moving

frame was performed. With the expression of the deviation operator, the expression of the dissipative current densities could be obtained, under the assumption that the macroscopic fields vary little on the microscopic scale and that the correlation functions rapidly decay to zero. Finally, the expression of the dissipative hydrodynamic equations and the Green-Kubo formulas for the transport coefficients were obtained quite naturally within the formalism. We stress the important role of the intrinsic symmetries of the system, in particular its isotropy, as well as the consequences of the time-reversal symmetry giving rise to the Casimir-Onsager reciprocal relations.

The results obtained are remarkable for several reasons. First, macroscopic properties have been expressed on the basis of the quantum dynamics of the particles, thus implying a more fundamental understanding of the hydrodynamic equations as well as of the transport coefficients. At the macroscopic scale, these coefficients are sometimes abstract. Their microscopic definition lifts the veil. Secondly, the results are analytical results of nonequilibrium statistical mechanics. Thirdly, the results obtained are valid for quantum systems. We note that the entropy production and the form of the dissipative hydrodynamic equations are the same than in the classical approach, justifying the validity of the local equilibrium statistical operator formalism and the expansion in powers of the gradients.

It is clear that the scope of the formalism is wide and that a multitude of other systems are amenable to study. The first idea would be to propose a generalisation, still in the quantum framework, to systems with broken continuous symmetries, such as crystalline solids or liquid crystals. To do this, it would be sufficient to identify a local order parameter, its associated conservation equation, and construct the corresponding local equilibrium statistical operator. These systems may no longer be isotropic, thus giving rise to tensorial transport properties. Another natural extension would be to obtain thermoelectric transport coefficients. It would then be necessary to deal with an electrically conducting material with a temperature gradient. A local electric charge density would have to be considered and its corresponding conservation equation identified.

The work also inspires many other open questions. A reformulation of this formalism on the basis of nonrelativistic quantum field theory would be interesting. As it was (partially) shown in this work, microscopic local conservation equations can be obtained within this formalism. Could the hydrodynamic equations and the Green-Kubo formulas be expressed in this same formalism too? We may also question the relativistic approach. Can the formalism of this work be extended to obtain relativistic hydrodynamic equations? Possible relativistic transport coefficients? Finally, we wonder about the numerical implementation of the quantum Green-Kubo formulas. How should this be done? To what extent are the corrections brought by the quantum formulas important compared to the classical ones? Also, how formulating hydrodynamics and understanding transport properties in the case of one and two-dimensional spaces?

APPENDIXES

APPENDIX A. CONSERVATION OF ANGULAR MOMENTUM

The pressure tensor will appear many times throughout this work. It therefore seems important to justify one of its most important property: it is a symmetric tensor. To do this, the balance equation for the angular momentum density is obtained. Since the system is isolated, it is not subjected to external forces that could cause it to rotate. Imposing the conservation of the angular momentum will imply the symmetry of the pressure tensor.

The angular momentum density is defined by $\mathbf{l} \equiv \mathbf{r} \times \rho \mathbf{u}$. A balance equation for $\mathbf{l}$ can be obtained by taking the cross product of $\mathbf{r}$ with the conservation equation for the momentum density (1.22) given by

$$\partial_t(\rho \mathbf{u}) + \nabla \cdot (\rho \mathbf{u} \mathbf{u} + \mathbf{P}) = 0. \quad (\text{A.1})$$

Using that $\mathbf{r}$ is an independent coordinate, $\mathbf{r} \times \partial_t(\rho \mathbf{u}) = \partial_t \mathbf{l}$. Let us look at the second and third terms.

Second term:

$$\begin{aligned} [\mathbf{r} \times \nabla \cdot (\rho \mathbf{u} \mathbf{u})]_i &= \varepsilon_{ijk} r_j (\nabla \cdot (\rho \mathbf{u} \mathbf{u}))_k \\ &= \varepsilon_{ijk} r_j (\partial_l (\rho u_l u_m \delta_{mk})) \\ &= \partial_l (\rho u_l \varepsilon_{ijk} r_j u_k) - \varepsilon_{ijk} \rho u_l u_k \partial_l r_j \\ &= \nabla \cdot (\rho \mathbf{u} [\mathbf{r} \times \mathbf{u}]_i) - \underbrace{\rho [\mathbf{u} \times \mathbf{u}]_i}_{=0}. \end{aligned} \quad (\text{A.2})$$

Third term:

$$\begin{aligned} [\mathbf{r} \times \nabla \cdot \mathbf{P}]_i &= \varepsilon_{ijk} r_j (\nabla \cdot \mathbf{P})_k \\ &= \varepsilon_{ijk} r_j \partial_l P_{lk} \\ &= \partial_l (\varepsilon_{ijk} r_j P_{kl}^\top) - \varepsilon_{ijk} P_{jk} \\ &= \partial_l (\varepsilon_{ljk} r_j P_{ki}^\top)^\top - \varepsilon_{ijk} P_{jk}. \end{aligned} \quad (\text{A.3})$$

In vector form, the balance equation for the angular momentum density thus reads

$$\partial_t \mathbf{l} + \nabla \cdot (\rho \mathbf{u} (\mathbf{r} \times \mathbf{u}) + (\mathbf{r} \times \mathbf{P}^\top)^\top) = -\boldsymbol{\varepsilon} : \mathbf{P}. \quad (\text{A.4})$$

The term $\boldsymbol{\varepsilon} : \mathbf{P}$ acts as a source term and must vanish in order to ensure the conservation of the angular momentum density. In components,

$$\boldsymbol{\varepsilon} : \mathbf{P} = \varepsilon_{ijk} P_{jk} \stackrel{!}{=} 0 \quad (\text{A.5})$$

implying that $\mathbf{P} = \mathbf{P}^\top$ is symmetric since the Levi-Civita symbol is antisymmetric.

Nonetheless, under certain circumstances, the balance equation for the angular momentum density must be considered in its own right. These are, for example, rotating systems or systems for which the particles carry an intrinsic angular momentum (spin).

APPENDIX B. ELEMENTARY THERMODYNAMICS

This appendix is to be seen as a quick reminder of some thermodynamic relations that will appear many times throughout this work.

Fundamental & Euler thermodynamic relations. The first and second laws of thermodynamics can be combined to give the fundamental thermodynamic relation (also Gibbs relation) for multicomponent systems,

$$dE = TdS - pdV + \sum_{k=1}^c \mu_k dN_k \quad (\text{B.1})$$

which is valid for systems at equilibrium. $E = E(S, V, N_1, \dots, N_c)$ is the total internal energy. Each term is constructed as a product of an intensive quantity (T, p, μ_k) with an extensive one (S, V, N_k). We can identify

$$T = \left(\frac{\partial E}{\partial S} \right)_{V, N_1, \dots, N_c} \quad (\text{B.2})$$

$$p = - \left(\frac{\partial E}{\partial V} \right)_{S, N_1, \dots, N_c} \quad (\text{B.3})$$

$$\mu_k = \left(\frac{\partial E}{\partial N_k} \right)_{S, V, \{N_j\}_{j \neq k}}. \quad (\text{B.4})$$

The extensive property of E implies that

$$E(\lambda S, \lambda V, \lambda N_1, \dots, \lambda N_c) = \lambda E(S, V, N_1, \dots, N_c) \quad \text{for } \lambda > 0. \quad (\text{B.5})$$

Consequently, E is an homogeneous function of first degree and with the use of the Euler's theorem,

$$E(S, V, N_1, \dots, N_c) = \left(\frac{\partial E}{\partial S} \right)_{V, N_1, \dots, N_c} S + \left(\frac{\partial E}{\partial V} \right)_{S, N_1, \dots, N_c} V + \sum_{k=1}^c \left(\frac{\partial E}{\partial N_k} \right)_{S, V, \{N_j\}_{j \neq k}} N_k \quad (\text{B.6})$$

yielding the Euler relation

$$E = TS - pV + \sum_{k=1}^c \mu_k N_k. \quad (\text{B.7})$$

Here, the chemical potentials has the unit $\text{J} \cdot \#^{-1}$ and can be modified to have unit $\text{J} \cdot \text{kg}^{-1}$.^a In such case, the fundamental thermodynamic relation (B.1) and the Euler relation (B.7) take the form

$$dE = TdS - pdV + \sum_{k=1}^c \mu_k dM_k \quad (\text{B.8})$$

^aUsing molar mass for example: if μ has $\text{J} \cdot \#^{-1}$ unit and $\tilde{\mu}$ has $\text{J} \cdot \text{kg}^{-1}$ unit, then $\mu = \tilde{\mu} \frac{M}{N_A}$.

$$E = TS - pV + \sum_{k=1}^c \mu_k M_k \quad (\text{B.9})$$

where $\tilde{\mu}$ was renamed μ for simplicity. In this work, we take the freedom to switch from one definition to the other without further notice.

The fundamental thermodynamic and Euler relations can be written in terms of the specific quantities (per unit mass) or in terms of the densities (per unit volume).

Defining the specific quantities

$$\begin{cases} E = eM \\ S = sM \\ V = \frac{1}{\rho}M \\ M_k = c_k M, \end{cases} \quad (\text{B.10})$$

the Euler relation now reads

$$e = Ts - \frac{p}{\rho} + \sum_{k=1}^c \mu_k c_k. \quad (\text{B.11})$$

Using (B.8),

$$\begin{aligned} Mde + edM &= TMds + TsdM - pMd\frac{1}{\rho} - \frac{p}{\rho}dM + \sum_{k=1}^c \mu_k c_k dM + \sum_{k=1}^c \mu_k M dc_k \\ \Leftrightarrow M\left(de - Tds + p d\frac{1}{\rho} - \sum_{k=1}^c \mu_k dc_k\right) &= (dM)\left(-e + Ts - \frac{p}{\rho} + \sum_{k=1}^c \mu_k c_k\right) \end{aligned} \quad (\text{B.12})$$

where the right-hand side vanishes as a consequence of (B.11). Hence,

$$de = Tds + \frac{p}{\rho^2}d\rho + \sum_{k=1}^c \mu_k dc_k \quad \text{or} \quad ds = \frac{1}{T}de - \frac{p}{\rho^2 T}d\rho - \sum_{k=1}^c \frac{\mu_k}{T}dc_k. \quad (\text{B.13})$$

Now, defining the densities

$$\begin{cases} E = \mathfrak{e}V \\ S = \mathfrak{s}V \\ N_k = n_k V, \end{cases} \quad (\text{B.14})$$

the Euler relation reads,

$$\mathfrak{e} = T\mathfrak{s} - p + \sum_{k=1}^c \mu_k n_k \quad (\text{B.15})$$

and the fundamental thermodynamic relation reads,

$$d\mathfrak{e} = Td\mathfrak{s} + \sum_{k=1}^c \mu_k dn_k. \quad (\text{B.16})$$

All these relations hold in the frame moving with the fluid element at equilibrium. Nonetheless, they can be extended to the laboratory frame where the center of mass of the fluid element moves at the velocity $\mathbf{u}$. If $\mathfrak{e}_0$ and $\mathfrak{e}$ denote respectively the energy density in the frame moving at velocity $\mathbf{u} = 0$ (that is the frame moving with the fluid element) and $\mathbf{u}$ (that is the laboratory

frame), one has $\epsilon = \epsilon_0 + \frac{1}{2}\rho\mathbf{u}^2$ and with the same notation $\mu_k = \mu_{k0} - \frac{1}{2}m_k\mathbf{u}^2$. For instance, in the laboratory frame, equation (B.15) becomes

$$\epsilon = T\mathfrak{s} - p + \sum_{k=1}^c \mu_k n_k + u^a g^a \quad (\text{B.17})$$

with the momentum density $g^a \equiv \rho u^a$. On the other hand, equation (B.16) becomes

$$d\epsilon = T d\mathfrak{s} + \sum_{k=1}^c \mu_k dn_k + u^a dg^a. \quad (\text{B.18})$$

Gibbs-Duhem relation & Massieu density. Taking the differential of (B.17) and comparing with (B.18), the Gibbs-Duhem relation can be obtained in the laboratory frame

$$dp = \mathfrak{s}dT + \sum_{k=1}^c n_k d\mu_k + g^a du^a. \quad (\text{B.19})$$

Also, we have

$$\beta p = \frac{1}{k_B} \mathfrak{s} - \beta \epsilon + \beta \sum_{k=1}^c \mu_k n_k + \beta u^a g^a \quad (\text{B.20})$$

with $\beta \equiv \frac{1}{k_B T}$ the inverse temperature, so that

$$\begin{aligned} d(\beta p) &= \beta dp + p d\beta \\ &= \frac{1}{k_B} d\mathfrak{s} - \beta d\epsilon - \epsilon d\beta + \sum_{k=1}^c n_k d(\beta \mu_k) + \beta \sum_{k=1}^c \mu_k dn_k + g^a d(\beta u^a) + \beta u^a dg^a \\ &= \frac{1}{k_B} d\mathfrak{s} - \beta (T d\mathfrak{s} + \sum_{k=1}^c \mu_k dn_k + u^a dg^a) - \epsilon d\beta + \sum_{k=1}^c n_k d(\beta \mu_k) + \beta \sum_{k=1}^c \mu_k dn_k + g^a d(\beta u^a) \\ &\quad + \beta u^a dg^a \\ &= -\epsilon d\beta + \sum_{k=1}^c n_k d(\beta \mu_k) + g^a d(\beta u^a). \end{aligned} \quad (\text{B.21})$$

Here βp is the Massieu density. Also

$$dp = -\frac{(\epsilon + p)}{\beta} d\beta + \sum_{k=1}^c \frac{n_k}{\beta} d(\beta \mu_k) + \frac{g^a}{\beta} d(\beta u^a) \quad (\text{B.22})$$

which will be used later.

APPENDIX C. CONSERVATION OF THE TOTAL ORBITAL ANGULAR MOMENTUM

In this appendix, we show that $[\hat{L}^a, \hat{H}] = 0$ with $\hat{\mathbf{L}} \equiv \sum_{i=1}^N \mathbf{r}_i \times \hat{\mathbf{p}}_i$. The general relations

$$[\hat{A}\hat{B}, \hat{C}] = [\hat{A}, \hat{C}]\hat{B} + \hat{A}[\hat{B}, \hat{C}] \quad (\text{C.1})$$

$$[\hat{A}, \hat{B}\hat{C}] = [\hat{A}, \hat{B}]\hat{C} + \hat{B}[\hat{A}, \hat{C}] \quad (\text{C.2})$$

will be used. We have

$$\begin{aligned} [\hat{L}^a, \hat{H}] &= \varepsilon^{abc} \sum_{i=1}^N [r_i^b \hat{p}_i^c, \hat{H}] \\ &= \varepsilon^{abc} \sum_{i=1}^N \sum_{j=1}^N \frac{1}{2m_j} [r_i^b \hat{p}_i^c, \hat{p}_j^2] + \frac{\varepsilon^{abc}}{2} \sum_{i=1}^N \sum_{j=1}^N \sum_{k(\neq j)}^N [r_i^b \hat{p}_i^c, V_{jk}]. \end{aligned} \quad (\text{C.3})$$

The two terms are expected to vanish separately.

First term:

$$\begin{aligned} \varepsilon^{abc} \sum_{i=1}^N \sum_{j=1}^N \frac{1}{2m_j} [\hat{r}_i^b \hat{p}_i^c, \hat{p}_j^2] &= \varepsilon^{abc} \sum_{i=1}^N \frac{1}{2m_i} [r_i^b, \hat{p}_i^2] \hat{p}_i^c \\ &= \varepsilon^{abc} \sum_{i=1}^N \frac{1}{2m_i} \{ [r_i^b, \hat{p}_i^d] \hat{p}_i^d \hat{p}_i^c + \hat{p}_i^d [r_i^b, \hat{p}_i^d] \hat{p}_i^c \} \\ &= \varepsilon^{abc} \sum_{i=1}^N \frac{1}{2m_i} \{ i\hbar \delta^{bd} \hat{p}_i^d \hat{p}_i^c + i\hbar \delta^{bd} \hat{p}_i^d \hat{p}_i^c \} \\ &= i\hbar \varepsilon^{abc} \sum_{i=1}^N \frac{1}{m_i} \hat{p}_i^b \hat{p}_i^c = 0 \end{aligned} \quad (\text{C.4})$$

as the Levi-Civita symbol is antisymmetric while $\hat{p}_i^b \hat{p}_i^c$ is symmetric.

Second term:

$$\begin{aligned} \frac{\varepsilon^{abc}}{2} \sum_{i=1}^N \sum_{j=1}^N \sum_{k(\neq j)}^N [\hat{r}_i^b \hat{p}_i^c, V_{jk}] &= \frac{\varepsilon^{abc}}{2} \sum_{i=1}^N \sum_{j=1}^N \sum_{k(\neq j)}^N r_i^b [\hat{p}_i^c, V_{jk}] \\ &= \frac{i\hbar}{2} \varepsilon^{abc} \sum_{i=1}^N \sum_{k(\neq i)}^N r_i^b F_{ik}^c - \frac{i\hbar}{2} \varepsilon^{abc} \sum_{j=1}^N \sum_{i(\neq j)}^N r_i^b F_{ji}^c = 0. \end{aligned} \quad (\text{C.5})$$

The last equality stands as the forces are central by hypothesis 4 in §2.3, so that $F_{ij}^a = -\frac{dV}{dr}(r_{ij}) \frac{r_{ij}^a}{r_{ij}}$, cf. (D.29). Hence $r_i^b F_{ji}^c$ is symmetric in b and c .

APPENDIX D. LOCAL CONSERVATION EQUATIONS

Relations (2.46), (2.47), (C.1) and (C.2) will be extensively used in what follows. Some of the calculations are heavy but the author wanted to give full details to fully understand the structure of the conservation equations for the several local densities.

Mass density. The Heisenberg equation (2.12) reads

$$i\hbar \partial_t \hat{\rho}(\mathbf{r}) = [\hat{\rho}(\mathbf{r}), \hat{H}]. \quad (\text{D.1})$$

One has,

$$\begin{aligned}
 [\hat{\rho}(\mathbf{r}), \hat{H}] &= \sum_{i=1}^N \frac{1}{2m_i} [\hat{\rho}(\mathbf{r}), \hat{p}_i^a \hat{p}_i^a] + \frac{1}{2} \sum_{i=1}^N \sum_{j(\neq i)}^N [\hat{\rho}(\mathbf{r}), V_{ij}] \\
 &= \frac{1}{2} \sum_{i=1}^N \sum_{k=1}^N \frac{m_k}{m_i} [\delta(\mathbf{r} - \mathbf{r}_k), \hat{p}_i^a \hat{p}_i^a] + \frac{1}{2} \sum_{i=1}^N \sum_{k=1}^N \sum_{j(\neq i)}^N m_k \underbrace{[\delta(\mathbf{r} - \mathbf{r}_k), V_{ij}]}_{=0 \text{ (trivially)}}.
 \end{aligned} \tag{D.2}$$

Developing,

$$\begin{aligned}
 \sum_{i=1}^N \sum_{k=1}^N \frac{m_k}{m_i} [\delta(\mathbf{r} - \mathbf{r}_k), \hat{p}_i^a \hat{p}_i^a] &= \sum_{i=1}^N \sum_{k=1}^N \frac{m_k}{m_i} \{ [\delta(\mathbf{r} - \mathbf{r}_k), \hat{p}_i^a] \hat{p}_i^a + \hat{p}_i^a [\delta(\mathbf{r} - \mathbf{r}_k), \hat{p}_i^a] \} \\
 &= i\hbar \sum_{i=1}^N \{ \partial_i^a \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_i^a + \hat{p}_i^a \partial_i^a \delta(\mathbf{r} - \mathbf{r}_i) \} \\
 &= -i\hbar \partial^a \sum_{i=1}^N \{ \hat{p}_i^a \delta(\mathbf{r} - \mathbf{r}_i) + \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_i^a \}
 \end{aligned} \tag{D.3}$$

where we have make use that $-\partial_i^a \delta(\mathbf{r} - \mathbf{r}_i) = \partial^a \delta(\mathbf{r} - \mathbf{r}_i)$. Hence,

$$[\hat{\rho}(\mathbf{r}), \hat{H}] = -i\hbar \partial^a \hat{g}^a(\mathbf{r}) \tag{D.4}$$

and the conservation equation for the mass local density is

$$\partial_t \hat{\rho} + \partial^a \hat{j}_\rho^a = 0 \tag{D.5}$$

with $\hat{j}_\rho^a \equiv \hat{g}^a$ the mass current density. The $\mathbf{r}$ and t dependence was left.

Particle density. The Heisenberg equation (2.12) reads

$$i\hbar \partial_t \hat{n}_k(\mathbf{r}) = [\hat{n}_k(\mathbf{r}), \hat{H}]. \tag{D.6}$$

Restricting ourselves to the non-trivial commutator, one has

$$\begin{aligned}
 [\hat{n}_k(\mathbf{r}), \hat{H}] &= \sum_{i \in \mathcal{S}_k} \frac{1}{2m_i} [\hat{n}_k(\mathbf{r}), \hat{p}_i^a \hat{p}_i^a] \\
 &= -i\hbar \partial^a \frac{1}{2} \sum_{i \in \mathcal{S}_k} \left\{ \frac{\hat{p}_i^a}{m_i} \delta(\mathbf{r} - \mathbf{r}_i) + \delta(\mathbf{r} - \mathbf{r}_i) \frac{\hat{p}_i^a}{m_i} \right\}
 \end{aligned} \tag{D.7}$$

directly with (D.3). The conservation equation for the particle local density is

$$\partial_t \hat{n}_k + \partial^a \hat{j}_{n_k}^a = 0 \tag{D.8}$$

with

$$\hat{j}_{n_k}^a \equiv \frac{1}{2} \sum_{i \in \mathcal{S}_k} \left\{ \frac{\hat{p}_i^a}{m_i} \delta(\mathbf{r} - \mathbf{r}_i) + \delta(\mathbf{r} - \mathbf{r}_i) \frac{\hat{p}_i^a}{m_i} \right\} \tag{D.9}$$

the particle current density.

Momentum density. The Heisenberg equation (2.12) reads

$$i\hbar \partial_t \hat{g}^a(\mathbf{r}) = [\hat{g}^a(\mathbf{r}), \hat{H}]. \quad (\text{D.10})$$

One has,

$$\begin{aligned} [\hat{g}^a(\mathbf{r}), \hat{H}] = & \sum_{i=1}^N \sum_{k=1}^N \frac{1}{4m_i} [\hat{p}_k^a \delta(\mathbf{r} - \mathbf{r}_k), \hat{p}_i^b \hat{p}_i^b] + \sum_{i=1}^N \sum_{k=1}^N \frac{1}{4m_i} [\delta(\mathbf{r} - \mathbf{r}_k) \hat{p}_k^a, \hat{p}_i^b \hat{p}_i^b] \\ & + \frac{1}{4} \sum_{i=1}^N \sum_{j(\neq i)}^N \sum_{k=1}^N [\hat{p}_k^a \delta(\mathbf{r} - \mathbf{r}_k), V_{ij}] + \frac{1}{4} \sum_{i=1}^N \sum_{j(\neq i)}^N \sum_{k=1}^N [\delta(\mathbf{r} - \mathbf{r}_k) \hat{p}_k^a, V_{ij}]. \end{aligned} \quad (\text{D.11})$$

First and second term:

$$\sum_{i=1}^N \sum_{k=1}^N [\hat{p}_k^a \delta(\mathbf{r} - \mathbf{r}_k), \hat{p}_i^b \hat{p}_i^b] = i\hbar \sum_{i=1}^N \hat{p}_i^a \left\{ \partial_i^b \delta(\hat{\mathbf{r}} - \hat{\mathbf{r}}_i) \hat{p}_i^b + \hat{p}_i^b \partial_i^b \delta(\hat{\mathbf{r}} - \hat{\mathbf{r}}_i) \right\} \quad (\text{D.12})$$

$$\sum_{i=1}^N \sum_{k=1}^N [\delta(\mathbf{r} - \mathbf{r}_k) \hat{p}_k^a, \hat{p}_i^b \hat{p}_i^b] = i\hbar \sum_{i=1}^N \left\{ \partial_i^b \delta(\hat{\mathbf{r}} - \hat{\mathbf{r}}_i) \hat{p}_i^b + \hat{p}_i^b \partial_i^b \delta(\hat{\mathbf{r}} - \hat{\mathbf{r}}_i) \right\} \hat{p}_i^a. \quad (\text{D.13})$$

Third and fourth term:

$$\begin{aligned} \sum_{i=1}^N \sum_{j(\neq i)}^N \sum_{k=1}^N [\hat{p}_k^a \delta(\mathbf{r} - \mathbf{r}_k), V_{ij}] &= i\hbar \sum_{i=1}^N \sum_{j(\neq i)}^N \sum_{k=1}^N F_{ij}^a (\delta_{ik} - \delta_{jk}) \delta(\mathbf{r} - \mathbf{r}_k) \\ &= 2i\hbar \sum_{i=1}^N \sum_{k(\neq i)}^N F_{ki}^a \delta(\mathbf{r} - \mathbf{r}_k) \end{aligned} \quad (\text{D.14})$$

$$\sum_{i=1}^N \sum_{j(\neq i)}^N \sum_{k=1}^N [\delta(\mathbf{r} - \mathbf{r}_k) \hat{p}_k^a, V_{ij}] = 2i\hbar \sum_{i=1}^N \sum_{k(\neq i)}^N \delta(\hat{\mathbf{r}} - \hat{\mathbf{r}}_k) F_{ki}^a. \quad (\text{D.15})$$

Combining all yields,

$$\begin{aligned} [\hat{g}^a(\mathbf{r}), \hat{H}] = & -i\hbar \partial^b \sum_{i=1}^N \frac{1}{4m_i} \left\{ \hat{p}_i^a \delta(\hat{\mathbf{r}} - \hat{\mathbf{r}}_i) \hat{p}_i^b + \hat{p}_i^a \hat{p}_i^b \delta(\hat{\mathbf{r}} - \hat{\mathbf{r}}_i) + \delta(\hat{\mathbf{r}} - \hat{\mathbf{r}}_i) \hat{p}_i^a \hat{p}_i^b + \hat{p}_i^b \delta(\hat{\mathbf{r}} - \hat{\mathbf{r}}_i) \hat{p}_i^a \right\} \\ & + \frac{i\hbar}{2} \sum_{i=1}^N \sum_{j(\neq i)}^N \left\{ F_{ij}^a \delta(\hat{\mathbf{r}} - \hat{\mathbf{r}}_i) + \delta(\hat{\mathbf{r}} - \hat{\mathbf{r}}_i) F_{ij}^a \right\}. \end{aligned} \quad (\text{D.16})$$

The last term can be written as

$$\frac{i\hbar}{2} \sum_{i=1}^N \sum_{j(\neq i)}^N F_{ij}^a \left\{ \delta(\hat{\mathbf{r}} - \hat{\mathbf{r}}_i) - \delta(\hat{\mathbf{r}} - \hat{\mathbf{r}}_j) \right\} \quad (\text{D.17})$$

where the action-reaction principle $F_{ij}^a = -F_{ji}^a$ was used and the labels i and j interchanged. Until the second half of the 1960s, a Taylor expansion of the Dirac delta distribution was used to develop the $\delta(\hat{\mathbf{r}} - \hat{\mathbf{r}}_i) - \delta(\hat{\mathbf{r}} - \hat{\mathbf{r}}_j)$ term. This was in particular used by H. Mori in 1958 [33]. It was later remarked [24, 34] that such approximation was not necessary to use and that an exact formula could be obtained. To do so, let us consider the position $\mathbf{r}_i$ and $\mathbf{r}_j$ of the particles labelled i and j and a smooth curve $\mathbf{R}_{ij}(\lambda)$ such that

$$\begin{aligned} \mathbf{R}_{ij} : [0, 1] &\rightarrow \mathbb{R}^3 \\ \lambda &\mapsto \mathbf{R}_{ij}(\lambda) \end{aligned} \quad (\text{D.18})$$

which satisfies $\mathbf{R}_{ij}(0) = \mathbf{r}_j$ and $\mathbf{R}_{ij}(1) = \mathbf{r}_i$.

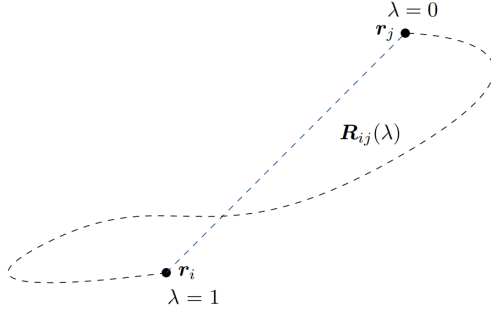


Figure 5.1: Smooth curve/straight line going from position $\mathbf{r}_j$ to $\mathbf{r}_i$.

Then,

$$\begin{aligned} &\delta(\mathbf{r} - \mathbf{r}_i) - \delta(\mathbf{r} - \mathbf{r}_j) \\ &= \delta(\mathbf{r} - \mathbf{R}_{ij}(1)) - \delta(\mathbf{r} - \mathbf{R}_{ij}(0)) \\ &= \int_0^1 d\lambda \frac{d}{d\lambda} \delta(\mathbf{r} - \mathbf{R}_{ij}(\lambda)) \\ &= -\nabla \cdot \int_0^1 d\lambda \frac{d\mathbf{R}_{ij}}{d\lambda} \delta(\mathbf{r} - \mathbf{R}_{ij}(\lambda)). \end{aligned} \quad (\text{D.19})$$

For a straight line between $\mathbf{r}_i$ and $\mathbf{r}_j$, $\mathbf{R}_{ij} = \lambda \mathbf{r}_i + (1 - \lambda) \mathbf{r}_j$ and one has

$$\delta(\mathbf{r} - \mathbf{r}_i) - \delta(\mathbf{r} - \mathbf{r}_j) = -\nabla \cdot \int_0^1 d\lambda \mathbf{r}_{ij} \delta(\mathbf{r} - \mathbf{R}_{ij}(\lambda)) \quad (\text{D.20})$$

with $\mathbf{r}_{ij} \equiv \mathbf{r}_i - \mathbf{r}_j$. We therefore define for convenience the quantity

$$D(\mathbf{r}; \mathbf{r}_i, \mathbf{r}_j) \equiv \int_0^1 d\lambda \frac{d\mathbf{R}_{ij}}{d\lambda} \delta(\mathbf{r} - \mathbf{R}_{ij}(\lambda)). \quad (\text{D.21})$$

Finally,

$$\begin{aligned} [\hat{g}^a(\mathbf{r}), \hat{H}] &= -i\hbar \partial^b \sum_{i=1}^N \frac{1}{4m_i} \{ \hat{p}_i^a \delta(\hat{\mathbf{r}} - \hat{\mathbf{r}}_i) \hat{p}_i^b + \hat{p}_i^a \hat{p}_i^b \delta(\hat{\mathbf{r}} - \hat{\mathbf{r}}_i) + \delta(\hat{\mathbf{r}} - \hat{\mathbf{r}}_i) \hat{p}_i^a \hat{p}_i^b + \hat{p}_i^b \delta(\hat{\mathbf{r}} - \hat{\mathbf{r}}_i) \hat{p}_i^a \} \\ &\quad - \frac{i\hbar}{2} \partial^b \sum_{i=1}^N \sum_{j(\neq i)}^N F_{ij}^a D^b(\mathbf{r}; \mathbf{r}_i, \mathbf{r}_j) \end{aligned} \quad (\text{D.22})$$

so that the conservation equation for the momentum local density reads

$$\partial_t \hat{g}^a + \partial^b \hat{f}_{g^b}^a = 0 \quad (\text{D.23})$$

with

$$\begin{aligned} \hat{f}_{g^b}^a &\equiv \sum_{i=1}^N \frac{1}{4m_i} \{ \hat{p}_i^a \delta(\hat{\mathbf{r}} - \hat{\mathbf{r}}_i) \hat{p}_i^b + \hat{p}_i^a \hat{p}_i^b \delta(\hat{\mathbf{r}} - \hat{\mathbf{r}}_i) + \delta(\hat{\mathbf{r}} - \hat{\mathbf{r}}_i) \hat{p}_i^a \hat{p}_i^b + \hat{p}_i^b \delta(\hat{\mathbf{r}} - \hat{\mathbf{r}}_i) \hat{p}_i^a \} \\ &\quad + \frac{1}{2} \sum_{i=1}^N \sum_{j(\neq i)}^N F_{ij}^a D^b(\mathbf{r}; \mathbf{r}_i, \mathbf{r}_j) \end{aligned} \quad (\text{D.24})$$

the momentum current density. As appearing so far in (D.23), the derivative term cannot be written as the divergence of the tensor $\hat{j}_{g^b}^a$. To solve this problem, we have to show that the momentum current density is a symmetric tensor $\hat{j}_{g^b}^a = \hat{j}_{g^a}^b$. To do this, let us decompose $\hat{j}_{g^b}^a$ as

$$\hat{j}_{g^b}^a = \hat{j}_{g^b}^a \text{ kin.} + \hat{j}_{g^b}^a \text{ pot.} \quad (\text{D.25})$$

with

$$\hat{j}_{g^b}^a \text{ kin.} \equiv \sum_{i=1}^N \frac{1}{4m_i} \left\{ \hat{p}_i^a \delta(\hat{\mathbf{r}} - \hat{\mathbf{r}}_i) \hat{p}_i^b + \hat{p}_i^a \hat{p}_i^b \delta(\hat{\mathbf{r}} - \hat{\mathbf{r}}_i) + \delta(\hat{\mathbf{r}} - \hat{\mathbf{r}}_i) \hat{p}_i^b \hat{p}_i^a + \hat{p}_i^b \delta(\hat{\mathbf{r}} - \hat{\mathbf{r}}_i) \hat{p}_i^a \right\} \quad (\text{D.26})$$

$$\hat{j}_{g^b}^a \text{ pot.} \equiv \frac{1}{2} \sum_{i=1}^N \sum_{j(\neq i)}^N F_{ij}^a D^b(\mathbf{r}; \mathbf{r}_i, \mathbf{r}_j). \quad (\text{D.27})$$

The kinetic part is symmetric without any further justification. If one assumes that $\mathbf{R}_{ij}(\lambda)$ describes a straight line between the points $\mathbf{r}_i$ and $\mathbf{r}_j$ as described above, then

$$\hat{j}_{g^b}^a \text{ pot.} = \frac{1}{2} \sum_{i=1}^N \sum_{j(\neq i)}^N F_{ij}^a r_{ij}^b \int_0^1 d\lambda \delta(\mathbf{r} - \mathbf{R}_{ij}(\lambda)). \quad (\text{D.28})$$

The force exerted on the particle i due to the particle j is given by^b

$$F_{ij}^a = -\frac{\partial V_{ij}}{\partial r_i^a} = -\frac{\partial V(r_{ij})}{\partial r_i^a} = -\frac{dV}{dr}(r_{ij}) \frac{\partial r_{ij}}{\partial r_i^a} \quad (\text{D.29})$$

with $r_{ij} = \|\mathbf{r}_i - \mathbf{r}_j\| = \sqrt{(r_i^b - r_j^b)(r_i^b - r_j^b)}$ so that $\frac{\partial r_{ij}}{\partial r_i^a} = \frac{r_{ij}^a}{r_{ij}}$. Accordingly

$$\hat{j}_{g^b}^a \text{ pot.} = -\frac{1}{2} \sum_{i=1}^N \sum_{j(\neq i)}^N \frac{dV}{dr}(r_{ij}) \frac{r_{ij}^a r_{ij}^b}{r_{ij}} \int_0^1 d\lambda \delta(\mathbf{r} - \mathbf{R}_{ij}(\lambda)) \quad (\text{D.30})$$

which is also clearly symmetric.

Now, the conservation equation for the momentum local density can be written in its final form

$$\partial_t \hat{g}^a + \partial^b \hat{j}_{g^a}^b = 0. \quad (\text{D.31})$$

Energy density. The Heisenberg equation (2.12) reads

$$i\hbar \partial_t \hat{\epsilon}(\mathbf{r}) = [\hat{\epsilon}(\mathbf{r}), \hat{H}]. \quad (\text{D.32})$$

Explicitly, the local energy density reads

$$\hat{\epsilon}(\mathbf{r}) = \sum_{k=1}^N \frac{1}{4m_k} \left\{ \hat{p}_k^a \hat{p}_k^a \delta(\mathbf{r} - \mathbf{r}_k) + \delta(\mathbf{r} - \mathbf{r}_k) \hat{p}_k^a \hat{p}_k^a \right\} + \frac{1}{4} \sum_{k=1}^N \sum_{l(\neq k)}^N \left\{ V_{kl} \delta(\mathbf{r} - \mathbf{r}_k) + \delta(\mathbf{r} - \mathbf{r}_k) V_{kl} \right\}. \quad (\text{D.33})$$

^bStrictly speaking, a different potential V_{kl} should be considered for particles of different species, cf. hypothesis 3 in §2.3. Here, the potential is assumed, for simplicity, to be the same for all particle species. In any case, this does not change the conclusion of the calculation.

One has,

$$\begin{aligned}
 [\hat{e}(\mathbf{r}), \hat{H}] = & \sum_{i=1}^N \sum_{k=1}^N \frac{1}{8m_i m_k} \left\{ [\hat{p}_k^a \hat{p}_k^a \delta(\mathbf{r} - \mathbf{r}_k), \hat{p}_i^b \hat{p}_i^b] + [\delta(\mathbf{r} - \mathbf{r}_k) \hat{p}_k^a \hat{p}_k^a, \hat{p}_i^b \hat{p}_i^b] \right\} \\
 & + \sum_{i=1}^N \sum_{j(\neq i)}^N \sum_{k=1}^N \frac{1}{8m_k} \left\{ [\hat{p}_k^a \hat{p}_k^a \delta(\mathbf{r} - \mathbf{r}_k), V_{ij}] + [\delta(\mathbf{r} - \mathbf{r}_k) \hat{p}_k^a \hat{p}_k^a, V_{ij}] \right\} \\
 & + \sum_{i=1}^N \sum_{k=1}^N \sum_{l(\neq k)}^N \frac{1}{8m_i} \left\{ [V_{kl} \delta(\mathbf{r} - \mathbf{r}_k), \hat{p}_i^b \hat{p}_i^b] + [\delta(\mathbf{r} - \mathbf{r}_k) V_{kl}, \hat{p}_i^b \hat{p}_i^b] \right\} \quad (D.34)
 \end{aligned}$$

where only non-trivial commutators have been written. First and second term:

$$\sum_{i=1}^N \sum_{k=1}^N [\hat{p}_k^a \hat{p}_k^a \delta(\mathbf{r} - \mathbf{r}_k), \hat{p}_i^b \hat{p}_i^b] = i\hbar \sum_{i=1}^N \hat{p}_i^a \hat{p}_i^a \left\{ \partial_i^b \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_i^b + \hat{p}_i^b \partial_i^b \delta(\mathbf{r} - \mathbf{r}_i) \right\} \quad (D.35)$$

$$\sum_{i=1}^N \sum_{k=1}^N [\delta(\mathbf{r} - \mathbf{r}_k) \hat{p}_k^a \hat{p}_k^a, \hat{p}_i^b \hat{p}_i^b] = i\hbar \sum_{i=1}^N \left\{ \partial_i^b \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_i^b + \hat{p}_i^b \partial_i^b \delta(\mathbf{r} - \mathbf{r}_i) \right\} \hat{p}_i^a \hat{p}_i^a \quad (D.36)$$

Third and fourth term:

$$\sum_{i=1}^N \sum_{j(\neq i)}^N \sum_{k=1}^N [\hat{p}_k^a \hat{p}_k^a \delta(\mathbf{r} - \mathbf{r}_k), V_{ij}] = 2i\hbar \sum_{k=1}^N \sum_{l(\neq k)}^N \left\{ F_{kl}^a \hat{p}_k^a + \hat{p}_k^a F_{kl}^a \right\} \delta(\mathbf{r} - \mathbf{r}_k) \quad (D.37)$$

$$\sum_{i=1}^N \sum_{j(\neq i)}^N \sum_{k=1}^N [\delta(\mathbf{r} - \mathbf{r}_k) \hat{p}_k^a \hat{p}_k^a, V_{ij}] = 2i\hbar \sum_{k=1}^N \sum_{l(\neq k)}^N \delta(\mathbf{r} - \mathbf{r}_k) \left\{ F_{kl}^a \hat{p}_k^a + \hat{p}_k^a F_{kl}^a \right\}. \quad (D.38)$$

Fifth and sixth term:

$$\begin{aligned}
 \sum_{i=1}^N \sum_{k=1}^N \sum_{l(\neq k)}^N [V_{kl} \delta(\mathbf{r} - \mathbf{r}_k), \hat{p}_i^b \hat{p}_i^b] = & i\hbar \sum_{k=1}^N \sum_{l(\neq k)}^N \left\{ (F_{kl}^b \hat{p}_l^b - F_{kl}^b \hat{p}_k^b + \hat{p}_l^b F_{kl}^b - \hat{p}_k^b F_{kl}^b) \delta(\mathbf{r} - \mathbf{r}_k) \right. \\
 & \left. - V_{kl} \partial^b (\hat{p}_k^b \delta(\mathbf{r} - \mathbf{r}_k) + \delta(\mathbf{r} - \mathbf{r}_k) \hat{p}_k^b) \right\} \quad (D.39)
 \end{aligned}$$

$$\begin{aligned}
 \sum_{i=1}^N \sum_{k=1}^N \sum_{l(\neq k)}^N [\delta(\mathbf{r} - \mathbf{r}_k) V_{kl}, \hat{p}_i^b \hat{p}_i^b] = & i\hbar \sum_{k=1}^N \sum_{l(\neq k)}^N \left\{ \delta(\mathbf{r} - \mathbf{r}_k) (F_{kl}^b \hat{p}_l^b - F_{kl}^b \hat{p}_k^b + \hat{p}_l^b F_{kl}^b - \hat{p}_k^b F_{kl}^b) \right. \\
 & \left. - \partial^b (\hat{p}_k^b \delta(\mathbf{r} - \mathbf{r}_k) + \delta(\mathbf{r} - \mathbf{r}_k) \hat{p}_k^b) V_{kl} \right\}. \quad (D.40)
 \end{aligned}$$

Combining all yields,

$$\begin{aligned}
 & [\hat{e}(\mathbf{r}), \hat{H}] \\
 = & -i\hbar \partial^b \sum_{i=1}^N \frac{1}{8m_i^2} \left\{ \hat{p}_i^a \hat{p}_i^a (\delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_i^b + \hat{p}_i^b \delta(\mathbf{r} - \mathbf{r}_i)) + (\delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_i^b + \hat{p}_i^b \delta(\mathbf{r} - \mathbf{r}_i)) \hat{p}_i^a \hat{p}_i^a \right\} \\
 & + \frac{i\hbar}{8} \sum_{i=1}^N \sum_{j(\neq i)}^N \left\{ \left(F_{ij}^a \left(\frac{\hat{p}_i^a}{m_i} + \frac{\hat{p}_j^a}{m_j} \right) + \left(\frac{\hat{p}_i^a}{m_i} + \frac{\hat{p}_j^a}{m_j} \right) F_{ij}^a \right) \delta(\mathbf{r} - \mathbf{r}_i) + \delta(\mathbf{r} - \mathbf{r}_i) \left(F_{ij}^a \left(\frac{\hat{p}_i^a}{m_i} + \frac{\hat{p}_j^a}{m_j} \right) + \left(\frac{\hat{p}_i^a}{m_i} + \frac{\hat{p}_j^a}{m_j} \right) F_{ij}^a \right) \right\} \\
 & - i\hbar \partial^a \sum_{i=1}^N \sum_{j(\neq i)}^N \frac{1}{8m_i} \left\{ V_{ij} (\hat{p}_i^a \delta(\mathbf{r} - \mathbf{r}_i) + \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_i^a) + (\hat{p}_i^a \delta(\mathbf{r} - \mathbf{r}_i) + \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_i^a) V_{ij} \right\}. \quad (D.41)
 \end{aligned}$$

Now, the first term on the third line can be doubled and labels i and j permuted in the twin term

$$\begin{aligned}
 & \frac{i\hbar}{8} \sum_{i=1}^N \sum_{j(\neq i)}^N \left(F_{ij}^a \left(\frac{\hat{p}_i^a}{m_i} + \frac{\hat{p}_j^a}{m_j} \right) + \left(\frac{\hat{p}_i^a}{m_i} + \frac{\hat{p}_j^a}{m_j} \right) F_{ij}^a \right) \delta(\mathbf{r} - \mathbf{r}_i) \\
 &= \frac{i\hbar}{16} \sum_{i=1}^N \sum_{j(\neq i)}^N \left(F_{ij}^a \left(\frac{\hat{p}_i^a}{m_i} + \frac{\hat{p}_j^a}{m_j} \right) + \left(\frac{\hat{p}_i^a}{m_i} + \frac{\hat{p}_j^a}{m_j} \right) F_{ij}^a \right) \delta(\mathbf{r} - \mathbf{r}_i) \\
 &\quad - \frac{i\hbar}{16} \sum_{i=1}^N \sum_{j(\neq i)}^N \left(F_{ij}^a \left(\frac{\hat{p}_i^a}{m_i} + \frac{\hat{p}_j^a}{m_j} \right) + \left(\frac{\hat{p}_i^a}{m_i} + \frac{\hat{p}_j^a}{m_j} \right) F_{ij}^a \right) \delta(\mathbf{r} - \mathbf{r}_i) \\
 &= \frac{i\hbar}{16} \sum_{i=1}^N \sum_{j(\neq i)}^N \left(F_{ij}^a \left(\frac{\hat{p}_i^a}{m_i} + \frac{\hat{p}_j^a}{m_j} \right) + \left(\frac{\hat{p}_i^a}{m_i} + \frac{\hat{p}_j^a}{m_j} \right) F_{ij}^a \right) (\delta(\mathbf{r} - \mathbf{r}_i) - \delta(\mathbf{r} - \mathbf{r}_j)) \\
 &= -\frac{i\hbar}{16} \sum_{i=1}^N \sum_{j(\neq i)}^N \left(F_{ij}^a \left(\frac{\hat{p}_i^a}{m_i} + \frac{\hat{p}_j^a}{m_j} \right) + \left(\frac{\hat{p}_i^a}{m_i} + \frac{\hat{p}_j^a}{m_j} \right) F_{ij}^a \right) \partial^b D^b(\mathbf{r}; \mathbf{r}_i, \mathbf{r}_j) \tag{D.42}
 \end{aligned}$$

so that finally,

$$\begin{aligned}
 [\hat{e}(\mathbf{r}), \hat{H}] &= -i\hbar \partial^b \sum_{i=1}^N \frac{1}{8m_i^2} \left\{ \hat{p}_i^a \hat{p}_i^a (\delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_i^b + \hat{p}_i^b \delta(\mathbf{r} - \mathbf{r}_i)) + (\delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_i^b + \hat{p}_i^b \delta(\mathbf{r} - \mathbf{r}_i)) \hat{p}_i^a \hat{p}_i^a \right\} \\
 &\quad - \frac{i\hbar}{16} \sum_{i=1}^N \sum_{j(\neq i)}^N \left(F_{ij}^a \left(\frac{\hat{p}_i^a}{m_i} + \frac{\hat{p}_j^a}{m_j} \right) + \left(\frac{\hat{p}_i^a}{m_i} + \frac{\hat{p}_j^a}{m_j} \right) F_{ij}^a \right) \partial^b D^b(\mathbf{r}; \mathbf{r}_i, \mathbf{r}_j) \\
 &\quad - \frac{i\hbar}{16} \sum_{i=1}^N \sum_{j(\neq i)}^N \partial^b D^b(\mathbf{r}; \mathbf{r}_i, \mathbf{r}_j) \left(F_{ij}^a \left(\frac{\hat{p}_i^a}{m_i} + \frac{\hat{p}_j^a}{m_j} \right) + \left(\frac{\hat{p}_i^a}{m_i} + \frac{\hat{p}_j^a}{m_j} \right) F_{ij}^a \right) \\
 &\quad - i\hbar \partial^a \sum_{i=1}^N \sum_{j(\neq i)}^N \frac{1}{8m_i} \left\{ V_{ij} (\hat{p}_i^a \delta(\mathbf{r} - \mathbf{r}_i) + \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_i^a) + (\hat{p}_i^a \delta(\mathbf{r} - \mathbf{r}_i) + \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_i^a) V_{ij} \right\}. \tag{D.43}
 \end{aligned}$$

The conservation equation for the energy local density reads

$$\partial_t \hat{e} + \partial^a \hat{j}_e^a = 0 \tag{D.44}$$

with

$$\begin{aligned}
 \hat{j}_e^a &\equiv \sum_{i=1}^N \frac{1}{8m_i^2} \left\{ \hat{p}_i^b \hat{p}_i^b (\delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_i^a + \hat{p}_i^a \delta(\mathbf{r} - \mathbf{r}_i)) + (\delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_i^a + \hat{p}_i^a \delta(\mathbf{r} - \mathbf{r}_i)) \hat{p}_i^b \hat{p}_i^b \right\} \\
 &\quad + \frac{1}{16} \sum_{i=1}^N \sum_{j(\neq i)}^N \left(F_{ij}^b \left(\frac{\hat{p}_i^b}{m_i} + \frac{\hat{p}_j^b}{m_j} \right) + \left(\frac{\hat{p}_i^b}{m_i} + \frac{\hat{p}_j^b}{m_j} \right) F_{ij}^b \right) D^a(\mathbf{r}; \mathbf{r}_i, \mathbf{r}_j) \\
 &\quad + \frac{1}{16} \sum_{i=1}^N \sum_{j(\neq i)}^N D^a(\mathbf{r}; \mathbf{r}_i, \mathbf{r}_j) \left(F_{ij}^b \left(\frac{\hat{p}_i^b}{m_i} + \frac{\hat{p}_j^b}{m_j} \right) + \left(\frac{\hat{p}_i^b}{m_i} + \frac{\hat{p}_j^b}{m_j} \right) F_{ij}^b \right) \\
 &\quad + \sum_{i=1}^N \sum_{j(\neq i)}^N \frac{1}{8m_i} \left\{ V_{ij} (\hat{p}_i^a \delta(\mathbf{r} - \mathbf{r}_i) + \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_i^a) + (\hat{p}_i^a \delta(\mathbf{r} - \mathbf{r}_i) + \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_i^a) V_{ij} \right\}. \tag{D.45}
 \end{aligned}$$

The energy current density can be rewritten in a more condensed form using $\hat{e}_i$,

$$\begin{aligned} j_e^a = & \sum_{i=1}^N \frac{1}{4m_i} \left\{ \hat{e}_i \left(\delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_i^a + \hat{p}_i^a \delta(\mathbf{r} - \mathbf{r}_i) \right) + \left(\delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_i^a + \hat{p}_i^a \delta(\mathbf{r} - \mathbf{r}_i) \right) \hat{e}_i \right\} \\ & + \frac{1}{16} \sum_{i=1}^N \sum_{j(\neq i)}^N \left(F_{ij}^b \left(\frac{\hat{p}_i^b}{m_i} + \frac{\hat{p}_j^b}{m_j} \right) + \left(\frac{\hat{p}_i^b}{m_i} + \frac{\hat{p}_j^b}{m_j} \right) F_{ij}^b \right) D^a(\mathbf{r}; \mathbf{r}_i, \mathbf{r}_j) \\ & + \frac{1}{16} \sum_{i=1}^N \sum_{j(\neq i)}^N D^a(\mathbf{r}; \mathbf{r}_i, \mathbf{r}_j) \left(F_{ij}^b \left(\frac{\hat{p}_i^b}{m_i} + \frac{\hat{p}_j^b}{m_j} \right) + \left(\frac{\hat{p}_i^b}{m_i} + \frac{\hat{p}_j^b}{m_j} \right) F_{ij}^b \right). \end{aligned} \quad (\text{D.46})$$

Mass density (on the basis of quantum field theory). As usual, we have that $i\hbar \partial_t \hat{\rho}(\mathbf{r}) = [\hat{\rho}(\mathbf{r}), \hat{H}]$. The commutator $[\hat{\rho}(\mathbf{r}), \hat{H}]$ can be decomposed in two parts,

$$[\hat{\rho}(\mathbf{r}), \hat{H}] = [\hat{\rho}(\mathbf{r}), \hat{H}_0] + [\hat{\rho}(\mathbf{r}), \hat{H}_{\text{int}}] \quad (\text{D.47})$$

with (2.70). One shows without any difficulty that $[\hat{\rho}(\mathbf{r}), \hat{H}_{\text{int}}] = 0$. Now,

$$[\hat{\rho}(\mathbf{r}), \hat{H}_0] = m \int d\mathbf{r}' [\hat{\psi}^\dagger(\mathbf{r}) \hat{\psi}(\mathbf{r}), \hat{\psi}^\dagger(\mathbf{r}') \hat{h}_0 \hat{\psi}(\mathbf{r}')]. \quad (\text{D.48})$$

Using (C.1) and (C.2), we have that

$$\begin{aligned} & [\hat{\psi}^\dagger(\mathbf{r}) \hat{\psi}(\mathbf{r}), \hat{\psi}^\dagger(\mathbf{r}') \hat{h}_0 \hat{\psi}(\mathbf{r}')] \\ &= [\hat{\psi}^\dagger(\mathbf{r}), \hat{\psi}^\dagger(\mathbf{r}') \hat{h}_0 \hat{\psi}(\mathbf{r}')] \hat{\psi}(\mathbf{r}) + \hat{\psi}^\dagger(\mathbf{r}) [\hat{\psi}(\mathbf{r}), \hat{\psi}^\dagger(\mathbf{r}') \hat{h}_0 \hat{\psi}(\mathbf{r}')] \\ &= [\hat{\psi}^\dagger(\mathbf{r}), \hat{\psi}^\dagger(\mathbf{r}') \hat{h}_0 \hat{\psi}(\mathbf{r}')] \hat{\psi}(\mathbf{r}) + \hat{\psi}^\dagger(\mathbf{r}') [\hat{\psi}^\dagger(\mathbf{r}), \hat{h}_0 \hat{\psi}(\mathbf{r}')] \hat{\psi}(\mathbf{r}) \\ &\quad + \hat{\psi}^\dagger(\mathbf{r}) [\hat{\psi}(\mathbf{r}), \hat{\psi}^\dagger(\mathbf{r}') \hat{h}_0 \hat{\psi}(\mathbf{r}')] + \hat{\psi}^\dagger(\mathbf{r}) \hat{\psi}^\dagger(\mathbf{r}') [\hat{\psi}(\mathbf{r}), \hat{h}_0 \hat{\psi}(\mathbf{r}')] \\ &= -\frac{\hbar^2}{2m} \hat{\psi}^\dagger(\mathbf{r}') [\hat{\psi}^\dagger(\mathbf{r}), \nabla'^2 \hat{\psi}(\mathbf{r}')] \hat{\psi}(\mathbf{r}) - \frac{\hbar^2}{2m} \hat{\psi}^\dagger(\mathbf{r}) \nabla'^2 \hat{\psi}(\mathbf{r}') \delta(\mathbf{r} - \mathbf{r}') - \frac{\hbar^2}{2m} \hat{\psi}^\dagger(\mathbf{r}) \hat{\psi}^\dagger(\mathbf{r}') [\hat{\psi}(\mathbf{r}), \nabla'^2 \hat{\psi}(\mathbf{r}')] \\ &= \frac{\hbar^2}{2m} \hat{\psi}^\dagger(\mathbf{r}') (\nabla'^2 \delta(\mathbf{r} - \mathbf{r}')) \hat{\psi}(\mathbf{r}) - \frac{\hbar^2}{2m} \hat{\psi}^\dagger(\mathbf{r}) \nabla'^2 \hat{\psi}(\mathbf{r}') \delta(\mathbf{r} - \mathbf{r}') - \frac{\hbar^2}{2m} \hat{\psi}^\dagger(\mathbf{r}) \hat{\psi}^\dagger(\mathbf{r}') \nabla'^2 [\hat{\psi}(\mathbf{r}), \hat{\psi}(\mathbf{r}')] \\ &= \frac{\hbar^2}{2m} \hat{\psi}^\dagger(\mathbf{r}') (\nabla'^2 \delta(\mathbf{r} - \mathbf{r}')) \hat{\psi}(\mathbf{r}) - \frac{\hbar^2}{2m} \hat{\psi}^\dagger(\mathbf{r}) \nabla'^2 \hat{\psi}(\mathbf{r}') \delta(\mathbf{r} - \mathbf{r}'). \end{aligned} \quad (\text{D.49})$$

Hence,

$$\begin{aligned} & m \int d\mathbf{r}' [\hat{\psi}^\dagger(\mathbf{r}) \hat{\psi}(\mathbf{r}), \hat{\psi}^\dagger(\mathbf{r}') \hat{h}_0 \hat{\psi}(\mathbf{r}')] \\ &= -\frac{\hbar^2}{2} \int d\mathbf{r}' \left\{ -\hat{\psi}^\dagger(\mathbf{r}') (\nabla'^2 \delta(\mathbf{r} - \mathbf{r}')) \hat{\psi}(\mathbf{r}) + \hat{\psi}^\dagger(\mathbf{r}) \nabla'^2 \hat{\psi}(\mathbf{r}') \delta(\mathbf{r} - \mathbf{r}') \right\} \\ &= -\frac{\hbar^2}{2} \int d\mathbf{r}' \left\{ -\nabla'^2 \hat{\psi}^\dagger(\mathbf{r}') \delta(\mathbf{r} - \mathbf{r}') \hat{\psi}(\mathbf{r}) + \hat{\psi}^\dagger(\mathbf{r}) \nabla'^2 \hat{\psi}(\mathbf{r}') \delta(\mathbf{r} - \mathbf{r}') \right\} \\ &= \frac{\hbar^2}{2} \left\{ \nabla^2 \hat{\psi}^\dagger(\mathbf{r}) \hat{\psi}(\mathbf{r}) - \hat{\psi}^\dagger(\mathbf{r}) \nabla^2 \hat{\psi}(\mathbf{r}) \right\} \\ &= \frac{\hbar^2}{2} \nabla \cdot \left\{ \nabla \hat{\psi}^\dagger(\mathbf{r}) \hat{\psi}(\mathbf{r}) - \hat{\psi}^\dagger(\mathbf{r}) \nabla \hat{\psi}(\mathbf{r}) \right\} + \frac{\hbar^2}{2} \left\{ -\nabla \hat{\psi}^\dagger(\mathbf{r}) \cdot \nabla \hat{\psi}(\mathbf{r}) + \nabla \hat{\psi}^\dagger(\mathbf{r}) \cdot \nabla \hat{\psi}(\mathbf{r}) \right\} \\ &= \frac{\hbar^2}{2} \nabla \cdot \left\{ \nabla \hat{\psi}^\dagger(\mathbf{r}) \hat{\psi}(\mathbf{r}) - \hat{\psi}^\dagger(\mathbf{r}) \nabla \hat{\psi}(\mathbf{r}) \right\}. \end{aligned} \quad (\text{D.50})$$

Finally,

$$\begin{aligned}
 \partial_t \hat{\rho}(\mathbf{r}) &= -\frac{i\hbar}{2} \nabla \cdot \{ \nabla \hat{\psi}^\dagger(\mathbf{r}) \hat{\psi}(\mathbf{r}) - \hat{\psi}^\dagger(\mathbf{r}) \nabla \hat{\psi}(\mathbf{r}) \} \\
 &= \frac{i\hbar}{2} \nabla \cdot \{ \hat{\psi}^\dagger(\mathbf{r}) \nabla \hat{\psi}(\mathbf{r}) - \nabla \hat{\psi}^\dagger(\mathbf{r}) \hat{\psi}(\mathbf{r}) \} \\
 &= \frac{1}{2} \nabla \cdot \{ -\hat{\psi}^\dagger(\mathbf{r}) (-i\hbar \nabla) \hat{\psi}(\mathbf{r}) - (i\hbar \nabla) \hat{\psi}^\dagger(\mathbf{r}) \hat{\psi}(\mathbf{r}) \} \\
 &= -\frac{1}{2} \nabla \cdot \{ \hat{\psi}^\dagger(\mathbf{r}) [\hat{\mathbf{p}} \hat{\psi}(\mathbf{r})] + [\hat{\mathbf{p}} \hat{\psi}(\mathbf{r})]^\dagger \hat{\psi}(\mathbf{r}) \}
 \end{aligned} \tag{D.51}$$

so that

$$\partial_t \hat{\rho}(\mathbf{r}) + \nabla \cdot \hat{\mathbf{g}}(\mathbf{r}) = 0 \tag{D.52}$$

as expected.

APPENDIX E. LEGENDRE TRANSFORM

In this appendix, the concept of Legendre transform is quickly reviewed, in particular, functional Legendre transform.

Multidimensional Legendre transform. Let $y = y(x)$ be a one variable smooth function over an interval $I \subset \mathbb{R}$ and $p(x) = \frac{dy}{dx}(x)$ its derivative at point x . The idea of a Legendre transformation is to be able to use p as an independent variable without losing any information on the function y . If and only if $\frac{d^2y}{dx^2} \neq 0$, $p = p(x)$ can be inverted (*i.e.*, $x = x(p)$). Indeed, if $\frac{d^2y}{dx^2} = \frac{dp}{dx} = 0$, p does not depend of x .

The Legendre transform $\psi = \psi(p)$ of the function $y(x)$ is defined by,

$$\begin{aligned}
 \psi(p) &= \left(px - y(x) \right) \Big|_{x=x(p)} \\
 &= px(p) - y(x(p)).
 \end{aligned} \tag{E.1}$$

As an aside remark, we notice that the Legendre transform exists for functions that are at least $C^1(I, \mathbb{R})$.

The idea is similar for functions of several real variables. Let $f = f(\mathbf{x})$ be a function of n real variables over a subset $U \subset \mathbb{R}^n$. Let us define $p_1 = \frac{\partial f}{\partial x_1}(\mathbf{x}), \dots, p_n = \frac{\partial f}{\partial x_n}(\mathbf{x})$. If the determinant of the Hessian matrix $\frac{\partial^2 f}{\partial x_i \partial x_j}$ is not equal to zero, then the Legendre transform of f is defined as

$$\begin{aligned}
 \psi(p_1, p_2, \dots, p_n) &= \left(\sum_{i=1}^n p_i x_i - f(x_1, \dots, x_n) \right) \Big|_{x_i=x_i(\mathbf{p})} \\
 &= \sum_{i=1}^n p_i x_i(\mathbf{p}) - f(x_1(\mathbf{p}), \dots, x_n(\mathbf{p})).
 \end{aligned} \tag{E.2}$$

Functional Legendre transform. These definitions can naturally be extended to the case where the function f depends on a continuous number of variables, *i.e.* for functionals. Let $F = F[f]$

be a functional of f and let us define $g(x) = \frac{\delta F[f]}{\delta f(x)}$ the functional derivative of F . If $\frac{\delta^2 F}{\delta f(x) \delta f(x')} \neq 0$, one can invert the previous relation as $f = f(g(x))$. The Legendre transform $G[g]$ of F is then given by

$$\begin{aligned} G[g] &= \left(\int dx g(x) f(x) - F[f] \right) \Big|_{f=f(g)} \\ &= \int dx g(x) f(g(x)) - F[f(g)]. \end{aligned} \quad (\text{E.3})$$

APPENDIX F. TRANSFORMATION OF THE ENERGY DENSITY & CURRENT DENSITY

In this appendix, the calculations of the section §4.2 are developed in detail.

Transformation of the energy density. With the definition of the energy local density, it suffices to calculate the transformation of the $\hat{p}_i^2$ operator. We have,

$$\begin{aligned} (\hat{p}_i^a)^2 &\rightarrow (\hat{p}_{i0}^a + m_i u_i^a)^2 \\ &= (\hat{p}_{i0}^a)^2 + m_i \hat{p}_{i0}^a u_i^a + m_i u_i^a \hat{p}_{i0}^a + m_i^2 (u_i^a)^2. \end{aligned} \quad (\text{F.1})$$

Hence,

$$\begin{aligned} \hat{e} &\rightarrow \hat{e}_0 + \frac{1}{4} \sum_{i=1}^N \left\{ \hat{p}_{i0}^a u_i^a \delta(\mathbf{r} - \mathbf{r}_i) + \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_{i0}^a u_i^a + u_i^a \hat{p}_{i0}^a \delta(\mathbf{r} - \mathbf{r}_i) + \delta(\mathbf{r} - \mathbf{r}_i) u_i^a \hat{p}_{i0}^a \right\} \\ &\quad + \frac{1}{2} \sum_{i=1}^N m_i (u_i^a)^2 \delta(\mathbf{r} - \mathbf{r}_i) \\ &= \hat{e}_0 + \frac{1}{4} \sum_{i=1}^N \left\{ \hat{p}_{i0}^a u_i^a \delta(\mathbf{r} - \mathbf{r}_i) + \delta(\mathbf{r} - \mathbf{r}_i) u_i^a \hat{p}_{i0}^a - i\hbar \delta(\mathbf{r} - \mathbf{r}_i) \partial_i^a u_i^a + \hat{p}_{i0}^a u_i^a \delta(\mathbf{r} - \mathbf{r}_i) + i\hbar \partial_i^a u_i^a \delta(\mathbf{r} - \mathbf{r}_i) \right. \\ &\quad \left. + \delta(\mathbf{r} - \mathbf{r}_i) u_i^a \hat{p}_{i0}^a \right\} + \frac{1}{2} \sum_{i=1}^N m_i (u_i^a)^2 \delta(\mathbf{r} - \mathbf{r}_i) \\ &= \hat{e}_0 + \hat{g}_0^a u^a + \frac{1}{2} \hat{\rho} u^2 \end{aligned} \quad (\text{F.2})$$

Transformation of the energy current density. For the sake of simplicity, let us take back the expression of the energy current density,

$$\begin{aligned} \hat{j}_e^a &= \sum_{i=1}^N \frac{1}{4m_i} \hat{e}_i \left\{ \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_i^a + \hat{p}_i^a \delta(\mathbf{r} - \mathbf{r}_i) \right\} \\ &\quad + \frac{1}{16} \sum_{i=1}^N \sum_{j(\neq i)}^N \left\{ F_{ij}^b \left(\frac{\hat{p}_i^b}{m_i} + \frac{\hat{p}_j^b}{m_j} \right) + \left(\frac{\hat{p}_i^b}{m_i} + \frac{\hat{p}_j^b}{m_j} \right) F_{ij}^b \right\} D^a(\mathbf{r}; \mathbf{r}_i, \mathbf{r}_j) + \text{h.c.} \end{aligned} \quad (\text{F.3})$$

Let us first look how the first term transforms.

$$\sum_{i=1}^N \frac{1}{4m_i} \hat{e}_i \left\{ \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_i^a + \hat{p}_i^a \delta(\mathbf{r} - \mathbf{r}_i) \right\} + \text{h.c.}$$

$$\begin{aligned}
 & \rightarrow \sum_{i=1}^N \frac{1}{8m_i} \left\{ 2\hat{e}_{i0} + \hat{p}_{i0}^b u_i^b + u_i^b \hat{p}_{i0}^b + m_i (u_i^b)^2 \right\} \left\{ \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_{i0}^a + \hat{p}_{i0}^a \delta(\mathbf{r} - \mathbf{r}_i) + 2m_i \delta(\mathbf{r} - \mathbf{r}_i) u_i^a \right\} + \text{h.c.} \\
 & = \sum_{i=1}^N \frac{1}{4m_i} \hat{e}_{i0} \left\{ \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_{i0}^a + \hat{p}_{i0}^a \delta(\mathbf{r} - \mathbf{r}_i) \right\} + \frac{1}{2} \sum_{i=1}^N \hat{e}_{i0} \delta(\mathbf{r} - \mathbf{r}_i) u_i^a \quad (\text{F.4})
 \end{aligned}$$

$$+ \sum_{i=1}^N \frac{1}{8m_i} \hat{p}_{i0}^b u_i^b \left\{ \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_{i0}^a + \hat{p}_{i0}^a \delta(\mathbf{r} - \mathbf{r}_i) \right\} + \sum_{i=1}^N \frac{1}{8m_i} u_i^b \hat{p}_{i0}^b \left\{ \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_{i0}^a + \hat{p}_{i0}^a \delta(\mathbf{r} - \mathbf{r}_i) \right\} \quad (\text{F.5})$$

$$+ \frac{1}{4} \sum_{i=1}^N \hat{p}_{i0}^b u_i^b \delta(\mathbf{r} - \mathbf{r}_i) u_i^a + \frac{1}{4} \sum_{i=1}^N u_i^b \hat{p}_{i0}^b \delta(\mathbf{r} - \mathbf{r}_i) u_i^a \quad (\text{F.6})$$

$$+ \frac{1}{8} \sum_{i=1}^N (u_i^b)^2 \left\{ \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_{i0}^a + \hat{p}_{i0}^a \delta(\mathbf{r} - \mathbf{r}_i) \right\} + \sum_{i=1}^N \frac{m_i}{4} (u_i^b)^2 \delta(\mathbf{r} - \mathbf{r}_i) u_i^a + \text{h.c.} \quad (\text{F.7})$$

Let us now look how the second term transforms.

$$\begin{aligned}
 & \frac{1}{16} \sum_{i=1}^N \sum_{j(\neq i)}^N \left\{ F_{ij}^b \left(\frac{\hat{p}_i^b}{m_i} + \frac{\hat{p}_j^b}{m_j} \right) + \left(\frac{\hat{p}_i^b}{m_i} + \frac{\hat{p}_j^b}{m_j} \right) F_{ij}^b \right\} D^a(\mathbf{r}; \mathbf{r}_i, \mathbf{r}_j) + \text{h.c.} \\
 & \rightarrow \frac{1}{16} \sum_{i=1}^N \sum_{j(\neq i)}^N \left\{ F_{ij0}^b \left(\frac{\hat{p}_{i0}^b}{m_i} + \frac{\hat{p}_{j0}^b}{m_j} + u_i^b + u_j^b \right) + \left(\frac{\hat{p}_{i0}^b}{m_i} + \frac{\hat{p}_{j0}^b}{m_j} + u_i^b + u_j^b \right) F_{ij0}^b \right\} D^a(\mathbf{r}; \mathbf{r}_i, \mathbf{r}_j) + \text{h.c.} \\
 & = \frac{1}{16} \sum_{i=1}^N \sum_{j(\neq i)}^N \left\{ F_{ij0}^b \left(\frac{\hat{p}_{i0}^b}{m_i} + \frac{\hat{p}_{j0}^b}{m_j} \right) + \left(\frac{\hat{p}_{i0}^b}{m_i} + \frac{\hat{p}_{j0}^b}{m_j} \right) F_{ij0}^b \right\} D^a(\mathbf{r}; \mathbf{r}_i, \mathbf{r}_j) \quad (\text{F.8})
 \end{aligned}$$

$$+ \frac{1}{16} \sum_{i=1}^N \sum_{j(\neq i)}^N \left\{ F_{ij0}^b (u_i^b + u_j^b) + (u_i^b + u_j^b) F_{ij0}^b \right\} D^a(\mathbf{r}; \mathbf{r}_i, \mathbf{r}_j) + \text{h.c.} \quad (\text{F.9})$$

All these terms must be combined together.

→ The first term on line (F.4) and the term on line (F.8), with their respective Hermitian conjugate, combine together to give $\hat{f}_{e_0}^a$.

→ The second term on line (F.4) and its Hermitian conjugate contribute to give

$$\begin{aligned}
 & \frac{1}{2} \sum_{i=1}^N \left\{ \hat{e}_{i0} \delta(\mathbf{r} - \mathbf{r}_i) u_i^a + u_i^a \delta(\mathbf{r} - \mathbf{r}_i) \hat{e}_{i0} \right\} \\
 & = \frac{1}{2} \sum_{i=1}^N \hat{e}_{i0} \delta(\mathbf{r} - \mathbf{r}_i) u_i^a + \frac{1}{2} u_i^a \sum_{i=1}^N \delta(\mathbf{r} - \mathbf{r}_i) \hat{e}_{i0} \\
 & = \frac{1}{2} \sum_{i=1}^N \left\{ \hat{e}_{i0} \delta(\mathbf{r} - \mathbf{r}_i) + \delta(\mathbf{r} - \mathbf{r}_i) \hat{e}_{i0} \right\} u_i^a \\
 & = \hat{e}_0 u^a \quad (\text{F.10})
 \end{aligned}$$

→ The two terms on line (F.5) will contribute to $\hat{f}_{\hat{s}_0^b}^a u^b$,

$$\sum_{i=1}^N \frac{1}{8m_i} \hat{p}_{i0}^b u_i^b \left\{ \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_{i0}^a + \hat{p}_{i0}^a \delta(\mathbf{r} - \mathbf{r}_i) \right\} + \sum_{i=1}^N \frac{1}{8m_i} u_i^b \hat{p}_{i0}^b \left\{ \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_{i0}^a + \hat{p}_{i0}^a \delta(\mathbf{r} - \mathbf{r}_i) \right\} + \text{h.c.}$$

$$\begin{aligned}
 &= \sum_{i=1}^N \frac{1}{4m_i} \left\{ \hat{p}_{i0}^b u_i^b \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_{i0}^a + \hat{p}_{i0}^b u_i^b \hat{p}_{i0}^a \delta(\mathbf{r} - \mathbf{r}_i) \right\} \\
 &\quad \sum_{i=1}^N \frac{1}{8m_i} [u_i^b, \hat{p}_{i0}^b] \left\{ \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_{i0}^a + \hat{p}_{i0}^a \delta(\mathbf{r} - \mathbf{r}_i) \right\} + \text{h.c.} \\
 &= \sum_{i=1}^N \frac{1}{4m_i} \left\{ \hat{p}_{i0}^b u_i^b \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_{i0}^a + \hat{p}_{i0}^b \hat{p}_{i0}^a u_i^b \delta(\mathbf{r} - \mathbf{r}_i) + \hat{p}_{i0}^b [u_i^b, \hat{p}_{i0}^a] \delta(\mathbf{r} - \mathbf{r}_i) \right\} \\
 &\quad + \sum_{i=1}^N \frac{i\hbar}{8m_i} \left\{ \partial_i^b u_i^b \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_{i0}^a + \partial_i^b u_i^b \hat{p}_{i0}^a \delta(\mathbf{r} - \mathbf{r}_i) \right\} + \text{h.c.} \\
 &= \sum_{i=1}^N \frac{1}{4m_i} \left\{ \hat{p}_{i0}^b u_i^b \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_{i0}^a + \hat{p}_{i0}^b \hat{p}_{i0}^a u_i^b \delta(\mathbf{r} - \mathbf{r}_i) + i\hbar \hat{p}_{i0}^b \partial_i^a u_i^b \delta(\mathbf{r} - \mathbf{r}_i) \right\} \\
 &\quad + \sum_{i=1}^N \frac{i\hbar}{8m_i} \left\{ \partial_i^b u_i^b \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_{i0}^a + \hat{p}_{i0}^a \partial_i^b u_i^b \delta(\mathbf{r} - \mathbf{r}_i) + i\hbar \partial_i^a \partial_i^b u_i^b \delta(\mathbf{r} - \mathbf{r}_i) \right\} + \text{h.c.} \\
 &= u^b \sum_{i=1}^N \frac{1}{4m_i} \left\{ \hat{p}_{i0}^b \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_{i0}^a + \hat{p}_{i0}^b \hat{p}_{i0}^a \delta(\mathbf{r} - \mathbf{r}_i) \right\} + i\hbar \partial^a u^b \sum_{i=1}^N \frac{1}{4m_i} \hat{p}_{i0}^b \delta(\mathbf{r} - \mathbf{r}_i) \\
 &\quad + i\hbar \partial^b u^b \sum_{i=1}^N \frac{1}{8m_i} \left\{ \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_{i0}^a + \hat{p}_{i0}^a \delta(\mathbf{r} - \mathbf{r}_i) \right\} - \hbar^2 \partial^a \partial^b u^b \sum_{i=1}^N \frac{1}{8m_i} \delta(\mathbf{r} - \mathbf{r}_i) + \text{h.c.}
 \end{aligned} \tag{F.11}$$

Including the Hermitian conjugate part, the first term is the kinetic term of $\hat{f}_{\hat{g}_0^b}^a u^b$. To recover the full expression of $\hat{f}_{\hat{g}_0^b}^a u^b$, the potential part must be subtracted. The second term can be written as follows,

$$\begin{aligned}
 &i\hbar \partial^a u^b \sum_{i=1}^N \frac{1}{4m_i} \left\{ \hat{p}_{i0}^b \delta(\mathbf{r} - \mathbf{r}_i) - \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_{i0}^b \right\} \\
 &= i\hbar \partial^a u^b \sum_{i=1}^N \frac{1}{4m_i} \left\{ \hat{p}_{i0}^b \delta(\mathbf{r} - \mathbf{r}_i) - \hat{p}_{i0}^b \delta(\mathbf{r} - \mathbf{r}_i) + i\hbar \partial_i^b \delta(\mathbf{r} - \mathbf{r}_i) \right\} \\
 &= -\frac{\hbar^2}{4} \partial^a u^b \partial^b \sum_{i=1}^N \frac{1}{m_i} \delta(\mathbf{r} - \mathbf{r}_i).
 \end{aligned} \tag{F.12}$$

The third term is equal to zero because

$$i\hbar \partial^b u^b \sum_{i=1}^N \frac{1}{8m_i} \left\{ \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_{i0}^a + \hat{p}_{i0}^a \delta(\mathbf{r} - \mathbf{r}_i) \right\} - i\hbar \partial^b u^b \sum_{i=1}^N \frac{1}{8m_i} \left\{ \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_{i0}^a + \hat{p}_{i0}^a \delta(\mathbf{r} - \mathbf{r}_i) \right\} = 0. \tag{F.13}$$

The fourth term reads

$$-\hbar^2 \partial^a \partial^b u^b \sum_{i=1}^N \frac{1}{8m_i} \delta(\mathbf{r} - \mathbf{r}_i) - \hbar^2 \partial^a \partial^b u^b \sum_{i=1}^N \frac{1}{8m_i} \delta(\mathbf{r} - \mathbf{r}_i) = -\frac{\hbar^2}{4} \partial^a \partial^b u^b \sum_{i=1}^N \frac{1}{m_i} \delta(\mathbf{r} - \mathbf{r}_i). \tag{F.14}$$

→ The two terms on line (F.6) contribute to $\hat{g}_0^b u^b u^a$. Indeed,

$$\frac{1}{4} \sum_{i=1}^N \hat{p}_{i0}^b u_i^b \delta(\mathbf{r} - \mathbf{r}_i) u_i^a + \frac{1}{4} \sum_{i=1}^N u_i^b \hat{p}_{i0}^b \delta(\mathbf{r} - \mathbf{r}_i) u_i^a$$

$$= \frac{1}{4} \sum_{i=1}^N \hat{p}_{i0}^b \delta(\mathbf{r} - \mathbf{r}_i) u^a u^b + \frac{1}{4} \sum_{i=1}^N \hat{p}_{i0}^b \delta(\mathbf{r} - \mathbf{r}_i) u^a u^b + \frac{i\hbar}{4} \sum_{i=1}^N \partial_i^b u_i^b \delta(\mathbf{r} - \mathbf{r}_i) u^a. \quad (\text{F.15})$$

If the Hermitian conjugate is included, the last term in $i\hbar$ vanishes and we get

$$\begin{aligned} & \frac{1}{4} \sum_{i=1}^N \hat{p}_{i0}^b u_i^b \delta(\mathbf{r} - \mathbf{r}_i) u_i^a + \frac{1}{4} \sum_{i=1}^N u_i^b \hat{p}_{i0}^b \delta(\mathbf{r} - \mathbf{r}_i) u_i^a + \text{h.c.} \\ & = \hat{g}_0^b u^b u^a. \end{aligned} \quad (\text{F.16})$$

→ The first term on line (F.7) contributes to $\frac{1}{2} \hat{g}_0^a u^2$. To see this, we have that

$$\begin{aligned} & \frac{1}{8} \sum_{i=1}^N (u_i^b)^2 \{ \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_{i0}^a + \hat{p}_{i0}^a \delta(\mathbf{r} - \mathbf{r}_i) \} + \text{h.c.} \\ & = \frac{1}{4} \sum_{i=1}^N \{ \delta(\mathbf{r} - \mathbf{r}_i) \hat{p}_{i0}^a + \hat{p}_{i0}^a \delta(\mathbf{r} - \mathbf{r}_i) \} u^2 + \frac{1}{8} \sum_{i=1}^N [(u_i^b)^2, \hat{p}_{i0}^a] \delta(\mathbf{r} - \mathbf{r}_i) + \frac{1}{8} \sum_{i=1}^N \delta(\mathbf{r} - \mathbf{r}_i) [\hat{p}_{i0}^a, (u_i^b)^2] \\ & = \frac{1}{2} \hat{g}_0^a u^2. \end{aligned} \quad (\text{F.17})$$

→ Finally, the last term on line (F.7) trivially gives the contribution $\frac{1}{2} \hat{\rho} u^2 u^a$.

At the end of the day, combining all the contributions, we get

$$\hat{j}_e^a \rightarrow \hat{j}_{e0}^a + \hat{e}_0 u^a + \hat{j}_{g0}^a u^b + \hat{g}_0^b u^b u^a + \frac{1}{2} (\hat{g}_0^a + \hat{\rho} u^a) u^2 - (\hat{\Omega}^a + \hat{\Omega}'^a) \quad (\text{F.18})$$

with

$$\begin{aligned} \hat{\Omega}^a & \equiv \frac{1}{2} \sum_{i=1}^N \sum_{j(\neq i)}^N F_{ij0}^a D^b(\mathbf{r}; \mathbf{r}_i, \mathbf{r}_j) u^b - \frac{1}{4} \sum_{i=1}^N \sum_{j(\neq i)}^N \{ (u_i^b + u_j^b) F_{ij0}^b \} D^a(\mathbf{r}; \mathbf{r}_i, \mathbf{r}_j) \\ & = \frac{1}{2} \sum_{i=1}^N \sum_{j(\neq i)}^N F_{ij0}^b \left(u^b - \frac{1}{2} (u_i^b + u_j^b) \right) D^a(\mathbf{r}; \mathbf{r}_i, \mathbf{r}_j) \end{aligned} \quad (\text{F.19})$$

where the symmetry of the potential part of the momentum current density was used, *cf.* (D.30) (valid only when the quantity $D^a(\mathbf{r}; \mathbf{r}_i, \mathbf{r}_j)$ is restricted to a straight line between $\mathbf{r}_i$ and $\mathbf{r}_j$). Also,

$$\begin{aligned} \hat{\Omega}'^a & \equiv \frac{\hbar^2}{4} \partial^a u^b \partial^b \sum_{i=1}^N \frac{1}{m_i} \delta(\mathbf{r} - \mathbf{r}_i) + \frac{\hbar^2}{4} \partial^a \partial^b u^b \sum_{i=1}^N \frac{1}{m_i} \delta(\mathbf{r} - \mathbf{r}_i) \\ & = \frac{\hbar^2}{4} \partial^b \{ (\partial^a u^b) \sum_{i=1}^N \frac{1}{m_i} \delta(\mathbf{r} - \mathbf{r}_i) \}. \end{aligned} \quad (\text{F.20})$$

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